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In Search of New Economic Cooperation Models Between Hong Kong and the Greater Bay Area

尋找香港與大灣區的新經濟合作模式

Strategic Public Policy Research Report

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Executive Summary

1. Abstract of the research

Hong Kong and the Pearl River Delta (PRD) have collaborated intensively and effectively since China's economic reform and opening up in the late 1970s. However, the once-successful market-driven economic cooperation model of 'Front Shop, Back Factory' encountered severe challenges in the face of the rapidly rising economic power of PRD and the declining role of Hong Kong as its bridge to the world in the last two decades, and the more recent turbulence in the global political and economic landscapes, such as the China-United States (US) trade war and the Coronavirus 2019 (COVID-19) pandemic.

This Strategic Public Policy Research Study aims to search for new economic cooperation models between Hong Kong and the Greater Bay Area (GBA) and presents policy recommendations based on the research findings. The five key research objectives are as follows: 1) to comprehensively review recent development trends of the PRD and its economic cooperation with Hong Kong; 2) to examine the competitiveness of the PRD with Yangtze River Delta (YRD) in the new economic environment of China and the world; 3) to explore what and where industries need to be promoted in Hong Kong and the PRD to enhance its competitiveness; 4) to examine new economic cooperation models between Hong Kong and the PRD in the new local and global economic environment; and 5) to identify opportunities and barriers of these new economic cooperation models. To achieve these objectives, we first review socioeconomic development, high-tech innovation, and cross-jurisdictional synergy of world-leading bay areas, and thus draw lessons for the development of the Guangdong-Hong Kong -Macao Greater Bay Area. A SWOT analysis of Hong Kong in fostering new economic development of the GBA under the "One Country, Two Systems" framework is then carried out. Of the traditional Four Key Industries, namely financial services, tourism, trading and logistics, and professional and producer services, only financial services and professional services are the industries that Hong Kong still has an edge, in addition to new high-tech industry that is rapidly developing in the PRD. By identifying the strengths, weakness, opportunities and threats of Hong Kong, we examine four potential economic cooperation models: (1) Model 1 - Hong Kong becoming a science and technology centre in the GBA; (2) Model 2 - Hong Kong riding

on increasing affluence of Mainland consumers and investors; (3) Model 3 - Hong Kong leading the GBA to go global and driving new economic growth from enterprise internationalisation; and (4) Model 4 - Continuously utilising Hong Kong to provide producer services for Mainland enterprises, setting up 'front desk' in the GBA and improving access to customer groups.

Model 1 is analysed by examining the high-tech development in Guangdong and the innovation performance and participation of Hong Kong in the high-tech industrial development in Dongguan and Shenzhen. The Annual Survey of Industrial Firms, patent analysis, firm-level questionnaire survey and in-depth interviews in the PRD are used for the analysis. For Model 2, Hong Kong's producer service performance on stock investment, insurance financing and importation is analysed. Hong Kong's overseas investment and international holding capacity and its contribution to the construction of the Belt and Road are examined for Model 3. Hong Kong's equity and bond financing and trade and logistics performance are evaluated for Model 4.

This study began in 15 April 2018, and since then numerous events have affected Hong Kong and the GBA. It has been heavily affected by the social unrest in 2019 and then by COVID-19 in 2020 to the present. We have exerted our best efforts to complete this research project despite these severe obstacles, especially in carrying out fieldworks and questionnaire surveys in the Pearl River Delta. The summary of the major findings of the study are as follows:

1.1 Comparative Study of Bay Areas

Our comparative analysis with other major Bay Areas in the world -- New York, San Francisco and Tokyo -- suggests that the sheer size of the GBA stands out, although its gross domestic product (GDP) per capita still lags behind. Economic, geographical and political transformation in the American and Japanese bay areas all started with the growth of port functions and related manufacturing activities, followed by the dominance of advanced producer services. Eventually, the bay areas transformed into global and national hubs of high-tech industries. This development path appears to be the direction that GBA is aiming for.

Hong Kong has always functioned as an important gateway connecting the Chinese economy and the rest of the world under the "One Country, Two Systems". The greatest

strength of Hong Kong lies in software (i.e. from its world-leading universities, excellent education system, global talents, intellectual property rights protection, abundant venture capital, transparent regulatory system, professional conduct rather than hardware, which is complementary to Hong Kong's partners across the border.

1.2 Model 1 - Hong Kong becoming a science and technology centre in the GBA

The degree of high-tech development considerably varies among the Guangdong cities. Those in the Eastern PRD have a higher percentage of high-tech industrial firms. Such percentage increases as the distance to ports and Hong Kong decreases. The PRD lags behind the YRD in both newly granted and total patents. Moreover, PRD has much fewer patents held by universities as well as patents jointly held by universities and enterprises. Rather, patents in the PRD are more concentrated within and across industries.

Recently, the PRD has departed remarkably from the export-oriented and labour-intensive development model during the early reform period towards an emerging high-tech industrial development model equipped with a regional and almost-complete supply system and mixed target market. Intra-firm organisational linkages between parent firms in Hong Kong and branch offices in the PRD form critical channels for labour mobility, technology transfer and producer service provision. Hong Kong's key roles in facilitating high-tech industrial development in the PRD now mainly relate to market expansion, international operation, corporate financing and producer service provision, whereas its previous advantages in investment and entrepreneurship, technological support and labour supply are declining.

Domestic-funded firms in the PRD mainly use services and technology in the PRD and other parts of Mainland China. They rarely use Hong Kong's services and technology primarily because of Hong Kong's higher cost in comparison to that in Mainland China, insufficient alignment with Mainland China's market demands, limited penetration into the awareness of business communities in the PRD, unawareness of Hong Kong's producer services and technology achievements, and Hong Kong residents' low intension to work in the PRD. Although Hong Kong- and other overseas-funded high-tech firms also mainly use services and technology in the PRD, they still use some Hong Kong services and technology, but not a lot. Reasons for using Hong Kong services include effectiveness to meet service demands and good understanding of international markets and rules. Reasons for using Hong Kong

technology include effectiveness to meet technological demands, alignment with international standards, and competitive advantages compared to technologies in Mainland China. Hong Kong labour forces have been hired mainly by Hong Kong-funded firms ◦ Most domestic-funded firms never hire Hong Kong labour forces. Reasons for hiring Hong Kong labour force include international vision and good understanding of international rules, strong capability to solve problems, and responsible and dedicated working attitude. Reasons for not hiring Hong Kong labour force include high labour cost, low intention to work in Mainland China, and unfamilarity with Mainland China.

1.3 Model 2 - Hong Kong riding on increasing affluence of Mainland consumers and investors

Mainland residents who wish to participate in overseas investments have limited channels. At present, the Shenzhen-Hong Kong Stock Connect and Shanghai-Hong Kong Stock Connect under the Connectivity Mechanism are the main channels. Hong Kong H-shares concentrate on the financial and information technology industries. Mainland A-shares are more dispersed and cover a broader range of industries. Mainland investors concentrate on private enterprises in the Hong Kong stock market, specifically in biotechnology and information technology industries. In turn, Hong Kong and overseas investors primarily invest in the manufacturing industries in A-shares. Thus, the two stock markets have complementary advantages and can bring incremental funds to each other.

Majority of the source of income of the Hong Kong insurance industry comes from Mainland residents. The best sellers are individual life businesses, centred and annualised non-linked insurance. Before 2016, the premium income from the Mainland significantly increased, then later declined sharply due to foreign exchange control. Thus, the business composition changed subsequently. In 2018, the proportion of annualised insurance from the Mainland proliferated, while that of single premium insurance declined. Hong Kong insurers have advantages of high investment returns, high employee and customer quality, low insurance premiums, wide coverage and tax and debt avoidance. Nevertheless, the disadvantages and risks are fluctuating returns, high exchange rate risks, troublesome claim-making and increasingly intensive competition from Mainland insurers.

The growth rate of Hong Kong imports from overseas to the Mainland has slowed down in recent years. These imports are mostly manufactured intermediate products (e.g. electrical machinery, telecommunication and sound recording equipment), that are commonly transported by land to the PRD. Consumption goods imported from Hong Kong to the Mainland hold little shares. These goods include food, luxury products and clothing, from trading partners such as the United States (US), France and Chile. Compared with other ports in the Mainland, Hong Kong's imported product sources and types are more diversified. In addition, Hong Kong has a distinct advantage in merchandising luxury goods.

1.4 Model 3 - Hong Kong leading the GBA to go global and driving new economic growth from enterprise internationalisation

For this model, we compare the M&A events in Hong Kong and the Mainland. In terms of industry distribution, Hong Kong's acquirers are mostly from financial industries (accounting for over 50%) and have low percentages of high-tech firms. By comparison, the Mainland's acquirers are more diverse in industry and have high percentage of high-tech firms. Hong Kong's acquirees have a wider range of industries and less concentrated than the Mainland's acquirees. In terms of geographical location, Hong Kong's acquirees are more internationalised and have a high proportion of targets in the Belt and Road countries and regions. By contrast, most of the Mainland's acquirees are local firms. In addition, the overseas acquirees of Hong Kong are mostly from Southeast Asian countries, while that of the Mainland are mainly from European companies. In terms of efficiency on acquisitions, Hong Kong enterprises have a high degree of completion, showing a higher professionalism and maturity in M&A. By studying subsidiaries of Mainland and Hong Kong enterprises, we find that the latter is much higher than that of the former, showing a high degree of internationalisation. Overseas subsidiaries of Hong Kong enterprises are highly concentrated in the Commonwealth countries and regions, and their location preferences are clear. Mainland enterprises have no apparent regional preferences. Among the acquirers targeting the companies in the Belt and Road, Hong Kong has a larger share than the Mainland.

1.5 Model 4 - Continuously utilising Hong Kong to provide producer services for Mainland enterprises

We analyse the listed companies in Hong Kong and find that more than half are private enterprises in the Mainland. Over the years, listings in the Hong Kong exchange have been dominated by financial and real estate companies while some high-tech companies (e.g. biomedical) are rare. Their share is far behind the New York Stock Exchange and Nasdaq. In terms of valuation, the Hong Kong market has a high level of understanding of the Mainland finance, new energy and consumer goods industries, but not of other industries. In terms of debt financing, the demand for Mainland enterprises is enormous, especially on infrastructure construction projects in the GBA. Bond financing for green projects had a huge demand during the industrial upgrading of the Guangdong-Hong Kong-Macao Greater Bay Area. However, these areas have no uniform green definition.

The import and export shares of the YRD in China's total is on the rise, while those of the PRD is on the decline. Overall, the geographic location of companies exporting through Hong Kong shows an expansion trend from the southeast coast to the inland areas, and the proportion of state-owned enterprises has been declining in recent years. Hong Kong-funded enterprises also gradually moved inland, where the growth of their exports come from.

2. Summary of Policy Implications and Recommendations

Based on the current findings on the opportunities and obstacles facing the four new economic cooperation models, the following policy recommendations for building bridges and removing barriers between Hong Kong and the GBA are proposed.

2.1 Model 1 - Hong Kong becoming a science and technology centre in the GBA

For Model 1, we propose the following policy recommendations : 1) fortify and enhance higher education and innovation capacity; 2) emulate the US Bayh-Dole Act to provide more incentives for university researchers to engage in technology transfer; 3) adjust the assessment criteria of the University Grants Committee for funding universities to prompt more socially value-added research projects; 4) attract, train and cluster world top talents, scholars and entrepreneurs for Hong Kong as an Innovation Centre; 5) promote Hong Kong's high-technology innovations to PRD high-tech firms;

6) further enhance Hong Kong's R&D environment by increasing public expenditure on R&D, giving more resource to the Research Grants Council for strengthening Hong Kong's basic research in order to develop applied research, and improving the mechanism of universities to encourage scholars to commercialize their research results; 7) develop 0-1 and 1-100 start-up industries in Hong Kong and high-tech firms in the PRD; 8) facilitate 'Front Desks, Back Offices' to obtain producer service businesses in the PRD; and 9) implement smart "just-in-time" GBA business and high-tech visa and cross-border counters to facilitate Mainland businessmen to come to Hong Kong to do business and understand our high-tech innovation and Mainland scholars and researchers to carry out collaborative research in developing Hong Kong as a GBA international innovation hub.

2.2 Model 2 - Hong Kong riding on increasing affluence of Mainland consumers and investors

For Model 2, we propose the following policy recommendations : 1) further relax product restrictions and investment quotas in the security market under the Connectivity Mechanism, and broaden existing trading channels that aim to optimise asset allocation of investors from both Hong Kong and the Mainland; 2) attract more high-quality enterprises from the Mainland to list on the Hong Kong Stock Exchange and more talents that are familiar with the Mainland industries; 3) introduce more talents who are familiar with Mainland enterprises and securities markets to increase the investment return on Hong Kong stocks; 4) vigorously develop RMB insurance products and improve the investment capacity of assets; 5) appropriately adjust product types to develop annualised non-linked products (such as permanent life insurance); 6) actively participate in the development of the GBA and deepen the insurance cooperation between Guangdong, Hong Kong and Macao; 7) go beyond serving consumers in the GBA, but also consider the broad market of the entire Mainland, especially the second- and third-tier cities and less developed regions; 8) adopt a more diversified mode of transportation and thereby further develop in markets outside of the GBA; 9) vigorously explore new product types and trading partners, especially expanding on products with high demand in the Mainland; and 10) introduce talents who are familiar with the Mainland consumption market and encourage local talents in Hong Kong to work and be trained in the GBA.

2.3 Model 3 - Hong Kong leading the GBA to go global and driving new economic growth from enterprise internationalisation

For Model 3, we propose the following policy recommendations : 1) adjust industrial policies to encourage a more diversified industrial development pattern; 2) Hong Kong to commit to nurture high-tech industries and enhance the competitiveness of science and technology; 3) Hong Kong to commit to financial technology innovation to ensure the status and advantages of international financial centres; 4) broaden Hong Kong participation in international economic cooperation and competition in the construction of the 'Belt and Road'; and 5) the Hong Kong government needs to encourage companies to expand into a broader overseas market and strengthen their international advantages.

2.4 Model 4 - Continuously utilising Hong Kong to provide producer services for Mainland enterprises

For Model 4, we propose the following policy recommendations : 1) Hong Kong Financial Services Industry must strengthen understanding of the GBA companies, especially private ones, to meet their development status and financing needs: 2) improve the current listing system and encourage high-quality companies from diverse industries to list in Hong Kong; 3) attract talents in all fields; 4) Hong Kong to support the GBA companies in offshore debt financing; 5) Hong Kong needs to communicate and cooperate with the Mainland to clearly define green standards; 6) follow the Hong Kong enterprises to the Mainland and establish more connection poles in the PRD region by using the 'Front Desks, Back Offices' model; and 7) Hong Kong to embrace the demands of Mainland private enterprises.

報告摘要

1. 研究摘要

自 1970 年代末期中國改革開放以來，香港和珠江三角洲（珠三角）的合作日趨緊密和高效。但是，過去依賴市場力量驅動下取得成功的“前店後廠”區域合作模式自 2000 年代以來面臨日益嚴峻的挑戰，難以為繼的態勢盡顯。一方面，珠三角經濟持續快速發展、國際化水準迅速提高，香港作為連接珠三角和世界橋樑的作用正在減弱；另一方面，近年來世界政治經濟格局的不穩定性加劇，中美貿易戰和新冠肺炎疫情的全球大流行很大程度上改變了港珠合作的目標、動力和機制。因此，本研究以尋找新時期香港與粵港澳大灣區之間新經濟合作模式為研究目標，並且在研究發現的基礎上提出政策建議。

本策略性公共政策研究專案有五個主要研究目標：1）全面回顧珠三角近年來的發展趨勢以及與香港的經濟合作；2）分析對比在中國和全球新經濟環境下珠三角和長三角的經濟競爭力；3）探索哪些產業應該在珠三角和香港的什麼地點佈局以提升其競爭力；4）研究新時期香港與珠三角之間的新經濟合作模式；5）探討這些新經濟合作模式的機遇與障礙。為達成以上研究目標，我們首先分析了其他世界級灣區在社會經濟發展、高新科技創新、跨區域協同方面的特徵，從而為粵港澳大灣區發展提供借鑒啟示，之後分析了香港在推動粵港澳大灣區經濟發展中的優勢、劣勢、機遇與挑戰 (SWOT)。香港傳統四大支柱產業 - 金融服務業、旅游業、貿易和物流業，以及專業及生產性服務業，目前只有金融服務業和專業及生產性服務業和在珠三角發展迅速的高新科技工業上仍然保持競爭優勢。在此基礎上，這研究提出了 4 個具有潛力的經濟合作模式。他們分別是：模式一：香港成為粵港澳大灣區的科技創新中心；模式二：香港借力內地日益富裕的消費者和投資者；模式三：香港帶動大灣區走向全球化和推動企業國際化；模式四：以 “前台後室” 的模式持續向內地企業輸出香港的生產性服務業。

基於工業企業年度調查、專利資料和企業問卷調查及深度訪談，我們研究了廣東省的高新科技發展、香港的創新表現以及香港參與到東莞和深圳的高新科技製造業發展以驗證模式一。對香港生產性服務業表現的分析，包括股票投資、保險和進口等，用於驗證模式二。香港的海外投資和國際支撐能力以及對於一帶一路建設的貢獻用於驗證模式三。香港的股權和債券融資表現以及貿易和物流表現用於驗證模式四。

我們的研究始於 2018 年 4 月 15 日，但是期間香港和大灣區發生了一系列重大事件，例如香港 2019 年的社會運動和 2020 年新冠肺炎疫情的全球大流行，嚴重影響了我們的研究進度和安排。在多重阻礙和困境下，我們盡最大努力開展了在珠三角的實地考察、問卷調查和深度訪談，完成了研究項目。 研究的主要發現如下。

1.1 不同灣區的比較研究

對紐約、三藩市和東京灣區的比較研究發現粵港澳大灣區在土地和人口規模上佔據優勢，但是人均 GDP 仍然明顯落後。美國和日本幾大灣區的經濟、地理和政治轉型均始於港口功能以及相關的製造業經濟活動，之後逐漸轉向生產性服務業以及最終成長為國家和全球的高新科技產業中心。這也正是粵港澳大灣區未來發展路徑的方向。

香港一直作為中國經濟連接世界的門戶。香港的優勢主要體現在其“軟環境”，例如世界級大學、出色的教育體系、全球尖端的人才、智慧財產權保護、充足的風險投資、透明的管制體系，專業的操守，而珠三角具備的生產要素和產業體系等“硬環境”恰恰與香港是互補的。

1.2 模式一：香港成為粵港澳大灣區的科技創新中心

廣東省高新科技產業發展水準具有顯著的空間差異，珠三角東部相比西部具有更高比重的高新科技企業。隨著與港口和香港的距離增加，高新科技企業的數量比重逐漸降低。珠三角在新授權專利和總專利數量上都落後於長三角，並且大學以及大學和企業共同申請專利的數量上珠三角也明顯少於長三角。珠三角的專利主要由企業申請。

珠三角近年來已經由改革開放初期的外向型、勞動密集型發展模式逐步轉向高新科技產業發展模式，開始具備了區域完整的供應體系和海內外混合的目標市場。香港總部和珠三角分支機構之間的企業內部聯繫構築了勞動力流動、技術轉移和生產性服務業供給的關鍵管道。香港推動珠三角高新科技產業發展的關鍵作用在於市場拓展、國際化運營、企業融資和生產性服務業供給，以前在投資創業、技術支持和勞動力供給層面的優勢正在削弱。

內資企業主要使用珠三角和內地其他區域的服務和高新科技, 很少使用香港的服務和技術，主要由於香港的成本較高、與內地市場的需求脫節、對內地商業模式的滲透不足，不瞭解香港的生產性服務和高新科技，以及香港居民較低意願到珠三角就業和對內地情況和市場不熟悉等原因。港資和其他外資企業很

多也使用珠三角的服務和高新科技，但仍會使用香港的服務和高新科技，然而比重不高。使用香港服務的原因是能夠有效滿足服務需求和對國際市場與規則較好的理解。使用香港高新科技的原因是能夠有效滿足技術需求、與國際標準對接以及相比內地科技有競爭優勢。僱用香港勞動力的主要是港資企業。大多數參與調查的內資企業從未僱用過香港的勞動力。僱用香港勞動力的原因在於他們的國際視野和對國際規則的理解、有能力解決問題和敬業負責的工作態度。沒有僱用香港勞動力的原因在於高成本、低意願到內地工作以及不熟悉內地情況。

儘管香港在研發上具有優勢，例如擁有國際領先的大學等，但是我們在珠三角高新科技企業的調研中，發現他們沒有利用香港科研的優勢，應用香港科技是有限的。雖然港資和其他外資企業會使用多些香港的科技，但是內資企業則較少使用香港的科技。內地企業不運用香港科技的主要原因包括：相比內地有更高的成本、與內地市場需求之間脫節、缺乏對內地商業模式的滲透，導致內地缺乏對香港科技的瞭解、以及珠三角本地高新科技的快速發展與香港形成競爭。香港科技與珠三角內外資企業需求之間的脫節情況已經出現，極需關注。

1.3 模式二：香港借力內地日益富裕的消費者和投資者

內地居民雖有意參與海外投資，卻缺少相應途徑。深港股票市場交易互聯互通機制與滬港股票市場交易互聯互通機制試點為目前主要的互聯互通機制。香港的 H 股主要包括金融與資訊科技行業，而內地的 A 股則涵蓋更廣泛的行業。內地投資者投資的港股以民企為主，行業集中於生物科技和資訊技術行業，而香港及境外投資者投資的 A 股以製造業為主，相信 A 股與 H 股優勢互補，能相互帶來增量資金。

香港保險業主要的收益來自內地業務額，其中最大的業務額主要為個人人壽業務，特別是年付非投資相連型保險。2016 年前來自內地的業務額增幅較大，隨後由於外匯管制急劇下降，且業務結構隨之改變，2018 年內地業務的年付業務快速增長，而整付業務大幅下跌。香港保險公司具有多方面優勢，如服務全球化、投資收益高、從業人員和客戶素質高、保費低、保額高、保障範圍廣、避稅避債等優勢與機遇，但也存在收益波動和匯率風險大、理賠麻煩及內地同業競爭日益激烈等劣勢和風險。

香港從海外進口到內地的近年來增速放緩且主要通過陸路運輸到珠三角地區，以轉口貿易和工業製成品（電動機械、通訊、錄音及音響設備和儀器等）

為主。消費品占香港幫助內地進口的比重較小，以食品、奢侈品和衣物等為主，交易夥伴有美國、法國、智利等，產品來源和種類較內地更多元化，在生產性消費品貿易上仍具優勢。此外，香港在奢侈品銷售方面具備顯著優勢。

1.4 模式三：香港帶動大灣區走向全球化和推動企業國際化

我們對中國內地與香港企業的收購模式和收購後表現進行了比較。從收購方的行業分佈看，金融業占香港企業一半以上，而高新科技行業占比較低，行業分佈較單一；內地企業收購方的行業則更多樣，高新科技行業占比較高。香港被收購方行業更廣泛且均衡，而內地被收購方的行業則較集中。從被收購方區位地點來看，香港收購方的收購目標分佈在海外國家或地區的比例高，在中國以外的“一帶一路”國家和地區的收購目標比內地多，而內地收購方的收購目標多在內地。香港收購的企業比內地多，大部分在東南亞國家，而內地主要收購歐洲企業。在收購效率上，香港收購方並購完成度高於內地，表現更加專業成熟。通過分析內地和香港企業全球子公司的分佈情況，我們發現香港企業的海外子公司占比遠高於內地企業，國際化程度高。香港企業海外子公司主要分佈在英聯邦國家及成員國的海外屬地，區位偏好明顯，而內地企業的收購時則無明顯區位偏好。香港企業分佈在“一帶一路”國家和地區的子公司占其全球子公司（含內地及香港）的比例高於內地。

1.5 模式四：以"前台後室"的模式持續向內地企業輸出香港的生產性服務業

通過分析在香港上市的公司，我們發現內地企業在港上市數目中民營企業占半數以上。多年來，香港資本市場上市公司多集中于金融、房地產業，而生物醫療、科技公司稀缺，其占比遠落後於紐約交易所及納斯達克。在估值方面，香港市場對內地金融、新能源、消費品行業的瞭解程度較高，但是其他行業仍舊不足。債務融資方面，內地企業需求大，尤其大灣區建設基建項目的需要。在大灣區產業升級過程中，綠色項目的綠色債券融資需求大，但是兩地對於綠色的定義並不一致。

長三角進出口份額呈上升趨勢，而珠三角進出口份額呈下降趨勢。總體來看，通過香港出口的企業位置呈現出從東南沿海向內陸地區擴展的趨勢，且其中國有企業的比例近年來不斷下降。此外，港資企業逐漸向內地轉移，而香港投資企業的出口增長源於內陸。

以“前台後室”的新經濟合作模式在大灣區設立“前台”以更有效地接近內地客戶，從而在香港“後室”的辦公室，用香港優質專業，有國際聯繫的生產性服務業為在大灣區“前台”接單後的內地客戶提供服務，以“前台後室”的新經濟合作模式為大灣區的廠商提供生產性服務，在大灣區設立“前台”以便更效地接近內地客戶。

2. 研究項目對政策建議的摘要

基於研究的主要發現，以下提出香港未來與大灣區的四種經濟合作模式的機遇與障礙的政策建議。

2.1 模式一：香港成為粵港澳大灣區的科技創新中心

就模式一而言，我們建議：1）加強高等教育的創新能力；2）參考美國的《貝多法案》（Bayh-Dole Act），讓發明者完全獲得大學和政府資助發明的所有智慧財產權，提高大學研究人員從事技術轉移的動力；3）大學教育資助委員會（University Grants Committee）調整給大學劃撥經費的考核評價標準，鼓勵更多從事對社會有影響力的研究；4）吸引和培養頂尖人才，使香港作為世界科研人才、創新創業人才的集聚地；5）讓珠三角的高新科技企業瞭解香港的創新科技；6）進一步加強香港研究與開發環境，增加研發經費的公共支出水平，投入多些資源給香港研究資助局，以加強香港的基礎研究來帶動應用研究，和優化香港的大學鼓勵學者把研究成果商品化；7）發展香港從 0 到 1 及從 1 到 100 的初創工業及鼓勵推動港商在珠三角設廠和分公司，加強與香港創新高新科技和生產性服務業的聯系；8）促進 “前台後室” 的模式，以取得珠三角和國內其他地區之生產性服務業務；9）利用高新科技提供網上即時商務簽證、在出入境口岸設立《大灣區商貿科技專用櫃檯和通道》，以方便和加強大灣區內地廠商到香港去認識和深入瞭解香港高新科技的成果和促進內地學者和科研人員到香港做合作研究，促進粵港澳國際創新科技中心的建設。

2.2 模式二：香港借力內地日益富裕的消費者和投資者

就模式二而言，我們建議：1）進一步放寬 “互聯互通” 機制中股票交易的產品和投資門檻限制，拓寬現有交易管道，優化兩地投資者的資產配置；2）吸引內地優質企業赴港上市以多元化港股的行業，引進更多熟悉內地市場的人才；3）建議引進更多熟悉內地企業和證券市場的人才，提高港股的投資收益率；

4）大力發展人民幣保險產品，加強人民幣資產投資能力；5）適當調整產品種類，發展年付非投資相連類保險產品（如永久人壽保險）；6）積極參與大灣區發展，深化粵港澳保險業合作；7）在大灣區以外，香港也應考慮整個內地市場，尤其是二、三線城市未開拓的地區；8）採取更多元化的貿易運輸方式，進一步開發大灣區以外的市場；9）積極探索新的進口產品種類與交易夥伴，尤其是內地需求旺盛的產品；10）引進熟悉內地消費市場的人才，鼓勵香港本地的人才走出去，在大灣區學習就業積累經驗。

2.3 模式三：香港帶動大灣區走向全球化和推動企業國際化

就模式三而言，我們建議：1）調整工業政策，鼓勵更多元化的發展模式；2）堅定培育高新科技產業，提升科技競爭力；3）發展金融科技創新，維持國際金融中心的地位與優勢；4）在 “一帶一路” 建設中提升參與國際經濟合作與競爭能力；5）鼓勵企業擴寬海外市場的投資，加強其國際優勢。

2.4 模式四：持續向內地企業輸出香港的生產性服務業，在大灣區設立“前台”以更效地接近內地客戶

就模式四而言，我們建議：1）香港金融服務業要加強對大灣區企業，特別是民營企業的瞭解，滿足其發展狀況及融資的需求；2）完善上市制度，鼓勵多元化行業的優質企業來港上市；3）廣泛地吸吶各行各業的相關人才；4）香港支援大灣區企業進行離岸債務融資；5）香港需要和內地溝通合作，統一清晰界定綠色標準；6）以 “前台後室” 的新經濟合作模式，在大灣區設立 “前台” 以更效地接近內地客戶和建立更多商業網絡，從而使在香港 “後室” 的辦公室，用優質專業具國際網路的生產性服務，為在大灣區 “前台” 接來的生意服務; 7）香港需要更多地瞭解並迎合內企業的需求。

Chapter 1
From 'World Factory' to Greater Bay Area: *In Search of New Economic Cooperation Models Between Hong Kong and the Greater Bay Area*

1.1 Introduction

City-regions are increasingly seen as key drivers of the global economy (Scott, 2001; Hall and Pain, 2006; Harrison and Hoyler, 2015). For example, as suggested by Florida et al. (2008), the 40 largest city-regional economies produce approximately two-thirds of the global economic output with only 18% of the world's population. In addition to debates regarding their exact nature and agglomeration dynamics (Burger and Meijers, 2016), city-regions attract extensive attention from policy circles (Storper, 2013; Harrison and Hoyler, 2015).

As the 'pioneer' in globalization, marketization, and urbanization of China, the Pearl River Delta (PRD) has developed into one of the most important and dynamic mega city-regions in the country and in East Asia. Characterised by an 'exo-urbanisation' (Sit and Yang, 1997), the development of the city-regional economy of the PRD highlights the importance of its economic synergy with Hong Kong. By relocating its manufacturing activities to the adjacent PRD to make use of low-cost land and labour, Hong Kong has developed the 'Front Shop, Back Factory' (FSBF) regional collaboration model with the PRD in the 1980s and 1990s. With such a win-win labour partnership, the PRD has become the famous 'world factory' while Hong Kong has concentrated on office development to support the factories in the PRD and transformed the city's economy into a service-based one. The FSBF model that blends the comparative advantages of the two sides of the border has generated great economic benefits at the city-regional scale, leading to joint economic prosperity of Hong Kong and the PRD. However, since the mid-2000s, the distinctive regional economic cooperation model has shown a weakening effect and has started facing steep global and local challenges. Hong Kong's leading role as the regional service centre has gradually diminished with the rise of producer service in Mainland China and the shrinking markets in Europe and North America. The export-oriented PRD has encountered issues such as rising production costs and breakneck competition from other Chinese mega city-regions, most notably the Yangtze River Delta (YRD). Furthermore, as China emerges as the World's second largest economy and transforms

from a receiver of foreign direct investment (FDI) to an emerging major investor, the PRD may no longer seem to be spearheading China's development. In the changing context of global-local dynamics, the traditional comparative advantages between Hong Kong and the PRD have eroded, and their previous 'front shop' and 'back factory' functions are both undergoing significant transformation.

Given the strategic importance of the Hong Kong-PRD in the national economy, economic cooperation and synergy between the two places increasingly attracts policy attention. Recently, the importance of this regional cooperation is re-emphasised by China's central government in its drive to develop the Guangdong-Hong Kong-Macao Greater Bay Area (GBA) (Figure 1.1). The growth trajectory of Hong Kong over the past decades indicates that the future of its economy hinges upon the competitiveness of, and its integration with, the PRD under "One Country, Two Systems". The GBA scheme offers considerable opportunities for Hong Kong to promote its economy that has suffered from the declining effect of the FSBF model. However, while efforts have been taken to gain insights on the transformation of the FSBF model (Yeh and Xu, 2006; Meyer et al., 2012; Schiller et al., 2015), the role(s) that Hong Kong can perform in the new regional context remains unclear. As such, examining the evolving economic relationship between Hong Kong and the PRD is necessary to capture new processes underlying the ongoing regional restructuring and the changing dynamics of the global production networks (GPNs). On this basis, we search for new cooperation models that can benefit the sustainable development of the mega city-regional economy of the GBA.

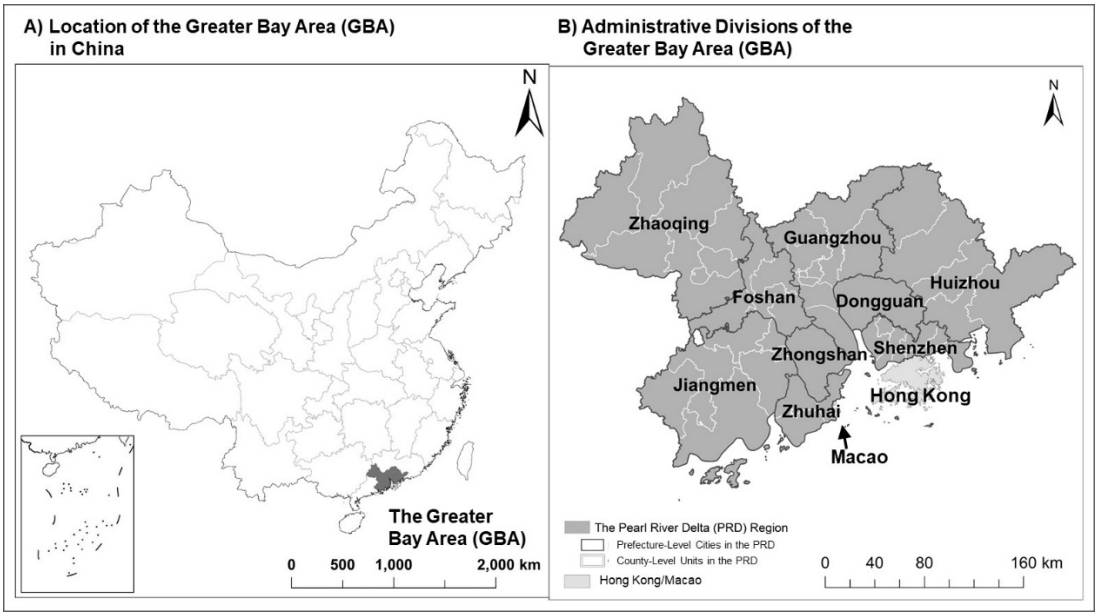


Figure 1.1 Location of the Guangdong-Hong Kong-Macao Greater Bay Area

Note: The Guangdong-Hong Kong-Macao Greater Bay Area (GBA comprises nine cities of the PRD plus the two Special Administrative Regions of Hong Kong and Macao (9+2).

1.2 Development of the PRD and its Economic Cooperation with Hong Kong

1.2.1 Development of the PRD and the Weakening of the FSBF Model since the Mid-2000s

By virtue of its geographical and ethnical proximity to Hong Kong, the PRD became the most attractive destination for investment from the other side of the border soon after China's opening and reform in 1978. With a distinctive entry mode, namely, processing and compensation trade (*sanlai yibu or* "three-plus-one" trading mix; Lin, 2006), Hong Kong and the PRD have made full use of their respective comparative advantages and have formed the well-known FSBF development model in the 1980s and 1990s. By relocating its manufacturing plants and operations to the 'back factory' of the PRD, Hong Kong concentrates on the 'upper end of industrial economy' (Sit and Yang, 1997, p. 656; Cheng and Zheng, 2001), functioning as the 'front shop' and transforming its economy towards a service-based one. By 2000, the share of employment in the manufacturing sector drastically declined to approximately 10% while that in the service sector accounted for nearly 80% (Census and Statistics Department, HKSAR Government,2001). Producer services that facilitate the global organisation of production have become the pillar sector of the Hong Kong economy, while Hong Kong's industries maintained its competitiveness by relocating

manufacturing functions to the lower-cost PRD. In turn, the PRD achieved integration with the GPNs via Hong Kong's market connections and institutional advantages. Such 'an outstanding case of the globalization process in the world' (Shen, 2003, p. 14; Shen, 2008) has triggered the emergence of the mega-city region of the PRD, leading to its own and Hong Kong's drastic economic and urban growth.

However, the development of the export-oriented industrialisation in the PRD has encountered serious difficulties amidst the evolving global and domestic business environments since the mid-2000s (Yang, 2012). As a consequence, the close relationship between Hong Kong and the PRD under the FSBF model has come under challenge (Yeh and Xu, 2006). On the basis of analysis on cross-border socio-economic interactions since the 1990s, Yang (2006) argues that the FSBF model is fading away since the late 1990s and 2000s. Liao and Chan (2011) observe that several Hong Kong manufacturing firms in the PRD have been forced to move outside the PRD since the 2000s. In recognition of the weakening complementary relationship between Hong Kong's producer services and the PRD's manufacturing, Hong Kong signed the Closer Economic Partnership Arrangement (CEPA) with the Mainland in 2003 to strengthen the cross-border interactions by further removing institutional obstacles. CEPA is expected to have greater impacts on the service sector and 'contribute more to the modernization of the country under CEPA' (Yang, 2004, p. 109). However, an evaluation of this effect on the service trade from Hong Kong to the Mainland indicates that CEPA-induced service trade is not significant except for traditional service industries such as tourism and logistics (Yan, 2016).

Hong Kong and the PRD are confronted by a changing global and regional business circumstance that is significantly transforming their economic collaboration (Yeh and Xu, 2006). Firstly, the successful integration of the PRD into the GPNs via Hong Kong is highly related to its low costs and abundant supply of labour. However, the catching-up of the inland regions and the central subsidies for peasants have reduced the attraction of the eastern coastal region for migrant workers (Chan and Buckingham, 2008; Liao and Chan, 2011). The lack of a large amount of low-cost labour is eroding the comparative advantages of the PRD in labour-intensive industries (Liao and Chan, 2011). On the other hand, the wage rates in the PRD has increased rapidly as the economy develops. The rising labour costs as well as decreasing availability of labour has been the most important factor leading to the recent relocation of Hong Kong

manufacturing firms (Liao and Chan, 2011). Secondly, the shrinking overseas market after the 2008 global financial crisis has delivered a devastating blow to the export-oriented industrialisation in the region. The survey conducted by the Federation of Hong Kong Industries (FHKI) (2007) indicated a high reliance of Hong Kong manufacturing firms in the PRD on external markets: of the surveyed firms, 52.4% were engaged in exports and 21.5% were involved in both exports and domestic sales. According to the information released by Guangdong Small and Medium Enterprise Bureau, suffering from the unprecedented financial crisis, 7,148 enterprises in the province closed down in 2008: 1,464 in Dongguan, 956 in Zhongshan, 709 in Zhuhai, 704 in Shenzhen and 526 in Foshan. With the worldwide economic recession and the shrinking demand, the external market can no longer perform a role as important as it used to in the regional development. Thirdly, with China's continuous rise in the global economic map, the geography of consumption has changed and the integration with the GPNs no longer has the only dynamics of regional development of the PRD. Next to the United States, China has become the second largest goods importer since 2009 (CSSB, 2010). A large domestic market may enable domestic forms to go 'national and integrating with the national economy' (Wei, 2009). As Wei (2011) finds out, the huge domestic market of China is critical for upgrading and restructuring the clothing industry in Wenzhou. Tapping into the domestic market has also attributed to the recent redistribution of foreign investment in China (Yang, 2009). Fourth, the dramatic expansion of producer services in the PRD, as a salient outcome of economic restructuring of the region, has reduced the need to utilise producer services from Hong Kong. The relocation of manufacturing from Hong Kong led to rapid industrialisation in the PRD in the 1980s and 1990s (Figure 1.2). Over the past two decades, however, service sector has noticeably expanded, accounting for 57.1% of the total GDP by 2019. In addition to the shift from the agricultural and manufacturing sectors to services, restructuring and upgrading has also occurred within the service sector in the PRD. Producer services in the PRD has shown rapid growth, with the industry employment increasing dramatically from 3.77 to 6.97 million persons over a 10-year period from 2003 to 2013 for work units in urban areas and towns (GDSB, 2004, 2014). The catching-up of the PRD in producer service development, as an alternative source of producer services for local manufacturing, has undermined the importance of Hong Kong in providing producer services and weakened their industrial cross-border linkage (Yeh, 2005). Last but not the least, policy transformation from the central and local

governments has dramatically changed the flexible environment for export-oriented processing industries. At the early development of the PRD, to increase the possibility of inserting the GPNs, urban entrepreneurialism was characterized by "local initiatives, aggressive market deregulation, and race-to-the-bottom territorial competition" (Chien and Wu, 2011, p.22). Regional policy emphasis and governance form may evolve with economic growth (Chien and Wu, 2011). Meanwhile, with market reorientation from external to domestic sales, policies are also reformulated to favour domestic firms over foreign investors (Liao and Chan, 2011). The tightening regional institutional setting has further posed challenges to export-oriented and FDI-induced development.

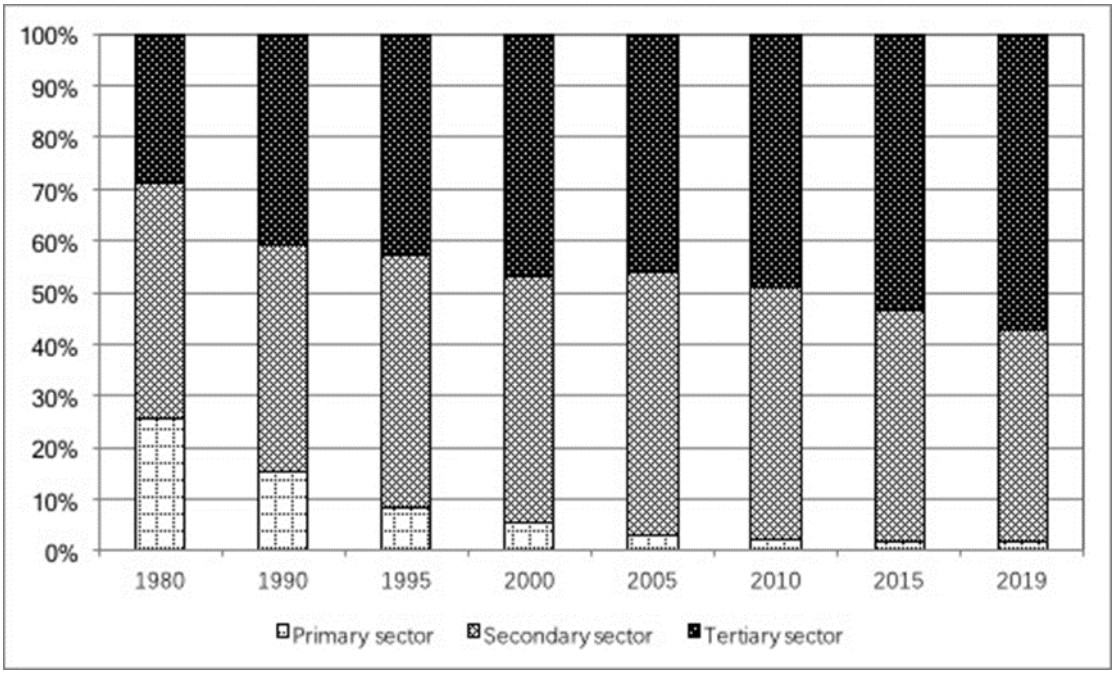


Figure 1.2 GDP of the PRD, 1980-2019

Source: GDSB, 2020

The present PRD highly differs from before. In the last 40 years since the economic reform of China in 1978, PRD has evolved into a super mega city region (Yeh and Chen, 2020) through different waves of urban development and economic transition (Yeh et al., 2015a). In addition to the development of high-tech industries, PRD has also established its own regional producer service linkages (Yeh et al., 2015b).

As Humphrey and Schmitz (2002) note, when a region increases its level of competencies and capabilities, opportunities to develop high-end sectors and perform new functions arise. Under the FSBF model, export-oriented manufacturing is the

dominant activity driving the regional development of the PRD. Since the 2000s, as a result of regional restructuring, high-end activities such as advanced manufacturing and producer services have grown rapidly. The sectoral and functional upgrading in the PRD, together with the market reorientation, has dramatically reduced the cross-border interactions. The effect of the FSBF cooperation model cannot be maintained and the importance of Hong Kong in the development of the city-regional economy appears to decrease in the changing era of globalisation. However, Hong Kong is still competitive over the PRD in terms of innovation capability and producer service development. In addition, Hong Kong enjoys unique institutional advantages under the 'One country, Two systems' framework. Although the traditional manufacturing-services linkages between the PRD and Hong Kong is weakening due to the technical development in the PRD and the expanding domestic market, the new regional context has also opened room for Hong Kong to perform new functions and build up new cross-border linkages and connections.

1.2.2 Guangdong-Hong Kong-Macao Greater Bay Area Initiatives

In 2014, the Chinese State Council proposed the National Plan on New Urbanisation (2014-2020) to achieve a 'new-type' of urbanisation and identify city-regions as the 'cornerstone' and 'major form' (Liu et al., 2016). The central government enthusiastically supports the integrated development between Hong Kong and the PRD as one of the leading mega-city regions in China. *The Outline of the Plan for the Reform and Development of the PRD (2008-2020)* formulated by the National Development and Reform Committee in December 2008 emphasized the collaboration in producer services between Hong Kong and the PRD. In the *12th Five-Year Plan (2011-2015)*, further collaboration between Hong Kong and the Mainland, especially between Hong Kong and Guangdong under the CEPA framework, is emphasised. In March 2017, Premier Li Keqiang promulgated the Guangdong-Hong Kong-Macao Greater Bay Area development in the 12th National Congress's Report. Later, in February 2019, the State Council issued *The Outline Development Plan for Guangdong-Hong Kong-Macao Greater Bay Area* ('*GBA Outline Development Plan 2019*' hereafter), which has placed the importance of Hong Kong-PRD economic cooperation to new heights. The *GBA Outline Development Plan 2019* proposed to make use of Hong Kong's advantageous producer services and research to facilitate the regional upgrading of the GBA. Such

development is not new to Hong Kong. As early as 2001, Mr. Tung Chee-hwa, the then Chief Executive of the HKSAR Government, has already attempted to promote economic cooperation between Hong Kong and the PRD in his Policy Address in 2001. The Greater Pearl River Delta Business Council was set up in March 2004 with prominent businessmen in Hong Kong to provide advice to the government on economic cooperation between Hong Kong and the PRD. In January 2003, the Pearl River Delta Panel was set up in the Central Policy Unit, which was later renamed to Pan-PRD Panel with the development of Pan-PRD regional cooperation by Mr. Zhang Dejiang, the then Guangdong Party Secretary, in July 2003. However, apart from the many studies and reports, such as the *Study on the Action Plan for Liveable Bay Area of the Pearl River Estuary* in 2010 and *Development Plan for City Clusters in the Guangdong-Hong Kong-Macao Bay Area* in 2017, economic cooperation between Hong Kong and the PRD has not considerable advanced except for the Individual Free Visitor Scheme and CEPA in 2003.

The *GBA Outline Development Plan 2019* is one of the latest policies attempts to foster economic synergies between Hong Kong and the PRD to capture economic agglomeration at the mega city-regional scale. The aim is to leverage the advantages of Hong Kong, Macao and Guangdong in developing the mega city-region into 'a vibrant world-class city cluster', 'a globally influential international innovation and technology hub', 'an important support pillar for the Belt and Road Initiative', 'a showcase for in-depth cooperation between the Mainland and Hong Kong and Macao' and 'a quality living circle for living, working and travelling' (see also 2022 Foundation, 2019). These goals can be achieved if the GBA improves regional innovation capability, develops a highly connected regional market and coordinates various production resources and factors. Substantial policy support to enhance such regional cooperation in innovation and facilitate effective and efficient cross-border flows of goods, capital, labour, information and technology is evident in *GBA Outline Development Plan 2019*. Major approaches as stipulated in the *GBA Outline Development Plan 2019* include Hong Kong's integration into the national innovation system; improved transport connectivity between Hong Kong and other cities in the region; Hong Kong's role as an international financial centre and a platform for the Belt and Road Initiative; greater convenience for Hong Kong residents to live and work in Mainland cities; and the development of new districts (e.g., Qianhai, Nansha, and Hengqin) to foster new modes

of synergies and coordination. In the newly issued plans for the Qianhai Shenzhen-Hong Kong Modern Service Industry Cooperation Zone on 6 September 2021, the total area involved is increased from 15 to over 120 square km, an 8-fold area expansion. The GBA initiatives aim to further remove the institutional barriers for Hong Kong to access the expanding Mainland market and enhance its strength as a free and open economy. This enables Hong Kong to reposition itself in the regional economy in the changing era of globalisation. However, while the industry system of the GBA has 'strong economic complementarity', the policy and regional plan to make use of Hong Kong's 'highly developed service sectors' and technological advantages to promote regional integration and agglomeration economy are less effective without a systematic understanding of the cross-border sectoral and spatial linkages. Is the decline of the FSBF model accompanied with newly emerging divisions of labour between Hong Kong and the PRD? In what way and to what extent can Hong Kong participate in the regional economy based on its current and potential competitiveness? The answers to these questions can help Hong Kong to capture the opportunities of the *GBA Outline Development Plan 2019* to develop its economy that is confronted with serious challenges in the new Chinese and global economic environment.

1.2.3 Recent Geographical-Economic Changes

While the *GBA Outline Development Plan 2019* initiatives have laid a solid institutional foundation for economic cooperation between Hong Kong and the PRD, the social unrest in Hong Kong in 2019 followed by the worldwide outbreak of the Coronavirus 2019 (COVID-19) since 2020 have brought about an unprecedented shock to not only the economy of Hong Kong but also its cross-border interaction. Figure 1.2 and Figure 1.3 show that the Hong Kong economy exhibited six consecutive quarters of negative economic growth in 2019 and 2020. The number of visitors from the Mainland of China, which has steadily increased since the mid-2000s, decreased significantly in 2019 and sharply in 2020 (Figure 1.4). The number of visitors from Mainland China in 2020 only reached about 2.7 million, merely 5.3% of that in 2018.

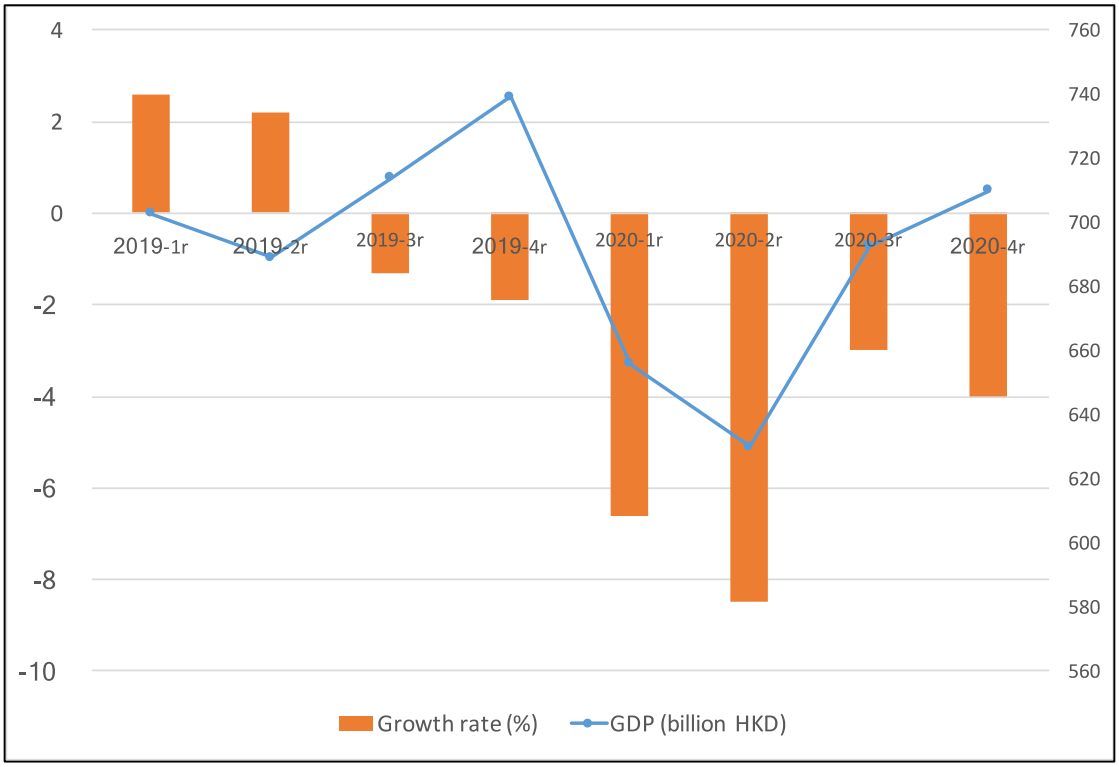


Figure 1.3 Hong Kong GDP, 2019-2020

Source: Census and Statistics Department, HKSAR Government, 2021.

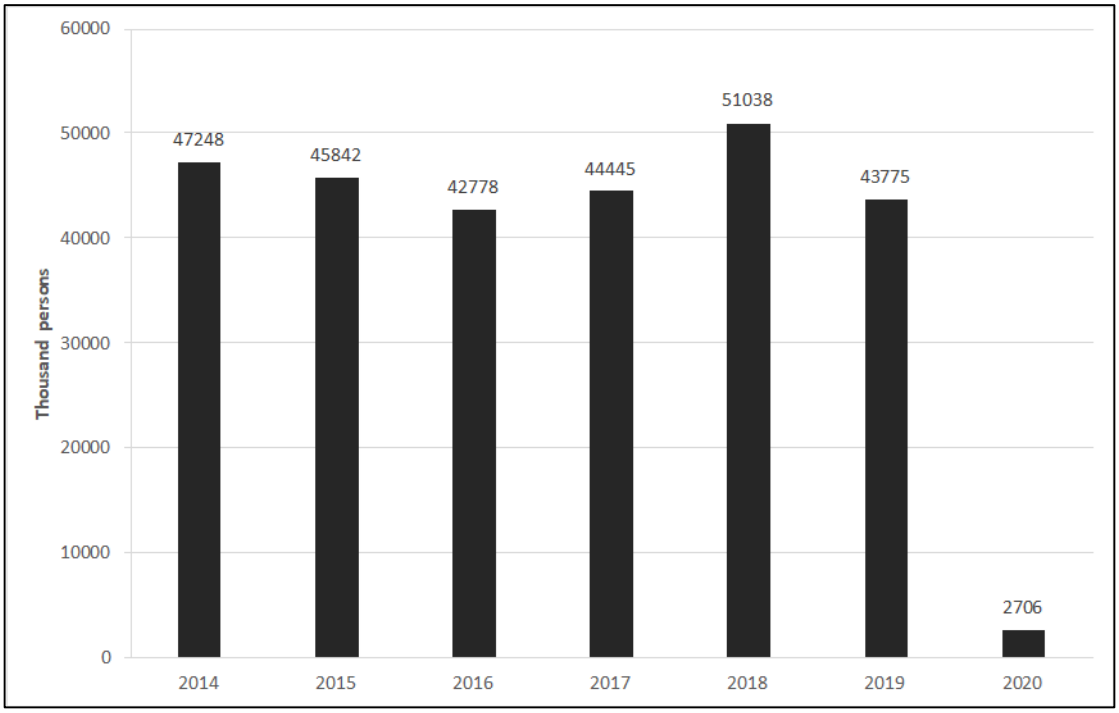


Figure 1.4 Number of visitors from Mainland China to Hong Kong, 2014-2020

Source: Census and Statistics Department, HKSAR Government, 2021.

By contrast, in facing the severe challenge of the epidemic, the Mainland economy has shown strong resilience. Despite a decline of 6.8% in the first quarter of 2020, the GDP soon showed a positive growth, flipping into 3.2% in the second quarter, 4.9% in the third quarter and 6.5% in the fourth quarter (CSSB, 2021). The steady and rapid rebound brought the GDP growth to 2.3%, enabling China to be the only major economy with positive economic growth in 2020. With a huge GDP that exceeds 100 trillion RMB and accounts for approximately 17% of the world's total, and by virtue of its capability to quickly control the epidemic spread, the Mainland has become a crucial engine of the global economy in the post-epidemic era. Thus, Hong Kong faces a pressing need to accelerate the pace of cooperation with the PRD to maintain its competitiveness in an increasingly turbulent global economic circumstance.

1.3 In Search of New Economic Cooperation Models Between Hong Kong and the GBA

1.3.1 Research Objectives

An important dimension of public policies is to shape the sectoral and territorial structure of the city-regional economies, such as where and what kind of industries to develop (Kemeney and Storper, 2015). This study attempts to identify new economic connections and labour relationships between Hong Kong and the PRD to operationalise the regional development plans for the GBA. The purposes of the study are twofold. Firstly and theoretically, the transforming regional division of labour between Hong Kong and the PRD is examined to contribute to the literature on the relationship between regional development and economic globalisation. For practical applications, new economic cooperation models between Hong Kong and the PRD are sought and opportunities and barriers for economic development and cooperation are identified to benefit policy formulation in fostering new cross-border synergies.

These two broad research objectives are further subdivided into the following research objectives :

1) To comprehensively review recent development trends of the Pearl River Delta and its economic cooperation with Hong Kong

2) To examine the competiveness of the PRD with Yangtze River Delta in the new economic environment of China and the world

3) To explore what and where industries need to be promoted in the PRD and Hong Kong to enhance its competiveness

4) To develop a new economic cooperation model between Hong Kong and the PRD in the new global economic environment

5) To identify opportunities and barriers of this new economic cooperation model

1.3.2 Research Questions

The following research questions are investigated to achieve the above research objectives and help the formulation of government policies in the PRD and Hong Kong to cope with the new changing spatial economies of China and the world and to capture the benefits of the GBA plan:

- Is the PRD competitive in the new spatial economy in China and the world?
- What lessons can be learnt from other Bay Areas and regions?
- What will be the new economic development model of the PRD?
- Can Hong Kong help the PRD to further develop, and what will be the new model of economic cooperation between Hong Kong and the GBA?
- What are the opportunities, barriers and facilitating policies and infrastructure that needs to be developed to foster such new economic development in the PRD and its economic cooperation model with Hong Kong?
- How can the cooperation, coordination and planning between Hong Kong and the PRD be further facilitated, especially with the growing Central Business Districts in the GBA, such as Zhujiang New District in Guangzhou and Qianhai in Shenzhen?

1.3.3 Research Methodologies

Both first-hand and secondary data sources are utilized to carry out qualitative as well as quantitative studies to achieve the above-mentioned research objectives. Fieldworks including on-site observation, individual- and firm-level questionnaire

surveys, and semi-structured in-depth interviews have been carried out in the GBA during 2018-2021. A field visit lasting for 5 days in the Tokyo Bay Area was done in July 2019. Extensive secondary datasets such as China's Annual Survey of Industrial Firms, China's patent data, China's firm-level custom data, datasets from the Insurance Authority of Hong Kong, HKEX data, and Orbis company database are collected and analysed using both quantitative and qualitative methods.

1.4 Summary

The comparative advantages of production systems may evolve over time (Jovanović, 2001). *Jovanović* (2001) reports that the comparative advantage and geography of production are dynamic concepts that evolve over time. With a unique institutional setting, the comparative advantages between Hong Kong and the PRD have led to a distinctive FSBF model in the 1980s and 1990s. However, since the mid-2000s, the changing global and local business environments has been diminishing the comparative advantage of the PRD in lower costs of labour and land, and that of Hong Kong in market connections with Europe and North America. The externally-driven mode of regional development has therefore experienced profound restructuring (Yeh and Xu, 2006; Yang, 2020). While the economic relationship under the FSBF model is eroding, economic cooperation between Hong Kong and the PRD is not yet obsolete because the new development and upgrading of the latter and the rise of the Mainland market extend the scope of collaboration based on the evolving cross-border complementarity.

The *GBA Outline Development Plan 2019* has provided substantial policy support to remove barriers in fusing economic complementarity of cities in the region and promotes integrated development at the mega city-regional scale. This finding is aligned with the *13th (2016-2020)* and *14th (2021-2025) National 5-Year Plan* with great emphasis on the development of 19 urban agglomerations for promoting economic development. The *Guangdong-Hong Kong-Macao Greater Bay Area* is one of the three key urban agglomerations, including Jing-Jin-Ji (Beijing-Tianjin-Hebei) and Yangtze River Delta, in the 14th National 5-Year Plan. The GBA has been elevated to the level of key national development.

Although its overseas market connection has decreased in importance, Hong Kong

still enjoys a free and open economy under the ‘One Country, Two Systems’ that can help industries in the PRD go beyond China and attract international firms for business. Its strength in scientific R&D and industrial innovation as well as in connecting with global innovation resources is of essential importance to the PRD that is urgently needed to shift its mode of growth from factor-driven to innovation-driven. Based on a careful examination, this study attempts to search for a new cooperation model that is suitable to the newly emerging and potential economic and labour relationship between Hong Kong and the PRD. The new roles that Hong Kong can play in the GBA are examined, and the difficulties and challenges in coordinating regional development are identified. The model of regional development is not static and fixed but evolves along with the changing global-local dynamics. The reconsideration of the FSBF model and research on the new synergies between Hong Kong and the PRD in the context of the *GBA Outline Development Plan 2019* can provide important policy reference to develop an internationally competitive first-class mega city-regional economy that has significant implications for economic development of Hong Kong, the GBA and China.

Chapter 2
Benchmarking the Greater Bay Area

2.1 Introduction

This chapter benchmarks the Greater Bay Area (GBA) against similar bays of New York, San Francisco and Tokyo. The bay areas of New York, San Francisco and Tokyo are chosen for the reason that they have (re)invented their economic bases (Sassen, 2011; Storper et al., 2015), a challenge that is currently faced by the GBA. The bay areas of New York, San Francisco and Tokyo are often considered as 'role models' for the GBA in policy narratives and media (e.g., Zhou, 2017; HKTDC, 2018). This chapter examines policy lessons that can be learnt from these world-leading bay areas, specifically regarding economic restructuring, city-regional coordination and high-tech innovation.

2.2 Research Methodology

Key dimensions of city-regional competitiveness are compared, including population and economic development, industrial division of labour, high-tech innovation and port and airport functions (e.g. Ni et al., 2021). The spatial boundaries of individual bay areas are defined based on a synthesis of related studies (Figure 2.1; e.g., The Pearl River Delta Spatial Planning Research Centre, 2018; HKTDC Research and Bay Area Council Economic Institute, 2021). Statistics of individual bay areas are consolidated from relevant official figures (e.g., population and land area), policy reports (e.g., port throughput), and rankings (e.g., list of top companies and universities). The identified patterns are subsequently discussed with the academic and policy literature in examining lessons that can be learnt from the development of the bay areas of New York, San Francisco and Tokyo.

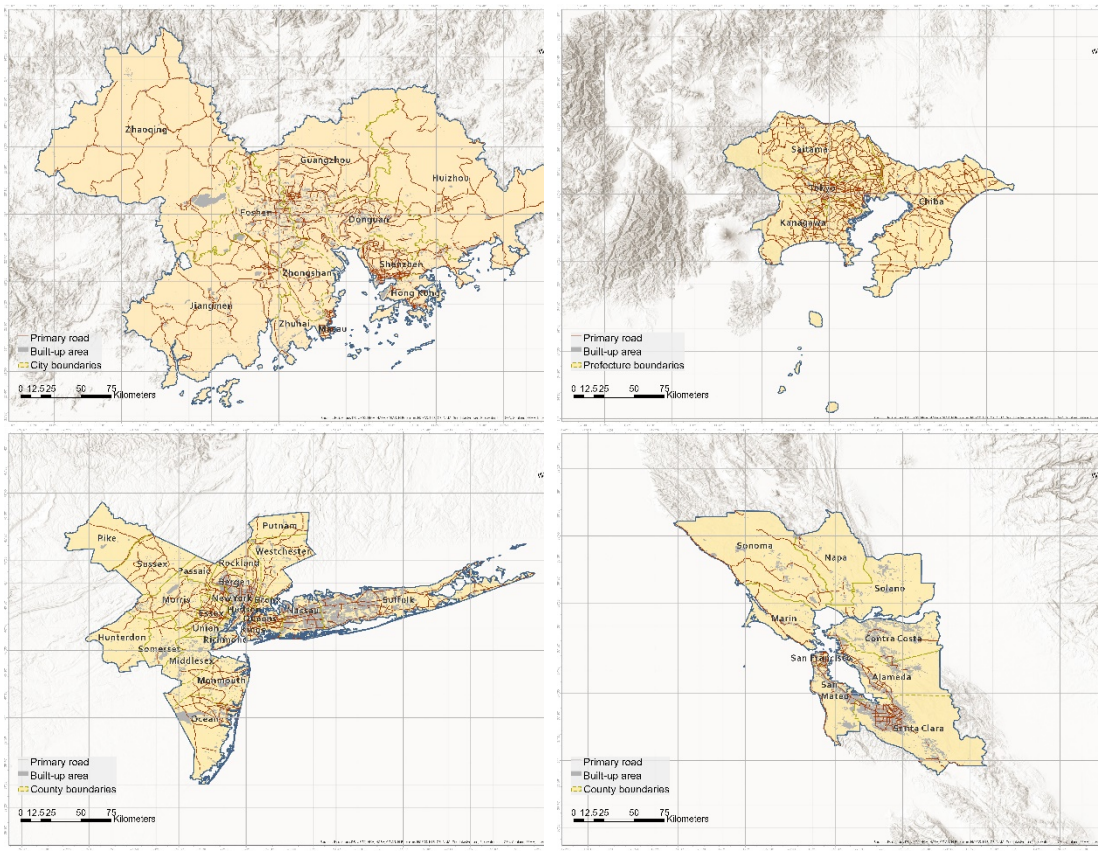


Figure 2.1 Spatial coverage of selected bay areas

Source: Esri, Airbus DS, USGS, NGA, NASA, CGIAR, N Robinson, NCEAS, NLS, OS, NMA, Geodatastyrelsen, Rijkswaterstaat, GSA, Geoland, FEMA, Intermap and the GIS user community

Note: Top left Greater Bay Area ;Top right Tokyo Bay Area; Bottom left New York Bay Area; Bottom right San Francisco Bay Area. The GBA covers Hong Kong, Macao and nine cities from Guangdong Province; Tokyo Bay Area in this study includes the Tokyo Metropolis and the prefectures of Chiba, Kanagawa, and Saitama; New York Bay Area is defined as New York-Newark-Jersey City, NY-NJ-PA Metropolitan Statistical Area; San Francisco Bay Area covers the nine-county region as defined by the Association of Bay Area Governments. These definitions are synthesized from slightly different definitions of regions used in various studies (e.g., The Pearl River Delta Spatial Planning Research Centre, 2018; HKTDC Research and Bay Area Council Economic Institute, 2021).

2.3 Benchmarking the GBA

2.3.1 Basic Population and Economic Conditions

The sheer size of the GBA stands out among the selected bay areas, although its GDP per capita still lags behind. By comparison with the benchmarks, the GBA has the largest land area, largest population and second-largest GDP (Table 2.1). However, the GDP per capita of the GBA in 2019 still falls behind those of other three bay areas.

Specifically, the GDP per capita of the GBA was approximately United States (US) $23,100 in 2019, accounting for a third of that in New York Bay Area and a fifth of that in San Francisco Bay Area (Table 2.1).

Table 2.1 Land areas, population and GDP in the four bay areas (2019)

	Area (km2)	**Population (1,000)**	**Population Density (persons per km^2)**	**GDP (billion US$)**	**GDP per capita (1,000 US$)**	**GDP to National GDP (%)**
New York Bay Area	17,435	20,147	1,155.5	1,573	78.1	8.2
San Francisco Bay Area	17,905	7,749	432.8	902	116.4	4.7
Tokyo Bay Area	13,376	36,728	2,745.8	1,694	46.1	34.2
Greater Bay Area	56,209	72,656	1,292.6	1,679	23.1	11.70

Notes: GDP and GDP per capita in the Tokyo Bay Area are from 2018.

Sources: US Census Bureau; Bureau of Economic Analysis, US Department of Commerce; Japan Statistic Year Book; Official websites of Tokyo Metropolis and the prefectures of Chiba, Kanagawa, and Saitama; Guangdong Statistical Yearbook; China Statistical Yearbook; Hong Kong in Figures; Macao Yearbook of Statistics

The GBA has large-scale businesses in multiple sectors. In the 2018 edition of *Fortune 500*, the GBA has the headquarters of nineteen large-scale companies (Table 2.2). Furthermore, these large-scale companies are distributed in various sectors such as finance, real estate and transportation. However, *Fortune 500* company headquarters in the other bay areas are often more specialised. For instance, *Fortune 500* companies the New York Bay Area concentrate mainly on the financial sector, while those in San Francisco Bay Area specialise in computers and information technology (Table 2.2).

Table 2.2 Fortune 500 companies headquartered in selected bay areas (2018)

	Number of Fortune Global 500 Company	**Key Sectors (Number of Companies)**
New York Bay Area	22	Financial and Insurance (12)
San Francisco Bay Area	9	Computer and Semiconductors (3); Information Technology (2)
Tokyo Bay Area	36	Financial and Insurance (10); Computer and Semiconductors (7); Transportation (6); Manufacturing (5)
Greater Bay Area	19	Financial and Insurance (3); Real Estate (3); Transportation (3)

Source: Fortune Global 500; Original sectors in the Fortune 500 are aggregated by the research team

The GBA is globally connected, most notably through international trade and investment and with Hong Kong as a key gateway (HSBC Group, 2016; see also Yu, 2021). In 2019, Hong Kong served as the largest source of realised foreign direct investments for Mainland China, with a cumulative value of US $1,195.5 billion and accounting for approximately 55% of Mainland China's total FDI volume (Trade and Industry Department, HKSAR Government, 2021). In 2020 Hong Kong functions as Mainland China's second largest export market, accounting for 10.5% (US $273.0 billion) of Mainland China's total exports (Trade and Industry Department, HKSAR Government, 2021).

The finance sector is increasing in significance in the GBA. According to the Global Financial Centres Index (GFCI) Ranking, the number of centres in the GBA increased from one (Hong Kong) in 2015 to three (Hong Kong, Shenzhen and Guangzhou) in 2018 (Figure 2.2).

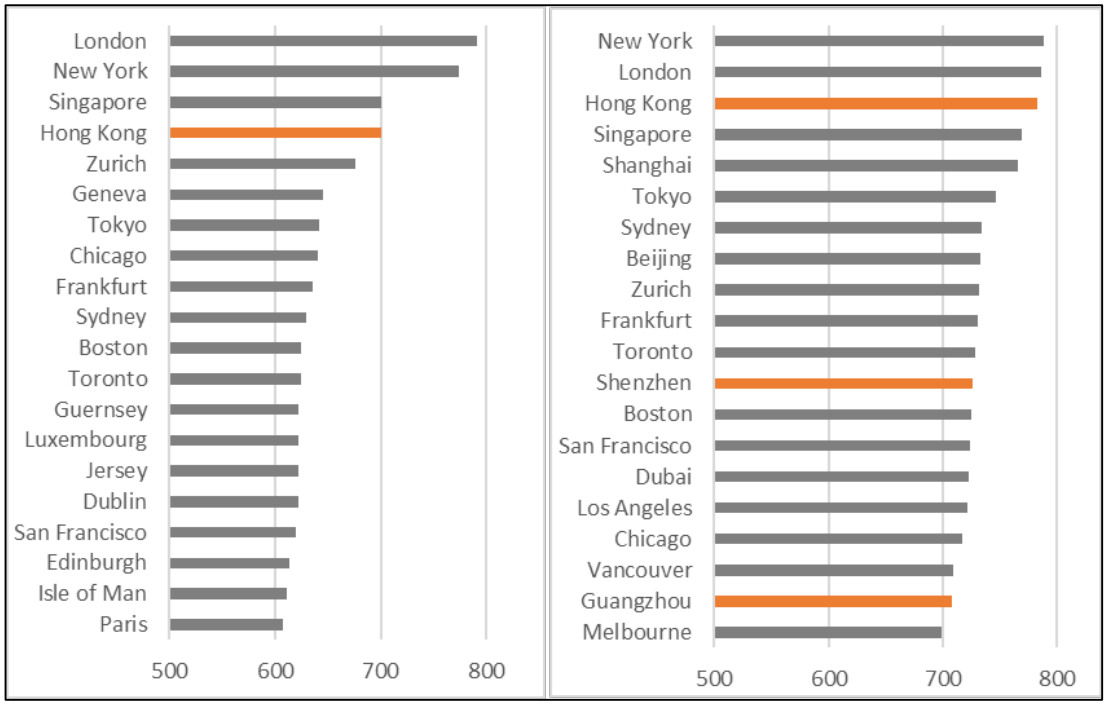


Figure 2.2 Top 20 GFCI Ranking of Global Financial Centre in 2008 (left) and 2018 (right)

Source: Global Financial Centres Index 24 - Long Finance, 2018 (https://www.longfinance.net/programmes/financial-centre-futures/global-financial-centres-index/)

However, the financial sector in the GBA may lack 'quality' and is not as 'global' as the GFCI ranking suggests. In terms of shares of foreign listed companies (Table 2.3), Hong Kong and Shenzhen stock exchanges lag behind stock exchanges in New York (New York Stock Exchange or NYSE and NASDAQ) and London. Specifically, although more than 20% of the firms listed on NYSE are foreign-listed companies, the same indicator is approximately 7% for Hong Kong.

Table 2.3 Market value and foreign listed companies of leading stock exchanges in 2018

Rank	Stock exchange	City	Market capitalisation (US$ billions)	Value of share trading (US$ billions)	Number of listed companies	Number of foreign listed companies	Percentage of foreign listed companies
1	New York Stock Exchange	New York	20,679.48	22,940.79	2,285	510	22.32%
2	Nasdaq-US	New York	9,756.84	43,656.06	3,058	436	14.26%
3	Japan Exchange Group Inc.	Tokyo	5,296.81	7,207.44	3,657	5	0.14%
4	Shanghai Stock Exchange	Shanghai	3,919.42	6,140.33	1,450	0	0.00%
5	Hong Kong Exchanges	Hong Kong	3,819.22	2,483.00	2,315	154	6.65%
6	Euronext	Amsterdam	3,730.40	2,263.20	1,208	149	12.33%
7	LSE Group	London	3,638.00	4,767.73	2,479	418	16.86%
8	Shenzhen Stock Exchange	Shenzhen	2,405.46	7,600.36	2,134	0	0.00%
9	BSE India Limited	Bombay	2,083.48	121.17	5,066	1	0.02%
10	National Stock Exchange of India Limited	Bombay	2,056.34	1,165.83	1,923	1	0.05%

Sources: Pan (2020) and World Federation of Exchanges Statistics

2.3.2 Industrial Division of Labour and Economic Output

Manufacturing and other secondary sectors still dominate in the GBA. Figure 2.3 illustrates that over 45% of employees in this area engaged in secondary industry in 2019. This proportion is higher than those of the bay areas of New York (13.7%), San Francisco (18.1%) and Tokyo (17.9%).

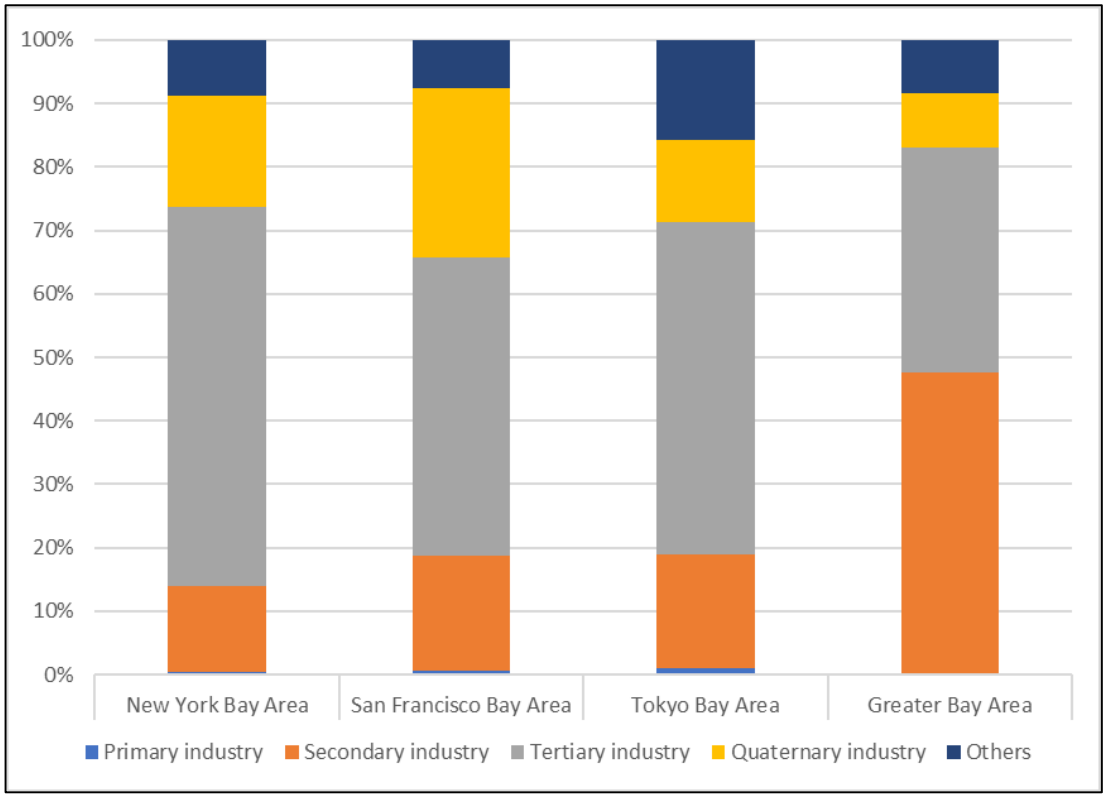


Figure 2.3 Employment by industrial sectors in the four bay areas (2019)

Sources: US Census Bureau Data 2020; e-Stat portal site of official statistics of Japan; Guangdong statistical Yearbook 2020; Macao Yearbook of Statistics 2020; Hong Kong Census and Statistics Department 2020

Note: Data of Tokyo Bay Area were for 2017

A synthesis of statistical yearbook of individual cities reveals that the manufacturing sector in the GBA contributed approximately a third of the total GDP, followed by wholesale and retail trade (13%) and finance and insurance (11%) in 2017. Still, the portion in education, finance and insurance, and information technology are increasing. This finding is consistent with the common observations that the GBA is evolving towards a more service-oriented economy (Wang et al., 2016; Yang, 2020).

2.3.3 High-Tech Innovation

Innovation is often considered a key to regional economic development (Storper, 2013; Storper et al., 2015). With the ascendance of the 'Fourth Industrial Revolution', innovation capacity in both production and service sectors are becoming increasingly crucial (Schwab, 2016). In this regard, the GBA may be still in the initial phase of high-tech innovation compared with Tokyo, New York and San Francisco bay areas.

According to the 2018 Forbes' edition of Forbes's The World's Most Innovative Companies, Tencent was the only innovative company in the GBA (Table 2.4). By contrast, there are more innovative companies in New York, San Francisco, and Tokyo. Specifically, the innovative companies in the San Francisco Bay Area were mostly in software and related services.

Within the GBA, Shenzhen and Hong Kong maintain relative advantages in innovation. Data from the Global Cities Initiative published by the Brookings Institution, Shenzhen show approximately three patent applications per thousand population for 2008-2012 (Table 2.5). For this indicator, Shenzhen ranked sixth among the 123 listed cities and immediately behind San Jose (9.42), San Diego (5.19), San Francisco (3.24), Boston (3.01) and Stuttgart (3.01). The same dataset points out that the venture capital per capita in the GBA lags behind those of New York and San Francisco (Table 2.5). As suggested by a Bay Area Council Economic Institute report, venture capital in San Francisco Bay Area is significant for commercializing innovation (Bay Area Council Economic Institute, 2019).

The competitive advantage of innovation may not have been fully reached and applied in the GBA. According to the Quacquarelli Symonds (QS) World University Rankings 2020 (Table 2.6), the GBA has five of the world's top 100 universities, all of which are from Hong Kong. However, in the 2019 Reuters Rankings of the World's Most Innovative Universities, no university in the GBA ranks among the top 100 (Reuters, 2019).

Table 2.4 Innovation companies in the productive sector in four bay areas

Top 100 Global Innovators by Clarivate Analysis 2018-2019	**New York Bay Area**	**San Francisco Bay Area**	**Tokyo Bay Area**	**Greater Bay Area**
Automotive			3	1
Chemicals & Cosmetics			6	
Household Goods			2	
Hardware & Electronics		6	9	
Manufacturing & Medical	2		6	
Telecommunications		1	1	1
Software and Service		1		
Aerospace and Defence			1	
Total	2	8	28	2
The World's Most Innovative Companies 2018 - Forbes	**New York Bay Area**	**San Francisco Bay Area**	**Tokyo Bay Area**	**Greater Bay Area**
Automobile & Component		1		
Pharmaceuticals, Biotechnology & Life Sciences	1			
Health Equipment & Service	1	2		
Software & Service	1	7	1	1
Media	1			
Capital Goods			1	
Foods, Beverage & Tobacco	1			
Retailing		1		
Household & Person Products	2	1	1	
Consumer Services			1	
Total	7	12	4	1

Source: Data combined from Top 100 Global Innovators by Clarivate Analysis (2018-2019), and The World's Most Innovative Companies - Forbes (2018)

Table 2.5 Innovation factors in four bay areas

	City	Patents per 1,000 Inhabitants, 2008-2012	Venture Capital per capita, US $ (ths.), 2006-2015
New York Bay Area	New York	0.76	1.71
San Francisco Bay Area	San Francisco San Jose	3.24 9.42	19.18 30.84
Tokyo Bay Area	Tokyo	2.49	0.02
Greater Bay Area	Hong Kong Guangzhou Shenzhen	0.25 0.36 2.99	0.10 0.04 0.11

Source: Data from Redefining Global Cities by the Brookings Institution

Table 2.6 Top universities in four bay areas

Bay areas	Top 100 Universities QS rankings 2020	The Times Higher Education 2020
New York Bay Area	Columbia University (18); New York University (39)	Columbia University (16); New York University (29)
San Francisco Bay Area	Stanford University (2); University of California, Berkeley (28)	Stanford University (4); University of California, Berkeley (13)
Tokyo Bay Area	The University of Tokyo (22); Tokyo Institute of Technology (58)	The University of Tokyo (36); Tokyo Institute of Technology (251-300)
Greater Bay Area	The University of Hong Kong (25); The Hong Kong University of Science and Technology (32); The Chinese University of Hong Kong (46); City University of Hong Kong (52); The Hong Kong Polytechnic University (91)	The University of Hong Kong (35); The Hong Kong University of Science and Technology (47); The Chinese University of Hong Kong (57); City University of Hong Kong (126); The Hong Kong Polytechnic University (171)

Sources: QS world university rankings and The Times Higher Education world university rankings

2.3.4 Port and Airport Functions

The GBA has multiple world-class ports and airports, showing its strengh in global transport and networking. Data from the American Association of Port Authorities shows that Guangzhou ranks as the world's third-largest cargo port whereas Shenzhen was the third largest container port in 2016 (Table 2.7). The Airport Traffic Report in 2017 shows that Hong Kong maintains the first rank in terms of airport trading and the eighth rank in terms of passenger transport in the world (Table 2.7). As for the total traffic volume of major ports and airports, the GBA is more advantageous. In 2016, the port cargo volume and container traffic reached 1,547.4 million tons and 66.6 million TEUs, respectively (Table 2.8). In addition, the total airport cargo volume and passenger numbers in the GBA in 2017 were 8.1 million tons and 194.5 million, respectively, higher than those of the three other bay areas (Table 2.8).

Table 2.7 Ports and airports ranking in the four bay areas

	Top 50 World Port Ranking-Cargo Volumes, 2016	**Top 50 World Port Ranking-Container Traffic, 2016**	**Top 50 International Airport Ranking-Cargo Volumes, 2017**	**Top 50 International Airport Ranking-Passenger Numbers, 2017**
New York Bay Area	New York/New Jersey (32)	New York/New Jersey (23)	New York Kennedy (23) Newark Liberty (36)	New York Kennedy (22) Newark Liberty (43)
San Francisco Bay Area	--	Oakland (60)	Oakland (48) San Francisco (49)	San Francisco (24)
Tokyo Bay Area	Chiba (24) Tokyo (47)	Tokyo (32) Yokohama (48)	Chiba Narita (8) Tokyo Haneda (22)	Tokyo Haneda (7) Chiba Narita (49)
Greater Bay Area	Guangzhou (3) Hong Kong (12) Shenzhen (19)	Shenzhen (3) Hong Kong (5) Guangzhou (7)	Hong Kong (1) Guangzhou (19) Shenzhen (24)	Hong Kong (8) Guangzhou (13) Shenzhen (34)

Source: Data from World Port Rankings - American Association of Port Authorities, 2016; Airport Traffic Report - The Port Authority of NY and NJ, 2017

Table 2.8 Traffic volume of ports and airports in the four bay areas

	Port Cargo Volumes, 2016 (million tons)	**Port Container Traffic, 2016 (million TEUs)**	**Airport Cargo Volumes, 2017 (million tons)**	**Airport Passenger Numbers, 2017 (million)**
New York Bay Area	121.0	6.3	2.02	132.3
San Francisco Bay Area	22.6	2.4	1.19	81.3
Tokyo Bay Area	365.5	7.1	3.7	117.1
Greater Bay Area	1,547.4	66.6	8.1	194.5

Sources: Data from World Port Rankings, US Port Rankings - American Association of Port Authorities, 2016; Airport Traffic Report - The Port Authority of NY and NJ, 2017; American Federal Aviation Administration; Port of Chiba; Government of Macao Statistics and Census; Guangdong Statistic Yearbook 2017; Port Industry Statistics

Ports and airports in the GBA face internal competitions. However, integrating and coordinating port and airport developments between cities is challenging, with issues such as repeated constructions (Yang, 2005; Homosombat et al., 2016). For example, both Guangzhou and Shenzhen are assigned to develop international aviation hubs. In 2019, the passenger volume of Guangzhou Airport surpassed that of Hong Kong Airport (Figure 2.4).

More recently, Hong Kong International Airport has been significantly affected by the Coronavirus 2019 (COVID-19) pandemic. In 2020, the annual passenger volume was 8.8 million, accounting for only 12% of that in 2019 (Airport Authority Hong Kong, 2021). By contrast, the Guangzhou International Airport ranked 1st globally with 43.76 million passengers while the Shenzhen International Airport ranked 5th with 37.92 million passengers (Airport Council International, 2019).

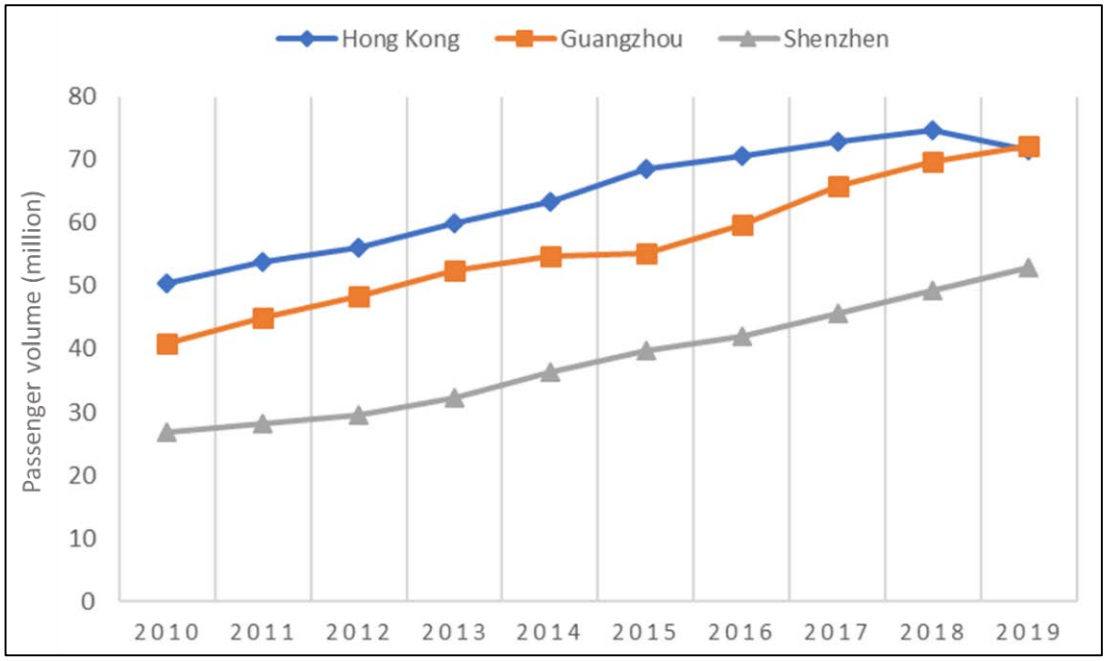


Figure 2.4 Passenger volume (million) of three major airports in the Greater Bay Area

Source: Data from Civil Aviation Department, HKSAR Government and Civil Aviation Administration of China

2.4 Lessons Learnt from New York, San Francisco and Tokyo Bay Areas

This section reviews lessons in economic (re)structuring, high-tech development and regional coordination in New York, San Francisco and Tokyo bay areas.

2.4.1 Evolving Economic Structures

While the New York, San Francisco and Tokyo bay areas have distinctive development trajectories, several commonalities can be found in their evolving economic structures (Yeh, 2006).

- Historically, the three bay areas have all experienced the rise and fall of port function and manufacturing sectors.
- As a response, the three bay areas have undergone economic restructuring through high-tech innovation and/or development of finance and producer services.
- In this restructuring, plans and policies are created to promote synergies, collaborations and coordinated development.
- Regional economic division of labour reflects both market forces such as decentralisation of industrial sectors and planned efforts such as government industrial strategies.

2.4.1.1 New York Bay Area

New York Bay Area has transitioned from manufacturing-oriented industries to service-oriented businesses. While manufacturing was the primary industry in New York before World War II, the city experienced manufacturing decline, often with factories relocating to peripheral areas (Godfrey, 1995). Since the 1970s, services activities in New York have increased dramatically and are regarded as a high policy priority, most notably highlighted in the third regional plan of New York (Yaro and Hiss, 1996). The geographical dispersal of economic activities and a series of regulatory changes also propelled the rise of finance and other advanced producer services (Yeh, 2006).

New York's role as one of the most important global financial and service centres is one of its key competitive advantages (SUNY Levin Institute, 2011). In addition to the financial and service industries, cultural and creative industries thrive in the New York Bay Area. For example, in 2019, New York City's fashion industries accounted for 6% of city's employment and contributed to a total wage of 10.9 billion (City of New York, 2021). Recent analyses have considered the resurgence of manufacturing in the New York area (see for example, Muessig, 2013), while the region has a substantive high-tech sector (most notably bioscience; LSCC Growth Commission, 2016). The New York Bay Area was significantly affected by the COVID-19 pandemic. For example, from June 2019 to June 2020, significant job loss took place in sectors such as leisure, hospitality, trade, and transportation (Milkman and Luce, 2020).

2.4.1.2 San Francisco Bay Area

The initial development and spatial form of the San Francisco Bay Area was significantly affected by manufacturing sectors (Walker, 2001). The area was more renowned for being the home to Silicon Valley (Saxenian, 1994). Federal expenditures including military contracts during and after the World War II helped boost the rise of the high-tech industries, such as aeronautics and semiconductors, in the San Francisco Bay Area (Heinrich, 2002; Schafran, 2013). New waves of high-tech industries such as computers, information technology and biotechnology also emerged in subsequent years (Munroe and Anguiano, 2003; Waslesh, 2010; Forman et al., 2016), with regional institutions such as the Bay Area Council having significant influence on fostering industrial transformation (e.g. Storper et al., 2015). Most notably, venture capital in

the Bay Area has grown rapidly since the 1990s, making the San Francisco Bay Area one of the world's largest pools of venture capital (Storper et al., 2015). Within the region, Silicon Valley and San Francisco largely become centres of innovation and service, respectively (Walker and Schafran, 2015). Recently, the San Francisco Bay Area may be positioned for a manufacturing revival due to technological advancement in 3D printing and robotics, and rising production costs in developing countries (Bay Area Council Economic Institution, 2016).

In the late 1990s, the high-tech industrial structure in the San Francisco Bay Area was transformed from 'manufacturing-based' to 'service-based' (Bay Area Economic Institute, 2016). Most notably, venture capital in the Bay Area has grown rapidly since the 1990s, making the San Francisco Bay Area the world's largest pool of venture capital. Since then, most notably, venture capital has grown rapidly and makes the San Francisco Bay Area the world's largest pool of venture capital. Within the region, Silicon Valley and San Francisco largely became centres of innovation and services, respectively (Walker and Schafran, 2015). In the 2000s, the San Francisco Bay Area has experienced manufacturing revival due to technological advancement such as 3D printing and robotics, rising productions in developing countries and active policy support (Bay Area Council Economic Institution, 2016). For example, employment in the manufacturing sector increased by 4.8% from 2010 to 2014 (Bay Area Council Economic Institution, 2016). More recently, the San Francisco Bay Area may be positioned for a manufacturing revival due to technological advancement in 3D printing and robotics, and rising production costs in developing countries (Bay Area Council Economic Institution, 2016).

2.4.1.3 Tokyo Bay Area

Tokyo's manufacturing industry developed rapidly after the second world war (Schmidt, 2017). The manufacturing industry started to decline in the 1970s and 1980s since production in Tokyo was often relocated to other regions in Japan and later to other countries, such as South Korea and China, with lower production costs (Harris, 1982; Matsumoto, 2009; Belderbos and Zou, 2006). Tokyo has been regarded as one of the global cities in the 1990s, as Japan became more integrated into and played a more important role in the global market (Drennan, 1996; Saito and Thornley, 2003; Matsumoto, 2009; Sassen, 2011). The global division of labour and industrial policies contribute to the rise of R&D activities and technological innovation in Tokyo

(National Research Council, 1997; Salazar-Xirinachs et al., 2014). In the 2000s, due to national government's policy for economic revitalisation, efforts have been put forward to upgrade the regional economy through high-tech innovation (Fujita and Hill, 2005). Still, more recent plans have been aiming to (re)developing the city into a leading global financial centre (Shirai, 2017).

The Tokyo Bay Area has competitive advantages in both producer services and high-tech sectors (Riew, 2007). Drawing references to individual US cities' role in their national urban system, Kamo (1991) suggests that Tokyo combines the roles of New York (e.g. finance and corporate headquarters), Washington (e.g. political centre), Silicon Valley (e.g. innovation) and Detroit (e.g. manufacturing) in the Japanese context (Yeh, 2006).

2.4.2 High-Tech Innovation in the Bay Areas

High-tech innovation in knowledge-based economy increasingly relies on interactive learning, systematic support and effective state intervention (Peck and Theodore, 2007; Asheim et al., 2019). As innovation capacities of New York, San Francisco and Tokyo Bay Areas are embedded within their respective national contexts (Nelson, 1993; Freeman, 1995; Motohashi, 2005), the analysis focuses on key dimensions of national innovation systems such as R&D spending, basic research, university-industry-government relations and venture capital (Ibata-Arens, 2008).

2.4.2.1 R&D spending

Sustained investments on R&D activities are necessary for innovation (Kor, 2006). In 1995, the share of R&D expenditure to total GDP in both the US and Japan was approximately 2.5%, compared with that of China at 0.6% (Figure 2.5). While China has significantly increased its R&D expenditure since the 1990s (Liu and White, 2001), the US and Japan have maintained a relatively high proportion of GDP as R&D expenditure in the past three decades.

The ratio of R&D expenditure to the overall GDP in key Mainland cities within the GBA is higher than China's national average, but may still fall behind that of the San Francisco Bay Area (Yang et al., 2020). According to Yang (2020, p. 109), R&D expenditure in the GBA reached US$ 35.1 billion in 2017, accounting for 2.36% of the total GDP in the region. Shenzhen continues to promote its R&D activities (Figure 2.6;

See also Yang, 2015) and increases its share of R&D expenditure to total GDP to more than 4%. However, the proportion of R&D expenditure within the total GDP in Hong Kong (less than 1%) is relatively low.

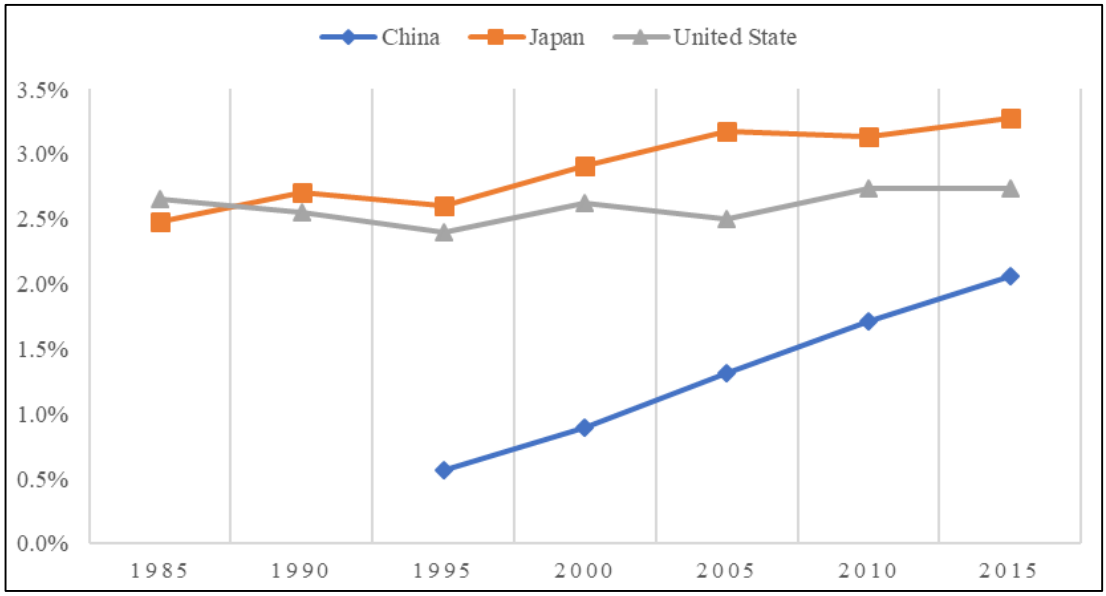


Figure 2.5 Percentage of R&D expenditure to GDP in Japan, the US and China

Source: Data calculated from Organisation for Economic Co-operation and Development

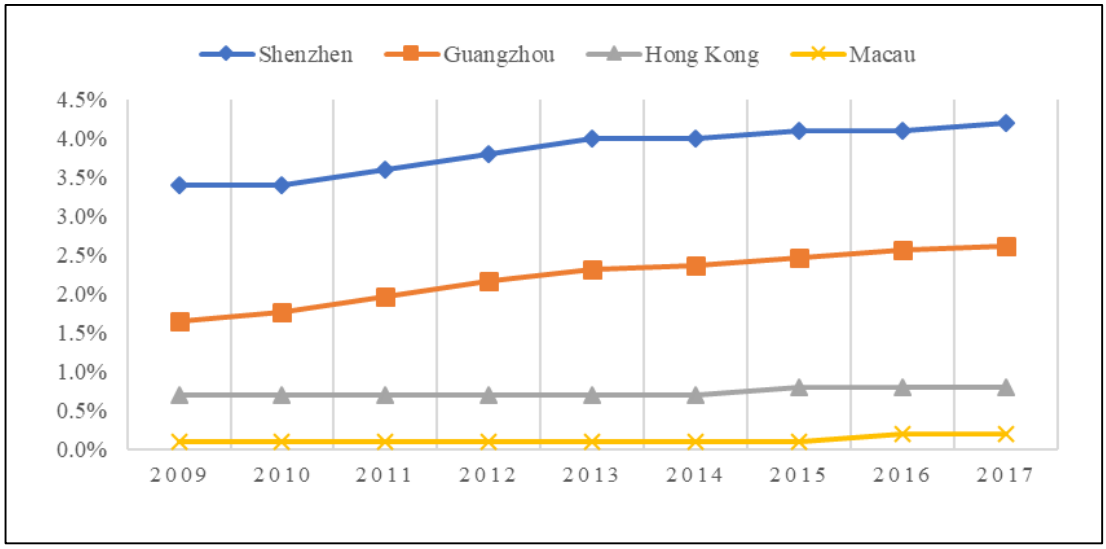


Figure 2.6 Percentage of R&D expenditure to GDP in major cities of the Greater Bay Area

Source: Data calculated from The World Bank, Guangdong Statistic Yearbook and Research on structure and effects of R&D expenditure in Shenzhen 2018

Furthermore, a certain level of R&D investments on basic research is necessary (e.g. Pavitt, 1991). In 2015, the share of basic research in total R&D expenditure was 11.91% in Japan and 16.99% in the US, in comparison with China at only 5.05% (Table

2.9). Even for Shenzhen, the leader of R&D investments in the GBA, spending on basic research was only 2.6% of the total R&D expenditure in 2019 (Figure 2.7). Notably, although R&D expenditure in Japan mainly comes from business enterprises (Fujita and Hill, 2005; Chokki et al., 2020), much of the private investment may be indirectly supported by the government (Fujita and Hill, 2005).

Table 2.9 Structure of R&D expenditure in Japan, the US, and China in 2015

	Japan	**U.S.**	**China**
Share of basic research	11.91%	16.99%	5.05%
Share of applied research	19.86%	19.62%	10.79%
Share of experimental development	63.69%	63.11%	64.16%
R&D expenditure by government	15.41%	25.25%	21.26%

Source: Calculated from Organisation for Economic Co-operation and Development (OECD) Statistics

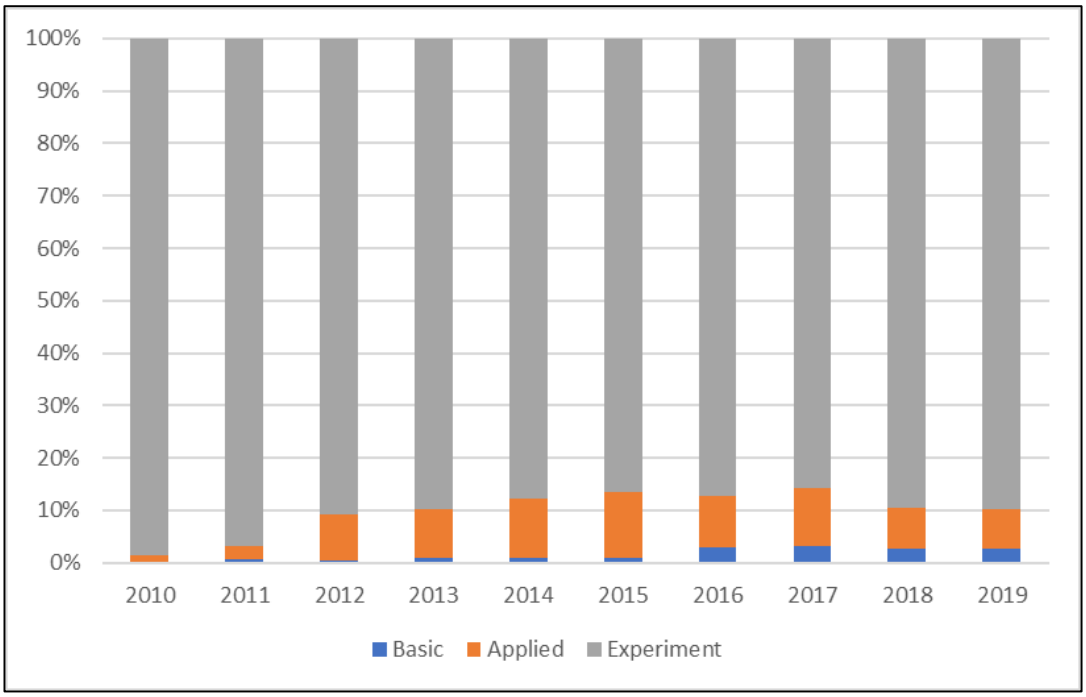


Figure 2.7 Structure of R&D expenditure in Shenzhen

Source: Shenzhen Statistical Yearbook, 2010-2019

2.4.2.2 Significance of Basic Research

Basic and applied research differ in their nature and effects (Akcigit et al., 2021). Tokyo and more generally the Japanese innovation model are initially seen to depend on applied research and experimental development (Bass, 1998) as well as benefiting from effectively using 'external technologies' (Mansfield, 1988). This view leads to questions regarding the long-term sustainability of such as innovation model. Recognising such a challenge and with the advent of technological revolutions and global financial crisis, Japanese firms now invests more in basic research under an open innovation model, aiming for transformational breakthroughs rather than incremental innovation (Okuyama et al., 2021).

In addition to changing behaviours of Japanese firms, the government also realised the significance of intensifying local R&D and university-industry linkages in the 1990s when the 'bubble economy' collapsed (Eshima, 2003). Since then, science and technology development have become a continuing national strategy aimed at improving the country's research capacity. For instance, the Science and Technology Basic Plan mostly focuses on basic research and the application of its resulting breakthroughs (Fujisue, 1998). In San Francisco Bay Area, advances in basic research contribute to more radical innovations and breakthroughs and to new industries, including rDNA and biotechnology (Feldman et al., 2015); circuit chips and information technology (Saxenian, 1994; Osman, 2020).

2.4.2.3 University-Industry-Government Relations

Innovation ecosystems in the US and Japan (often represented by Silicon Valley and Tokyo, respectively) have various differences because of their political, social and economic underpinnings (Fujita and Hill, 2005). For example, the innovation ecosystem has hallmarks of being entrepreneurial and linked with liberal markets, while Japan is often associated with a state 'big-push' model with (Cooke et al., 2007). Key actors within these innovation ecosystems involve industries, universities and governments, which are termed as the 'Triple Helix' (Etzkowitz and Zhou, 2017). Two modes of innovation collaboration are noticeable from the bay areas of Tokyo and San Francisco.

The former focuses on companies/industries as the bases of innovation and their leading roles in practical application and/or commercialisation of research results

(Motohashi, 2005). Public institutions and platforms such as ‘R&D consortia’ and the University Network for Innovation and Technology Transfer (UNITT) are often playing a facilitating role in this company/industry-centred mode of collaboration (Sakakibara, 1997; Whitley, 2002; Hishida, 2013).

The latter is more commonly seen in the San Francisco Bay Area as aiming to commercialise and market research. Often, many university spin-offs are empowered by the presence of vibrant venture capital, tolerant entrepreneurship environment, and dynamic institutional adaptations such as the launch of Bayh-Dole Act (Mowery et al., 2001; Florida et al., 2017). In this mode, university and industrial research labs are critical for boosting radically new technologies and products in the region (Feldman et al., 2015). Universities and research facilities are encouraged to share laboratory and human resources with the private sector (Zucker and Darby, 1996). Furthermore, various non-US companies seek to identify business partners in the San Francisco Bay Area and tap into more advanced technologies and local talent pools and research facilities (Zeller, 2010; c.f., Bathelt et al., 2004; Taylor et al., 2014). According to the San Francisco Bay Area Council Economic Institute (2014), companies from advanced economies such as Japan and the United Kingdom have a strong alliance with innovative resources in San Francisco Bay Area.

In the GBA, cross-border cooperation is critical for synergies between universities, research institutes and industries (Ma et al., 2020). Hong Kong can take the role of a scientific hub in the GBA and collaborate with high-tech industries in Mainland China, especially in Shenzhen (Wang et al., 2021). Changes in the regulatory framework such as patent rights, support for university spin-offs and incentives for patent commercialisation can all possibly foster such collaboration (Wang et al., 2021). Despite a series of policy initiatives for establishing a cross-border regional innovation system in recent years, effective innovative relationships among diverse actors in the GBA are yet to materialise (Yang et al., 2021).

Still, dynamic and mutually-reinforcing relations are developed between large corporations and small firms in Tokyo (Motohashi, 2005). Similarly, an effective innovation ecosystem consisting of interrelated large and small firms, venture capitalists, intellectual property specialist and entrepreneurs are located present in San Francisco Bay Area (Storper et al., 2015). In the GBA, large corporations often lead in patents and their commercialisation. For example, in 2016, according to data from

Shenzhen Statistical Bureau, enterprises above a designated size in Shenzhen contribute approximately 90% of the city's total R&D spending.

2.4.2.4 Venture Capital Investments

Venture capital is often positively associated with business creation, entrepreneurship and innovation (Florida et al., 2017). In the San Francisco Bay Area, venture capital firms take a leading role among diverse financial institutions (Osman, 2020). In Japan, the venture capital market is often considered to be not as mature as that in the US, with innovation in start-ups and products in Japan achieving comparatively lower performance (Nakano, 2017). This scenario may be due to the lack of entrepreneurism and mobility of talent within the ecosystem (National Research Council, 2010). Consequently, in recent years the Japanese government has removed the minimum capitalisation requirements and has encouraged investment companies to provide financial and managerial support (Palmer et al., 2018).

In the GBA, the venture capital market has grown significantly. Shenzhen leads the region in venture capital investment (Figure 2.8). The annual venture capital investment in Shenzhen in 2017-2019 was US $27.67 billion, while the that in Hong Kong is relatively low at US $6.3 billion.

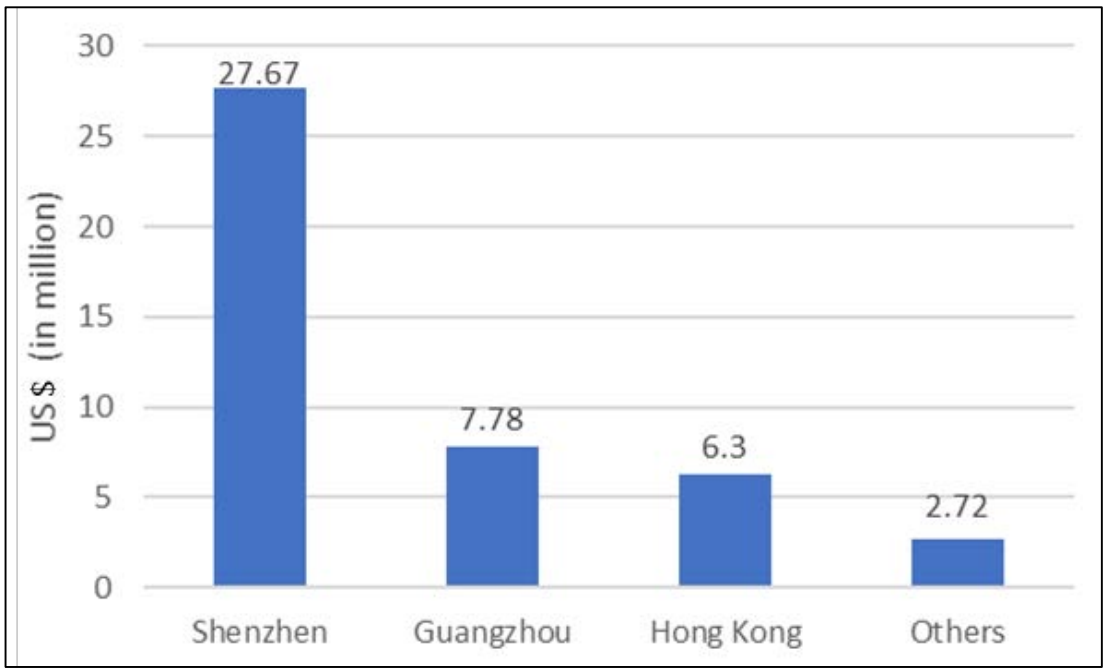


Figure 2.8 Annual venture capital investment in the Greater Bay Area in 2017-2019 (in US$ billion)

Source: China Venture Institute

2.4.3 Cross-jurisdictional Synergy, Coordination and Management

2.4.3.1 Coordination in Transportation

Effective regional governance and planning are crucial for the development of an integrated transportation system (Planning Department, HKSAR Government, 2009). Tokyo Bay Area adopts a more 'centralised' approach wherein the central government plays a significant role in coordinating and developing regional development, often through central ministries such as Ministry of Land, Infrastructure, and Transport and Ministry of Economy, Trade, and Industry (Kantor et al., 2012; HKTDC, 2018; Ohashi, 2018). This approach, however, does not preclude other informal coordinating mechanisms such as the Capital Region Summit (Kantor et al., 2012). In the New York Bay Area, despite having no unified regional level government, cross-jurisdictional entities are commonly created to oversee specific policy areas, the most notable of which is the Port Authority of New York and New Jersey (Rodrigue, 2004; Hamilton, 2012). In the San Francisco Bay Area, association of local governments exerts significance influence on regional planning and development. For example, the Association of Bay Area Governments (ABAG) and Metropolitan Transportation Commission (MTC) are key stakeholders for the development of public transportation (ABAG, 2021; MTC, n.d.).

By contrast, the GBA has not yet established an ordered port system and intense competition exists among cities (Planning Department, HKSAR Government, 2009). For instance, Shenzhen, Hong Kong and Guangzhou all have strategic goals as international maritime trade hubs (Choi, 2021). The integrated development of the regional transportation development is being guided by the *Framework Agreement on Deepening Guangdong-Hong Kong-Macao Cooperation in the Development of the Greater Bay Area 2017*. Several regional transportation projects have been completed/under construction, including the regional high speed rail links and multiple bridges crossing the Pearl River.

2.4.3.2 Coordination in Industrial Development

The Tokyo Bay Area also adopts a centralised approach to regional industrial and spatial development (e.g. Sorensen, 2001). For example, amidst the transformation towards a post-Fordist economy, polycentric urban patterns in the Greater Tokyo Area are often shaped by central and upper-level governments (Ohasi, 2018). In San

Francisco Bay Area, regional organisations such as Bay Area Council, MTC and ABAG have been instrumental in industrial transformation and development (Storper et al., 2015; Wong, 2013). In the San Francisco Bay Area, the three core cities (San Francisco, San Jose, and Oakland) have developed into different employment clusters (Walker and Schafran, 2015). Recent analyses of the New York metropolitan area have also highlighted the creation of community-based regional economic councils as key 'enabling policies' (LSCC Growth Commission, 2016). In the GBA, the development was less centrally coordinated and is rather in a more multi-scalar fashion. The challenges and difficulties in regional coordination are well documented (Yeh and Xu, 2008; Planning Department, HKSAR Government, 2009; J. Xu and Yeh, 2013).

2.4.3.3 Regional Planning and Institutions for Coordinated Development

Both government and non-government organisations contribute to the planning and development of city-regions (Harrison and Hoyler, 2015; Harrison et al., 2021). As noted above, the San Francisco Bay Area has a long string of regional planning and coordination entities (Dyble, 2008; Wong, 2013), ranging from ABAG and sector-specific districts/committees such as Bay Area Air Quality Management District, business associations such as the Bay Area Council to non-profit organisations such as the San Francisco Bay Area Planning and Urban Research Association. In addition to the cross-jurisdictional public entities such as the Port Authority, non-governmental institutions such as the Regional Plan Association have been active about regional development in the New York Bay Area (Parsons, 1994).

In the GBA, the Framework Agreement on Deepening Guangdong-Hong Kong-Macao Cooperation in the Development of the Greater Bay Area was signed in 2017 by the National Development and Reform Commission along with the governments of Guangdong, Hong Kong and Macao. In 2019, the Outline Development Plan for the Guangdong-Hong Kong-Macao Greater Bay Area was released. In addition to these government-led platforms, non-government/official mechanisms are necessary for regional coordination (Planning Department, HKSAR Government, 2009; see also Zhou, 2017).

2.5 Summary

Despite revolutionary advancements in global telecommunication and transportation over the recent decades, the uneven social and geographical distribution of population and economic activities has never been reduced. By contrast, income disparity between the rich and poor has widened and uneven economic development across the globe has intensified. The functioning of agglomeration economies, coupled with the attraction of favourable natural environment, close proximity to world-class universities and established economic infrastructure, has given rise to a new global economy anchored by several great bay areas in the northern hemisphere. The recent designation of the Guangdong- Hong Kong-Macao Greater Bay Area is a welcome and timely endeavour in response to a new trend of uneven global development with enormous potential.

This chapter situates the GBA in a new global economy and compares the region against established American and Japanese benchmarks. This comparative analysis clearly identifies the GBA to have the largest population and land mass. A critical assessment uncovers several characteristics that deserve special attention from planners and policy makers, including an economy dominated by manufacturing with lagging advanced services, shortage of global command and control functions, a high-tech industry single-dominated by telecommunications manufacturing and growing quantity with poor quality in terms of invention patents. These limitations associated with the GBA present planners and policy makers with gaps that need to be filled and opportunities for future development.

Chapter 3
Identifying Strengths, Weakness, Opportunities and Threats for Hong Kong to Foster a New Economic Development of the Greater Bay Area

3.1 Background Analysis

3.1.1 Research Methodology

This chapter reviews the strengths, weaknesses, opportunities, and threats for Hong Kong to develop new economic collaboration in the Greater Bay Area. It begins by highlighting key economic, political, and social dynamics at global, national and regional levels, most notably the trade war and COVID-19. The subsequent review synthesises ongoing debates regarding the repositioning of Hong Kong in the Guangdong-Hong Kong-Macao Greater Bay Area in the academic literature, policy reports, and media.

3.1.2 International level

This chapter reviews the strengths, weaknesses, opportunities and threats for Hong Kong in the context of developing the Guangdong- Hong Kong-Macao Greater Bay Area (GBA). At the international level, key background changes cause the decline of global economies amidst the Coronavirus 2019 (COVID-19) pandemic. In 2019, the global economic growth was 2.9%, approximately 0.7% lower than that in 2018 (International Monetary Fund or IMF, 2020a). Global investment and trade have also taken a heavy toll. According to the World Bank (2020a), the rate of global trade growth in 2019 has fallen to 1.4%, the weakest since the 2008 global financial and economic crisis. Similarly, the International Monetary Fund (IMF; 2020b) observes a historically deep recession of the global economy in 2020, again much worse than that in 2008.

The COVID-19 pandemic has affected global business, trade, tourism and consumption against a backdrop of deglobalisation (Bieber, 2020). According to the World Trade Organisation (WTO) (2020, 2021), the volume of global merchandise trade has declined by 5.3% in 2020 and now faces uneven recovery prospects. Actual recovery may depend on the effectiveness of public health measures (World Bank, 2020b). Travel bans and restrictions also affect global tourism. The assessment of the

United Nations (UN) WTO (UNWTO, 2020) shows that international arrivals declined by 74% in 2020. Furthermore, lockdowns during the pandemic have resulted in a decrease in consumption (Fernandes, 2020) and changes in relevant behaviours (Puttaiah et al., 2020).

The COVID-19 pandemic reorganises global production networks in a 'deglobalising' world. Lockdowns and travel bans and restrictions caused by the COVID-19 pandemic considerably interrupt global supply chain and production (Peng et al., 2021). For example, a survey by the Institute for Supply Management (2020) shows that significant portions of the supply chains of United States (US) companies are affected in the early months of the pandemic. Furthermore, several companies had considered relocating their facilities or suppliers in various countries or bring production home to reduce the related risks (Shih, 2020). In addition, in April 2020, governments such as in the US and Japan are deliberating/announcing policies to pay the costs of moving American and /Japanese firms out of China (Denyer, 2020) to reduce the excessive reliance on the latter (Rapoza, 2020).

De-globalisation goes beyond the economic realm during the COVID-19 pandemic. For example, lockdowns and travel restrictions are forecasted to have significant impacts on global higher education (Martel, 2020; Schwartz, 2020). Preliminary analyses in 2021 confirm new patterns and challenges in student mobility (e.g. double-digit percentage drop of international students in former popular destination countries such as Germany and the US) and in learning and teaching (Farnell et al., 2021).

Protectionism and trade tensions have been rising between major economies. The US-China trade war since 2018 stands out as an example (York, 2020) , with many rounds of negotiations (Wong and Koty, 2020; Bown, 2021). Still, although the Comprehensive Agreement on Investment between China and EU has been agreed in principle by the leaders of the EU Council in December 2020 (European Commission, 2020; Greiger, 2021), the agreement has not been ratified and signed. As a gateway for global capital into China and one of the EU's most important economic partners in Asia (Malmström, 2018), Hong Kong could strengthen its role in channelling investment and trade between Mainland China and the EU (Sin, 2019).

3.1.3 National level

Due to pressures from domestic economic restructuring and international trade and geopolitical tensions in recent years, the growth of the Chinese economy has been slowing down despite still being one of the world's fastest-growing economies (Reuters, 2020). For example, while the annual GDP growth of China still leads among key economies, its annual growth rate has gradually declined in the past decade (Figures 3.1 and 3.2).

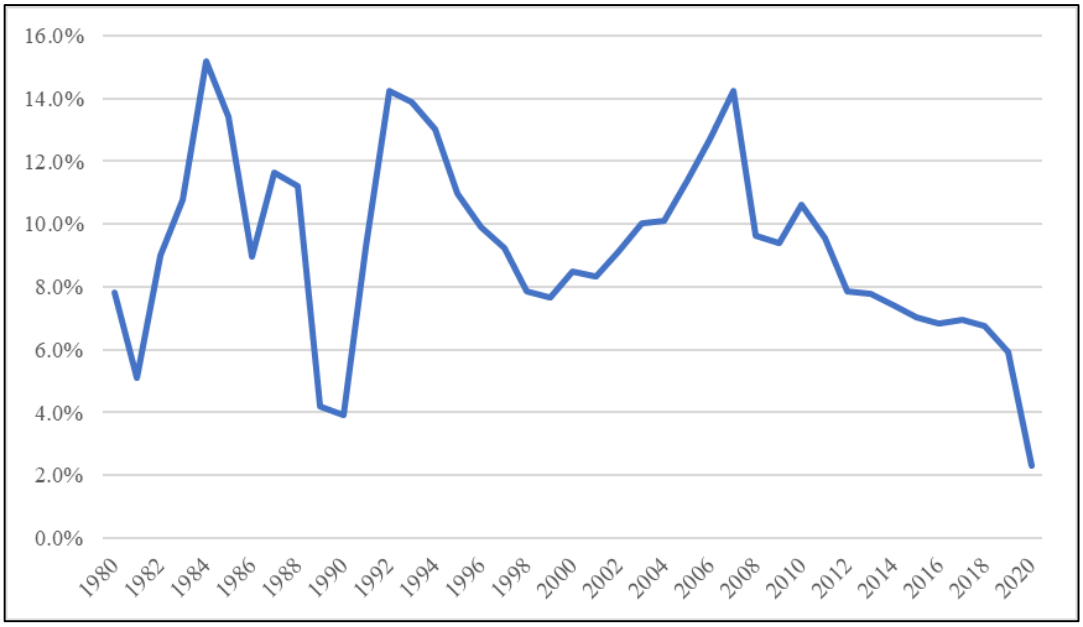


Figure 3.1 GDP growth rate (%) in China, 1980-2020

Source: World Bank

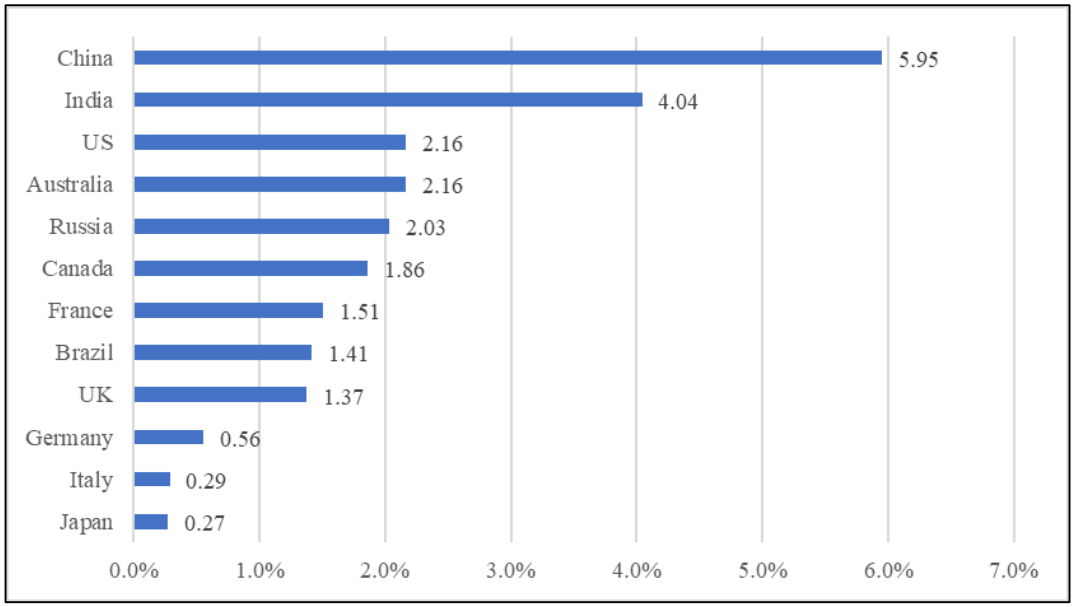


Figure 3.2 Comparison in annual growth rate (%) between selected countries in 2019

Source: World Bank

China's export-oriented economic model, which has sustained its rapid growth in the past 40 years, faces external challenges (Gereffi, 2009; Zhu and Pickles, 2014; Morrison, 2018). China's labour productivity is relatively low and the economy's advantages are in labour-intensive, low-tech sectors (Xiao, 2019; National Bureau of Statistics, 2018). The rising labour costs in China may force labour-intensive industries to move to other developing countries or move back to advanced economies (Davis, 2016). For example, labour costs in China may already be higher than those in other major developing economies (Szmigiera, 2021). The situation is further affected by recent changes such as trade wars and pandemics (Lau ,2020).

China has the world's largest industrial system that can help the national economy recover from the pandemic (State Taxation Administration, 2021). According to the Ministry of Industry and Information Technology, China may be the only country that has all of the listed industrial categories in the UN (Huang, 2019). Furthermore, China ranks first worldwide in over 220 out of 500 key industrial products (Yang, 2019). Analyses from the UN Industrial Development Organisation (UNIDO, 2021) suggest that, amongst others: (1) the early effects from the pandemic caused a 6.8% drop in global manufacturing in 2020; (2) most countries are in gradual recovery; and (3) China's manufacturing has recovered significantly and is back to pre-pandemic level.

China plans to build a new development model, that s, 'dual circulation', to

highlight the significance of both domestic and international markets (Xinhua, 2020a, 2020b). In addition to the global market and export-oriented economies, this new model emphasises the importance of domestic markets,and sustainable economic growth (Xinhua, 2020b, 2020c). China has put forward actions to develop its 'green economy' in the 13th 5-Year Plan (Organisation for Economic Co-operation and Development or OECD, n.d.). More stringent environmental policies have been adopted in recent years (Development Research Centre of the State Council of China, 2017).

Recently, China has initiated a series of plans and initiatives, such as *National Innovation-Driven Development Strategy Outline* to strengthen its innovation capacity (Deloitte China Innovation Department, 2019). New policies have also emphasised the protection of intellectual property (Xinhua, 2019a). China is catching up in innovation investment and patent applications. In recent years, China has invested over 2% of its GDP on R&D, catching up with averages of OECD countries (Figure 3.3). Furthermore, according to data from the World Intellectual Property Indicators, patent applications processed by China's National Intellectual Property Administration has rapidly increased in the past two decades (Figure 3.4; World Intellectual Property Organisation, 2019).

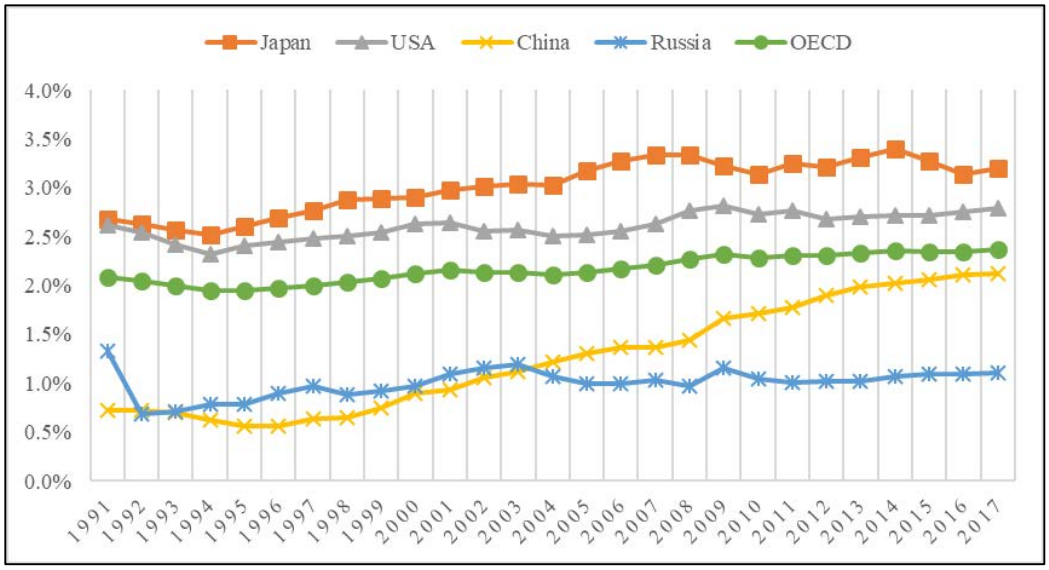


Figure 3.3 Total percentage of GDP on R&D activities in 2000-2017

Source: OECD Science and Technology Indicators Database, https://www.oecd-ilibrary.org/industry-and-services/gross-domestic-spending-on-r-d/indicator/english_d8b068b4-en

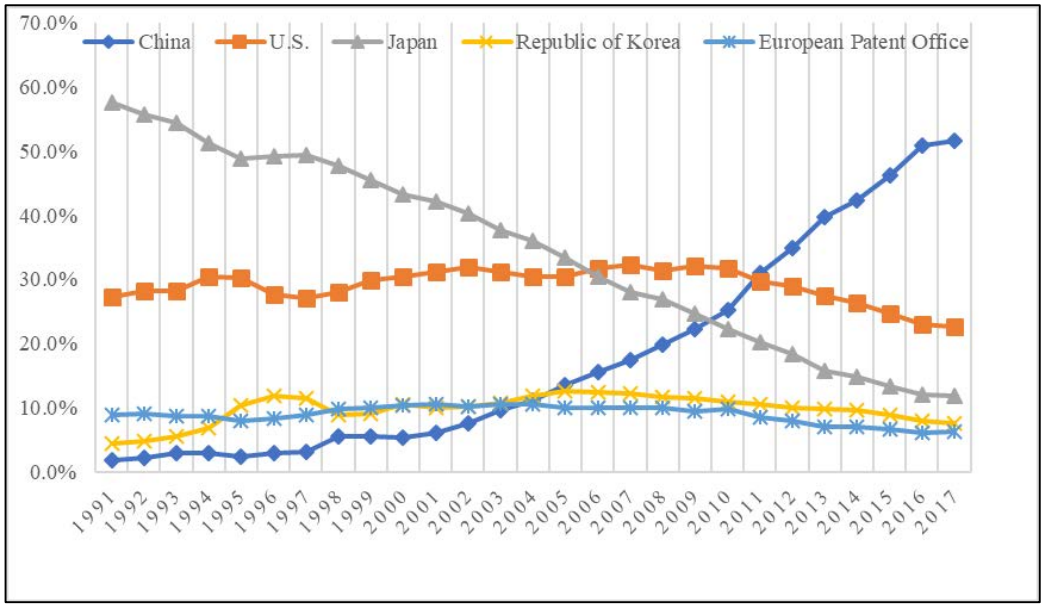


Figure 3.4 Trends in patent applications from five leading intellectual property offices 1991-2017

Source: Data from World Intellectual Property Organisation, https://www.wipo.int/publications/en/series/index.jsp?id=37

However, the rapidly increasing number of domestic patents in China may not necessarily correspond to a significant increase in innovation quality. For instance, performance in the filed Chinese triadic patent may potentially reflect quality (China Power, 2016; Fisch et al., 2017). According to the 2017 OECD Patent Statistics, China only had 3,792 triadic patents in 2016, accounting for approximately 7.8% of those from OECD countries (48,853), 29.5% of those from the US (12,873) and 21.7% of those from Japan (17,489).

3.1.4 Regional level

Hong Kong can spearhead the region's economic and technological development due to its innovation capacity, sophisticated market, and transparent regulatory framework (HKTDC Research and Bay Area Council Economic Institute, 2021; Office of the Government Economist, HKSAR, 2020). For example, high-tech industries from Mainland Chinese cities may tap into capital and technological knowledge in and through Hong Kong (see more examples in HKTDC Research and Bay Area Council Economic Institute, 2021). In the past years, Hong Kong has endured social unrest, economic slowdown and the COVID-19 pandemic. Due to social unrest, in the third and fourth quarters of 2019, visitors to Hong Kong declined by 26% and 50.5% on a year-on-year basis, respectively, while similarly retail sales dropped by 19.5% and

24.1%, respectively, from the previous year (HKSAR government, 2020a). On an annual basis, Hong Kong's GDP also declined to 8.9% in the first quarter of 2020 (Census and Statistics Department, HKSAR Government, 2020).

Export-oriented industries in the PRD may be hit by the COVID-19 pandemic as well as the ongoing deglobalisation. The foreign trade volume of Guangdong province declined by 11.8% in the first quarter of 2020 (Guangdong Statistical Bureau, 2020). For the same period, its GDP decreased by 6.7% (Guangdong Statistical Bureau, 2020; Zhang, 2021). At present, Hong Kong may be recovering from the pandemic and have first-mover advantages with its public health measures and vaccination programme (Curran et al., 2020; Yau, 2021). The mitigation of the pandemic's influence may contribute to economic recovery, specifically in tourism and business. For example, during the Tomb Sweeping Day holiday in April 2021, Mainland China experienced a 144.6% growth in the number of travellers and a 228.9% growth in tourism income compared with 2020 (People.cn, 2021).

3.2 Strengths of Hong Kong

3.2.1 Advantages in Technological Innovation

Hong Kong has advantages in its quality research environment, given its quality research environments, leading universities and research institutions and talent pool (Innovation and Technology Bureau, HKSAR Government, 2018). Specifically, Hong Kong ranks 14th in the 2018 Global Innovation Index (Dutta et al., 2018) and the 11th in the 2018 Institute for Management Development (IMD) Digital Competitiveness ranking (Institute for Management Development, 2018a; HKTDC, 2021).

Universities in Hong Kong are among the top in various global rankings, with five universities occupying top 100 positions in the 2020 QS World University Rankings. In particular, science and engineering subjects in the universities of Hong Kong also occupy respectable positions (Innovation and Technology Bureau, HKSAR, 2018). Hong Kong has an educated talent pool, potentially contributing to its development as a technology and innovation centre. More than 80% of the population in Hong Kong has received secondary school education or above in 2017, while statistics from the Education Bureau, HKSAR Government suggest that the proportion of population with post-secondary education has been approximately 30%. In addition, strategic programmes are put forth to attract global talent (Research Office of Legislative

Council Secretariat, 2020; see however, Hong Kong X Foundation, 2017). Still, in the 2018 IMD World Talent Ranking, Hong Kong had a respectable 18th (out of 63) rank (Institute for Management Development, 2018b).

Sound protection of intellectual property facilitates the R&D environment of Hong Kong. For example, under the 'One Country, Two Systems' framework, the intellectual property law system aligns closely with WTO Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS; Intellectual Property Department, HKSAR Government, 2019; 2020). Collaborations are formed between Hong Kong and Mainland China despite their separate intellectual property systems (Li, 2017). Hong Kong's intellectual property law system covers various domains (e.g., copyrights and patents) and has an effective enforcement mechanism (Intellectual Property Department, HKSAR, 2019).

3.2.2 Free Market and Efficient Financial System

Financial and free markets in Hong Kong may support its economic and technological development. Hong Kong ranks as one of the world's most free economy (Heritage Foundation, 2018; Sum, 2019; Ventura, 2020) More recently, the HKSAR government has introduced further changes to make profit tax rate attractive for enterprises (KPMG International Limited or KPMG, 2018a; Inland Revenue Department, HKSAR Government, 2021; see also Innovation and Technology Bureau, HKSAR Government, 2018).

According to the GFCI Ranking Hong Kong ranks third among the world's international financial centres in 2018 (Z/Yen Partners, 2018) and has a large cluster of financial institutions. (Value Walk, 2020; Hong Kong Monetary Authority, HKSAR Government, 2021). Furthermore, Hong Kong topped global Initial Public Offering (IPO) market in 2018 (KPMG, 2018b). Apart from its significance in the global market, Hong Kong serves as a gateway to Mainland China, in part propelled by the Closer Economic Partnership Arrangement between the two jurisdictions for overcoming trade barriers and providing market access (Tsang, 2019; Yang, 2016).

3.2.3 Transparent Regulatory Framework

Hong Kong ranks 16th among the 113 countries or territories in 2018 in the World Justice Project's 'Rule of Law' index (World Justice Project, 2018). The legal system

provides investors with assurance of enforceable contracts, and protection on intellectual property and personal data (Intellectual Property Department, HKSAR Government, 2020). In addition, Hong Kong is reputed for its low tolerance for corruption (Gong and Wang, 2013; Heritage Foundation, 2018). Hong Kong also has an extensive set of financial market regulations (Woo, 2015; See also Arner et al., 2010).

On 30 June 2020, the Hong Kong National Security Law is passed and signed. While this law may be considered necessary for the much-needed social stability (Lam, 2021; Chinese Consulate General in Kota Kinablu, Malaysia, 2020), there are concerns of freedom of speech and judicial independence (Mang and Chow, 2020; SCMP, 2020).

3.3 Weaknesses of Hong Kong

3.3.1 Lacking Governmental Support on Technology and Innovation

Hong Kong technology innovation is often associated with the lack of government interventions (Wang J., 2018). In the 1980s, the Hong Kong colonial government chose the laissez-faire approach when manufacturing industries were relocated, notably without industrial subsidies (Lau and Kuan, 1990; Lo et al., 2001). By contrast, technology and innovation in other 'Asian Dragon' economies such as Singapore benefit from active government support (Steinbock, 2017; Wang J., 2018).

While the Hong Kong government in recent years has targeted high-tech industries by establishing stakeholder committees, deliberating/implementing favourable policies, and developing key infrastructures (e.g. Science Park and Cyberport), the GDP contribution of the technology industry may not have experienced significant increases (Education Bureau, HKSAR Government, 2016; Hong Kong X Foundation, 2017; HKSAR Government, 2020b; see also Baark and So, 2006). The government support may not have been adequate (Kang and Jiang, 2020) and challenges remain large for innovations and start-ups (Hammond and Ruehl, 2020). For example, Hong Kong 2019 R&D expenditures have been at approximately 0.9% of its GDP (Census and Statistics Department, HKSAR Government, 2019), lower than those of Mainland China (2.23%), Japan (3.24%), South Korea (4.64%), Singapore (1.84%), and the US (3.06%) (OECD, n.d.; see also Deloitte Research, 2017).

3.3.2 Under-Developed Innovation Ecosystem and Lack of Collaboration between Different Local Actors

Overall, the development of ecosystems for technological innovation in Hong Kong lags behind those in other 'Asian Dragon' economics such as South Korea and Singapore (Ng, 2017). In particular, local actors/stakeholders within the innovation ecosystem may lack collaboration such as in technology transfer (Our Hong Kong Foundation, 2015; Hong Kong X Foundation, 2017). For example, according to the *Hong Kong Innovation Activities Statistics* released by the Census and Statistics Department, HKSAR Government (2014, 2019), the majority of business undertaking technology and innovation activities did not have collaboration arrangement.

The most vulnerable segment of this innovative system may be the lack of angel investors and venture capital for start-ups (Fung, D. 2021; see also Lau et al., 2013). For example, according to the Hong Kong X Foundation (2017), despite a sizeable amount of HK-based venture capital firms, the investment on local innovation start-ups was limited (See however, Ng, 2019).

3.3.3 Economic Instability Brought by Social Unrest

Social unrest in Hong Kong hit its economy hard amidst global geopolitical and economic tensions. For example, Hong Kong experienced an economic recession in the third quarter of 2019 (Au, 2019), while retail sales hit the lowest record in recent decades (Russolillo, 2019). Transportation and logistics especially the airport have also been significantly affected (Everstream, 2020).

Tourist numbers fell sharply after the social unrest. In the third quarter of 2019, tourism arrivals in Hong Kong decreased significantly compared with the same period in 2018 (Xinhua, 2019b; see also Figure 3.5). Various events had been cancelled or postponed (Yau, 2019). Statistics compiled by the Hong Kong Trade and Development Council show a significant decline in the trade between Hong Kong and Mainland China, the US and the EU (Tsang, A., 2019; HKSAR Government, 2021).

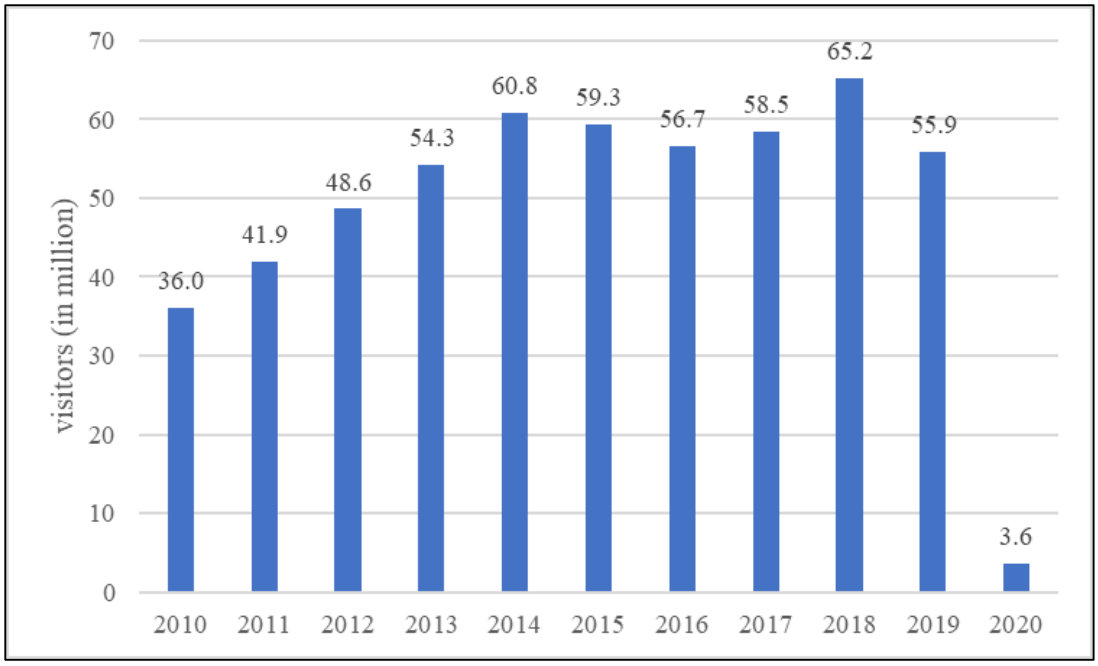


Figure 3.5 Total visitor arrivals in Hong Kong from 2010 to 2020 (in million)

Source: Hong Kong Tourism Board Annual Reports and Press Releases, https://www.discoverhongkong.com/eng/hktb/about/annual-report.html

3.4 Opportunities for Implementing the New Economic Models in Hong Kong

3.4.1 China's National Strategies of 'Going Global' and 'Dual Circulation'

In response to domestic and international challenges and with the aim to restructure its economy, the Chinese government has initiated new strategies such as 'Going Global' and 'Dual Circulation' (China Policy, 2017; Wang and Hu, 2017; Xinhua, 2020a).

Hong Kong can serve as a leading international platform for China's enterprises to raise capital and expand business with its high-end professional services, familiarity with global business culture and world-class hard and soft infrastructures (Chrisman et al., 2015; HKSAR Government, 2017; Chu, 2018; Meyer, 2020). Specifically, being one of the world's leading IPO markets (Lee, 2021), Hong Kong's securities markets can help companies from Mainland China to raise international capital (Pan and Brooker, 2014; Fang and Pan, 2021). Despite recent social and geopolitical events, Hong Kong's significant role as a global financial centre is difficult to replace (Li, 2020; Meyer, 2020). The legislation imposing sanctions on Chinese companies on the US stock markets for failing to comply with US regulatory requirements may provide new opportunities for the Hong Kong stock market (Kharpal, 2021; Michaels, 2020). This bill may potentially place such companies in conflict with China's state secrets

protection rules (Klein, 2020), which may subsequently launch secondary offerings outside the US. Several major Chinese tech companies have already launched secondary offerings in Hong Kong (Zhang, 2020).

Hong Kong can benefit from deeper and wider cooperation with Mainland China, especially within the GBA (Shen, 2021), including but not limited to the development of innovation corridors between Hong Kong, Macao and other cities in the PRD (Xinhua, 2021), collaborations in higher education (e.g. branch schools of Hong Kong's universities; Table 3.1) and of policy support to attract both overseas and Mainland investments (Pan, 2021).

Table 3.1 Branch schools in Mainland China of Hong Kong universities

University	Branch school location	Status	Official website
Hong Kong Baptist University (Beijing Normal Univeristy)	Zhuhai, Guangdong	Established in 2005	https://www.uic.edu.cn/
The Chinese University of Hong Kong	Shenzhen, Guangdong	Established in 2014	https://www.cuhk.edu.cn/
Hong Kong University of Science and Technology	Guangzhou, Guangdong	Established in 2019	https://hkust-gz.edu.cn/zh-hans
City University of Hong Kong	Dongguan, Guangdong	Preparation from April 2020	NA
The Hong Kong Polytechnic University	Foshan, Guangdong	Preparation from March 2020	NA
The Open University of Hong Kong	Zhaoqing, Guangdong	Preparation from April 2020	NA
The University of Hong Kong	Shenzhen, Guangdong	Announced in 2021	NA

Source: Information gathered from the websites of individual universities

3.4.2 Guangzhou-Shenzhen- Hong Kong Innovation Corridor

The GBA attempts to facilitate the development of the Guangzhou-Shenzhen-Hong Kong Innovation Corridor (Mu, 2019; Wu, 2019). Its planning, design and development have drawn reference to many of the world's leading innovative areas and aim to leverage on the region's capital, infrastructure, talent, and emerging innovation ecosystem (Greater Bay Insight, 2021; HKTDC Research and Bay Area Council Economic Institute, 2021; Information Office of the People's Government of Guangdong Province, 2017).

Hong Kong can become the gateway in this development (KPMG, 2018c). Familiar with both technology standards and actual application in the international markets, Hong Kong can serve as the bridge for data, technology and knowledge between Mainland China and the wider international community while fostering innovation locally with its talent pool (Baark and Sharif, 2006; See also Sigler et al., 2021 for a more general analysis highlighting Hong Kong's 'liaison' role in global corporate networks). For example, such initiatives may include the Hong Kong-

Shenzhen Innovation & Technology Park and joint innovation laboratories with leading international institutes (Leung, 2018; Innovation and Technology Bureau, HKSAR Government, 2018; also the more recent plan for a 'Northern Metropolis' emphasising technology and innovation as stipulated in HKSAR Government (2021a)).

3.4.3 Development of Transport and Information Infrastructures

Key projects such as Hong Kong-Zhuhai-Macao bridge and high-speed rails are critical for the connectivity between Hong Kong and other cities in the GBA (The Guangdong- Hong Kong-Macao Greater Bay Area Development Office, 2018a). According to the *Outline Development Plan for the Guangdong-Hong Kong-Macao GBA* (State Council of China, 2019), projects such as the Guangzhou-Shanwei Railway, Guangzhou-Maoming-Zhanjiang Railway and improved Beijing-Hong Kong-Macao Highway may further enhance the regional connectivity (Cyrill, 2018).

Advancements in information infrastructure (e.g. 5G) contribute to economic productivity and efficiency (World Economic Forum, 2020). The GSM Association (2021) predicts that the 5G shares of total mobile connections can respectively reach 47%, 33% and 69% in Mainland China, Hong Kong and Macao by 2025. A pre-COVID estimate (GSM Association, 2020) has suggested that the economic contribution of mobile technology in China is predicted to increase from $759 to $906 billion from 2019 to 2024. In particular, the GBA leads China's 5G development with its 5G industry cluster (China News, 2020). For example, Huawei and ZTE in Shenzhen are major producers of 5G equipment and technologies (Li, 2019; Gartner (2021) as cited in CRN 2021).

3.4.4 Changing Social Infrastructure in the GBA

Recent policies aim to develop a social infrastructure (e.g. tax, housing and education) in other cities of the GBA that may be attractive to Hong Kong residents and businesses. For example, Hong Kong residents working in other cities in the GBA can enjoy a preferential tax rate, significantly reducing the financial burden (Guangzhou Rencai Net, 2019). Moreover, in terms of buying properties in other cities in the GBA, Hong Kong residents enjoy the same policy treatment as local residents (MacauHub, 2019). Furthermore, children of Hong Kong residents who work in cities in Guangdong province can enjoy the same enrolment policies as local residents (The Guangdong-Hong Kong-Macao Greater Bay Area Development Office, 2018b).

Furthermore, provincial and local governments are launching policies to support start-ups by Hong Kong and Macao youth in the GBA with benefits, such as start-up funds, health insurance and housing allowance (The Guangdong-Hong Kong-Macao Greater Bay Area Development Office, 2018c; South Network News, 2019). Still, several policies are initiated to facilitate cross-border professional services, such as those related to the mutual recognition of professional qualifications and the pilot scheme to allow Hong Kong lawyers to practice civil and commercial laws in the GBA (Wong, 2020; Trade and Industry Department, HKSAR Government, 2003; Department of Justice, HKSAR Government, 2020). The mutual recognition of intellectual property rights between Hong Kong and Mainland China is also set on the agenda (State Council of China, 2019; Dong, 2021). Close partnership is also necessary amidst the COVID-19 pandemic. For example, central and Guangdong authorities have pledged to support Hong Kong, such as by providing stable supply of daily necessities and help the implementation of public health measures (Shanghai Guancha Net, 2020; People.cn, 2020; HKSAR Government, 2021b).

3.5 Threats for the New Economic Models in Hong Kong

3.5.1 Differences in the Financial, Legal and Political Institutions between Hong Kong and other GBA cities

Hong Kong and Macao do not share the same administrative, legal and financial system with other cities in the GBA under the 'One Country, Two Systems'. Different governance systems may slow down the exchanges of goods, information, and capital and raises the difficulties of collaboration between cities in the GBA (X. Wang, 2018).

Regional integration and cross-border governance are traditionally challenging (Planning Department, HKSAR Government, 2009; Yang and Li, 2013; Zhang et al., 2018; Yeh and Xu, 2006). A critical factor in tackling these issues may be improving socioeconomic and political integration within the GBA (Lui, 2015; Li et al., 2018; Li and Bray, 2007) with policy initiatives from both public and private stakeholders targeting cross-border school children (People's Daily, 2018), collaborations in finance, higher education and medical services (Yeh et al., 2015b; Lin, 2019) and eco-environmental co-governance system (Zhu, 2019).

3.5.2 Increasingly Homogenous Industrial Structures

Cities in the GBA are confronted with the challenge of increasingly homogenous industrial structures. On the one hand, high-tech and service sectors have grown rapidly in the GBA, especially in Shenzhen, with the high-tech industry contributed approximately 30% of GDP growth in Shenzhen in the first three quarters of 2018 (Zhang, 2018). On the other hand, homogenous high-tech and service industries may constrain the stable development of the GBA; a synthetic review of strategic plans of individual cities may reveal that these cities having similar plans for strategic industries such as information and communication technology and high-tech manufacturing (see also HKTDC, 2021).

With the rapid development of the service industry, Mainland Chinese cities in the GBA may depend less on services provided by Hong Kong companies (Ljungwall and Li, 2020). Figure 3.6 shows that the service sector accounts for more than 50% of the labour force in Shenzhen and Guangzhou, while the region as a whole remains a manufacturing stronghold. Specifically, these sectors in Guangzhou and Shenzhen provide services to nearby cities (Yeh et al., 2015b) potentially reducing the demands for those from Hong Kong.

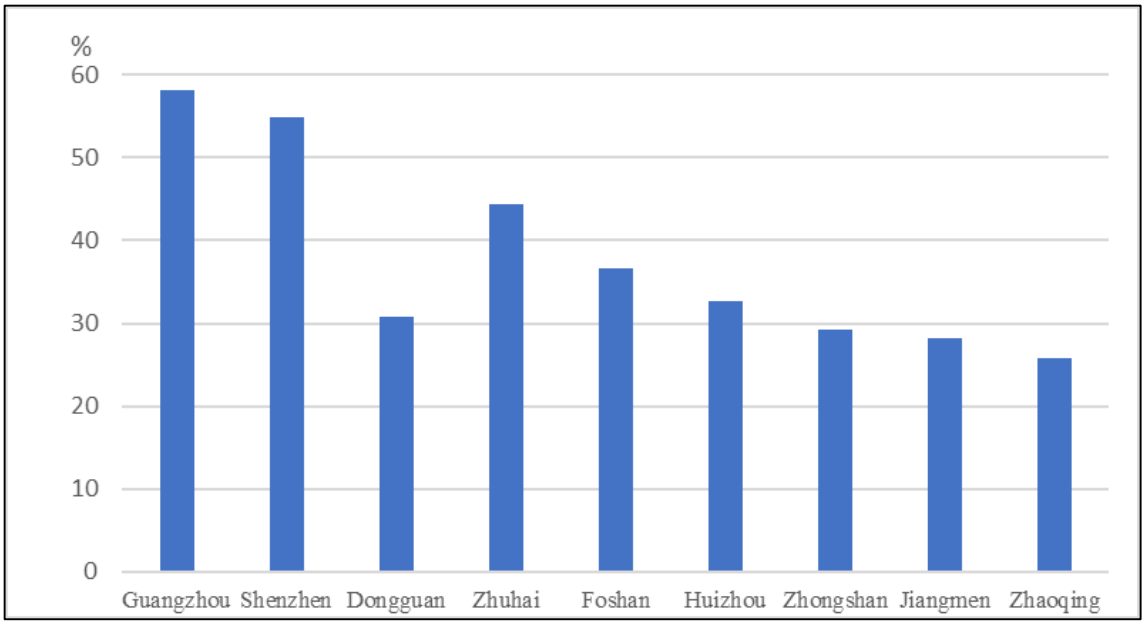


Figure 3.6 Proportion of employment in service sectors in Mainland cities of the GBA in 2016

Source: Guangdong Statistical Yearbook 2017

3.5.3 Rising Production Costs in the GBA

Rising production costs in the GBA may force the manufacturing companies to move to other developing countries (e.g. South Asian countries) or less developing regions in China. Rising labour costs in the PRD are often linked with demographic changes, ageing population and socio-economic upgrading of rural populations (Wang et al., 2020; see also, Tang and Hao, 2018). Specifically, according to a survey of Hong Kong-invested manufacturers, a majority of the companies faced labour shortages in 2018 (The Chinese Manufactures' Association of Hong Kong, 2020; see also Liao and Chan, 2011).

The relocation of manufacturing firms in the GBA may lead to a decrease in demand for Hong Kong's high-end producer service provider, which has been its role for companies in the GBA and other regions in Mainland China for several decades (Yeh et al., 2015b).

3.5.4 Trade War, COVID-19 Pandemic and Unstable Geopolitical Environment

The unstable geopolitical environment may constrain Hong Kong's development as a transport, technology, and service centre. As a major gateway for Mainland China's imports and exports, Hong Kong may be hit hard by the increased tariffs on Chinese goods and the US removal of preferential trade and economic treatments afforded by Hong Kong's special status (Martin, 2020). The trade war not only affects trade itself but also presents implications for technology; for example, technology transfer between Hong Kong and the US may be subject to further scrutiny and high-tech development costs may rise as significant portions of telecommunications equipment in Hong Kong are imported from the US (Tsang and Leung, 2020). Technological firms from the Mainland, such as Huawei and ZTE, have already been targeted by the US (Mozur and Swanson, 2018; Strumpf and Ferek, 2020). Furthermore, the effects of geopolitical and economic changes on firms' decision to enter/remain in Hong Kong have received much attention (Lyons and Yoon, 2021).

As mentioned above, the COVID-19 pandemic posits significant challenges of recovery to all economies worldwide and may accelerate the trend of deglobalisation (Susskind et al., 2020). As an international hub, Hong Kong may be particularly affected in this deglobalising process (Bishop and Roberts, 2020; see however, Banafa, 2020).

3.6 Summary

For decades, Hong Kong has always functioned as an important gateway and spearhead for global capital to enter China's Mainland and for the Chinese economy reaching out to the world. While the geographic location of Hong Kong remains unchanged, the geopolitical environment in which Hong Kong is embedded has experienced profound and unprecedented changes. The rapid economic transformation of the PRD and rapid rise of the Chinese economy as a global power with its own international network and the Belt and Road Initiative have effectively phased out the old model of 'Front Shop, Back Factory' upon which Hong Kong has depended to build its partnership with its hinterland over the PRD in the 1980s and 1990s. Hong Kong has to reposition itself in the rapid changing local, regional and national economic environment with the continuing trade war and Sino-American geopolitical tension under the 'One Country, Two System'. Against this backdrop, it is a high time for Hong Kong to seriously re-evaluate its strengths, weakness, opportunities and threats to reposition itself in the framework of the Guangdong-Hong Kong-Macao Greater Bay Area and the *GBA Outline Development Plan 2019*.

Chapter 4
Hong Kong as an Innovation Hub

4.1 Introduction

We are ushered in the fourth industrial revolution, which is characterised by the integration of physical, digital and biological spheres. Several technology breakthroughs have marked the current technology waves, including robotics, artificial intelligence (AI), nanotechnology, quantum computing, biotechnology, the Internet of Things (IoT) and fully autonomous vehicles (AV). The world's advanced economy are taking the development of AI as a major national strategy to strengthen national competitiveness and accelerate to introduce plans and strategies to attract and incentivise top talents and scientists. China, as a developing country, spares no efforts to catch up with the technology race. For example, in 2017, the Chinese State Council announced the 'Issuance of the Next Generation Artificial Intelligence Development Plan', outlining the national strategy to apply artificial intelligence (AI) in a wide range of fields. AI in education, medical care, pensions, environmental protection, urban operations, judicial services and other fields. In 2015, the plan *Made in China 2025* was released by Chinese Premier Li Keqiang to upgrade the Chinese industry. The Guangdong government has since implemented a series of fiscal and tax policies to echo that goal, announcing plans to upgrade the manufacturing sector on the west bank of the Pearl River, which has been regarded as pilot demonstration city clusters in line with *Made in China 2025*. They include cities of Zhuhai, Foshan, Zhongshan, Jiangmen, Yangjiang and Zhaoqing, as well as the Shunde district. They are regarded as pilot demonstration city clusters in line with *Made in China 2025*.

One of the key goals in the *Framework Agreement on Deepening Guangdong-Hong Kong-Macao Cooperation in the Development of the Greater Bay Area* 2017 and the *Outline Development Plan for the Guangdong-Hong Kong-Macao Greater Bay Area 2019* is to develop a world-class cluster of cities with a leading role in advanced manufacturing and innovation.

In this chapter, we examine the high-tech development of the cities in the GBA, comparison of innovation performance between PRD and YRD and Hong Kong's advantage in universities and research institutions.

4.2 Research Methodology

In this chapter, we examine the determinants for city-industry growth using China's Annual Survey of Industrial Firms, as well as China's patent data. We attempt to examine the variations in innovations across cities and industries. We use the Annual Survey of Industrial Firms of China from 1998 to 2013 to examine the development of high-tech manufacturing firms in Guangdong province. The dataset is reliable as it is released by the National Bureau of Statistics of China and it is compiled as census data for industrial sector. It has rich information on manufacturing firms' basic information such as company name, address and industry, as well as accounting information such as profit, cost, and taxation. Fifteen years of data also allows us to examine the development and evolution of the manufacturing sector in China. In this chapter, we attempt to analyse the allocation and development of high-tech manufacturing firms in Guangdong and examine if Hong Kong has played a critical role in Guangdong's high-tech development and whether it will continue to benefit Guangdong's high-tech development.

We also use the Chinese patent data to match the Annual Survey of Industrial Firms to see the innovation performance of the manufacturing firms. We combine the two datasets by matching the company names. We compare the innovation performance of manufacturing firms in Pearl River Delta and Yangtze River Delta and study the characteristics of the innovation performance of Guangdong's manufacturing industries.

4.3 High-tech Development in Cities of Guangdong Based on Annual Survey of Industrial Firms

The growth of China's manufacturing has been remarkable since the economic reform and opening up policy in 1978. Guangdong, the province that has benefited from the policy the most, emerged as a cluster of labour-intensive and low-tech manufacturing firms. In recent years, however, Guangdong has experienced a miraculous increase in high-tech firms particularly in cities such as Shenzhen, Dongguan and Huizhou. A tremendous variation in the degree of high-tech development among the Guangdong cities. We also find that cities closer to Hong Kong have larger percentage of high-tech firms and skill labour. We believe that trade openness of Guangdong through Hong Kong affects the high-tech development in the cities near Hong Kong. The possible reason is that Hong Kong has less import

restriction and have better access to high-tech intermediate goods from the world market. When trading with Hong Kong, cities near Hong Kong have lower transportation costs and enjoy tax benefits. Therefore, comparing with cities farther away from Hong Kong, cities near Hong Kong have better access to high-tech intermediate goods for high-tech development. Moreover, with lower transportation costs, high-tech products exporters near Hong Kong have cost advantages comparing with cities farther away from Hong Kong. There is a large literature studying the effect of globalisation and trade liberalisation in the developing countries. As Goldberg and Pavcnik (2007) summarises, globalisation can affect the demand of high skilled workers and skills premiums through several channels. Firstly, outsourcing or global production sharing result in increase in demand for high killed workers in developing countries. Trade openness allows developed countries to outsource the production of intermediate goods to developing countries, causing increase in demand for skilled labour. Feenstra and Hanson (1997) examine the impact of outsourcing on wage inequality in the case of Mexico while Hsieh and Woo (2005) study the case of Hong Kong. Secondly, prices of capital goods decline because of trade liberalisation and the import of capital goods increases. The increases of capital goods in the form of machines, office equipment and capital goods are accompanied by increases in skilled labours (Acemoglu, 2003; Pavani, 2003). Thirdly, the utilisation of international capital inflow increases the need for skilled workers. Cragg and Epelbaum (1996) and Behrman et al. (2000) provide support for this argument. Fourthly, exposure to world market induces the competitions. Firms avoid competitions by hiring more skilled labour or upgrading existing labour to improve firm productivity and product quality (Kaplanand Verhoogen, 2006; Melitz, 2003). These four channels all can be applied to Guangdong. Since the reform and opening up, Guangdong has been an important base of outward processing production and the largest supplier of Hong Kong. In turn, Hong Kong has shifted its low-value-added productions to Guangdong, and keep industries with greater skill content and with higher value-added. Though the production shifted to Guangdong are considered low value-added from the perspective of Hong Kong, but high value-added from the perspective of Guangdong. The production attracts labour flowing into Guangdong and at the same time raises the skills level of existing labour. The proximity to Hong Kong gives cities in Guangdong the advantage of being closed to world market, advanced access to foreign capital and cutting-edge production technology, which all increase the demand of skilled and unskilled labour.

4.3.1 Geography of High-tech Share in Guangdong

Guangdong is considered as an important trade hub of China. In 2019, Guangdong has imported RMB 407 billion worth of goods and services, while its exports have reached RMB 629 billion. These figures account for 19.5% and 25.1% of that year's national total, respectively. For Guangdong's exports, 24.9% consists of re-exports via Hong Kong. The United States (US) is Guangdong's top export destination, comprising 16.1% of the provincial total. The European Union (EU) follows closely behind as the second largest export destination, with 15.6% of the provincial total. Guangdong's top export destinations in Asia are Japan (3.9%), India (3.0%) and South Korea (2.7%). The top three import origins of Guangdong are the Association of Southeast Asian Nations (ASEAN) (18.8%), Taiwan (15.6%) and South Korea (11.2%). Guangdong's imports and exports are characterised by high-tech products. In 2019, Guangdong has exported $219.3 billion worth of high-tech products (34.8% of the provincial total), among which $160.1 billion are computer and communication-related products (25.4% of the provincial total). In the same year, Guangdong has imported $221.9 billion worth of high-tech products (54.5% of the provincial total), among which 34.1% were electronics and 12.2% were computer and communication products.

Guangdong cities with lower trade costs, as measured by their geographic distances to Hong Kong and other ports, have a higher percentage of high-tech industrial firms and skilled labour, according to the data of the Chinese Annual Survey of Industrial Firms. In this study, we categorise high- or low-tech firms by industry code,[1] then calculate the percentage of high-tech industrial firms for each city in Guangdong. Figure 4.1 shows that cities in the PRD have the highest percentage of high-tech industrial firms. In addition, high-tech percentage increases as the distance to Hong Kong decreases. Moreover, the high-tech industries spread eastwards instead of westwards. Figures 4.2 and 4.3 show that the city of Guangzhou was in the lead in terms of gross manufacturing output among other Guangdong cities, with predominantly labour-intensive industries. Guangzhou is the capital of and the most populated city in the province, as well as one of the three largest cities in China with various prestigious universities. Yet, Shenzhen, Dongguan and Huizhou, three smaller cities located to the east, surpass Guangzhou in the percentage of high-tech firms in the recent 10 years.

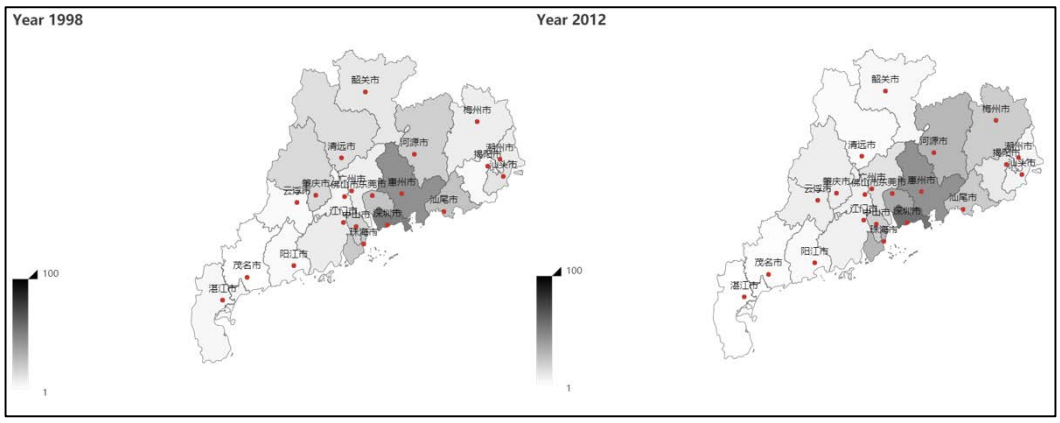


Figure 4.1 Percentage of high-tech industrial firms of cities in Guangdong

Source: Chinese Annual Survey of Industrial Firms

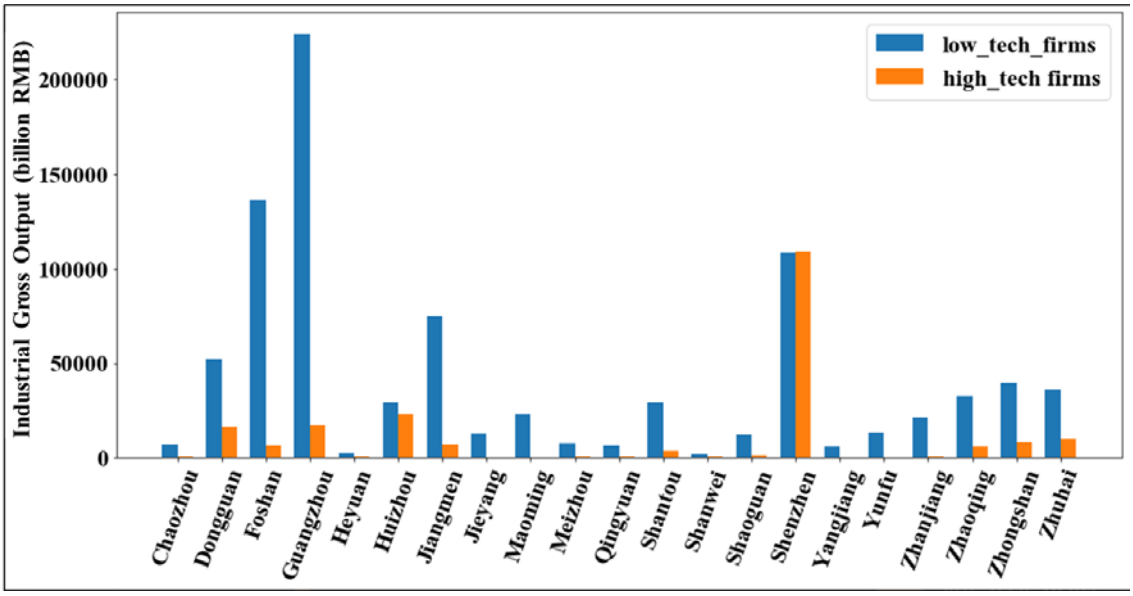


Figure 4.2 Industrial gross output produced by high- and low-tech firms located in cities in Guangdong in 1998

Source: Annual Survey of Industrial Firms (1998)

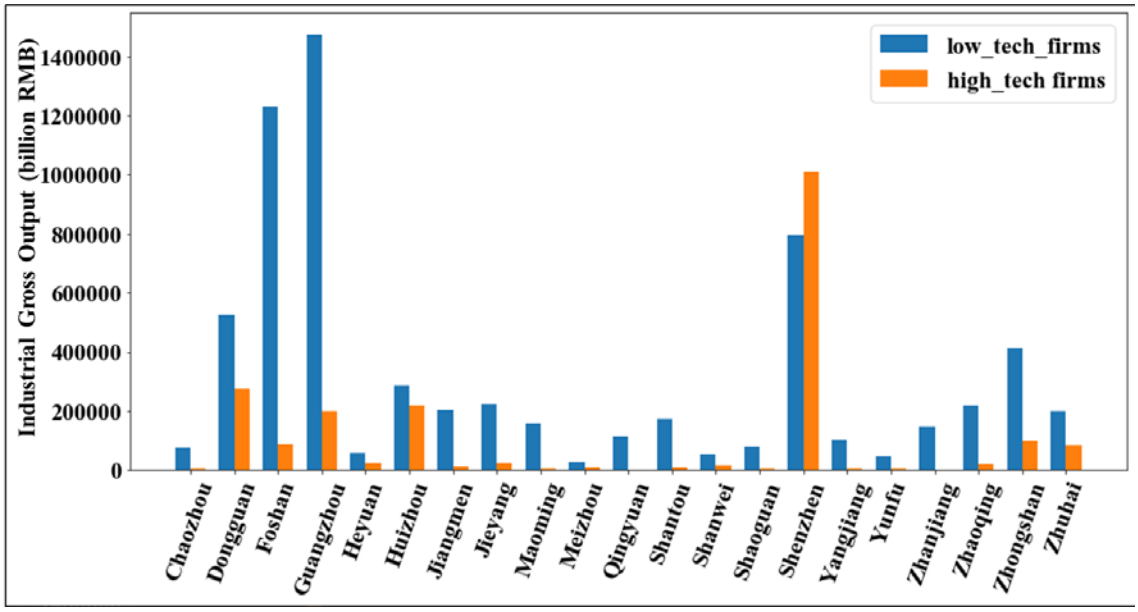


Figure 4.3 Industrial gross output produced by high- and low-tech firms located in cities in Guangdong in 2012

Source: Annual Survey of Industrial Firms (2012)

4.3.2 Comparison of Eastern and Western PRD

The PRD area is separated by sea into the eastern and western parts. The eastern part (Shenzhen, Dongguan, Huizhou) is dominated by foreign capital, boasting the largest computer and electronics industry cluster in China. The western part (Foshan, Zhuhai, Zhongshan, Jiangmen) is dominated by local private capital and specialises in manufacturing electrical machinery and equipment. Figure 4.4 shows the manufacturing composition by gross output value of the cities on Eastern PRD (upper four of the figure) and Western PRD (lower four of the figure). In the former group, the largest proportion of production is in communication, computers and electronic equipment, contributor to high-tech industrial output. Productions of the shown Western PRD cities have more diverse composition.



Figure 4.4 Manufacturing Composition by Gross Output Value in Eastern PRD (upper four) and Western PRD (lower four)

Source: Guangdong Statistics Yearbook 2019

4.4 A Comparison of Innovation Performance of the PRD and YRD

4.4.1 Number of Patents

The number of published invention patents is an essential indicator of scientific and technological innovation capability. This study evaluates the innovation performance of the Pearl River Delta (PRD) region by comparing the patent numbers of the provinces and independent municipalities covering the PRD, Yangtze River Delta (YRD), and Beijing-Tianjin-Hebei mega city-region. These are Guangdong, Jiangsu, Zhejiang, Shanghai, Beijing, and Tianjin provinces, where regional economies are relatively advanced in the Mainland. Figure 4.5 shows the number of published

invention patents every year from 1985 to 2016 of provinces and independent municipalities mentioned above. The published number of invention patent holdings in all regions have noticeable improvement since China joined the WTO in 2001. After the financial tsunami in 2008, the growth in number is even more rapid.[2] The top three regions, in terms of inventions published in 2015, were Jiangsu, Guangdong and Beijing. At that time, the published number in the YRD, namely Jiangsu, Zhejiang and Shanghai, is 2.3 times more than that of the PRD, which is Guangdong. Figure 4.6 shows that the number of new patents in the PRD reach approximately one-third to half of that in the YRD. The PRD has been one of the most technologically innovative areas in China, second only to the Jiangsu region, and yet lags in both newly granted and total patents.

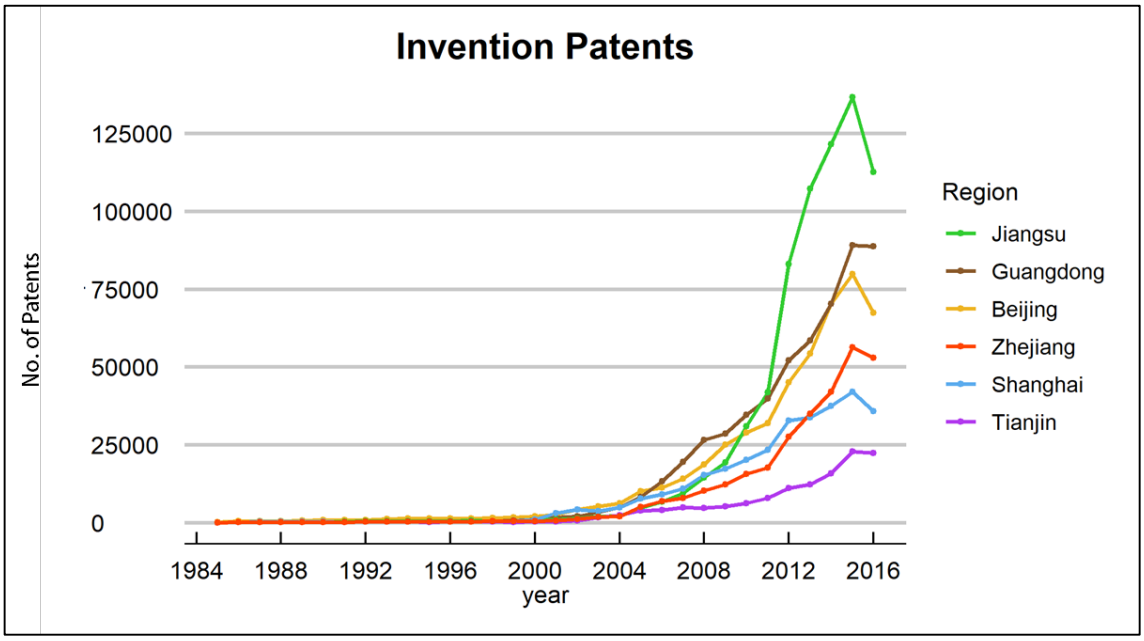


Figure 4.5 Number of published invention patent every year in regions of the PRD, YRD and Beijing-Tianjin-Hebei Urban Agglomeration from 1984 to 2016

Source: China Patent Database

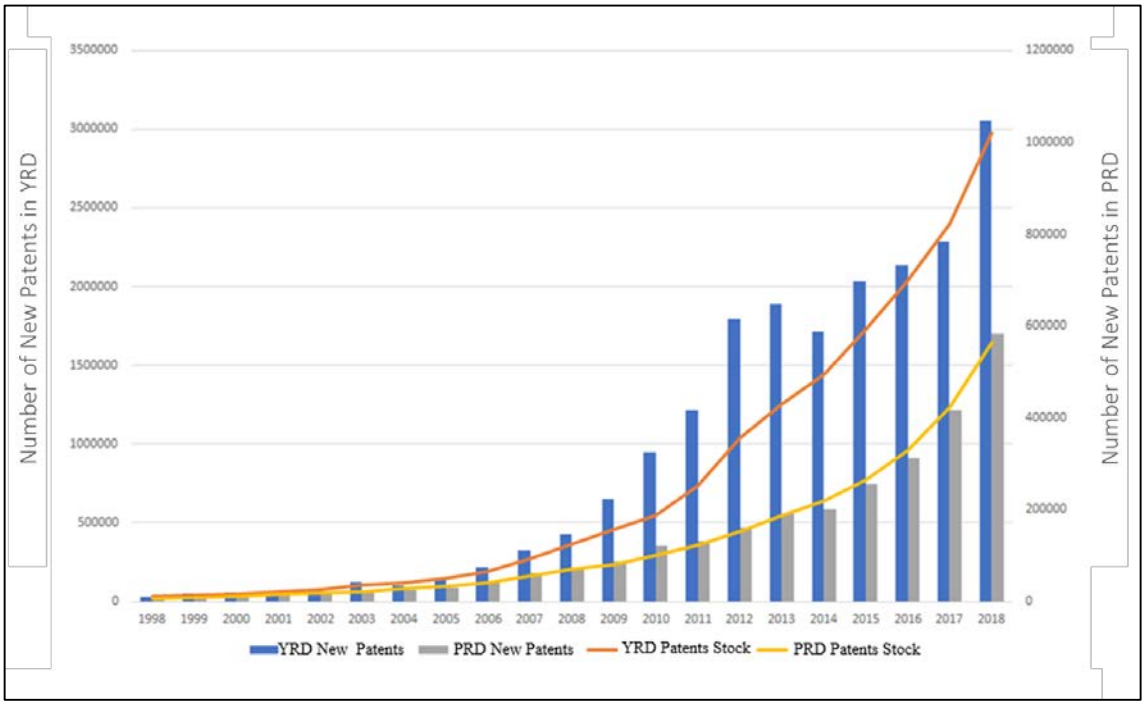


Figure 4.6 Number of new patents issued and patents stock in the YRD and PRD from 1998 to 2018

Source: China Patent Database

4.4.2 Patent Assignees

The composition patterns of regional invention patent assignees are also investigated. Two indicators, namely, shares of top five published patent applicant and Herfindahl-Hirschman Index (HHI)[3] are used to analyse the concentration of published invention patent applicants in the PRD, YRD and the Beijing-Tianjin-Hebei Urban Agglomeration regions.

Figures 4.7 and 4.8 show that the two indicators have comparable trends, and thus have a consistent depiction of concentration. The first graph of the total share of the top five applicants is used to illustrate the degree of published patent concentration in these regions. The concentration of applicants in Guangdong ranks second between 2003 and 2007, with the highest concentration in 2008-2014. After 2002, the concentration of Jiangsu is the lowest and indicates that Guangdong's innovation output is likely to come from a small number of enterprises, where Jiangsu has more applicants within institutions. The top five producers in Guangdong, Jiangsu, Shanghai, Zhejiang, Beijing and Tianjin in 1995, 2005 and 2015 are listed in Appendix B. The top five producers of major cities in Guangdong (Shenzhen, Guangzhou, Foshan, Dongguan) in 1995, 2005 and 2015 are listed in Appendix C.

Figure 4.9 shows that universities hold much fewer patents and the jointly held universities and enterprises in the PRD are likewise limited. Not only do universities apply for patents in the region start later, but its growth, although being steadily positive, is relatively slower.

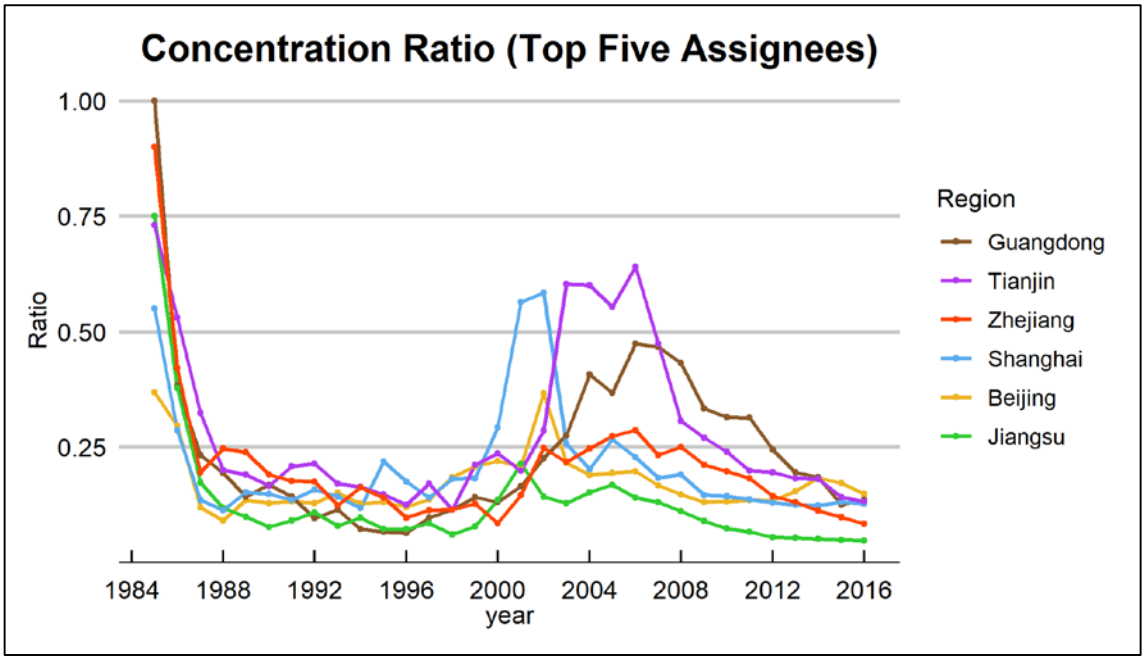


Figure 4.7 Applicant concentration of published invention patents in the PRD, YRD and Beijing-Tianjin-Hebei Urban Agglomerations from 1984 to 2016 - measured in accumulated percentage share of top five applicants

Source: China Patent Database

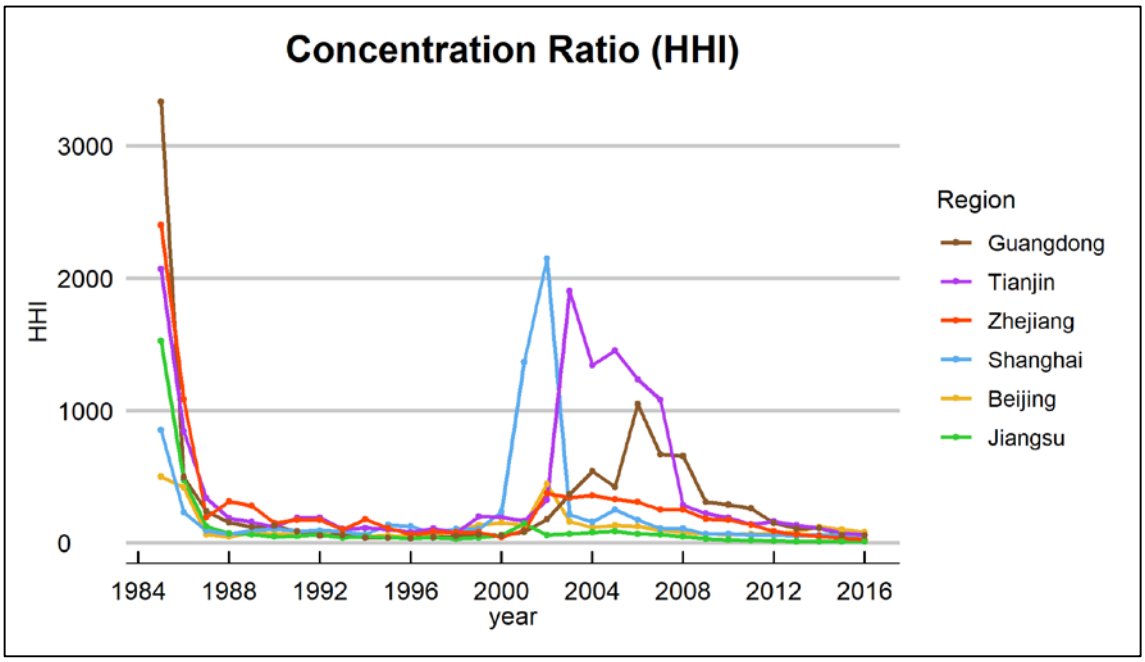


Figure 4.8 Applicant concentration of published invention patents in the PRD, YRD and Beijing-Tianjin-Hebei Urban Agglomerations from 1984 to 2016 - measured in HHI

Source: China Patent Database

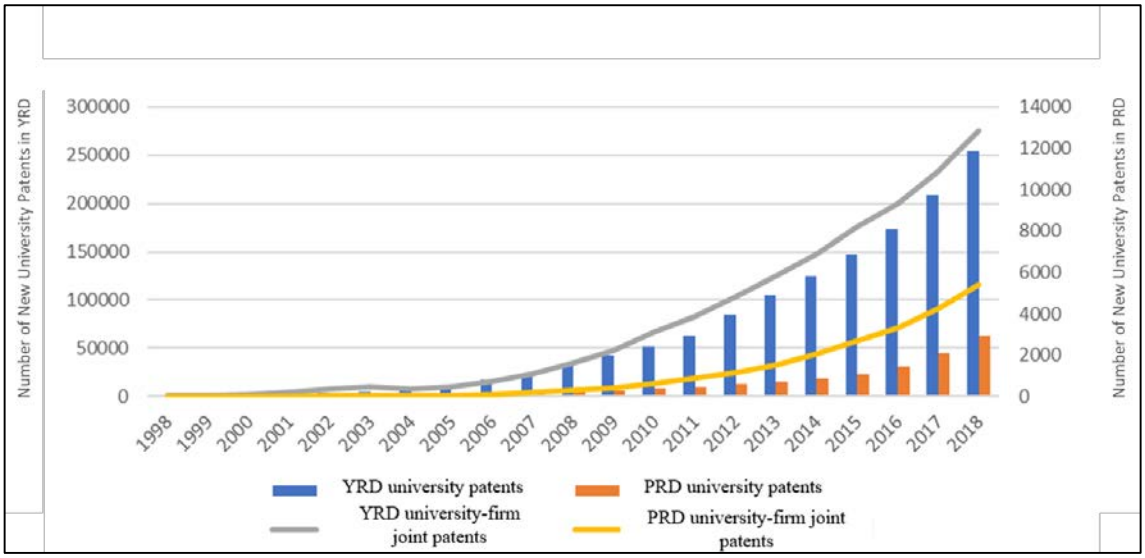


Figure 4.9 Number of university patents and university-firm joint patents issued in PRD and YRD from 1998 to 2018

Source: China Patent Database from 1998 to 2019

4.4.3 Industrial Distribution of Patents

Substantial potential risk occurs when the R&D concentrates in a few enterprises in Guangdong. When those enterprises with strong innovation capacity are under external shocks, the innovation core of the entire province is likely to be severely affected. The risk amplifies when applicants concentrate within a single industry. To further analyse the industry concentration of published invention patents, we match the Chinese patent database with that of industrial enterprises with designated size (1998-2013). Figure 4.10 shows the top 10 industries on the numbers of published invention patents in 1998-2013 in Guangdong. The Telecommunication Equipment Manufacturing industry has a much higher published number than that all other industries since 2000. Its growth rate was particularly rapid in 2005-2008, mainly driven from Huawei and ZTE in Shenzhen. The published number of Huawei Technologies Co., Ltd. proliferated from 1,700 patents in 2005 to 4,394 in 2008. ZTE Corp. increased from 2,359 to 6,212 between 2007 and 2008. Figure 4.11 shows the top 10 industries on numbers of published invention patents in 1998-2013 in Shenzhen, the top five industry matches precisely with those in Guangdong. The top five published patent number of Shenzhen is 88% that of Guangdong, clearly showing that the former drives the latter's innovation.

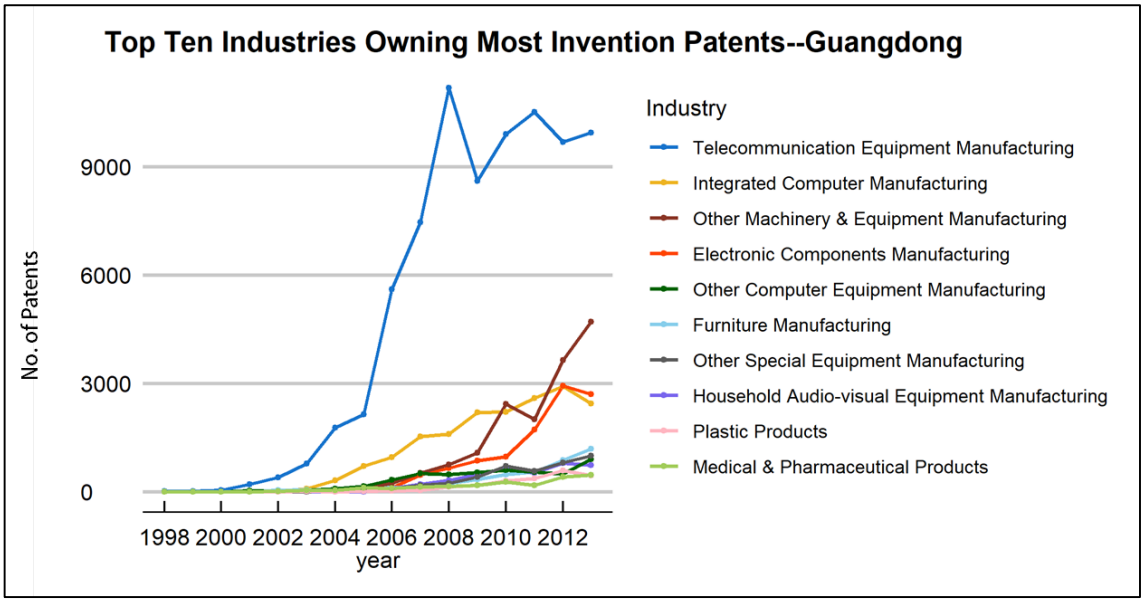


Figure 4.10 Number of published invention patents every year across the top 10 industries in Guangdong from 1998 to 2013

Source: China Patent Database and China Industrial Enterprise Database (1998-2013)

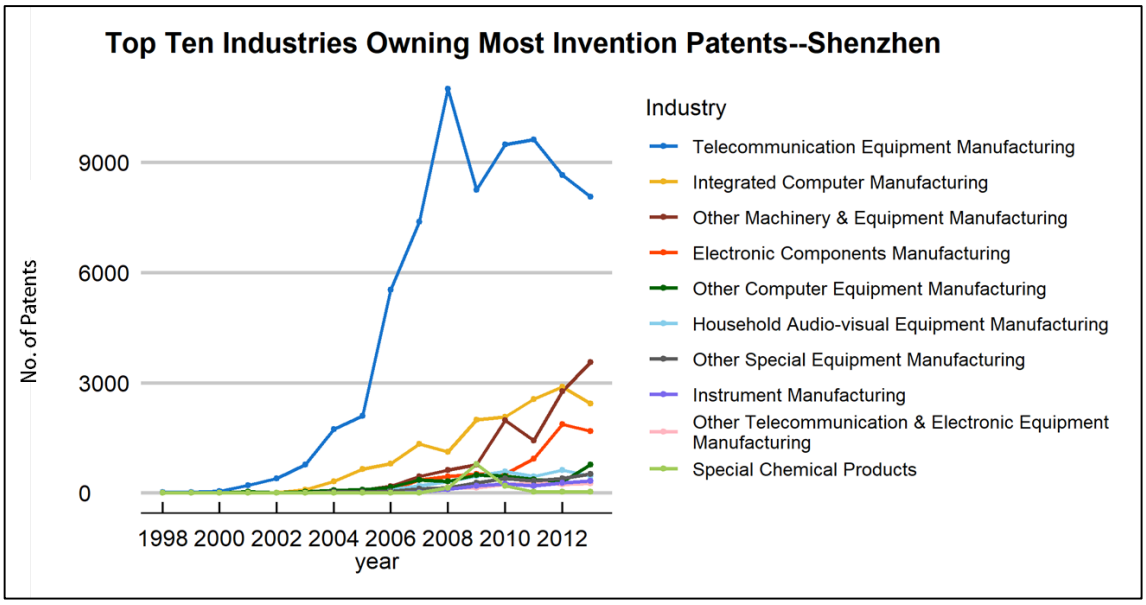


Figure 4.11 Number of published invention patents every year across the top 10 industries in Shenzhen from 1998 to 2013

Source: China Patent Database and China Industrial Enterprise Database (1998-2013)

The industrial concentration of published invention patent applicant across the PRD, YRD and Beijing-Tianjin-Hebei Urban Agglomeration regions on the top five shares are horizontally compared. Figure 4.12 shows the results. After 2001, the

industry concentration of Guangdong always ranked first or second. The top five industries of Guangdong accounted for over 80% of shares in 2003-2008. However, the sum of shares of the top five industries in Jiangsu, Zhejiang and Shanghai in YRD was consistently below 60% after 2002.

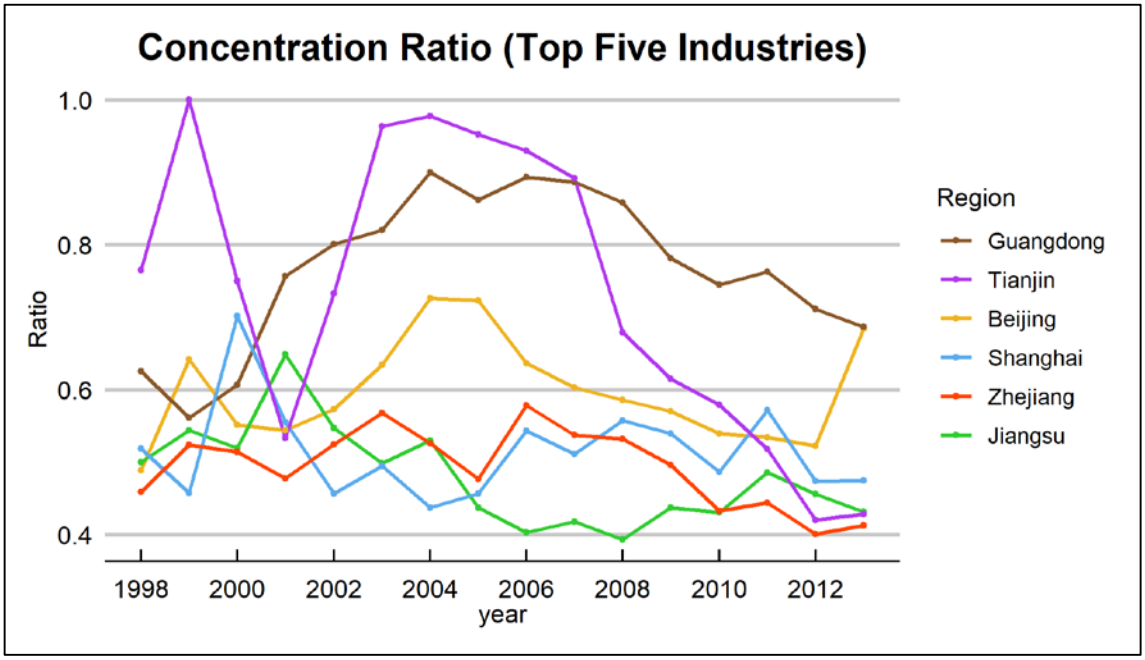


Figure 4.12 Industrial concentration of published invention patent applicants in the PRD, YRD and Beijing-Tianjin-Hebei Urban Agglomeration regions from 1998 to 2013 - share of top five applicants as the measure

Source: China Patent Database and China Industrial Enterprise Database (1998-2013)

Table 4.1 provides a list of the top five industries with the number of published invention patents and regional share through the year 2005 and 2013 across Guangdong, Jiangsu, Shanghai, Zhejiang, Beijing and Tianjin. The published number in Guangdong's top five industries in 2005 was as high as 86%, except for lower than the 95% from Tianjin, far exceeding other provinces and municipalities. In Guangdong, the proportion of the top five industries in 2013 was still the highest compared with the 68% of other regions. This finding reflects that Guangdong patents are concentrated not only in a few large enterprises but also in a few industries such as communications, computers and electronic components.

Table 4.1 List of the top five industries on numbers of published invention patents in the PRD, YRD and Beijing-Tianjin-Hebei Urban Agglomerations (2005, 2013)

Province/ Municipality	2005	2013
Guangdong	Telecommunication Equipment Manufacturing (2140; 57%) Integrated Computer Manufacturing (713; 19%) Other Computer Equipment Manufacturing (162; 4%) Other Machinery & Equipment Manufacturing (115; 3%) Medical & Pharmaceutical Products (114; 3%)	Telecommunication Equipment Manufacturing (9939; 32%) Other Machinery & Equipment Manufacturing (4708; 15%) Electronic Components Manufacturing (2707; 9%) Integrated Computer Manufacturing (2447; 8%) Household Appliance Manufacturing (1198; 4%)
Jiangsu	Medical & Pharmaceutical Products (121; 13%) Other Computer Equipment Manufacturing (101; 11%) Telecommunication Equipment Manufacturing (68; 7%) Other General Equipment Manufacturing (64; 7%) Synthetic Materials Manufacturing (46; 5%)	Other Special Equipment Manufacturing (3444; 11%) Other Machinery & Equipment Manufacturing (3201; 10%) Other General Equipment Manufacturing (3088; 10%) Electronic Components Manufacturing (2270; 7%) Instrument Manufacturing (1339; 4%)
Shanghai	Electronic Components Manufacturing (296; 19%) Household Audio-Visual Equipment Manufacturing (138; 9%) Telecommunication Equipment Manufacturing (114; 7%) Steel Rolling Processing Industry (83; 5%) Medical & Pharmaceutical Products (65; 4%)	Electronic Components Manufacturing (2072; 22%) Other Special Equipment Manufacturing (730; 8%) Other General Equipment Manufacturing (593; 6%) Other Machinery & Equipment Manufacturing (509; 5%) Electricity, Heat Production and Supply Industry (506; 5%)
Zhejiang	Medical & Pharmaceutical Products (118; 18%) Other General Equipment Manufacturing (66; 10%) Telecommunication Equipment Manufacturing (49; 8%) Other Special Equipment Manufacturing (41; 6%) Other Machinery & Equipment Manufacturing (35; 5%)	Other General Equipment Manufacturing (1468; 12%) Other Machinery & Equipment Manufacturing (1107; 9%) Other Special Equipment Manufacturing (1051; 9%) Telecommunication Equipment Manufacturing (720; 6%) Electronic Components Manufacturing (718; 6%)
Beijing	Integrated Computer Manufacturing (341; 38%) Telecommunication Equipment Manufacturing (146; 16%) Medical & Pharmaceutical Products (60; 7%) Other Special Equipment Manufacturing (55; 6%) Electronic Components Manufacturing (47; 5%)	Electricity, Heat Production and Supply Industry (5346; 39%) Electronic Components Manufacturing (1329; 10%) Integrated Computer Manufacturing (1253; 9%) Other Special Equipment Manufacturing (937; 7%) Telecommunication Equipment Manufacturing (586; 4%)
Tianjin	Household Appliance Manufacturing (1621; 90%) Basic Chemical Raw Material Manufacturing (28; 2%) Petroleum and Natural Gas Exploitation Industry (23; 1%) Medical & Pharmaceutical Products (20; 1%) Other Special Equipment Manufacturing (19; 1%)	Medical & Pharmaceutical Products (519; 14%) Other Special Equipment Manufacturing (383; 10%) Other Machinery & Equipment Manufacturing (278; 8%) Other Food Processing and Food Manufacturing (227; 6%) Other General Equipment Manufacturing (179; 5%)

Source: China Patent Database and China Industrial Enterprise Database (1998-2013)

Figure 4.13 shows the distribution of patents in the two delta regions according to industrial categories. Despite having the largest number of patents in both regions, the group ‘electrical machinery and equipment’ accounts for merely 14% in YRD while over 30% in PRD. The patents of the former are more widely distributed to different groups, including ‘General Equipment Manufacturing’, ‘Special Equipment Manufacturing’, ‘Metal Products’ and ‘Communication, Computers and Other Equipment’, each comprising approximately 10% of the total. By comparison, the industrial distribution of patents in PRD is apparently less diverse. There seem to be concentrations of R&D activities in the PRD on a few enterprises in selected industries (e.g., Huawei and ZTE). While such concentration may benefit from agglomeration and regional specialisation, it may lead to issues of over-reliance on such firms and industries.

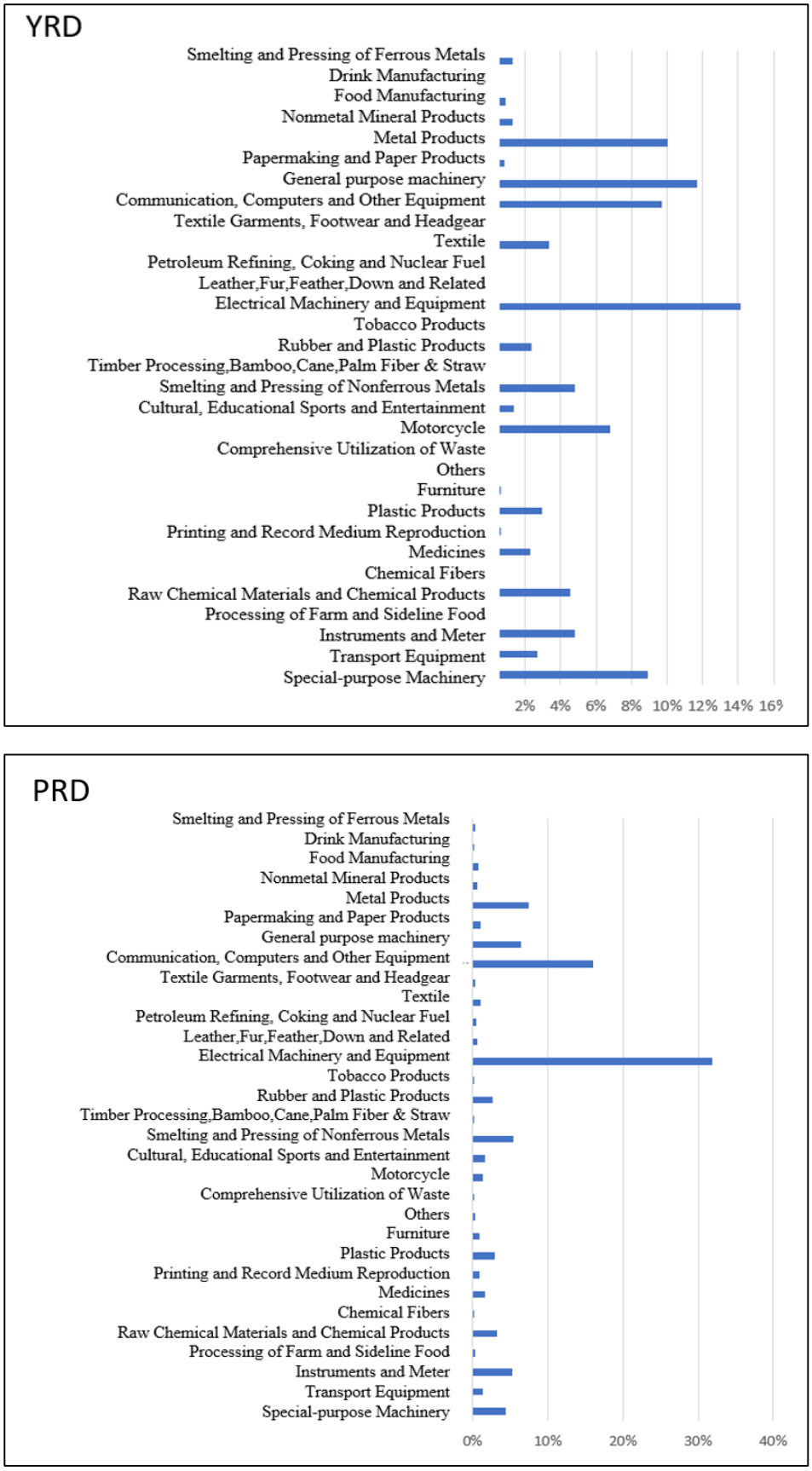


Figure 4.13 Distribution of patents by industries in the YRD and PRD

Source: China Patent Database and China Industrial Enterprise Database (1998-2013)

4.5 Hong Kong Innovation Capability

4.5.1 World Ranking of Hong Kong Universities

Universities are the bases and foundations of R&D activities, the hub of innovation. Hong Kong has four of the top 100 universities in the Quacquarelli Symonds (QS) World University Rankings 2019 (Innovation and Technology Bureau, HKSAR Government, 2018), with a high level of higher education and strong scientific and technological innovation capabilities. Guangdong has a strong manufacturing base and is known as the 'world factory' that can be used to carry out research and cooperate with Hong Kong universities. This cooperation can enhance rapid production industrialisation, and together with other cities in the GBA, forming a 'scientific research - outcome transformation - product development' chain of the scientific innovation system. Hong Kong can be established as a centre on attraction, cultivation and transportation of talents, levelling up its R&D capability and building the foundation for the GBA development.

The relationship between the quality of universities and promotion of innovation has been widely investigated. The productivity at academic institutions may be significantly affected by the design of intellectual property rights and associated incentive schemes (Lach and Schankerman, 2008). Government policies must target at facilitating more efficient licensing in those specific technology areas to promote innovation incentives (Galasso and Schankerman, 2015). Sharon Belenzon (2007) studied the incentive effects of the staff salary of university technology transfer offices and that of local development goals by the university patent licensing performance. The technology transfer office from private universities has higher chances to use incentive pay than public universities to incentivise their staff. With incentive pay, universities can generate 30%-40% more revenue per license.

In addition, universities with stronger local development goals have a 30% reduction in revenue per license but are more likely to be recognised by local (within state) start-ups. Universities can encourage researchers' involvement in innovation through three major directions.

Firstly, lower the rate of charge (Figure 4.14) for funded activities to increase the profits. Secondly, provide support to help university researchers protect the property rights of their innovations and offer more flexible choices on patent ownership. Thirdly,

aid researchers in securing better commercial value for their innovations through advertising and promotion. In short, local universities have a vital role in maximizing the innovation capability of Hong Kong by providing better incentives to university researchers.

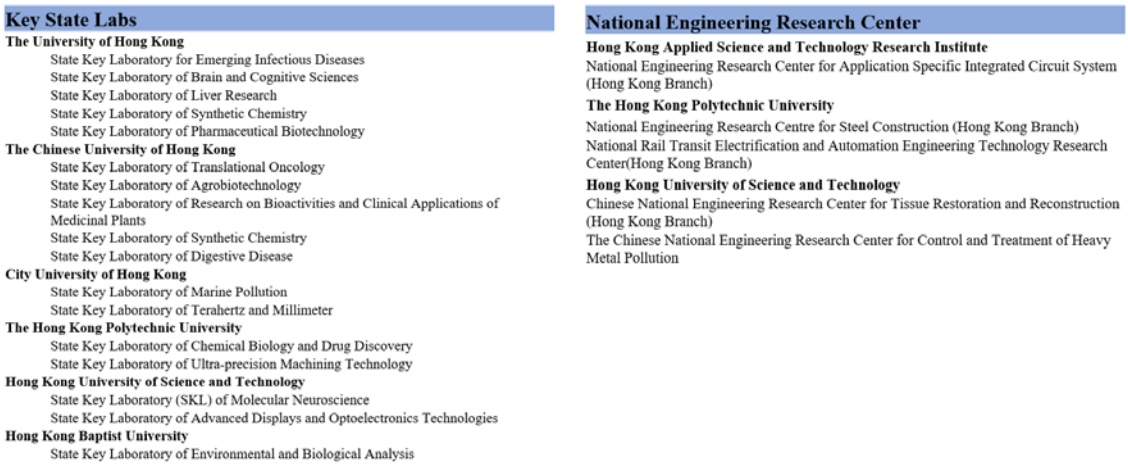
Name of Univesity	Non-UGC-funded Research grants/contracts
The Unviersity of Hong Kong	15%
The Chinese University of Hong Kong	15%-25%
The Hong Kong Unviersity of Science and Technology	15%
City University of Hong Kong	15%-16.5%
The Hong Kong Polytechnic University	15%
Hong Kong Baptist University	15%

Figure 4.14 Recovery rate for Non-UGC-funded research grants/contracts in major universities in Hong Kong

Source: Websites of the major unviersiteis

4.5.2 Key Research Institutes in Hong Kong

Remarkably, 16 State Key Laboratories are located in Hong Kong (as shown in Figure 4.15), whereas only 11 are in the whole of Guangdong Province. Meanwhile, the Chinese National Engineering Research Centres has six branches in Hong Kong and 23 in Guangdong. These numbers demonstrate Hong Kong's research capability, which can be of great assistance to the PRD in moving up the value chain and diversifying into other industries such as materials and biomedical sciences. By becoming a centre of technological innovations for this region (second leg for Hong Kong economy) and helping firms there to thrive, Hong Kong can also maintain its leading position in producer services, in particular the financial services (first leg for Hong Kong economy).



Key State Labs

The University of Hong Kong
- State Key Laboratory for Emerging Infectious Diseases
- State Key Laboratory of Brain and Cognitive Sciences
- State Key Laboratory of Liver Research
- State Key Laboratory of Synthetic Chemistry
- State Key Laboratory of Pharmaceutical Biotechnology

The Chinese University of Hong Kong
- State Key Laboratory of Translational Oncology
- State Key Laboratory of Agrobiotechnology
- State Key Laboratory of Research on Bioactivities and Clinical Applications of Medicinal Plants
- State Key Laboratory of Synthetic Chemistry
- State Key Laboratory of Digestive Disease

City University of Hong Kong
- State Key Laboratory of Marine Pollution
- State Key Laboratory of Terahertz and Millimeter

The Hong Kong Polytechnic University
- State Key Laboratory of Chemical Biology and Drug Discovery
- State Key Laboratory of Ultra-precision Machining Technology

Hong Kong University of Science and Technology
- State Key Laboratory (SKL) of Molecular Neuroscience
- State Key Laboratory of Advanced Displays and Optoelectronics Technologies

Hong Kong Baptist University
- State Key Laboratory of Environmental and Biological Analysis

National Engineering Research Center

Hong Kong Applied Science and Technology Research Institute

National Engineering Research Center for Application Specific Integrated Circuit System (Hong Kong Branch)

The Hong Kong Polytechnic University

National Engineering Research Centre for Steel Construction (Hong Kong Branch)

National Rail Transit Electrification and Automation Engineering Technology Research Center(Hong Kong Branch)

Hong Kong University of Science and Technology

Chinese National Engineering Research Center for Tissue Restoration and Reconstruction (Hong Kong Branch)

The Chinese National Engineering Research Center for Control and Treatment of Heavy Metal Pollution

Figure 4.15 List of Key State Laboratories and National Engineering Research Centres in Hong Kong

Source: University websites, compiled by the researchers

In terms of the R&D activities undertaken by research institutions, the innovation capability of Hong Kong can be amply demonstrated. In 2008, the total domestic R&D expenditure has grown over the years with a rising rate of increase. Since then, higher education sector has exceeded the business sector, becoming the most important R&D performing sector in Hong Kong. Figure 4.16 shows that the total domestic R&D reached 19,713 million HK dollars by 2016. Although its role of performing R&D is the smallest, the government shows the fastest growth, as per Figure 4.17. The R&D expenditure from government in 2016 is 914 million HK dollars, which is approximately 3.25 times of its expenditure in 2005. These figures demonstrate the importance and potential of research institutes in engaging in R&D.

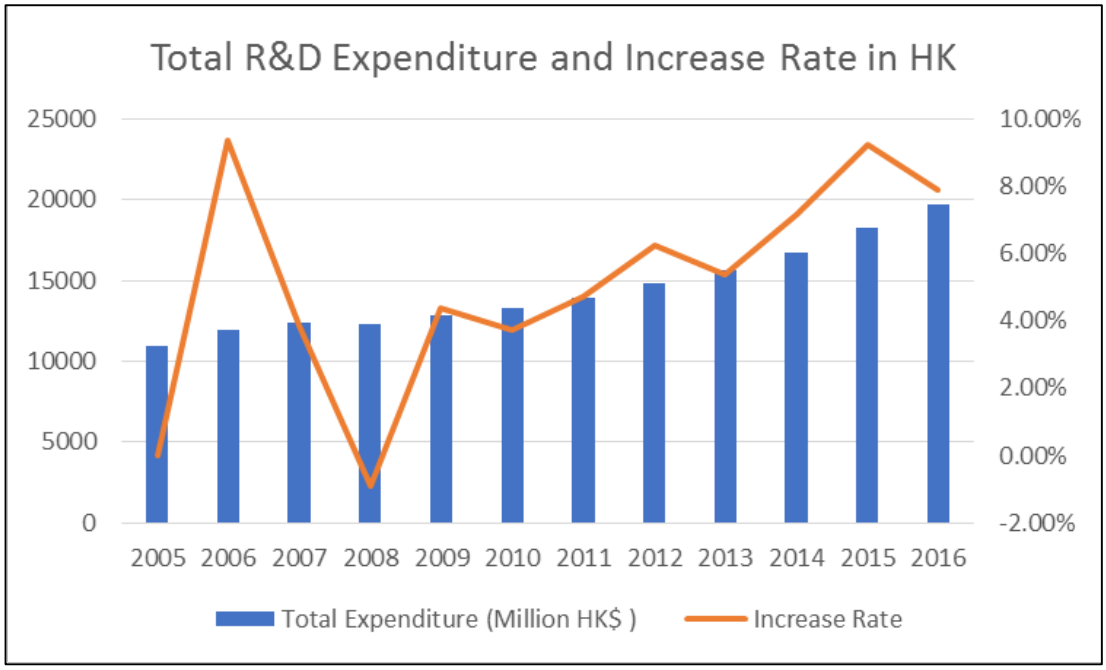


Figure 4.16 Total R&D Expenditure and Increase Rate in Hong Kong from 2005 to 2016

Source: C&SD Hong Kong SAR

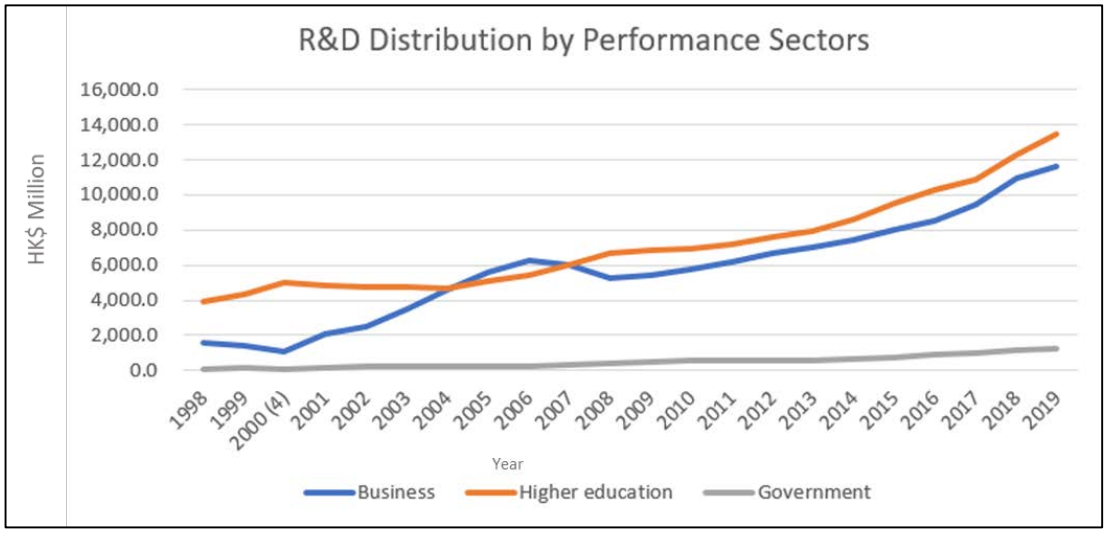


Figure 4.17 Distribution of R&D performed by different sectors from 2005 to 2016

Source: C&SD Hong Kong SAR

4.6 Summary

Our data analysis reveals a tremendous variation in the degree of high-tech development among the Guangdong cities. Cities in the Eastern PRD have a higher percentage of high-tech industrial firms. Moreover, the percentages of high-tech industries increase as the distance to ports and Hong Kong decreases.

The reason that the Eastern PRD has larger high-tech firms may be its proximity to Hong Kong, one of the most important trade hubs. These cities have better access to and can import high-tech material from the world markets. Therefore, cities in other parts of PRD with better geographical connections with Hong Kong can have more access to world markets. Several major bridges connecting Western and Eastern PRD are already in place or in plans to be built, including the Hong Kong-Zhuhai-Macao Bridge and the Shenzhen-Zhongshan Bridge. These connections are advantageous to Hong Kong, bringing down the transportation costs among cities and facilitating the flow of both goods and people. Moreover, the cooperation within the city clusters in the GBA can be enhanced.

By comparing the innovation performance between PRD and YRD, we reveal the following findings. Firstly, PRD lags behind YRD in both newly granted patents and total patents. Secondly, patents held by universities and those jointly held by universities and enterprises are much fewer in PRD. Thirdly, patents in PRD are more concentrated within and across industries. The concentration of PRD R&D activities in a few industries and in a few enterprises (such as Huawei, ZTE and Foxconn) may yield benefits associated with industrial agglomeration and regional specialisation, but may also lead to the danger of over-reliance on a few firms within a few industries, the success of which depends on cooperation from other regional and national economies. The recent sanction on ZTE and Huawei by the US government is a warning that must, be taken seriously. Meanwhile Hong Kong has several of the best universities in Asia that can help PRD move up the value chain and diversify into other industries such as materials sciences and biomedical sciences. Becoming a centre of technological innovations for the PRD (second leg for Hong Kong economy) and helping firms there to thrive can also help Hong Kong to maintain its leading position in producer services, in particular the financial services (first leg for Hong Kong economy).

End Notes

1 In 2002, the National Bureau of Statistics of China released a list of industry codes for industries that were considered high-tech. The high-tech industry is defined as an industry with a high R&D intensity, measured by the percentage of R&D expenses to revenue. The manufacturing of medicines, aeronautics, communication equipment, computers and other electronic equipment, medical equipment, instruments and meters are major high-tech industries on the list.

2 The slight decrease in the number of published invention patents in 2016 was due to incomplete collection of data collected in that year (data are only available until the announcement date before October 19).

3 The Herfindahl-Hirschman index refers to the sum of squares of the various market entities' total revenue shares or total asset shares, and is used to measure market competition, concentration, or dispersion of market entities given market shares. For example, in this study, the calculation method is the sum of squares of published invention patent shares on publications number of each applicant by the year. Value ranges in [0, 10000]. The greater value represents less competition and higher concentration in the market.

Chapter 5
Hong Kong's Role in High-tech Industrial Development in the Pearl River Delta: Evidence from Firm-Level Questionnaire Survey and In-depth Interviews in Dongguan and Shenzhen

5.1 Introduction

The Pearl River Delta (PRD) has been one of the most burgeoning regions in industrialisation and urbanisation in China and over the world since the late 1970s (Lin, 1997; Sit and Yang, 1997; Yeh, 2001), when the economic reform and opening-up policy was initiated to endow it with the 'one step ahead' advantage (Vogel, 1989). Together with rural reforms and product commoditisation in the 1980s, foreign capital, technology and information primarily from Hong Kong was highly engaged in PRD's rural industrialisation and modernisation, which was labelled as the widely-known 'Exo(genous)-Urbanisation' model (Sit and Yang, 1997, p. 647) or the 'PRD' model (Wei et al., 2009, p. 410). Benefiting from geographical proximity and social affiliations (e.g. pre-existing kinships, inter-personal trust and connections) (Lin, 2006), the PRD stood out as an ideal destination for manufacturing activities relocating from Hong Kong as the latter began to suffer from rising costs and shortage of land in the 1970s (Yeh, 1997). Consequently, a successful regional collaboration model named 'Front Shop, Back Factory' (FSBF) emerged (Shen, 2003), under which Hong Kong transformed into an international financial and trade centre and the PRD grew into a 'World Factory' (Yang and He, 2017).

Since the mid-2000s, however, the development model of the PRD that over-relies on low cost, seemingly limitless cheap rural surplus labour and robust overseas markets in developed countries has been increasingly struggling against the changing global economic conditions and stringent local institutional regulations (Zhou, 2015; Xue and Wu, 2015). Firstly, after economic development at a breakneck rate mainly driven by intensive input of various production and natural resources for almost three decades, the PRD now faces a deteriorating environment, shortage of land for construction, and ascending cost of labour and raw materials (Yang, 2012). Secondly, the global financial crisis rendered the downturn of overseas demand for Chinese products particularly in North America and Western Europe, which exerts much pressure on labour-intensive and export-oriented firms in the PRD competing on price instead of quality and relying on purchase orders from foreign companies (Xue and Wu, 2015; Zhu and Lan, 2016). Thirdly, a series of institutional adaptations were implemented to deal with the

sustainable development challenges steered by the state, including the following: cancellation or reduction of an export Value Added Tax rebate; expanded prohibited categories for processing and trade products, particularly in Eastern coastal regions; rigorous pollution-control requirements; and the implementation of the new Labour Contract Law stipulating more responsibilities of employers (Yang, 2012; Sharif and Huang, 2019). Against these developments, the PRD has no choice but to switch onto a new development path fuelled by higher value-added economic activities and moving towards a higher-end position within the global division of labour (Yeung and Lin, 2003; Yang, 2020).

The FSBF model, therefore, has been gradually ineffective in coping with this new wave of economic transition towards a knowledge- and technology-intensive economy of the PRD and Hong Kong (Yeh and Xu, 2006; Yang, 2020). The relocation of manufacturing activities from Hong Kong to the PRD had been almost completed by the mid-1990s (Tuan and Ng, 1995), and the subsequent focus of inter-regional collaboration since the handover of Hong Kong in 1997 and the execution of Mainland and Hong Kong Closer Economic Partnership Arrangement (CEPA) in 2003 unfolded into service investment and trade associated with increasingly intensive social interactions and traffic and information flows (Yang, 2006; Liu et al., 2016). On the one hand, the manufacturing plants invested by Hong Kong in the PRD progressively lost its competitive advantage due to both the failure of industrial upgrading in the home region (Yang, 2007) and continuously fierce competition from diversified sources of foreign capital and recently soaring domestic economic power (Lin, 2006; Yang, 2020). On the other hand, producer services, in which Hong Kong enjoy superior performances, have also been growing dramatically in the PRD to embrace the considerable market opportunities yielded by the economic transformation (Yeh et al., 2015b; Yang et al., 2018). As a consequence, the function of both the 'Shop' and the 'Factory' has weakened.

In response to the challenges facing the PRD and its collaboration with Hong Kong, the central government initiated the GBA (e.g. nine municipalities in the PRD and two SARs of Hong Kong and Macao) regional strategy in 2017 to better facilitate and coordinate new collaboration models among the places for developing a world first-class bay area, ideal for working, living and travelling (Lin, 2017). This strategy led to the promulgation of the *Outline Development Plan for Guangdong-Hong Kong-Macao Greater Bay Area* by the State Council in February 2019. The technological innovation,

institutional adjustments, and a new round of high-level opening-up are argued to be the fresh underpinnings for the GBA. Hong Kong has to move beyond as an investor and outsourcer by extending itself as a talent provider, technology producer and a 'window' for two-way opening-up for the PRD. Though critics of the expired collaboration FSBF model and assumptions of future collaboration models have proliferated in recent years (Shen and Luo, 2013; Yeh and Chen, 2020; Zhong and Su, 2019; Yang, 2020), little is known about the industrial and technological linkages between the technology-intensive manufacturing industries in the PRD and Hong Kong and what new potential collaboration models can be forged against the new wave of industrial upgrading under the GBA initiative (Li et al., 2021). Given this context, this study seeks to understand: (1) High-tech industrial development in the PRD and the reasons for its capability to move upward to such development; (2) Characteristics of high-tech industrial firms in the PRD by random sampling; (3) Hong Kong's participation in high-tech industrial development in the PRD; and (4) Views of high-tech industrial firms on Hong Kong roles.

5.2 Literature Review

5.2.1 Economic Development and Restructuring: Evolution of the Hong Kong-PRD Collaboration Models

From the historical perspective, the Hong Kong-PRD collaboration model can be separated into five stages with distinctive defining features. Prior to 1978, the two places had few interactions. After the Second World War, Hong Kong, as a British colony with the influx of refugees from China, rapidly developed into an industrial and export hub in Asia with labour-intensive sectors such as garments, footwear and electronic goods (Yeh and Xu, 2006). By contrast, the PRD was much less engaged in external associations and more self-sufficient under state socialism and central planning economy (Lin, 1997).

The economic reform and opening-up policy in 1978 set out new directions for the Hong Kong-PRD relationships (Sit and Yang, 1997; Yeh, 2001). After rapid industrialisation for nearly three decades, Hong Kong eagerly expected to turn up new space for continuous capital accumulation and indigenous industrial upgrading. In the neighbouring PRD, the 'first-move advantage' in opening-up in socialist China (Vogel, 1989) together with its abundant rural surplus labour, cheap land prices, flexible

business environment and half-hearted regulations, jointly made a favourable place for manufacturing activities in Hong Kong to lower cost and expand profit for higher value-added operations (Yeh, 1997). Cities that are closest to Hong Kong, such as Shenzhen and Dongguan, were the pioneers of embracing the influx of Hong Kong investments primarily with the entry form of processing and trade (i.e., *sanlai yibu*) to step onto rapid export-oriented industrialisation pathways (Lin, 2006; Yang, 2007). The FSBF model thus formed during this time period and lasted for almost two decades.

After Hong Kong's handover to China in 1997, the Hong Kong-PRD relationships have entered into a new phrase characterised with increasingly intensive service and social interactions beyond the industry domain alone. However, the FSBF model alarmingly became less effective and Hong Kong-PRD collaboration became more ambiguous and challenging in the 2000s (Yang, 2006; Shen and Luo, 2013). In particular, with the accomplishment of industrial relocation, a new round of transfer of low value-added services industries began to occur between Hong Kong and the PRD. Since the entrance of the World Trade Organisation (WTO) of China in 2001, the alluring massive domestic market instead of cost savings emerged as a more significant factor for foreign investment in the PRD (Yang, 2007; Zhou et al., 2011). Meanwhile, foreign direct investment (FDI) in the PRD, which heavily relied on Hong Kong in the 1980s and 1990s, had become more diversified, with sources extending into other 'Asian Tigers', Japan, and the US. (Lin, 2006; Z. Xu and Yeh, 2013). Along with the continuous injection of FDI from other sources, the domestic capital accumulation has also played an increasingly important role since the 2000s (Zhou, 2015; Yang, 2020). For example, Shenzhen has been experiencing sustained industrial upgrading and diversification driven by both FDI and domestic technology advancements, witnessing the emergence of domestic technological giants such as Huawei and ZTE in the electronics sector (Wang and Lin, 2013; Yang, 2015) and Tencent in the Internet sector (Coe and Yang, 2021). Since the mid-2000s, traditional labour-intensive and small-scale manufacturing firms had to further relocate to the Pan-PRD regions due to rising costs and institutional reforms (Liao and Chan, 2011; Yang, 2012). High value-added and technology-intensive firms have been transforming into more specialised and agglomerated around Shenzhen as a high-tech hub (Zhou et al., 2011; Yang, 2015). During this stage, amidst continuous economic transition and industrial upgrading, the role of Hong Kong declined and even disconnected to a certain extent with the production complex in the PRD (Yeh et al., 2015b). Shenzhen, the domestic innovation

hub, had taken off and exerted greater influence in the PRD through knowledge spillovers, entrepreneurship, industrial relocation and various forms of inter-firm linkages (Wang and Lin, 2013; Yang, 2020).

Over the past decade, under the dramatic impacts of the 2008 global financial crisis and severe challenges for regional sustainable development, the relationships between Hong Kong and the PRD have increasingly headed for new directions (Yeh and Xu, 2006; Shen and Luo, 2013; Zhou, 2015; Zhong and Su, 2019). From the Hong Kong side, two new avenues are repeatedly emphasised for future development and competitiveness construction, namely, the science- and technology-based development and a market re-direction into the domestic Mainland market of China under the 'dual circulation' national strategy. The HKSAR Government aims to deliver more support in R&D expenditure for firms, universities and public institutes, particularly in the cutting-edge technological domains (e.g. ICT, Artificial Intelligence, Bio-tech, and Robotics) that enjoy competitive advantage in research capacity, technological excellence and talent cultivation. Equally important, firms are being given aids for expanding their investments and market shares in Mainland China to overcome institutional and border obstacles under the 'One Country, Two Systems' framework. From the PRD side, industrial restructuring has been accelerated by policy initiatives such as the 'dual transfer', 'empty the cage for new birds', 'three olds transformation' and 'machine replaces manpower', under which labour-intensive industries are encouraged to relocate from the core PRD area to the relatively peripheral region (e.g. northern, western and eastern Guangdong province and also middle and western parts of China) (Xue and Wu, 2015; Yang and He, 2017; Sharif and Huang, 2019). High-tech and high value-added economies (i.e. new 'birds') are proactively fostered and attracted to fill into the 'cage' (i.e. the PRD) (Yang, 2012; Xue and Wu, 2015; Sharif and Huang, 2019). Market re-orientation is also taking place for firms in the PRD from the developed overseas market to the developing domestic market and emerging markets, especially in countries along the 'Belt and Road' (e.g. Southeast Asia and Africa) (Liu and Dunford, 2016; Yang, 2020). Different from the gradually ambiguous and increasingly competitive relationships in the previous stage, Hong Kong and the PRD are re-directing themselves to search for new opportunities for future collaborations and synergies. How new collaboration models between Hong Kong and the PRD in high-tech industrial fields, knowledge-intensive producer service domains, and market re-orientation to materialise remains to be seen (Figure 5.1).

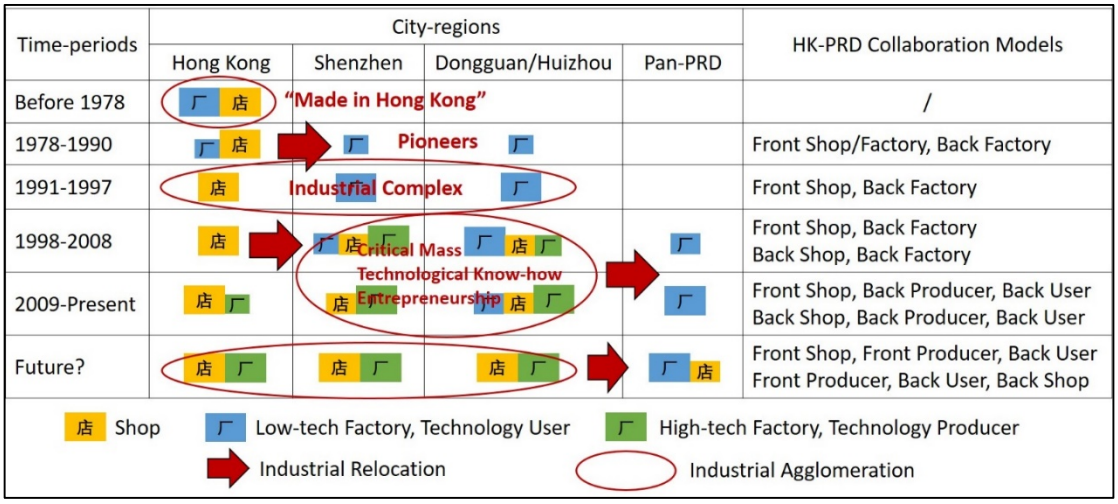


Figure 5.1 Evolution of the PRD-Hong Kong collaboration models

5.2.2 Evolutionary Industrial Development: Using Dongguan as an Example

With Dongguan as an example, a city that has been highly influenced by Hong Kong over the past four decades, the theoretical framework of evolutionary industrial development can be used to gain a better understanding of the industrial dynamics in the PRD and multi-scalar contributing actors associated with but not limited to Hong Kong. Scholars in evolutionary economic geography categorised industrial evolution into different types of path development (Isaksen and Jakobsen, 2017). For example, path renewal results from the recombination of local existing knowledge and skills with other local or non-local knowledge that empower entrepreneurship and innovation (Boschma and Frenken, 2011). Path creation refers to the growth of entirely new industries for a region or country that normally benefits from R&D activities or the introduction of external novel knowledge. Notably, industrial evolution in Dongguan may be driven by a hybrid of local endogenous branching (Boschma, 2017), importation of non-local knowledge and skills (Saxenian, 2006) and the combination of local and non-local dynamics (Yang, 2020).

Given the historical and geographical context of the PRD and Hong Kong, Dongguan was a typical case for 'Exo(genous)-Urbanisation' during the early reform period (Sit and Yang, 1997). In the wake of sustaining capital accumulation and institutional adaptation, sources of capital became more diversified and territorial embeddedness of industrial activities deepened (Yang and Liao, 2010). Localised

collective learning initiated by supplier clusters incrementally arose in the previous "satellite industrial platform" (Markusen, 1996, p. 298; Zhou et al., 2011). In the aftermath of the 2008 global financial crisis, despite the shutdown of thousands of firms, Dongguan surprisingly emerged as a resilient manufacturing city that rapidly recovered through market expansion and redirection (Zhou, 2015; Yang and Fu, 2017), spatial restructuring and state rescaling (Lin, 2006; Xue and Wu, 2015) and technological and industrial upgrading (e.g. the Songshan Lake Science and Technology Industrial Park) (Yang, 2020).

Understanding the underlying economic growth and internal industrial structure change of contemporary Dongguan requires more attention on tracing the origins of its new high-tech industries at multiple geographical scales (e.g. city, region, country, global). The strengthening of localised capacity in capital accumulation, specialised cluster formation, labour resource mobilisation and institutional adaptations in Dongguan have been well investigated (Xue and Wu, 2015; Zhang and Sun, 2019). Meanwhile, influences from Hong Kong and Taiwan have been changing towards high-tech fields (e.g. cooperation base for biotechnology industry on both sides of the Taiwan Straits in Dongguan), and the increasing spill-over effects from Shenzhen (e.g. relocation of Huawei R&D departments and mobile terminal headquarter into the Songshan Lake Science and Technology Industrial Park) (Yang, 2020). Building a multi-scalar framework to trace the origins of the new industrial activities in Dongguan is imperative to reveal the industrial evolution dynamics and contributing actors (Figure 5.2).

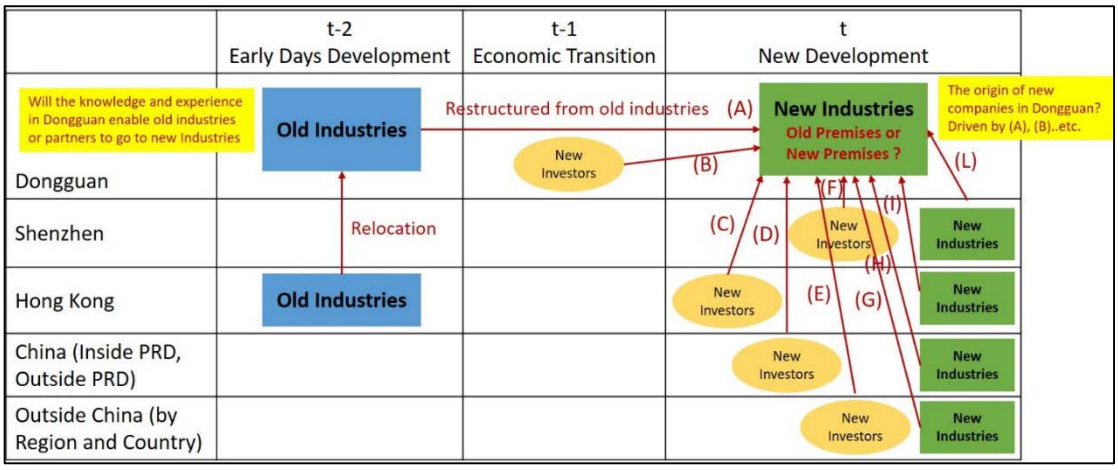


Figure 5.2 Evolutionary Industrial Development of Dongguan

5.2.3 Micro Firm-level Foundation of Industrial Restructuring

Grounded on the framework of evolutionary industrial development at the meso-level (e.g. industry), a micro-level (e.g. firm) analytical lens is more pertinent and operative for carrying out firm-level questionnaire survey. As highlighted by the evolutionary economic geography, industrial restructuring mainly results from knowledge spill-overs that lead to (re)combination of different sources of knowledge and the generation of new knowledge (Hassink and Shin, 2005; Asheim et al., 2011). The micro-foundation of industrial restructuring stresses more on firm heterogeneity in knowledge base, absorptive capacity and business strategies that draw the image unevenly and selectively (Giuliani, 2007). Firm-level interpretations of industrial restructuring deny that the knowledge spilled 'in the air' (Marshall, 1890) as public goods or 'black box' relying on geographical proximity, but feature certain channels and sources that must be spread (Grillitsch and Trippl, 2014). Four types of channels, including spin-offs, diversification, labour mobility and social networking, are widely considered to be conducive for industrial restructuring as the firm-level foundation (Boschma and Frenken, 2011). Spin-offs and diversification function as direct mechanisms whilst labour mobility and social networking perform as indirect ways that may later result in spin-offs and diversification (Tanner, 2014). Actors that likely link to new firms through knowledge linkages include universities and research institutes, suppliers, clients, competitors and professional service firms (Asheim et al., 2019). Moreover, attributes of associated actors, for example, components of capital, industrial sector, geographical location, can achieve a better understanding of the origin of firms as well as transformative industrial dynamics.

Drawing upon the framework of the micro-foundation of industrial restructuring, this study pays specific attention to the origin of firms in high-tech industries in the PRD by tracing the channels of spin-offs and firm diversification and involved actors at various geographical scales. For each firm to be investigated, its establishment can be attributed to entrepreneurs or other firms (i.e. subsidiaries) (Neffke et al., 2018). If a firm was set up by entrepreneurs, then the questions focus on whether the entrepreneur was experienced (i.e. has previously owned firms) or new (i.e. has not previously owned firms) (Klepper, 2007). Experienced entrepreneurs are further inquired about the attributes of previous firms they owned, including location, sector and motivations. New entrepreneurs are asked about the organisations they worked for before they set up this present firm. By comparison, if the firms were set up by other companies, their

parent firms are traced in terms of location, sector and motivations. The subsidiaries are separated into local and non-local ones. For non-local spinoffs, firm diversification and subsidiaries are not directly connected to regional pre-existing industrial activities, but are rather labelled as new start-ups. Local spin-offs, firm diversification and subsidiaries are directly linked to regional pre-existing industrial structure.

5.2.4 Perceived Advantages of Hong Kong in High-tech Industrial Development in the PRD

From the perspective of production resource input for high-tech industry development (e.g. land, labour, capital, entrepreneurship, technology, producer service), Hong Kong still possesses apparent advantages in various aspects compared with the PRD, which may lay the foundation for future collaborations in the new wave of high-tech industrial development in the GBA.

The PRD had low-cost and abundant land provision in early period of reforms, but after three decades' bottom-up *laissez-faire* development, now faces ascending land and labour costs and even scarce land for construction (Yeh and Li, 1999). Against the new trends of outward relocation, technological upgrading and specialised re-agglomerations in the PRD, Hong Kong is capable of initiating a round of investment of high value-added economic activities in technology R&D, knowledge-intensive business services and environment-friendly manufacturing activities in the former to meet the needs of the economic transition and to capture more market opportunities and shares in Mainland China.

Labour in the PRD was traditionally characterised as low-cost, low-skilled and plentiful, because of the large population size and rural reforms that released a considerable amount of rural surplus labour that freely flooded into the PRD since the late 1980s (Lin, 1997). In recent years, by reaching the Lewis Turning Point and the painful blow received by labour-intensive firms from the decline of overseas markets, the PRD has been facing shortage of low-cost labour, resulting in spatially selective shrinkage of a number of towns (Du and Li, 2017). Consequently, under the deliberate transformation of its labour structure, a transitional framework mixed of low- and high-skilled labour is emerging. Labour from Hong Kong generally has higher average education attainment than that of Mainland China. Therefore, Hong Kong has the potential to offer a pool of high-quality labour to raise efficiency and productivity of various sectors in the PRD. In the case of high-skilled talents, Hong Kong has world-leading experts in scientific and technological fields, and professional talents

specialised in business and managerial domains with international vision. These high-skilled talents from Hong Kong can boost the economic transition in the PRD not only by fuelling knowledge and technologies but also functioning as bridges linking the regional and global economies.

Capital injected into the PRD in the past was primarily from Hong Kong, but its sources have been increasingly diversified since the 2000s (Lin, 2006; Z. Xu and Yeh, 2013). Even if the PRD has undergone intensive capital accumulation and territorialisation sourced from various actors through multiple channels, Hong Kong still plays as one of its most important investors. High-tech industrial firms in the PRD may have difficulties in raising capital due to risks and uncertainties in R&D and product development. Hong Kong, as the global financial centre, can help these firms raise capital and offer professional financial services under the endowment of central government with more flexible institutional arrangements in high-tech firms' financing and overseas activities under the 'One Country, Two Systems' framework.

Entrepreneurship in the PRD initially came from Hong Kong under cross-border investment and penetration of firms, primarily in the form of *sanlai yibu*, which stimulated the indigenous entrepreneurship and the other foreign investments that followed (Lin, 2006; Z. Xu and Yeh, 2013). Entrepreneurship plays a crucial role in opening new paths of high-tech industry development, particularly by driving spin-offs from university professors and firm R&D engineers or exploring new fields of technologies and business (Florida et al., 2017). In comparison to the transitional institutional context in Mainland China (Yeh et al., 2015a; Wu, 2018), Hong Kong enjoys, under the 'One Country, Two Systems', a more established legal system in intellectual property protection, which ensures a better incubation environment for domestic and overseas high-tech start-ups and alignment with the international community in rules and standards. Accordingly, the PRD firms can thus functionally operate as a hinterland and technology-user for these start-ups to mature and apply their achievements to reap the commercial returns.

The PRD started with a poor technological basis in its early days has gradually raised its high-tech industries over the past decade, which, to a certain extent, may have surpassed Hong Kong. Although the level of urbanisation and industrialisation in the PRD have been elevated, its technological capacity is still under development, characterised by introducing and absorbing overseas technologies and concentrating on

‘follow-up’ innovation (Wang and Lin, 2013; Yang, 2015). Undoubtedly, Shenzhen has flourished as one of the innovation hubs in China and even in the world, driven by state support, relaxed entrepreneurial environment and influences from Hong Kong (Shen and Luo, 2013; Yang, 2015). A number of domestic technological giants, particularly in the electronics and Internet industries, such as Huawei, ZTE and Tencent, have paved the path for high-tech industrial development in Shenzhen (Zhou et al., 2011). Other key cities, such as Guangzhou, Dongguan and Foshan, are striving to reinvent themselves with the new high-tech economy. According to the Global Innovation Index in 2019 by World Intellectual Property Organization, Hong Kong-Shenzhen ranked second all over the world. This indicates the competency of the Hong Kong-Shenzhen complex and the complementarity in the cross-border regional innovation system between the two cities. Hong Kong and Shenzhen have much potential to carry out technological collaborations in cutting-edge science and technology R&D and institutional collaborations. Furthermore, Hong Kong should make better use of the technological hierarchy to transfer more technologies needed by other cities in the PRD to stimulate indigenous R&D and facilitate the technology adjustment and application towards the Mainland market.

Producer service input has become increasingly important in economic efficiency, market expansion and knowledge production (Yeh, 2005; Yang and Yeh, 2013). Tersiarisation combined with rising producer services have rapidly materialised in the PRD since the 2000s, with a number of established Central Business Districts and Free Trade Pilot Zones (Lin, 2004; Yeh and Chen, 2020). Producer services in Hong Kong, however, are still superior than its counterparts in the PRD in international linkages and service qualities (Yeh et al., 2015b). Firstly, producer service development in Hong Kong has a longer history than that in the PRD, which implies that Hong Kong can still offer professional services that highly rely on continuous accumulation of knowledge and experience with better quality to their customers, with competitive advantages in trade-, finance- and technology-related domains. Secondly, producer services in Hong Kong capture more accurate and timely information across the world, and are more skilled in handling international business matters and maintaining dialogues with international communities. Against the background of the Belt and Road initiative and China becoming the growing global investor, producer services in Hong Kong under the ‘One Country, Two Systems’ can provide more services to firms in the PRD (Table 5.1; Figure 5.3).

Table 5.1 Perceived advantages of HK in high-tech industry development in the PRD

Production Factors	PRD in Early Days	PRD in Present-day	Perceived Advantages of Hong Kong
Land	Cheap, Abundant	Rising cost, Scarce land for construction	Investment of high value-added and environment-friendly activities
Labour	Cheap, Abundant, Low-skilled	Rising cost, Shortage of cheap labour, A mix of low- and high-skilled labour	High-skilled labor, Labor specialized in business and managerial fields, Talents with international vision
Capital	Mainly from HK	Diversified sources	High-tech investment, Capital raising
Entrepreneurship	Mainly from HK	Diversified sources	High-tech subsidiaries, High-tech spin-offs
Technology	Weak	Upgrading	Technological collaboration Technology transfer
Producer Service	Mainly from HK	Diversified sources	Hong Kong producer services with better quality and specialization in overseas market (e.g., the Belt and Road countries)

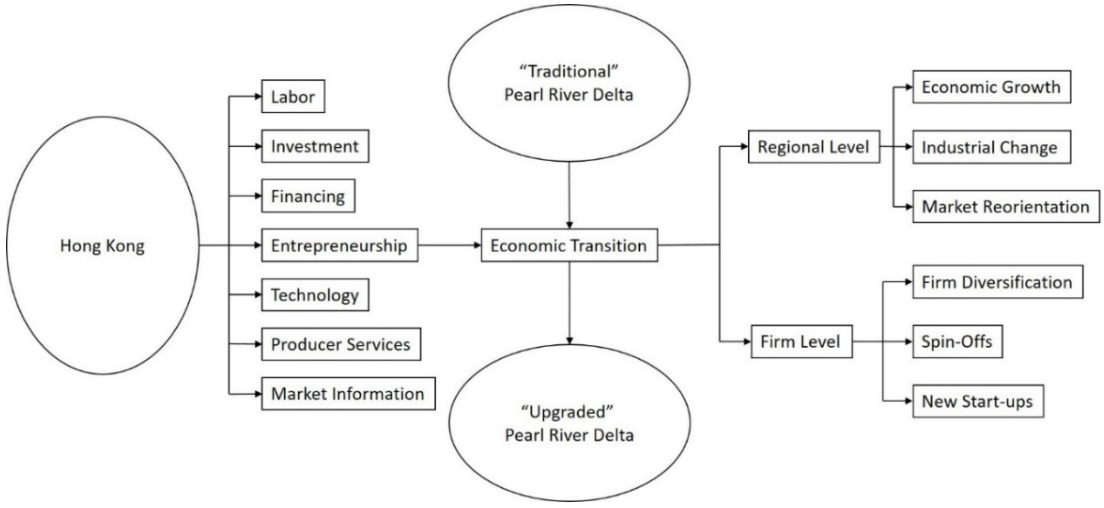


Figure 5.3 Perceived role of HK in the new wave of economic transition in the PRD

5.2.5 Potential Regional Collaboration Models between the PRD and Hong Kong for Future Development of the GBA

Evolving from the FSBF model towards the overseas market particularly in the developed countries, Hong Kong tends to invest more service activities and enhance 'Shop' functions in the PRD to seize the enlarging domestic market demands derived from domestic manufacturing and market re-orientation of Hong Kong-invested firms (Yang and Fu, 2017). In addition to the FSBF, the collaboration model between the PRD and Hong Kong can be reflected as the 'Front Shop, Back Shop, Back Factory' Model, with further geographical division of the 'Shop' function.

Under the economic transition towards a knowledge- and technology-intensive economy, collaboration objectives between the two places evolves from cost-reduction towards innovation (Storper, 1997; Li et al., 2020). Apart from being a centre of international finance, trade and service, Hong Kong can gain more weight as an international technological innovation centre. Technologies invested in Hong Kong and by Hong Kong-invested firms aim to be the engine for technological upgrade and economic transition in the PRD. Consequently, another potential new collaboration model is 'Front High-tech Producer, Back High-tech User' between Hong Kong and the PRD with respect to the cross-boundary innovation system.

Furthermore, the FSBF model may still make sense with the functional transformation of both 'Factory' and 'Shop'. On the one hand, the 'Factory' by no means remains the standardised workshops but shifts into higher value-added technology-based plants. The standardised 'Factory' can hardly survive in the PRD any longer, and needs to further move inland and to less developed regions in the Mainland (Yang and He, 2017). The 'Shop', on the other hand, can diversify into multiple market targets not limited to the developed world but also incorporate the Mainland and the emerging overseas market in the Belt and Road. The technology-equipped factories and multiple market targets shops can achieve a new balance between the PRD and Hong Kong based on mutual complementarity dynamically coupled with changing global and local dynamics (Yang, 2020; Figure 5.4).

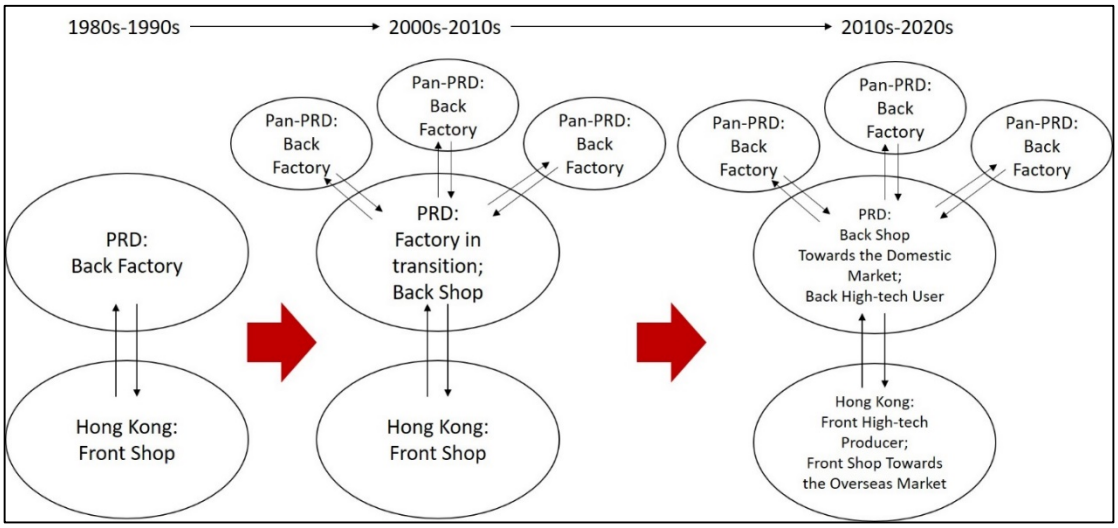


Figure 5.4 Potential collaboration models between the PRD and Hong Kong

5.3 Research Design

In recognition of the importance of tracing the micro-foundations (e.g. firm-level) in unveiling evolutionary industrial development in the PRD and its transformative collaboration patterns with Hong Kong, we need to carry out a firm-level questionnaire survey with in-depth interviews. The firm-level questionnaire survey not only helps to understand the new wave of high-tech industrial development in the GBA, but also to search for new economic cooperation models between Hong Kong and the GBA. The specific research design of the questionnaire survey is as follows.

5.3.1 Research Objectives

1) To examine high-tech industrial development in the PRD and the reasons for its capability to move upward
2) To identify the roles of Hong Kong in participating in the PRD's new wave of high-tech industrial development
3) To reveal the views of high-tech industrial firms in the PRD on Hong Kong roles

5.3.2 Research Questions

1) What are the characteristics of high-tech industrial development in the PRD?
2) What are the reasons for the PRD capability to move upwards to high-tech industrial development?
3) What are the roles of Hong Kong in participating into the PRD's new wave of high-tech industrial development?
4) What are the views of high-tech industrial firms in the PRD on Hong Kong roles?

5.3.3 Research Methodology

A professional survey firm with rich experience in high-technology industrial development and the GBA studies was commissioned, with the joint efforts of our research assistants, to carry out firm-level questionnaire survey in Dongguan and Shenzhen. The questionnaire survey lasted for five months from January to May 2021. Firms belonging to the two target sectors were randomly selected from the supplier's internal firm-level database. In addition to the firm-level questionnaire survey, 50 in-depth semi-structured interviews with firm senior managers, government officials, academic scholars and urban and industrial planners were carried out, with an average

interview time of 1 hour. The interviews focus on whether and why they use or not use Hong Kong knowledge and technological support in industrial development, and the barriers and potential opportunities for Hong Kong-PRD collaboration in the new wave of high-technology industrial development in the GBA. The interviews are used to enhance our understanding of the findings of the firm-level questionnaire survey.

5.3.4 Questionnaire Design

5.3.4.1 Questionnaire Structure

The structure of the questionnaires is as follows:

1) Part A. Firm attributes
2) Part B. Firm history
3) Part C. Locational factors
4) Part D. Supplier and market linkages
5) Part E. Technological sources and collaborations
6) Part F. Producer service linkages
7) Part G. Cooperation with Hong Kong
8) Part H. Market and innovation performances
9) Part I. Post-Pandemic business plan

5.3.4.2 Questions Designed in Each Section

1) Part A. Firm attributes

   Name, geographical location, ownership, source of investment, date of establishment, firm origin, number of employees, history of geographical relocation, industrial sector/product, form of products, form of production

2) Part B. Firm history

   Way of establishment, career experience of founder, motivation of entrepreneurship, attributes of parent firm, motivation of subsidiary investment, relocation history, previous geographical location

3) Part C. Location factors

   Proximity to suppliers/customers, cost, skilled labour, intellectual support, agglomeration economy, product service, infrastructure, firm history, intermediary agency, geographical close to Hong Kong

4) Part D. Supplier and market linkages

   Geographical location of suppliers and market areas, proportion of production value in each area

5) Part E. Technological sources and collaborations

   Sources of core technologies and geographical locations, channels for developing new products, geography of technological collaborations, geographical sources of R&D and managerial employees

6) Part F. Producer service linkages

   Types of external producer service input, types of producer service provider, geography of producer service input

7) Part G. Cooperation with Hong Kong

   Importance of investment, labour, finance, producer services, market information from Hong Kong, reasons for using or not using various inputs from Hong Kong

8) Part H. Market and innovation performances

   Sales revenue in 2019, sales revenue of new products, R&D expenditure, number of patents, and types of new products in the last three years (2017-2019)

9) Part I. Post-Pandemic business plan

   Geographical relocation, business strategy, development direction, market reorientation

5.3.5 Sampling Approach

As the key objectives of this study are to examine high-tech industrial development in the PRD and to identify the roles of Hong Kong in facilitating the new wave of industrial upgrading in the PRD, the sampling approach of the firm-level questionnaire survey must aim to answer these research questions. The selected sample firms must be high-tech industrial firms that both represent the newly emerging industrial upgrading trends in the PRD and those that most likely use Hong Kong's competitive knowledge and technological support. Hence, leading technology-intensive manufacturing industries in the PRD over the last decade, key technology-intensive industries to be fostered in the GBA, and technology-intensive domains with competitive advantages in Hong Kong are selected in this study.

5.3.5.1 Identifying Leading Technology-intensive Manufacturing Industries in the PRD in the 2010s

In the new wave of economic transition and industrial upgrading in the PRD since the mid-2000s, this study particularly focuses on growing technology-intensive industries instead of all sectors. The past FSBF model is rapidly fading away and loses its importance. Further knowledge is necessary about the rising industries in the PRD and what role can Hong Kong play in this new wave of economic development that is based on Industry 4.0. From the *Guangdong Statistical Yearbooks* (2012-2018), Manufacture of Communication Equipment, Computers and Other Electronic Equipment (SIC 39), Manufacture of Electrical Machinery and Equipment (SIC 38), Manufacture of Automobile (SIC 36), Manufacture of General-purpose Machinery (SIC 34), and Manufacture of Special-purpose Machinery (SIC 35) are five technology-intensive sectors with the largest Gross Industrial Output Value in the PRD during 2011-2017 (GDSB, 2012-2018; Figure 5.5)

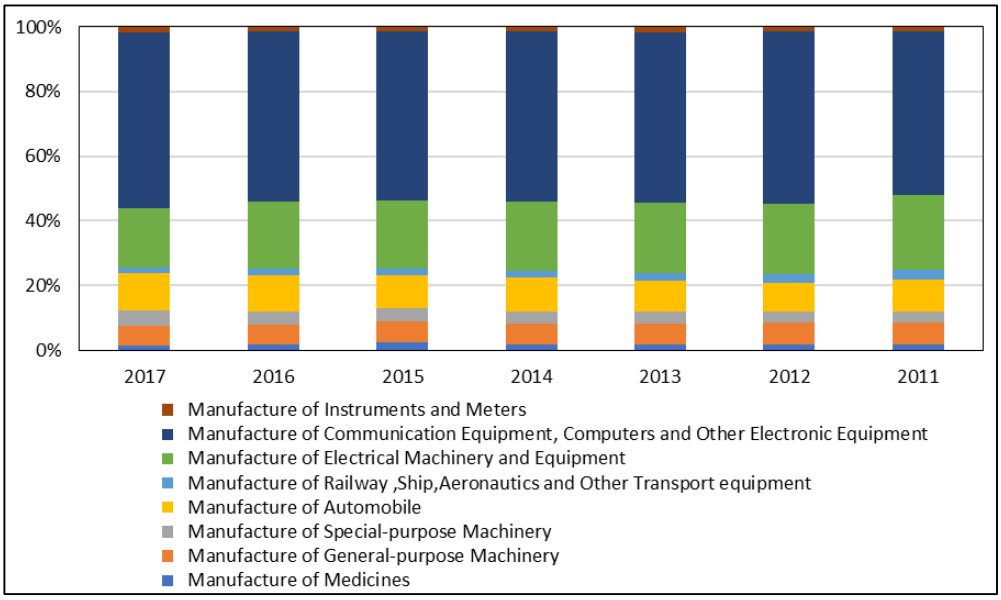


Figure 5.5 Gross Industrial Output Value of technology-intensive industries in the PRD, 2011-2017

Source: Guangdong Statistical Bureau, 2012-2018

The five leading technology-intensive industries in the PRD are concentrated in different cities (Figure 5.6). Manufacture of Communication Equipment, Computers and Other Electronic Equipment (SIC 39) is mainly concentrated in Shenzhen,

Dongguan and Huizhou. Manufacture of Electrical Machinery and Equipment (SIC 38) is concentrated in Foshan, Shenzhen and Dongguan. Manufacture of Automobile (SIC 36) is concentrated in Guangzhou, Foshan and Shenzhen. Manufacture of General-purpose Machinery (SIC 34) is concentrated in Foshan, Shenzhen and Dongguan. Finally, Manufacture of Special-purpose Machinery (SIC 35) is concentrated in Shenzhen, Foshan and Dongguan (Table 5.2).

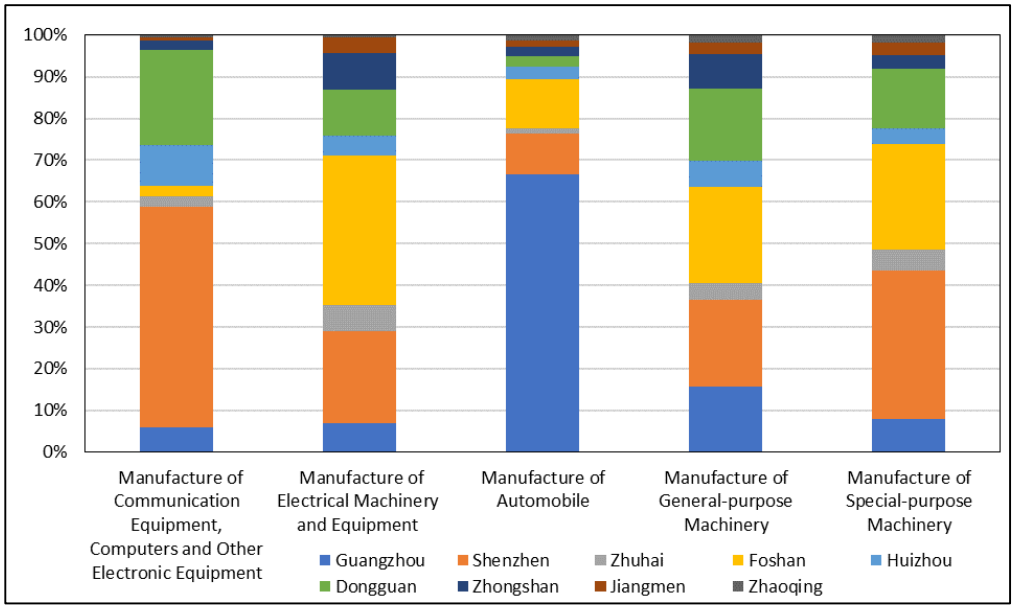


Figure 5.6 Gross Output Value of technology-intensive industries above designated size by city in 2017

Source: Guangdong Statistical Bureau, 2018

Table 5.2 Ranking of cities by the five leading technology-intensive industries in the PRD, 2017

City	Manufacture of Communication Equipment, Computers and Other Electronic Equipment (SIC 39)	Manufacture of Electrical Machinery and Equipment (SIC 38)	Manufacture of Automobile (SIC 36)	Manufacture of General-purpose Machinery (SIC 34)	Manufacture of Special-purpose Machinery (SIC 35)
Guangzhou	4	5	1	4	4
Shenzhen	1	2	3	2	1
Zhuhai	6	6	9	7	5
Foshan	5	1	2	1	2
Huizhou	3	7	4	6	6
Dongguan	2	3	5	3	3
Zhongshan	7	4	6	5	7
Jiangmen	8	8	7	8	8
Zhaoqing	9	9	8	9	9

Source: Guangdong Statistical Bureau, 2018

5.3.5.2 Technology-intensive Industries Highlighted by Recent Policy Initiatives

The *Outline Development Plan for the Guangdong-Hong Kong-Macao Greater Bay Area* that was promulgated in February 2019 highlighted the smart development of the manufacturing industry, with a particular focus on robotics, high-speed and high-precision processing equipment and intelligent equipment sets (Table 5.3). The *Outline* also highlighted supports for enterprises with capacities in system integration and intelligent equipment development. It aims to strengthen existing competitive advantages in fields such as equipment manufacturing, petrochemicals, and automobile.

Table 5.3 Industries highlighted in the Outline Development Plan for the Guangdong-Hong Kong-Macao Greater Bay Area, February 2019

Industries to be promoted	**China Industry Classification System**	**Technology-intensive industries with the largest Gross Output Value in the PRD**
Robotics	Manufacture of Other General-purpose Machinery (SIC 349)	Manufacture of General-purpose Machinery (SIC 34)
Robotics-related key components	Manufacture of small and special electric machine and components (SIC 381/3813); Manufacture of electric machinery (SIC 381); Power transmission and distribution and manufacture of control equipment (SIC 382)	Manufacture of Electrical Machinery and Equipment (SIC 38)
Intelligent equipment	Manufacture of Other General-purpose Machinery (SIC 349/3491); Manufacture of industrial automatic control system device (SIC 4011)	Manufacture of General-purpose Machinery (SIC 34)
Automobile	Manufacture of Automobile (SIC 36)	Manufacture of Automobile (SIC 36)
Home appliances	Manufacture of Home appliances (SIC 385)	Manufacture of Electrical Machinery and Equipment (SIC 38)
Electronic information	Manufacture of Communication Equipment, Computers and Other Electronic Equipment (SIC 39)	Manufacture of Communication Equipment, Computers and Other Electronic Equipment (SIC 39)

Source: Outline Development Plan for the Guangdong-Hong Kong-Macao Greater Bay Area

5.3.5.3 Technology-Intensive Industries with Competitive Advantage in Hong Kong

According to the Chief Executive's 2018 Policy Address, 'funding approval of $10 billion has been obtained from the Legislative Council for establishing two I&T clusters at the Hong Kong Science Park, with one focusing on healthcare technologies and the other on artificial intelligence and robotics technologies' (p. 34). Similarly, Hong Kong Trade Development Council (2021) repeatedly highlights that 'major research focus of Hong Kong start-ups includes information and communication technologies (ICT), software as a service (SaaS), Internet of things (IoT), data analysis, biotechnology (Bio-Tech), artificial intelligence (AI), robotics, virtual reality (VR) and augmented reality (AR), as well as new material. In terms of applications, fintech (financial technology), smart city and smart home, healthcare and big data applications are among the most popular sectors.

For R&D capabilities, human capital plays an essential role in ensuring innovation-oriented development. Universities in Hong Kong enjoy competitive advantages over their counterparts in the PRD, particularly in talent cultivation and R&D competence in several science and engineering subjects. As a result, Hong Kong universities can provide professional talents and transfer appropriate technologies to firms in the PRD.

5.3.5.4 Potential Industrial Application of Technology-Intensive Industries in Hong Kong

From the technological domain side, Hong Kong technology-intensive firms, mostly in R&D service sectors, are suggested to cultivate competitive advantages in ICT, Bio-Tech and AI. Accordingly, a wide range of industrial sectors can apply these cutting-edge technologies to renovate the products and increase value-added output and competitiveness (Table 5.4; Figure 5.7). Given the limited land space and high production cost, Hong Kong is unlikely to develop mass production plants extending from the R&D activities of those technologies in its own territory. The PRD, which is has a close location and intensive socio-economic linkages with Hong Kong, has the potential to function as the hinterland for Hong Kong technology-intensive service firms to provide applications to industrial plants to grasp commercial return and contribute to the competitiveness of the GBA as a whole. Notably, different

technologies may be applied to heterogeneous industrial sectors. For example, ICT can be used in the Manufacture of Computers (SIC 391), Communication Equipment (SIC 392), Electron Device (SIC 397) and of Electronic Components and Special Electronic Materials (SIC 398). Bio-tech can be used in Manufacture of Medicine (SIC 27) and of Medical Instruments and Equipment (SIC 358). AI can be used in the Manufacture of Other General-purpose Machinery (SIC 349), Electronics and electrical machinery equipment (SIC 356), Intelligent consumer equipment (SIC 396) and of Electron Device (SIC 397).

Table 5.4 Industrial sectors with technology and production advantages in the GBA

Technology-intensive services with competitive advantage in Hong Kong	**Potential industrial applications of technology-intensive services in Hong Kong**	**Industrial sectors promoted by policy initiatives in the PRD**	**Industrial sectors with competitive advantages in the PRD**
Information and communication technology (ICT)	Manufacture of Computers (SIC 391) Manufacture of Communication Equipment (SIC 392) Manufacture of Electron Device (SIC 397) Manufacture of Electronic Components and Special Electronic Materials (SIC 398)	Manufacture of Other General-purpose Machinery (SIC 349) Manufacture of electric machinery (SIC 381) Power transmission and distribution and manufacture of control equipment (SIC 382) Manufacture of industrial automatic control system device (SIC 4011) Manufacture of Automobile (SIC 36) Manufacture of Home appliances (SIC 385) Manufacture of Communication Equipment, Computers and Other Electronic Equipment (SIC 39)	Manufacture of Communication Equipment, Computers and Other Electronic Equipment (SIC 39) Manufacture of Electrical Machinery and Equipment (SIC 38) Manufacture of Automobile (SIC 36) Manufacture of General-purpose Machinery (SIC 34) Manufacture of Special-purpose Machinery (SIC 35)
Biotechnology (Bio-Tech)	Manufacture of Medicine (SIC 27) Manufacture of Medical Instruments and Equipment (SIC 358)		
Artificial intelligence (AI)	Manufacture of Other General-purpose Machinery (SIC 349) Manufacture of Electronics and electrical machinery equipment (SIC 356) Manufacture of Intelligent consumer equipment (SIC 396) Manufacture of Electron Device (SIC 397) Manufacture of industrial automatic control system device (SIC 4011)		

Source: Hong Kong Trade Development Council Research 2018; GDSB, 2012-2018; Outline Development Plan for the Guangdong-Hong Kong-Macao Greater Bay Area

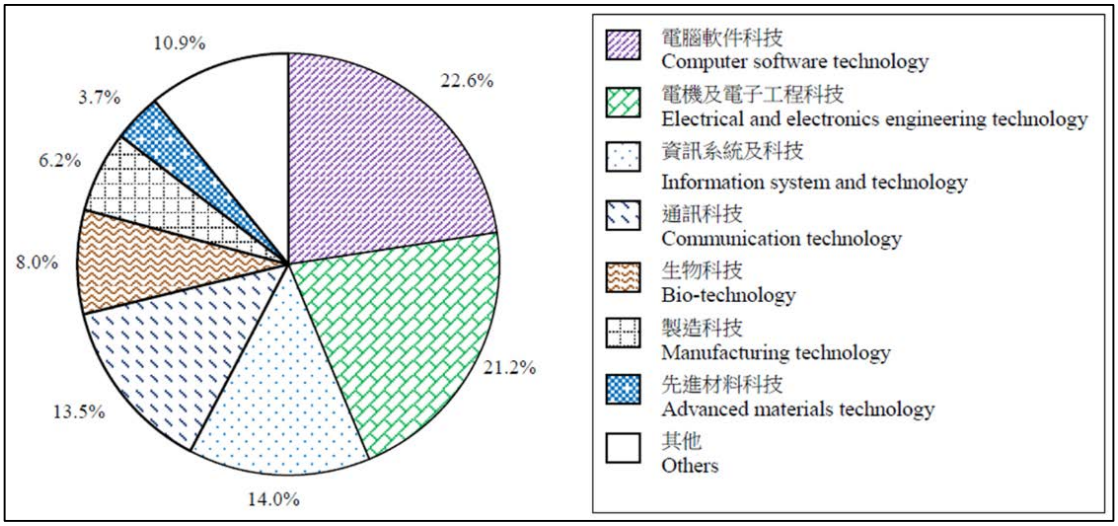


Figure 5.7 Distribution of total expenditure on in-house R&D activities in the business sector in 2018 by technology area

Source: Hong Kong Innovation Activities Statistics 2018, p. 6

5.3.5.5 Research Samples

The principle of sampling for the firm-based questionnaire survey is to identify firms in the technology-intensive sectors with leading status in the PRD in the 2010s and being the potential applicants of technologies with competitive advantages in Hong Kong (Figure 5.8). After listing all the three-digit industrial sectors under each two-digit sector with leading status in the PRD (Table 5.5), we continue to narrow them by filtering out three-digit level sectors that can apply technologies with competitive advantages in Hong Kong (Figure 5.8; Table 5.6), namely, ICT, Bio-tech, AI and Robotics. Using the above sample criteria, finally, intelligent manufacturing equipment using ICT, AI and Robotics and biomedical products using Bio-tech are selected as sample for the firm-level questionnaire survey in Dongguan and Shenzhen, where Hong Kong exerts the greatest influences on the PRD (see also Shen and Luo, 2013; Yeh and Chen, 2020).

A total of 363 high-technology industrial firms were interviewed, including 178 firms in Dongguan and 185 firms in Shenzhen, 194 intelligent equipment

manufacturing firms and 169 biomedical product firms and 122 domestic-funded, 117 Hong Kong-funded and 124 other overseas-funded firms.

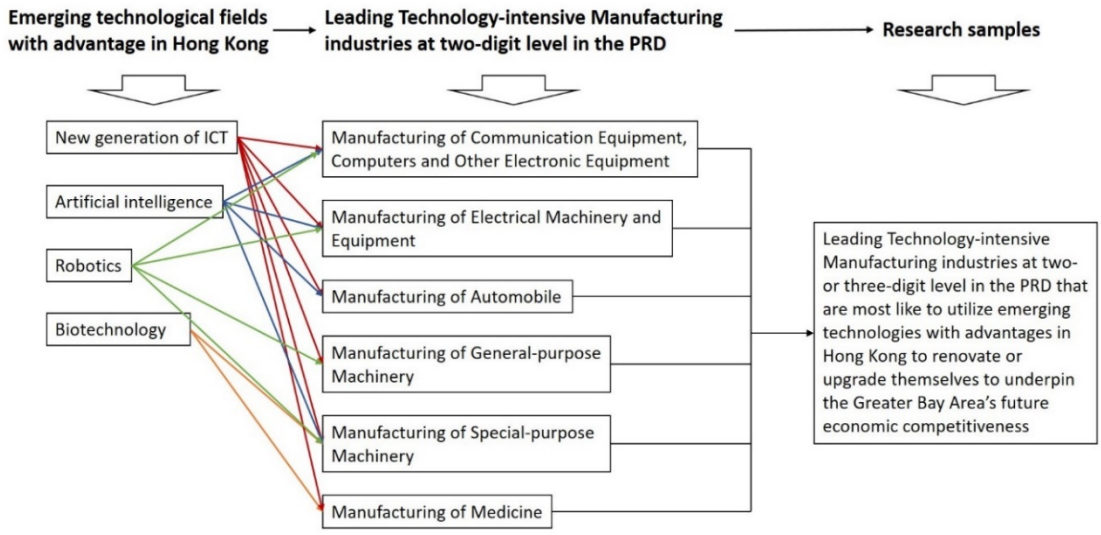


Figure 5.8 Sampling approach

Table 5.5 Industrial sectors at three-digit level marked in red to be surveyed

Leading technology-intensive manufacturing industries in the PRD in the 2010s		**Hong Kong technologies can be applied**
Two-digit level	**Three-digit level**	
SIC 27 Manufacture of Medicine 醫藥製造業	SIC 271 Manufacture of Chemical API 化學藥品原料藥製造	Bio-tech
	SIC 272 Manufacture of Chemical Preparation 化學藥品製劑製造	Bio-tech
	SIC 273 Processing of Chinese Herbal Pieces 中藥飲片加工	Bio-tech
	SIC 274 Manufacture of Proprietary Chinese Medicine 中成藥生產	Bio-tech
	SIC 275 Manufacture of Veterinary Drugs 獸用藥品製造	
	SIC 276 Manufacture of Biopharmaceutical Products 生物藥品製品製造	Bio-tech
	SIC 277 Manufacture of Sanitary Materials and Medical Supplies 衛生材料及醫藥用品製造	Bio-tech
	SIC 278 Manufacture of Pharmaceutical Auxiliary Materials and Packaging Materials 藥用輔料及包裝材料	Bio-tech
SIC 34 Manufacture of General Equipment 通用設備製造業	SIC 341 Manufacture of Boiler and Original Equipment 鍋爐及原動設備製造	
	SIC 342 Manufacture of Metal Processing Machinery 金屬加工機械製造	AI, Robotics
	SIC 343 Manufacture of Material Handling Equipment 物料搬運設備製造	AI, Robotics
	SIC 344 Manufacture of Pumps, Valves, Compressors and Similar Machinery 泵、閥門、壓縮機及類似機械製造	AI, Robotics
	SIC 345 Manufacture of Bearings, Gears and Transmission Parts **軸承、齒輪和傳動部件製造**	AI, Robotics
	SIC 346 Manufacture of Oven, Fan, Packaging and other Equipment 烘爐、風機、包裝等設備製造	
	SIC 347 Manufacture of Cultural and Office Machinery 文化、辦公用機械製造	
	SIC 348 Manufacture of General Parts **通用零部件製造**	AI, Robotics
	SIC 349 Manufacture of Other General Equipment 其他通用設備製造業	AI, Robotics
SIC 35 Manufacture of Special Equipment 專用設備製造業	SIC 351 Manufacture of Special Equipment for Mining, Metallurgy and Construction 採礦、冶金、建築專用設備製造	AI, Robotics
	SIC 352 Manufacture of Special Equipment for Chemical, Wood, and Nonmetal Processing 化工、木材、非金屬加工專用設備製造	AI, Robotics
	SIC 353 Manufacture of Special Equipment for Food, Beverage, Tobacco, and Feed Production 食品、飲料、煙草及飼料生產專用設備製造	AI, Robotics
	SIC 354 Manufacture of Special Equipment for Printing, Pharmaceutical, Daily Chemical, and Daily Necessity 印刷、製藥、日化及日用品生產專用設備製造	Bio-tech, Robotics
	SIC 355 Manufacture of Special Equipment for Textile, Garment and Leather Processing 紡織、服裝和皮革加工專用設備製造	AI, Robotics

	SIC 356 Manufacture of Special Equipment for Electronic and Electrical Machinery 電子和電工機械專用設備製造	ICT
	SIC 357 Manufacture of Special Equipment for Agriculture, Forestry, Animal Husbandry and Fishery 農、林、牧、漁專用機械製造	AI, Robotics
	SIC 358 Manufacture of Medical Instruments and Equipment 醫療儀器設備及器械製造	Bio-tech, Robotics
	SIC 359 Manufacture of Environmental Protection, Post, Social Public Services and Other Special Equipment 環保、郵政、社會公共服務及其他專用設備製造	AI, Robotics
SIC 38 Manufacture of Electrical Machinery and Equipment 電氣機械和器材製造業	SIC 381 Manufacture of Electric Motors 電機製造	AI, Robotics
	SIC 382 Manufacture of Power Transmission and Distribution and Control Equipment 輸配電及控制設備製造	AI, Robotics
	SIC 383 Manufacture of Wire, Cable, Optical Cable and Electrical Equipment 電線、電纜、光纜及電工器材製造	ICT
	SIC 384 Manufacture of Batteries 電池製造	ICT
	SIC 385 Manufacture of Household Electric Appliances 家用電力器具製造	
	SIC 386 Manufacture of Non-electric Household Appliances 非電力家用器具製造	
	SIC 387 Manufacture of Lighting Equipment 照明器具製造	
	SIC 389 Manufacture of other Electrical Machinery and Equipment 其他電氣機械及器材製造	
SIC 39 Manufacture of Communication Equipment, Computers and Other Electronic Equipment 計算機、通信和其他電子設備製造業	SIC 391 Manufacture of Computers 計算機製造	ICT
	SIC 392 Manufacture of Communication Equipment 通信設備製造	ICT
	SIC 393 Manufacture of Radio and Television Equipment 廣播電視設備製造	
	SIC 394 Manufacture of Radar and Supporting Equipment 雷達及配套設備製造	ICT
	SIC 395 Manufacture of Non-professional Audio-visual Equipment 非專業視聽設備製造	
	SIC 396 Manufacture of Intelligent Consumer Equipment 智能消費設備製造	AI, Robotics
	SIC 397 Manufacture of Electric Devices 電子器件製造	ICT
	SIC 398 Manufacture of Electronic Components and Special Electronic Materials 電子元件及電子專用材料製造	ICT
	SIC 399 Manufacture of Other Electric Devices 其他電子設備製造	AI

Sources: China Industrial Classification for National Economic Activities (2017) and China Industrial Classification for Strategic Emerging Industries (2018)

Table 5.6 Sample distribution by sector and capital source in each city (Dongguan and Shenzhen)

Leading technology-intensive manufacturing industries in the PRD that potentially use technological support with competitive advantages in Hong Kong		**Number of firms from different capital sources**	**Sample number**
Industrial Sector	**Four-digit Level Sector**		
Intelligent Manufacturing Equipment 智能裝備製造	SIC 3491 Manufacture of Industrial Robotics 工業機器人製造	Domestic: 30 Hong Kong, Macao, Taiwan: 30 Overseas: 30	90
	SIC 3492 Manufacture of Special Robotics 特殊作業機器人製造		
	SIC 3493 Manufacture of Additive Manufacturing Equipment 增材製造裝備製造		
	SIC 3961 Manufacture of Wearable Smart Devices 可穿戴智能設備製造		
	SIC 3962 Manufacture of Intelligent Vehicle Equipment 智能車載設備製造		
	SIC 3963 Manufacture of Intelligent Unmanned Aerial Vehicle 智能無人飛行器製造		
	SIC 3964 Manufacture of Service Robotics 服務消費機器人製造		
	SIC 4011 Manufacture of Industrial Automation System 工業自動控制系統裝置製造		
Biomedical Products 生物醫藥產品	SIC 2761 Manufacture of Biomedicine 生物藥品製造	Domestic: 30 Hong Kong, Macao, Taiwan: 30 Overseas: 30	90
	SIC 2762 Manufacture of Gene Engineer Pharmaceutical 基因工程藥物和疫苗製造		
	SIC 353 Manufacture of Special Equipment for Food, Beverage, Tobacco, and Feed Production 食品、饮料、烟草及饲料生产专用设备制造		
	SIC 3581 Manufacture of Medical Diagnostic, Monitoring and Therapeutic Equipment 醫療診斷、監護及治療設備製造		
	SIC 3582 Manufacture of Equipment and Instruments Used in Stomatology Department 口腔科用設備及器具製造		
	SIC 3583 Manufacture of Medical Laboratories and Medical Disinfection Equipment and Instruments 醫療實驗室及醫用消毒設備和器具製造		
	SIC 3584 Manufacture of Medical, Surgical, and Veterinary Instruments 醫療、外科及獸醫用器械製造		
	SIC 3585 Manufacture of Mechanical Treatment and Ward Nursing Equipment 機械治療及病房護理設備製造		
Total number in each city		90 x 2=180	
Total number in the PRD		180 x 2=360	

5.4 Research Findings

5.4.1 High-tech Industrial Development in the PRD

5.4.1.1 Economic Restructuring in the PRD since the 2000s

5.4.1.1.a Economic and Industrial Development

Since 2000, the PRD has seen a rapid and steady GDP growth. However, the growth rate has plateaued since 2012 (Figure 5.9). Meanwhile, the value-add of high-tech industries in the PRD continuously grows, already accounting for approximately 20% of the GDP in 2019 (Figure 5.10). The PRD is facing structural changes in industries, wherein the high-tech domain, a burgeoning industry, is gradually taking an indispensable part of the economies in the region.

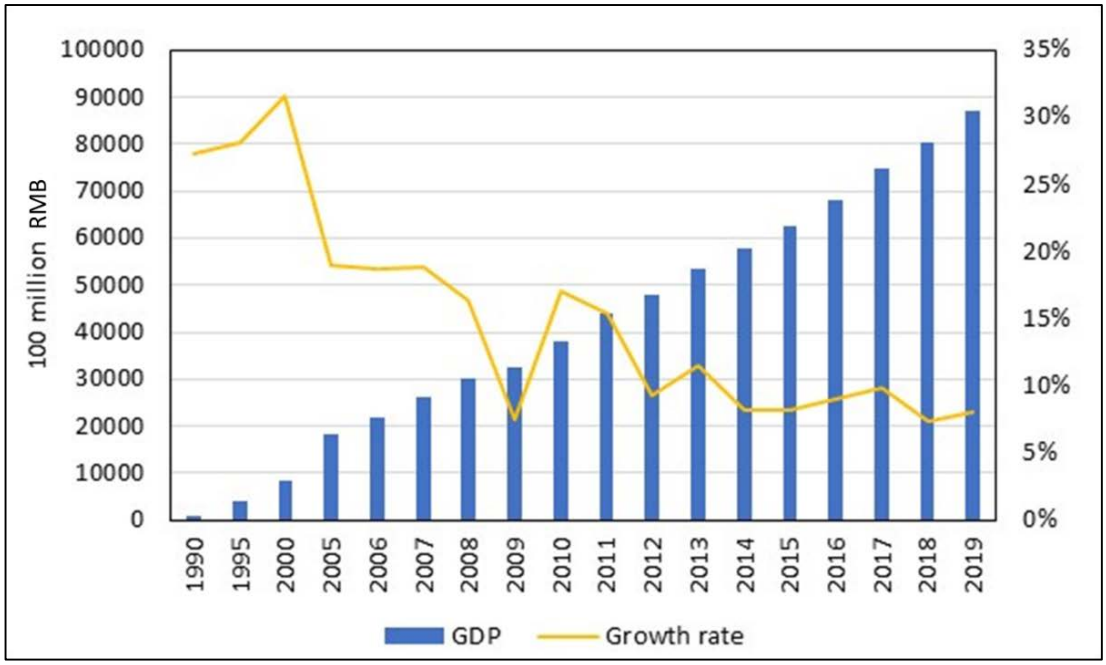


Figure 5.9 GDP and its growth rate in Guangdong since 1990

Source: Guangdong Statistical Year Book, 2020

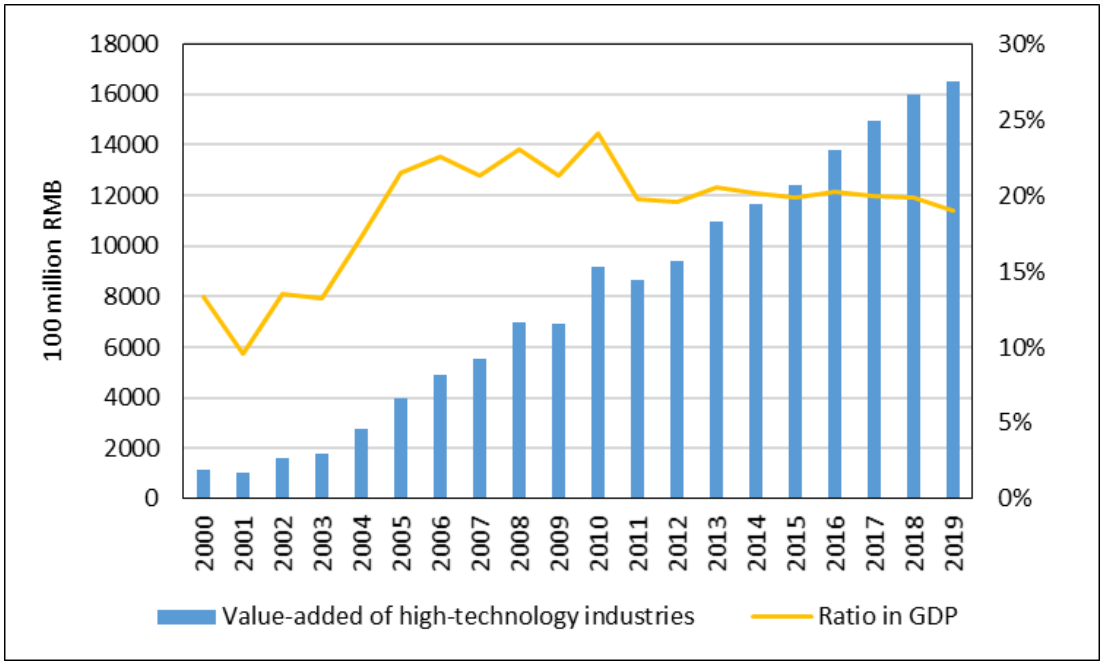


Figure 5.10 Value-added of high-tech industries in Guangdong since 2000

Source: Guangdong Statistical Year Book 2020

5.4.1.1.b Industrial Structural Change

The PRD has been a secondary industry-based region since the 1980s, being famous for its prosperous manufacturing-related industries. However, in 2012, the contribution rates of secondary industry initially dropped below 50%, and in 2019, only accounted for 33.6% of the entire three strata of industry (Figure 5.11). The secondary-industrial feature of the PRD is fading due to the rise of the tertiary industry and the economic knowledge and technological intensification. Since 2012, the tertiary industry has been growing steadily (Figure 5.11) and now accounts for the largest proportion of all industry domains in the PRD.

In addition, the export-oriented economy in the PRD, one of the largest coastal mega city-regions in China, is under transition. In the past few decades, the balance between exports and imports in this region has increased, but began to slightly decline after 2015 (Figure 5.12). The number of exports remains constant with a low increased ratio (Figure 5.12). This finding suggests that the PRD is transitioning from export to a mixture of export and domestic sales.

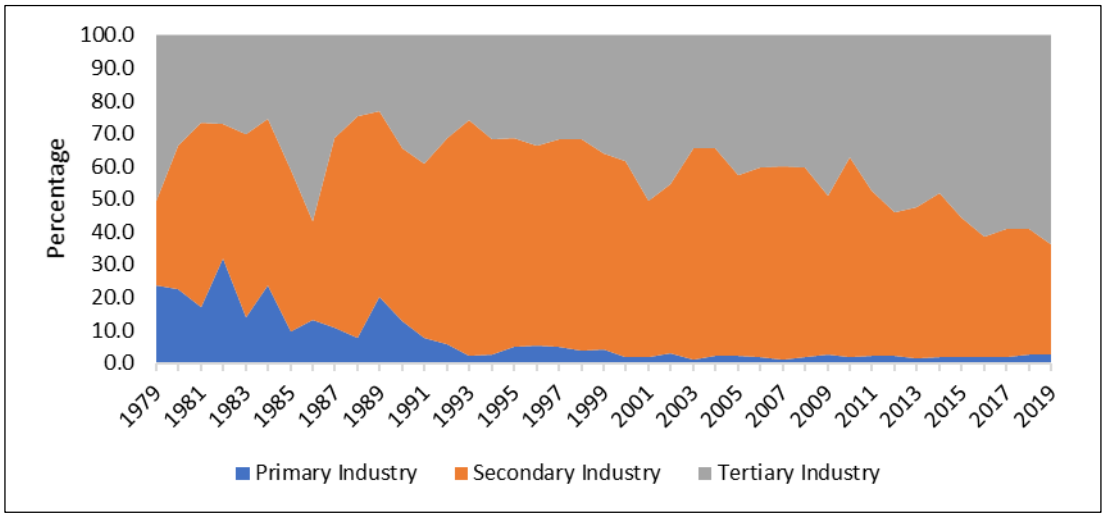


Figure 5.11 Contribution Rates of the Three Strata of Industry

Source: Guangdong Statistical Year Book 2020

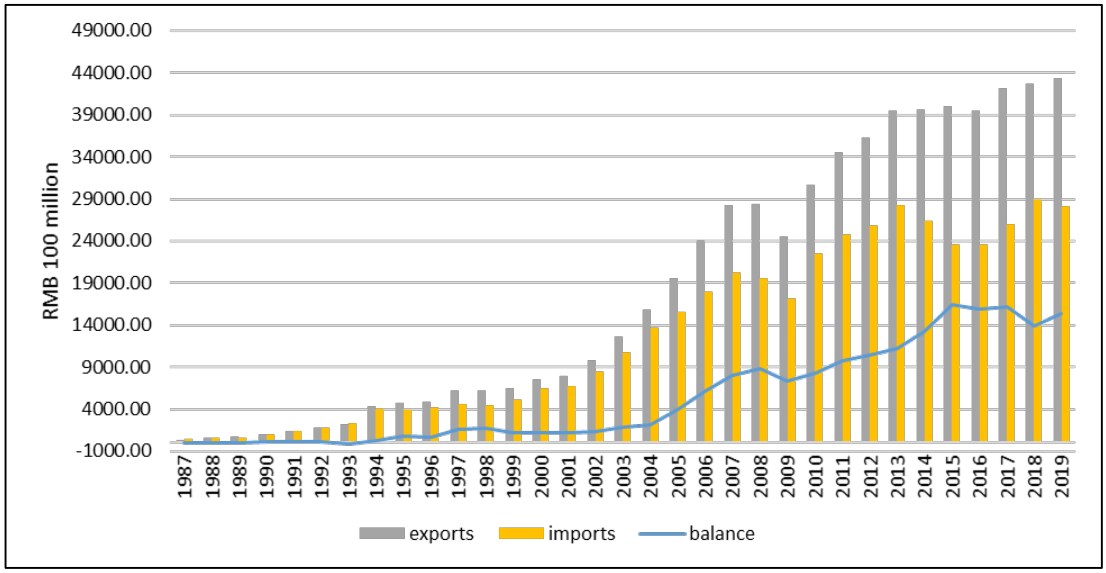


Figure 5.12 Total Value of Imports and Exports

Source: Guangdong Statistical Year Book 2020

Another apparent transition is the declining role of labour-intensive industries, such as textile since 2010. The same trend is also observed in the resource-intensive industry. By contrast, the growth of value-added of technology-intensive industry is robust and steady (Figure 5.13). This reflects a technological intensification transition in the PRD, coinciding with the government advocacy in developing high-tech industries.

Consider the biomedical and intelligent equipment industries in Shenzhen as examples. These two industries have an average annual growth rate of 10.4% and 32.8% (Figure 5.14). As in the report from Qianzhan and Huoshi Research Institute

(Huoshi Research Institute, 2019; Qianzhan Research Institute, 2020), Shenzhen already has 3,517 biomedical firms and 649 industrial robotics firms by 2018. The PRD is witnessing an unprecedented growth of high-tech industries.

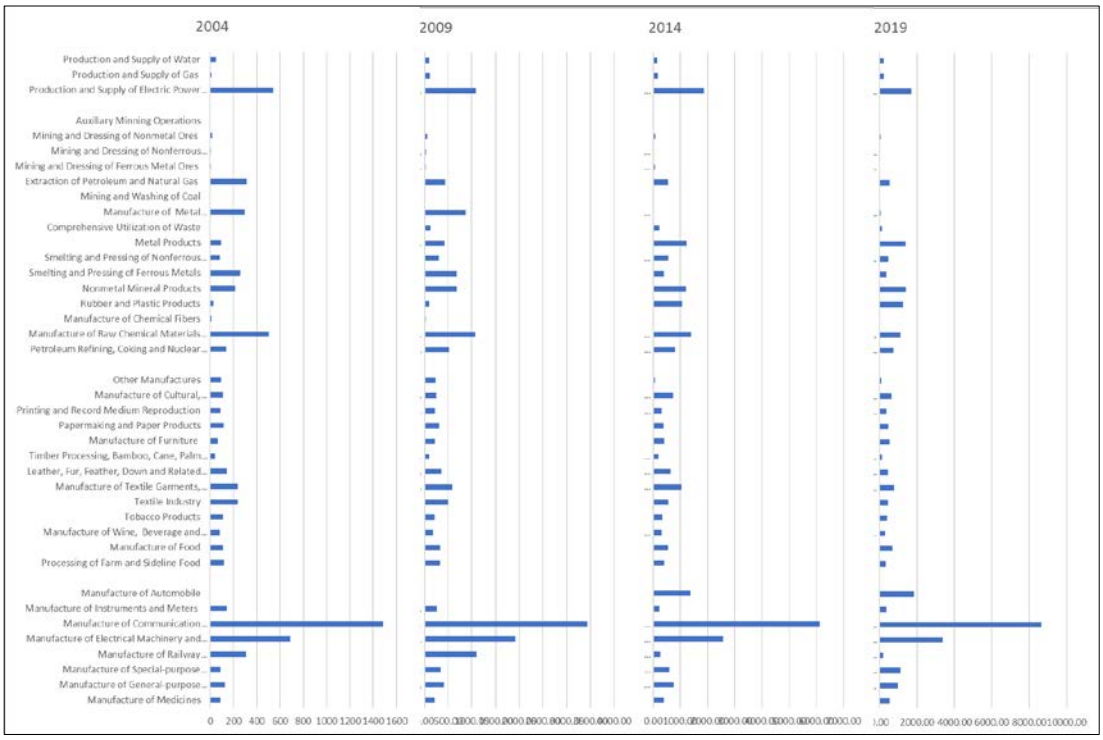


Figure 5.13 Value-added of Industry above Designated Size Grouped by Sector (2004-2019)

Source: Guangdong Statistical Year Book 2020

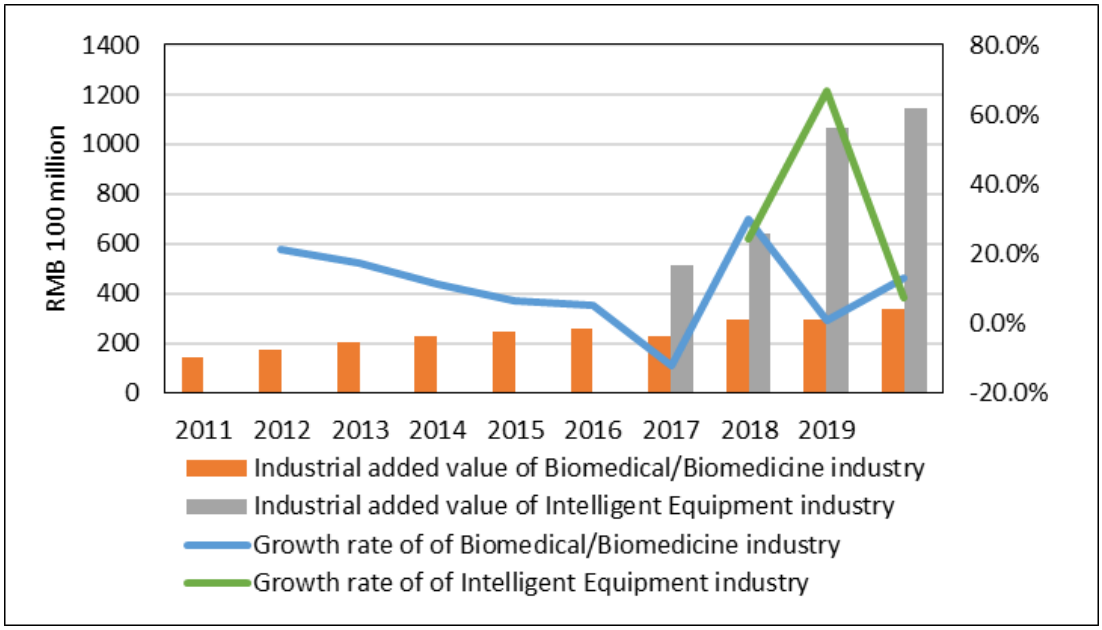


Figure 5.14 Biomedical/Biomedicine and Intelligent Equipment industries in Shenzhen during 2011-2019

Source: City Statistical Yearbooks of Shenzhen 2012-2020

5.4.1.2 Explanations on Economic Restructuring in the PRD since the 2000s

Economic restructuring is a dynamic, complex and continuous process taking place in any region. In economically advanced and rapidly changing regions such as the PRD in China, influential factors that drive regional economic restructuring are diverse and multidimensional. Under this context, this study attempts to explain the economic restructuring process in the PRD since the mid-2000s through different perspectives, namely, production resources and market structure, institutional context, GPNs and regional economic evolution.

5.4.1.2.a Changing Production Resources and Market Structure

Prior to the late 1970s when the economic reform and opening up in China was initially launched, the PRD was characterised by an overwhelmingly agricultural economy, since state investment preferred old heavy industrial bases and Western inland regions to Southeast coastal regions during the central planning period (Lin, 1997; Ma, 2002). Benefitting from its geographical proximity and cultural affinity to Hong Kong, and its 'one step ahead' position in China's economic reform and opening-up, the PRD embraced a rapid industrialisation that is primarily driven by export-oriented and labour-intensive industries relocated from Hong Kong since the late 1970s (Yeh, 1997; Lin, 1997). During the first two decades since the economic reform and opening-up, the PRD was able to attract considerable foreign investments because of its terribly low costs of land, labour force and business operation, together with robust market demands from developed countries in North America and Western Europe (Yeh, 2001). The PRD development model thus became one of the classical regional development models in post-reform China, labelled as the 'Exo(genous)-urbanisation' model or PRD model (Sit and Yang, 1997; Wei et al., 2009).

Nevertheless, the 'Exo(genous)-urbanisation' model, encountered serious challenges since the mid-2000s partially as a result of changing conditions of production resources and market structure faced by the PRD. Given the continuous capital accumulation, dramatic construction land expansion, deterioration of environment quality and changing rural-urban social structure in Mainland China, the cost of running business in the PRD—including the wage of labour force, land rental

price and various fees for environmental protection—remarkably elevated, raising intractable issues for the once-effective development model (Yeh and Li, 1999; Yang, 2012). Since the early 2000s, issues related to a shortage of low-wage migrant labour forces and increasingly intense labour-capital conflicts have been publicly exposed. Rural-urban migration then reached a turning point owing to most of rural surplus labour forces had been absorbed by industries in Chinese cities (Chan and Buckingham, 2008). Migrant labour in the PRD was striving to obtain a higher level of salary and living standard than in their previous employment. Meanwhile, the rapidly expanding land for construction exacerbated environmental quality and approached the limit of regional resource-bearing capacity in the PRD, which also ceased the sustainability of previously efficient development model. Furthermore, the destination market for PRD products has been redirected from an export-oriented one towards a more mixed one comprising both overseas and domestic markets (Zhou, 2015; Yang and Fu, 2017). Along with the appreciation of the RMB under the international pressure to China since the mid-2000s and the subsequent 2008 global financial crisis, overseas market demands for labour-intensive products in the PRD have massively decreased (Yang and He, 2017). Overall, the changing production resources and balancing market structure jointly caused the PRD's departure from the previous development model that primarily relied on export-oriented and labour-intensive industries and low-wage and low-skill migrant labour from inland regions.

5.4.1.2.b Institutional Context in Restructuring

During the early period in post-reform China, marketisation, globalisation, and decentralisation can properly summarise the institutional restructuring, specifically in a gradual manner (Wei, 2000). In this institutional restructuring, the PRD enjoyed 'one step ahead' in China (Vogel, 1989), resulting in its intensive overseas linkages through foreign direct investments and global trade, *laissez-faire* market environment, entrepreneurial local governments and even half-hearted regulatory framework for any activities that are beneficial to regional and national economic development (Xu and Yeh, 2005). The PRD opening created new space for foreign capital to achieve 'spatial fix' to retain capital accumulation and value capture (Zhou et al., 2011). The *laissez-faire* market environment was attractive for capital accumulation by market forces. Entrepreneurial local governments under fiscal decentralisation and political motivation formed solid coalitions between governments and business community in

the pursuit of interest (Xue and Wu, 2015). Moreover, half-hearted regulatory framework offered a flexible business climate and lowered transaction cost for carrying out business in the PRD. Under this institutional context, the PRD gradually became an energetic production base and so-called 'World Factory' to accommodate export-oriented and low-end industrial activities (Lin, 2006; Yang and He, 2017).

However, the institutional context has considerably changed since the early 2000s. Firstly, China entered the World Trade Organisation (WTO) in 2001, which to a greater extent aligned business rules and standards in the PRD with national and international ones. On top of the overseas market, the Chinese domestic market became more open and larger for exploration. Foreign investments newly arrived in the PRD were more interested in the booming domestic market (Yang, 2007). Secondly, after recognising the PRD's decreasing comparative advantages in developing processing and trade activities, the central government initiated a series of policies to change its institutions, such as the removal of tax rebate incentives and the augmentation of prohibited industrial categories for non-deposit processing trade (Yang, 2012). These top-down policy endeavours aimed to relocate low value-added processing and trade activities from coastal areas to central and western inland regions to achieve a geographically even development across China, while promoting higher value-added development in the coastal areas (Zhu and Pickles, 2014). Thirdly, the Guangdong provincial government promulgated the 'double relocation' strategy to facilitate the outward relocation of low-end, highly-polluted, low value-added industries and low-skill labour force into the less developed northern, western, and eastern Guangdong to foster high-end environment-friendly and high value-added industries and high-skill labour force in the PRD. The 'double relocation' strategy aimed to redistribute industrial activities at the provincial level and underpin industrial upgrading in the PRD as the core region of Guangdong (Lim, 2016). Pairs of industrial transfer parks and matched infrastructure were constructed in less developed regions in Guangdong to create space for relocated industries from the PRD. In summary, the changing institutional context further squeezed the profit room and survival possibilities for export-oriented labour-intensive industries in the PRD, which forced its economic entities to choose to move up, out or west (see also Yang, 2012; Yeh an Chen, 2020).

5.4.1.2.c Reconfiguration of GPNs

The rapid industrialisation and development of the PRD cannot be isolated from the intrusion of global economic forces and its insertion into global value chains and global production networks (Lin, 2006; Yeung, 2009). Since the late 1970s, China's economic reform and opening up and the PRD's 'one step ahead' in particular intersected well with the global industrial relocation trends from industrialised developed countries in the Global North to industrialising developing countries in the Global South and from newly industrializing economies in Asia (e.g. South Korea, Taiwan, Hong Kong, Singapore) to less developed countries (Chen, 2007). The PRD embraced the relocation of foreign-funded industries and therefore engaged smoothly into the Western flagship firm-led and Western market-oriented global production networks (GPNs). From the 1970s to early 2000s, the PRD grew rapidly into a 'World Factory' by participating in GPNs, primarily from the low-end position of supplying labour-intensive and low value-added products to flagship Western firms. During this period, a considerable majority of firms in the PRD were Original Equipment Manufacturers (OEMs) that took advantages of its low-cost labour and land in the international division of labour, but lack power in the GPNs (Yang and Liao, 2010; Liu, 2017).

Since the mid-2000s, however, globalisation reached a turning point in the organisation of global trade and investments (Yang, 2020). The former significance of labour-intensive products and low-skilled labour force in the GPNs declined. Technology-intensive products and high-skilled labour force gained much more weight in connecting GPNs in the new era, under new reverse globalisation trends, complicated geopolitical conflicts and technological breakthroughs worldwide (Yeung and Coe, 2015). Consequently, labour-intensive OEMs riding on low-cost and low-skilled labour force gradually decoupled from GPNs. By contrast, technology-intensive suppliers with R&D capacity and swift responding ability to downstream clients turned out more capable of engaging in GPNs (Yang, 2020).

On top of the changing nature of GPNs since the mid-2000s, restructuring in flagship players and target markets have also shifted. Firstly, flagship multinational corporations creating and governing GPNs are not only from Western developed economies but also from developing countries, such as several technological giants (e.g. Huawei, ZTE, Tencent) from China (Yang, 2020). Flagship multinational corporations from China became more powerful in their respective GPNs. Secondly, the target

market of GPNs moved from the ever-dominant overseas market towards the enlarging Chinese domestic market and emerging markets in newly industrialising economies in Southeast Asian and Africa. The shifting destination market remarkably restructured the organisation of GPNs and facilitated the interest redistribution among relevant stakeholders (Yang and Fu, 2017). For example, facing different kinds of market, upstream R&D activities require distinctive capabilities to deal with geographically sensitive market preferences, which may reduce the effectiveness of former flagship governors in the GPNs (Yeung and Coe, 2015). In other cases, despite having strong R&D capacity, several firms must rely on differentiated downstream distributers that control key channels to access the target market (Yang and Fu, 2017). Thus, the prominent reconfiguration of GPNs is highly coupled with economic restructuring in the PRD.

5.4.1.2.d Historical Accumulation and Regional Economic Evolution

For regional economic restructuring, endogenous dynamics is as important as exogenous forces, especially through the lens of recently booming evolutionary economic geography (Boschma and Lambooy, 1999; Boschma and Frenken, 2006; Grabher, 2009). Even though the PRD essentially resorted to exogenous dynamics, specifically global capitalist forces, to achieve the first round of capital accumulation and industrialisation, its subsequent economic restructuring cannot be attained without the historical accumulation over the previous decades. Considerable evidence shows that the PRD benefits enormously in capital, technologies, labour forces and modern management from the injection of FDI and intensification of global trade linkages (Lin, 2006; Zhao and Zhang, 2007; Yang and Liao, 2010; Z. Xu and Yeh, 2013). Following the arrival of various sources of foreign investments, indigenous economic forces have been fostered and cultivated. For example, Taiwanese electronics firms that initially relocated to the PRD only chose to use their own supply chain originating from Taiwan, but gradually began to select domestic firms that can satisfactorily meet their demands via ‘learning by doing’ (Yang and Liao, 2010). Emerging endogenous dynamics, although not yet considerable, derived from former exogenous forces plays a key role in the economic restructuring in the PRD (Yeh et al., 2020; Yang, 2020).

Regionally, the PRD has been transforming from an agglomeration of processing and trade (*sanlai yibu*) platform towards being a high-tech centre in China. The regional transformation is an outcome of the aggregation of micro-level evolutionary

mechanisms and city-level path creations. As for micro-level evolutionary mechanisms, many leading firms have developed from former processing and trade entities towards independent innovative players by consistent upgrading and transformation endeavours, thereby achieving product diversification and complication, such as Tencent and Huawei in Shenzhen. In Dongguan, several founders of high-tech firms are former employees in Hong Kong- and Taiwan-funded foreign companies. These founders have acquired industrial experience and practiced core technological crafts, which paved the way for their subsequent entrepreneurship and independent R&D input (Li et al., 2020). Analysis of leading firms listed in a brand new national-level industrial zone, the Songshan Lake Science and Technology Industrial Park in Dongguan, indicates that over 60% are spin-offs and sectoral diversified branch offices of incumbent firms in Dongguan and Shenzhen. Similarly, Guangzhou has been transforming from a traditional trade and service centre into a high-technology hub fostering bio-tech, AI and new-generation of information technology, under which its state-owned industrial forces accumulated in the planning economy stage and well-developed market apparatus play a pivotal role.

5.4.2 Characteristics of sample firms surveyed

5.4.2.1 Firm attributes

5.4.2.1.a Development Trends

From the number of firms established in each year (Figure 5.15), the majority have been founded since the mid-2000s, particularly in the aftermath of the 2008 global financial crisis. By the division of sample firms into domestic-, Hong Kong-, and overseas-funded ones, three waves of development are identified. Hong Kong-funded firms dominated in the 1990s, with a relatively longer history in the PRD. The second wave witnessed the rise of other overseas-funded firms since the early 2000s when China successfully entered the WTO. In the post-crisis era, the rise of domestic-funded firms as the third wave of development is highly apparent. Thus, the emergence of high-tech industrial development in the PRD has primarily taken place since the mid-2000s, which reflects the new economic dynamics switching from relying on exogenous foreign investments to fostering endogenous development (Yang, 2015).

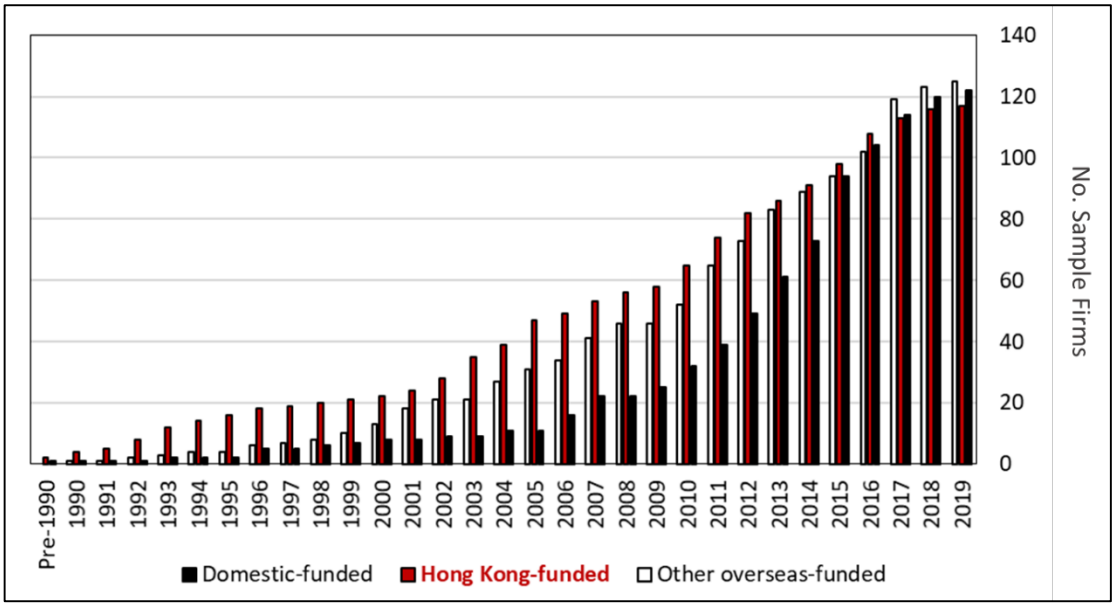


Figure 5.15 Accumulated number of sample firms founded in each year

5.4.2.1.b Types of Firms

The sample firms are primarily Wholly Foreign-Owned Enterprises, Private or Private Holding Enterprises and Sino-Foreign Joint Venture Enterprises. Sino-Foreign Contractual Enterprises, State-Owned or State Holding Enterprises and Other Types of Enterprises (e.g. Collectively-Owned Enterprises) account for a limited proportion of the sample (Figure 5.16). Together, Wholly Foreign-Owned Enterprises and Private or Private Holding Enterprises account for over 80% of all sample firms, reflecting the market forces as the increasingly key contribution to the development of high-tech industries in the PRD. The finding also mirrors the significant deviation from the dominance of processing and trade types of firms (e.g. *sanlai yibu*) and collectively-owned rural industrial firms during the early reform period (see also Lin, 2006).

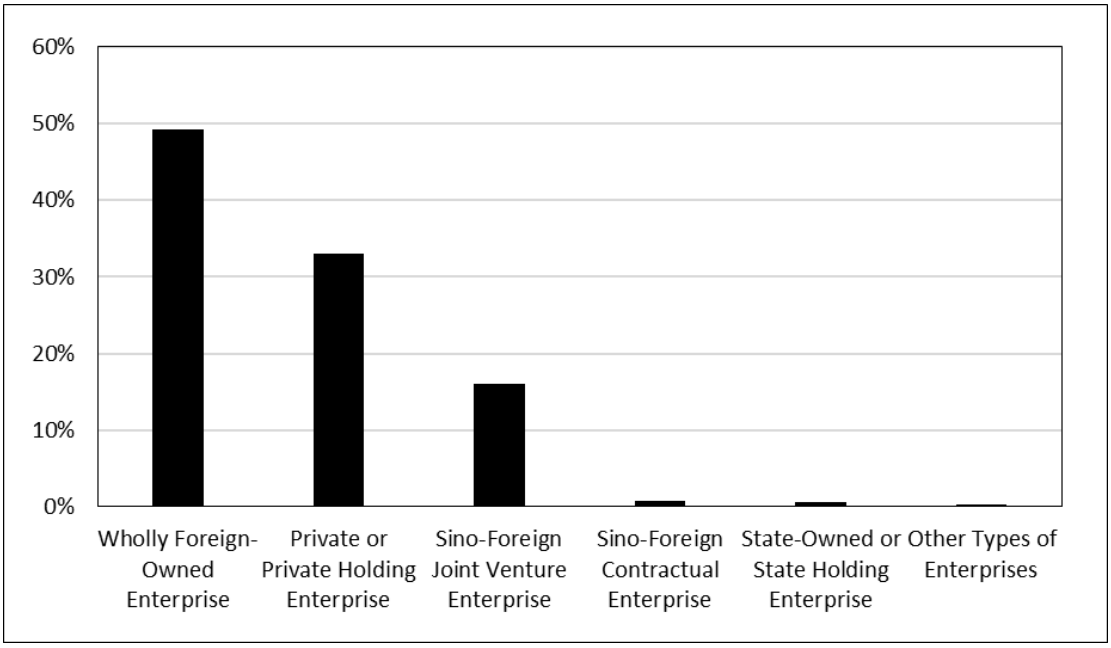


Figure 5.16 Types of sample firms

5.4.2.1.c Employment Scale

In terms of the employment scale of all sample firms, small-and-medium sized firms play a central part. Firms with 1-50, 51-100 and 101-500 employees account for over 85% of all samples (Figure 5.17). In terms of the employment scale of different types of firms, a divergency is observed among domestic-, Hong Kong-, and overseas-funded ones. In contrast to the other two types of firms, domestic-funded firms tend to have a smaller employment size, which indicates the endogenous development forces that are still in the early stages. Domestic-funded firms with 1-50 and 51-100 employees occupy approximately 60% of the total. Other overseas-funded firms have a more balanced employment structure. The distribution of firm quantity with 1-50, 51-100, 101-500, and above 500 employees is rather even in quantity. However, Hong Kong-funded firms tend to have a larger employment scale. For instance, firms with 1-50 employees only have less than 3% while those with 51-100, 101-500, and 501-100 employees account for over 90% of the total. This finding remarkably differs from Hong Kong investments in the PRD during the early reform period when most of the Hong Kong-funded firms were small-scale, labour-intensive and process-and-trade-oriented (Sit and Yang, 1997; Yeh and Xu, 2006).

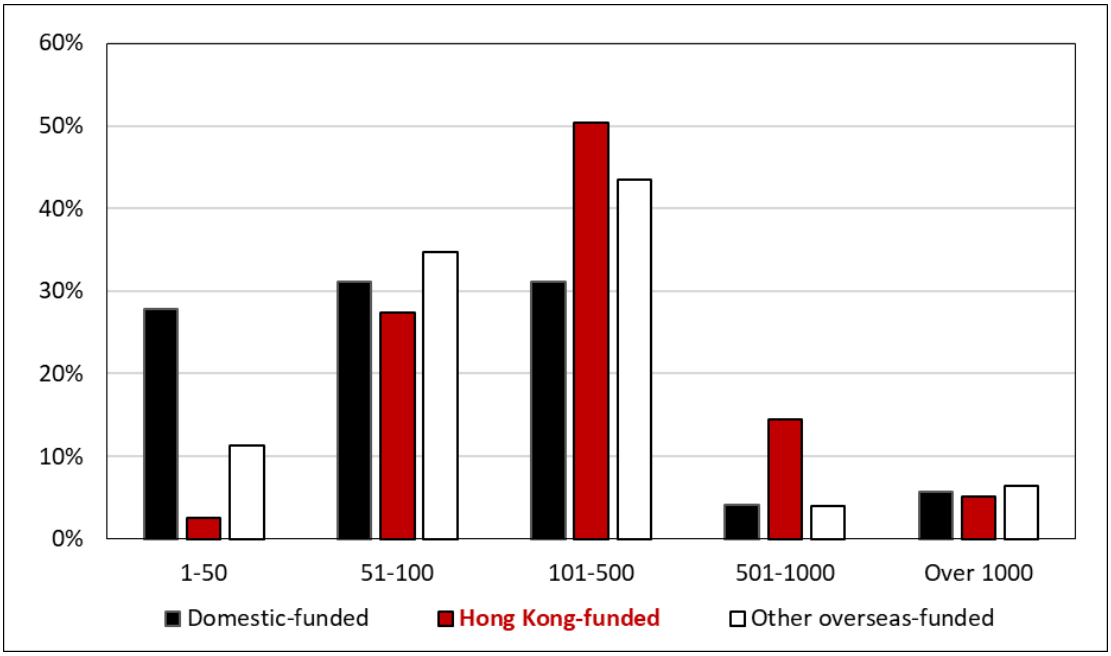


Figure 5.17 Employment scale of sample firms

5.4.2.1.d Types of Products

Over 65% of the sample firms generate final products instead of raw materials/components or intermediary items (Figure 5.18). This finding indicates a dramatic restructuring of the production networks in the PRD. The division of three types of firms shows that more domestic-funded firms than Hong Kong- and other overseas-funded firms are manufacturing final products (Figure 5.19). By contrast, more Hong Kong- and other overseas-funded firms engage in production activities of raw materials/components and semi-finished products. This finding shows that to a greater extent, Hong Kong- and other overseas-funded firms participate in the GPNs led by flagship enterprises, while domestic-funded firms strive to become the leading players in regional, national and GPNs.

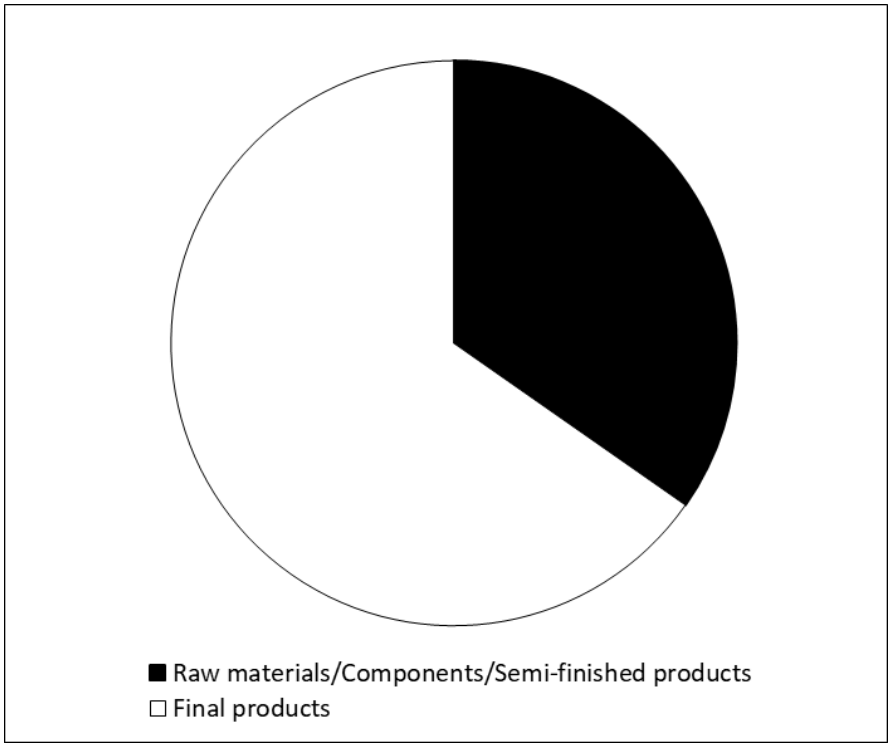


Figure 5.18 Product types of sample firms

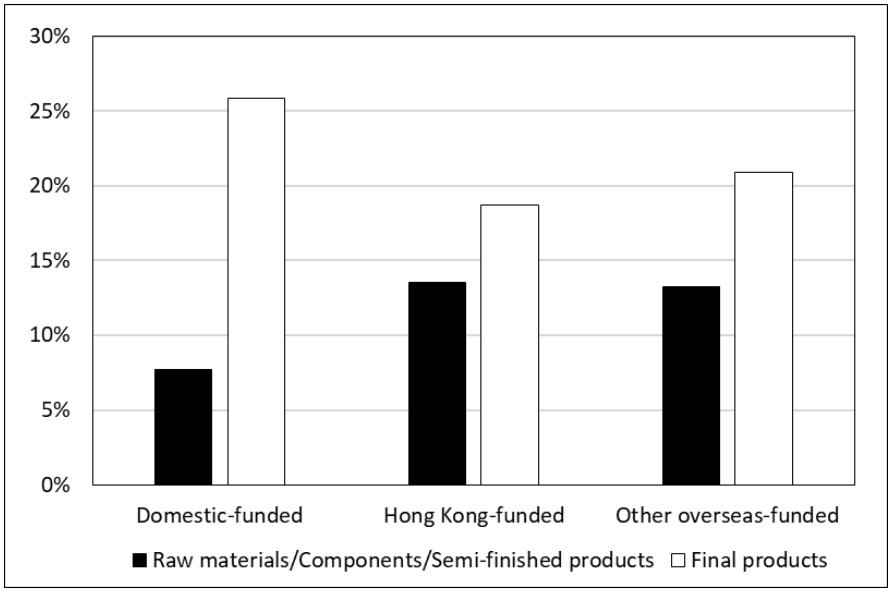


Figure 5.19 Product types of sample firms with different capital sources

5.4.2.1.e Production Firms

Over 80% of firms are already Original Design Manufacturers (ODM) and Original Brand Manufacturers (OBM) (Figure 5.20), which has considerably changed from earlier economic situations in the PRD in which processing and trade business entities and Original Equipment Manufacturers (OEM) were pervasive (Sit and Yang, 1997; Lin, 1997). Although OEM still occupy approximately 30% of the sample, the firms

are not exclusively OEMs but also simultaneously ODMs or OBMs as inferred by the percentages of their take up in all sample firms. This finding implies that industrial upgrading has been occurring in high-technology industrial sectors in the PRD, not only in the industrial level but also in the industrial chain level. From the three types of firms with different capital sources, the distribution pattern is more or less the same (Figure 5.21). The minor difference lies in the proportion of three types of firms in ODMs and OBMs, indicating the design and brand advantages of other overseas-funded firms over domestic-funded ones. Hong Kong-funded firms are in the intermediary position.

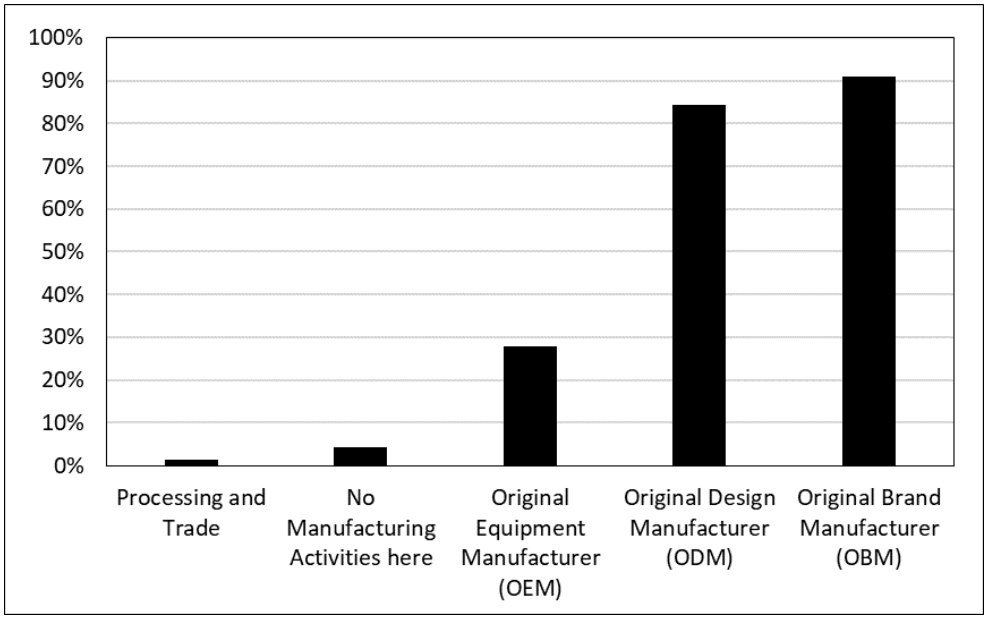


Figure 5.20 Production forms of sample firms

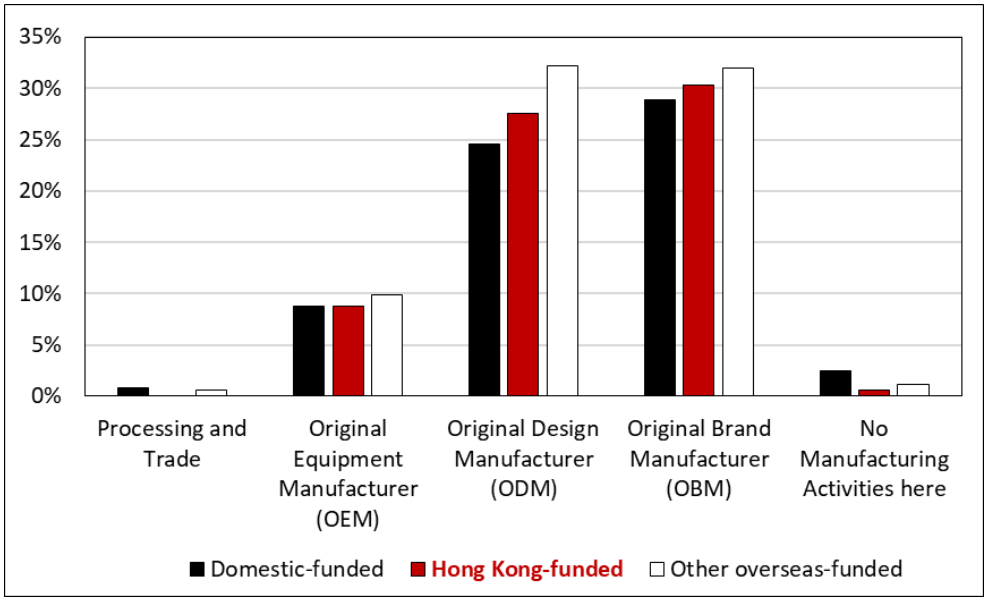


Figure 5.21 Production forms of sample firms with different capital sources

5.4.2.2 Locational Factors

The reasons for firms to be located in the PRD show the regional context and new development trends. The sample firms were asked to rate the importance of their reasons for choosing to locate in the PRD by picking out the top five out of 23 hypothesised reasons and provide a rating from 1 to 5, with 5 being the most important and 1 being the least important. From the average score of each locational factor (Figure 5.22), six reasons noticeably stand out: (1) Geographical proximity to goods suppliers; (2) Easy access to regional market; (3) Complete transport, communication, and information infrastructure; (4) Geographical agglomeration of firms in the same or related sectors; (5) Easy access to domestic market; and (6) Sufficient supply of technical/engineering workforce. Clearly, based on these research findings, the regional context of the PRD has shifted from low-cost and low-skilled labour force, sufficient land supply and flexible business climate towards a relatively complete regional supply system, an enlarging regional and domestic market, high-quality infrastructure provision, abundant high-skilled labour force, emerging agglomeration economies and institutionally improved business environment. Through rapid industrialisation and urbanisation over the past four decades, the regional context in the PRD has been remarkably restructured towards an increasingly advanced industrial system.

Further categorising locational factors according to firm capital sources, several differences among the three types of firms (Figure 5.23). Regional market, high-quality infrastructure and high-skilled labour force are of utmost importance to domestic-funded firms. Abundant suppliers, agglomeration economies and access to regional and domestic markets attract more attention from Hong Kong-fund and other overseas-funded firms. This finding indicates that domestic-funded firms prefer to take advantage of the high-quality production resources and rising regional market in the PRD. By comparison, Hong Kong- and other overseas-funded firms to a greater degree rely on increasingly complete regional supply system, agglomeration economies and multi-layer markets that they can more easily access by locating in the PRD.

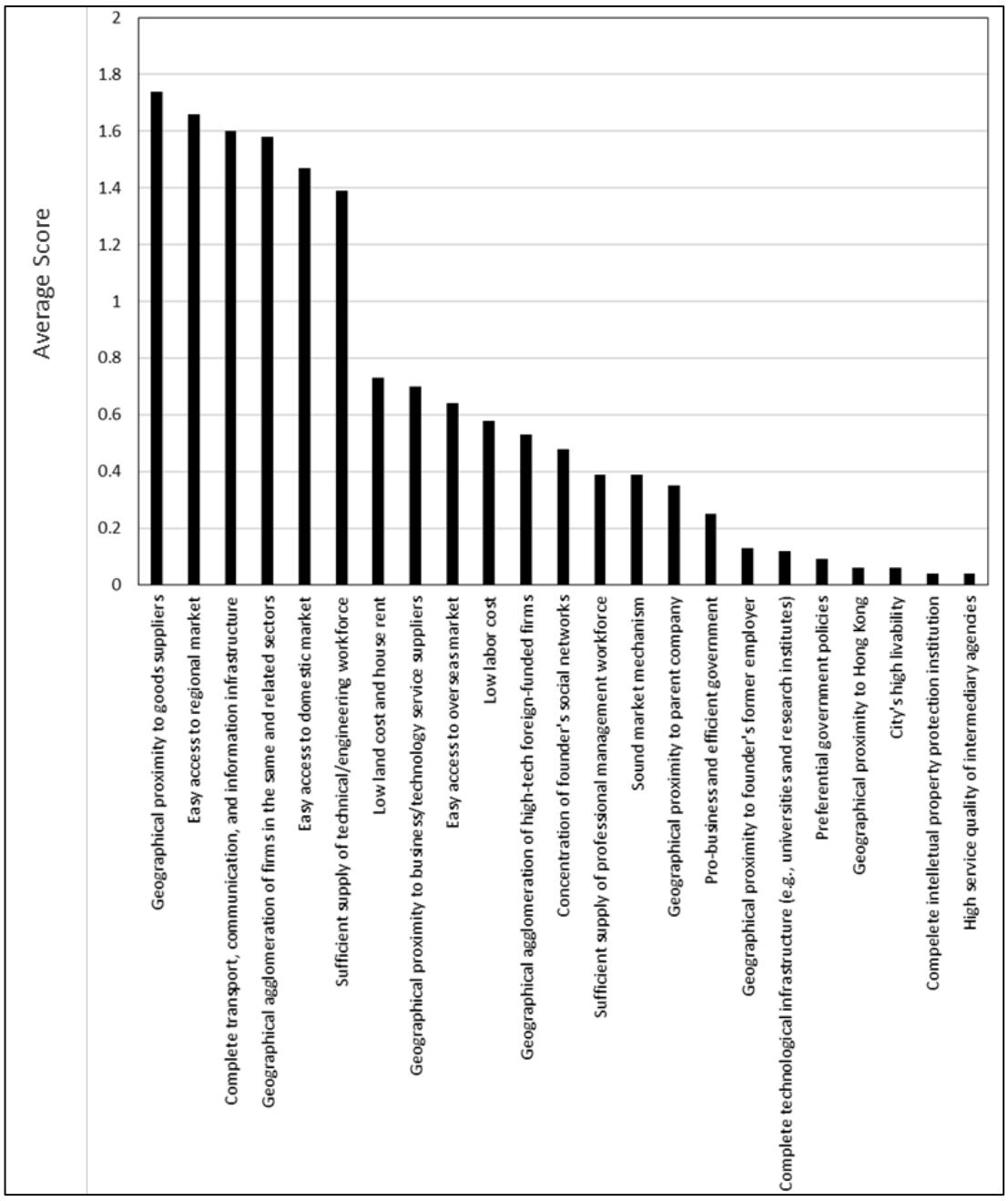


Figure 5.22 Locational factors of sample firms in the PRD

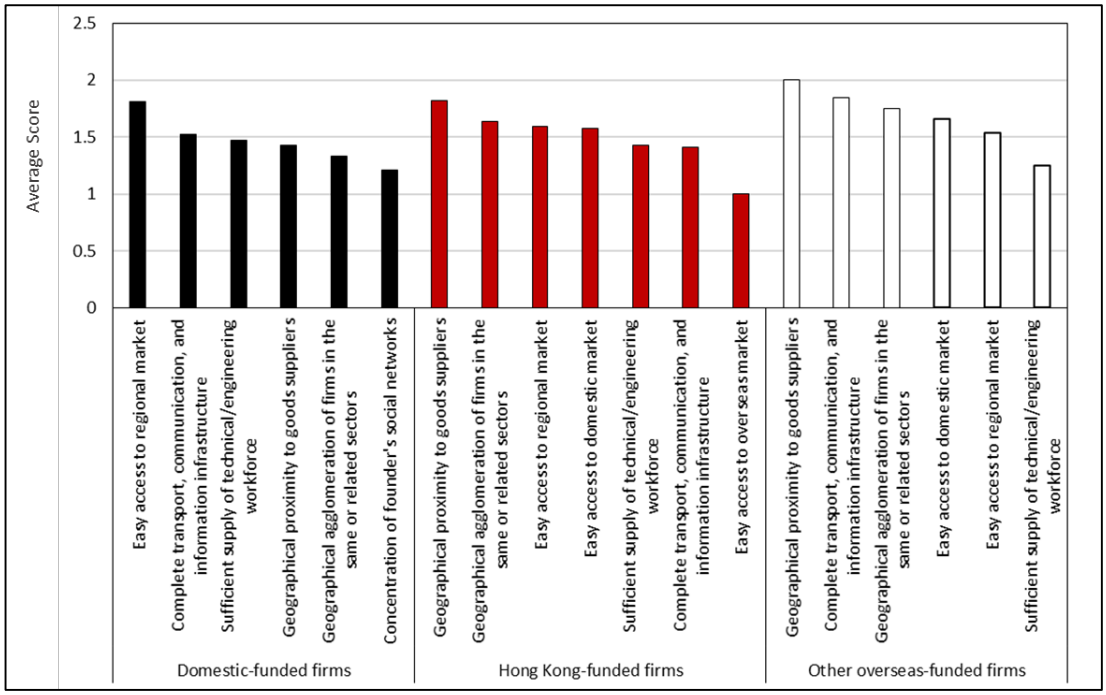


Figure 5.23 Locational factors of sample firms with different capital sources in the PRD

5.4.2.3 Market Linkages

The geographical distribution of suppliers for the sample firms is highly concentrated in the PRD (Figure 5.24). This concentration reflects the emergence of a regionally complete supply system for biomedical products and intelligent manufacturing equipment. Still, over 20% of suppliers are located in other regions in Mainland China. Overseas suppliers only contribute to less than 10% of total value. From the capital sources of suppliers (Figure 5.25), domestic-funded firms have become the primary suppliers for all types of firms. Although other overseas-funded firms occupy approximately 15% of the total, they are more inclined to supply other overseas-funded and Hong Kong-funded firms than domestic-funded ones.

The geographical distribution of market is a greater mixture of regional, domestic and overseas (Figure 5.26). The PRD regional market has jumped onto the largest market destination for the sample firms. Other regions in Mainland China and overseas countries/regions are quite similar with each other in market status. The capital composition of customers (Figure 5.27) shows that domestic-funded clients are most significant for sample firms, surpassing overseas-funded ones. This indicates the increasingly fierce competition between domestic- and overseas-funded firms in capturing the high-tech products market. Moreover, domestic firms have been

upgrading from the single role of being suppliers towards the dual role of being both suppliers and clients. By contrast, however, Hong Kong-funded firms share a rather limited portion in the market, mainly sourcing from Hong Kong- and other overseas-funded firms.

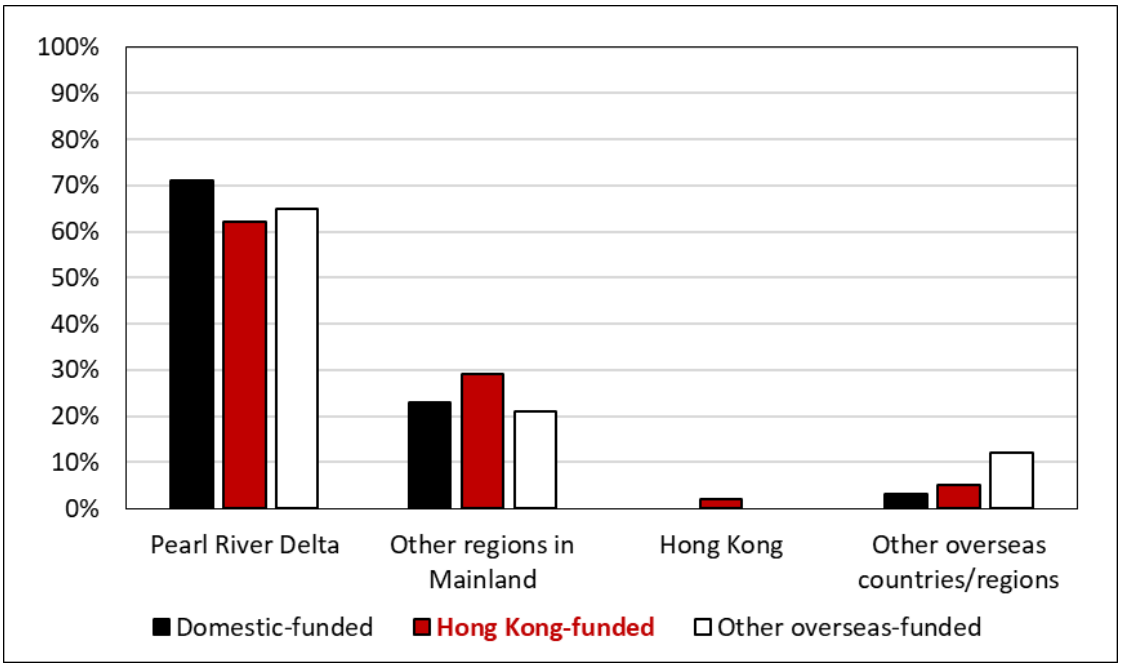


Figure 5.24 Geographical distribution of goods suppliers

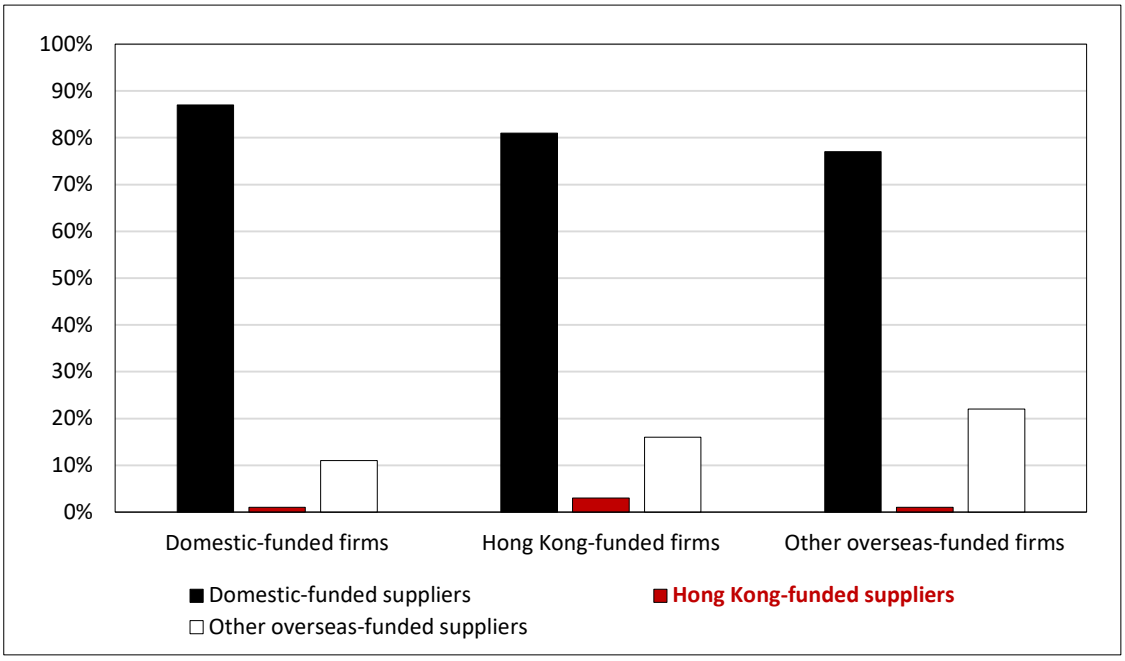


Figure 5.25 Distribution of suppliers from different capital sources

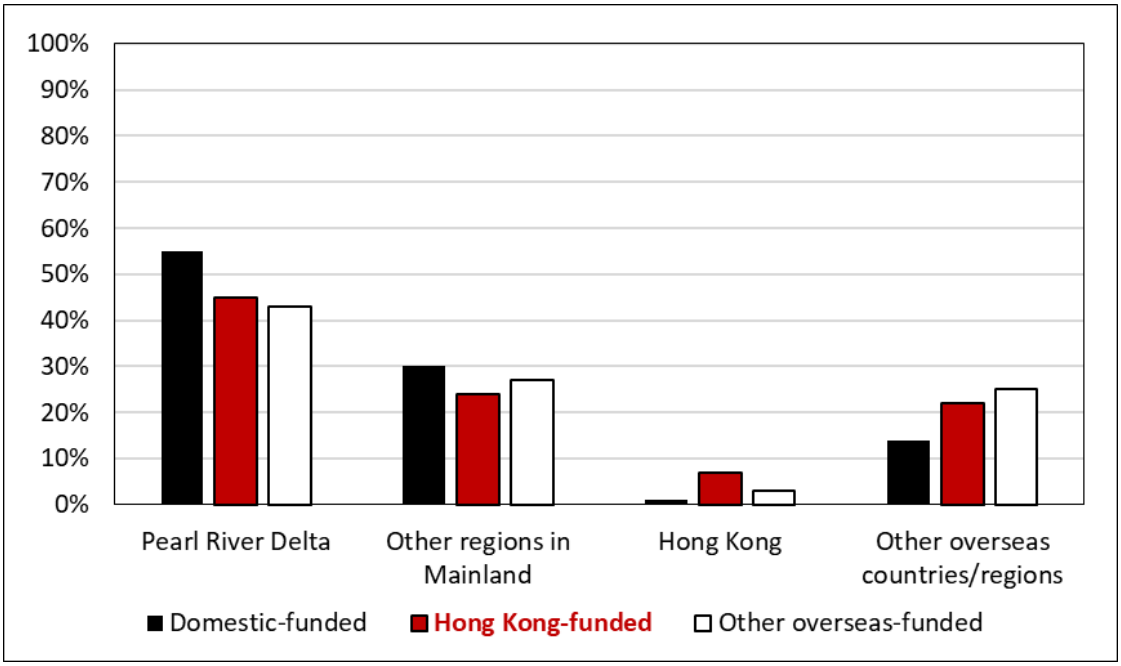


Figure 5.26 Geographical distribution of market

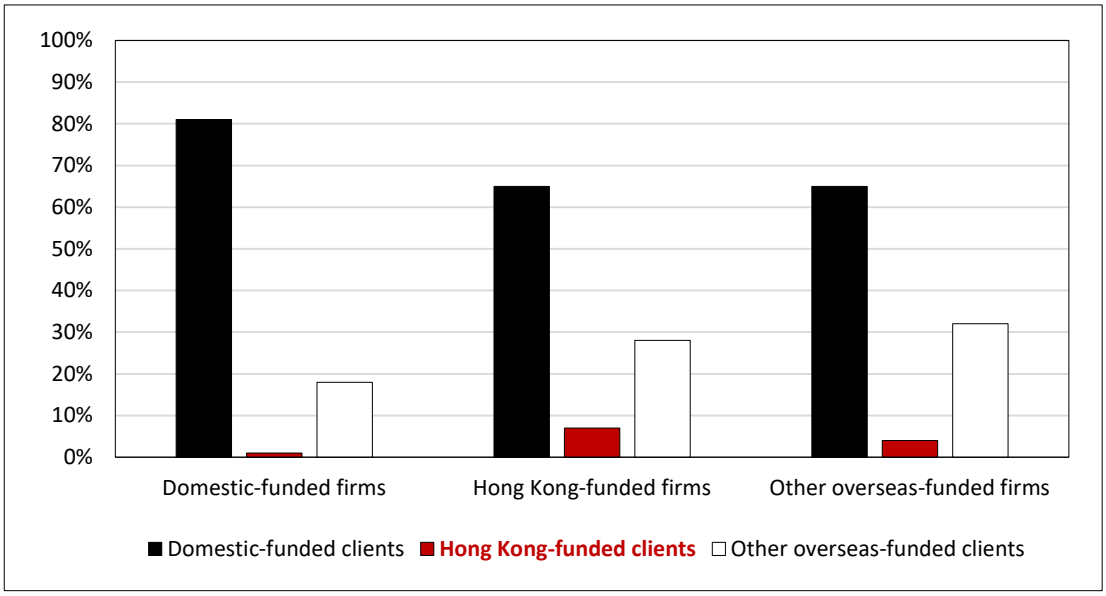


Figure 5.27 Distribution of clients with different capital sources

5.4.2.4 Market Performance

The geographical distribution of market of different types of firms (Figure 5.28) shows that Hong Kong- and other overseas-funded firms are more export-oriented than domestic-funded firms. Although the latter tends to work more on the domestic market, these firms generally have a smaller market turnover than Hong Kong- and other overseas-funded firms, revealing a rather fragmented domestic market structure and

competitive business environment. By contrast, Hong Kong- and other overseas-funded firms have larger market turnover for each firm, and thus perform better in the absolute market turnover.

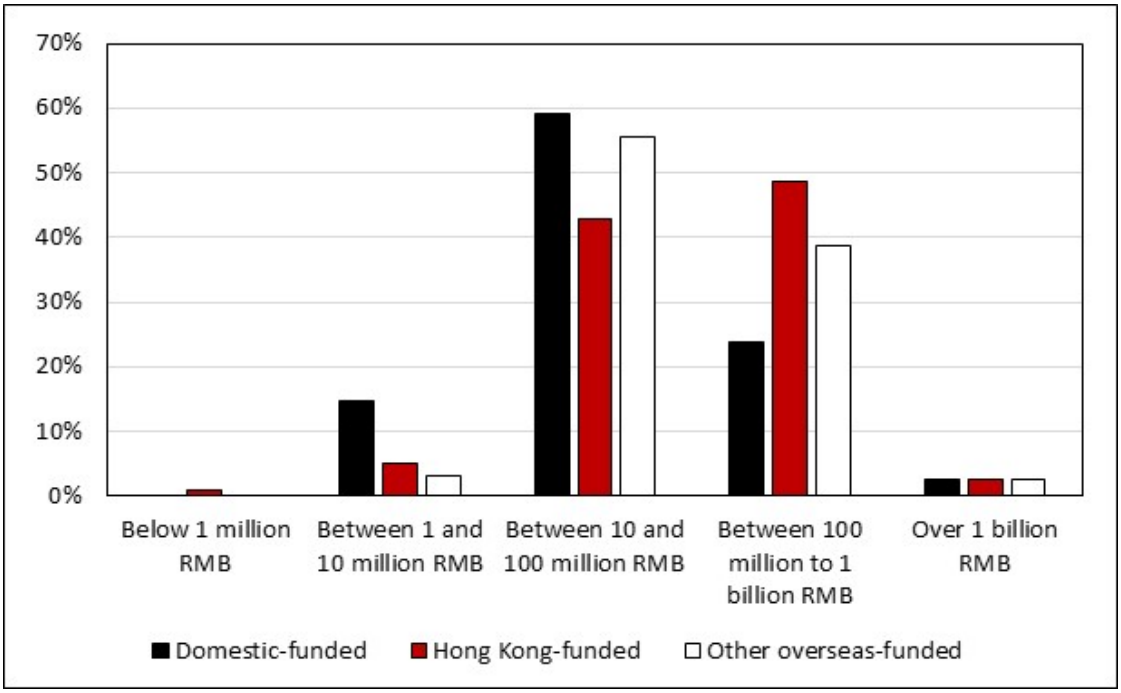


Figure 5.28 Gross income from sales of sample firms with different capital sources

5.4.2.5 Innovation performance

5.4.2.5.a Establishment of Independent Research Institute

Examination of whether firms have established independent research institute (Figure 5.29) can reveal the R&D intensity of firms to a certain extent. Overall, approximately 40% of the firms have independent research institutes and the other 60% have none. Clearly, high-tech industrial firms have internal differentiation in engaging in R&D activities. Among the three types, Hong Kong-funded firms perform the best with approximately 55% having established independent research institutes. Domestic- and other overseas-funded firms are almost similar in this respect.

5.4.2.5.b Ways to Develop New Products/Technologies

In terms of the most important means of developing new products or technologies (Figure 5.30), independent R&D has been the primary for domestic-funded firms. By contrast, other overseas-funded firms mainly rely on research outputs from their parent companies, which indicates their low-level embeddedness into the regional innovation

system. In between the two types, Hong Kong-funded firms rely upon both independent R&D and external research outputs from parent companies and other collaborators.

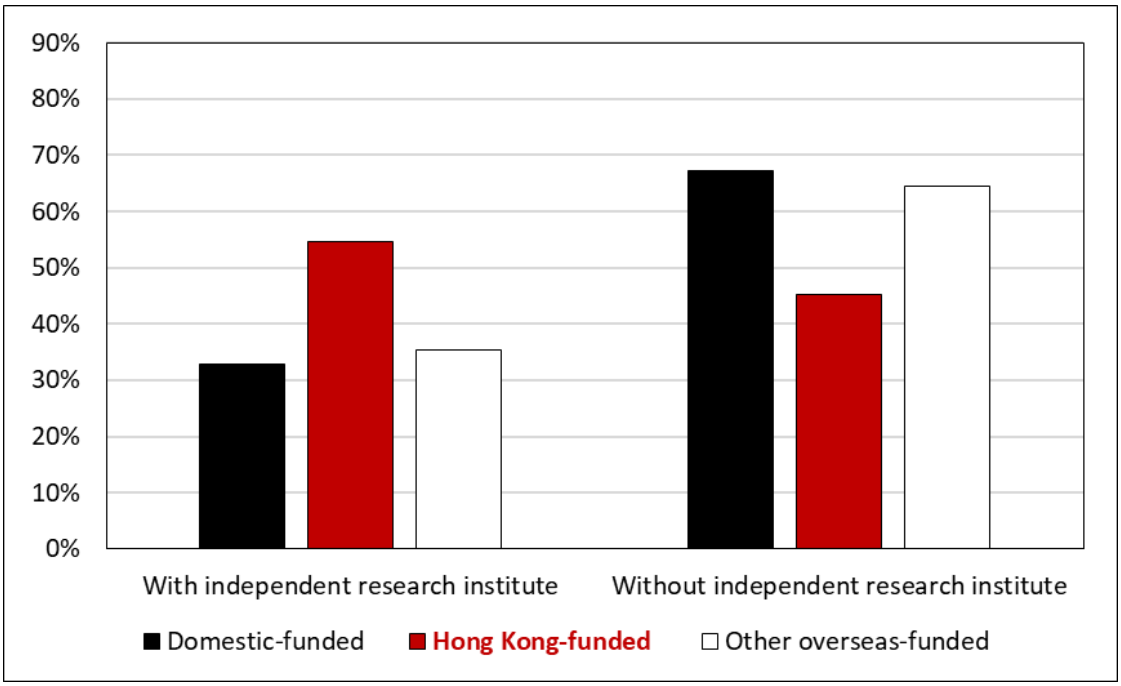


Figure 5.29 Establishment of independent research institutes

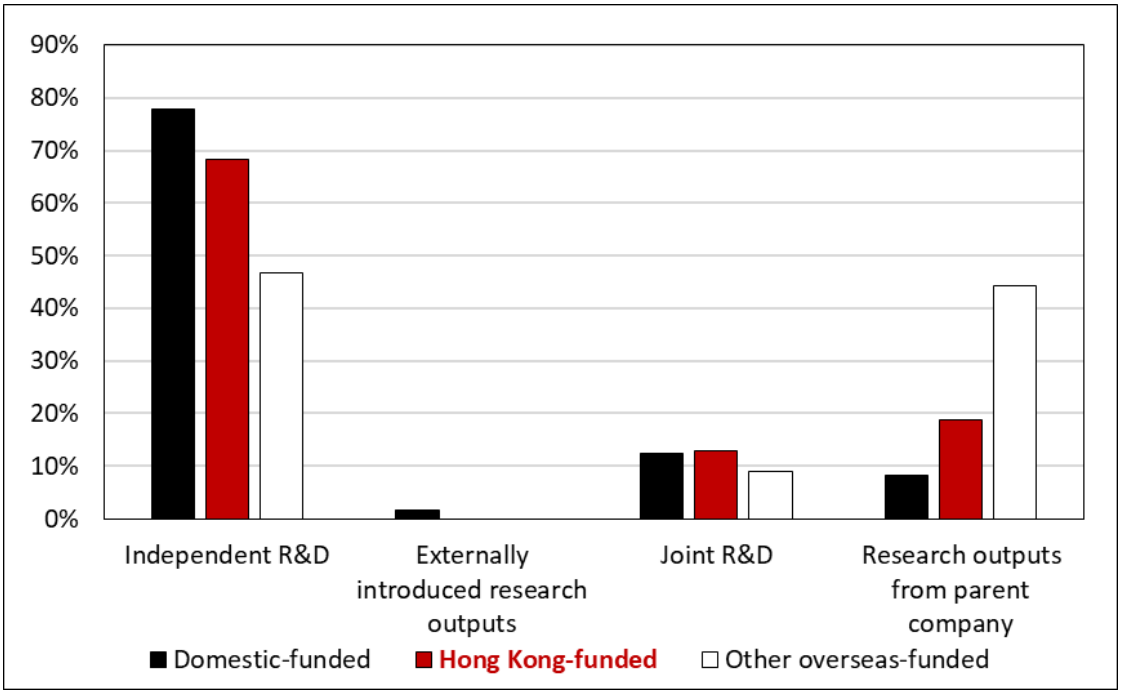


Figure 5.30 Most important way to develop new products/technologies

5.4.2.5.c Different Levels of New Products/Technologies

Different levels of new products/technologies developed by firms—namely, globally new, nationally new, firm-level new, or no new products/technologies—can measure how radical or incremental are the innovations that firms carry out (Figure 5.31). At present, the PRD still lacks globally cutting-edge technological achievements and products. Over half of the sample firms are developing firm-level new products/technologies, showing a sign of incremental innovation. Notably, the radical innovation towards the domestic market is still progressing, given that over 30% of firms are developing nationally new products/technologies.

Most of the globally new products/technologies are contributed by other overseas-funded firms (Figure 5.32). Domestic-funded firms tend to participate in incremental or follow-up innovation in developing firm-level new products/technologies. Hong Kong-funded firms are more capable of developing nationally new products/technologies. Despite being unable to compare with other overseas-funded firms in cutting-edge radical innovation, Hong Kong-funded firms perform better in taking technological advantages to compete in the domestic market.

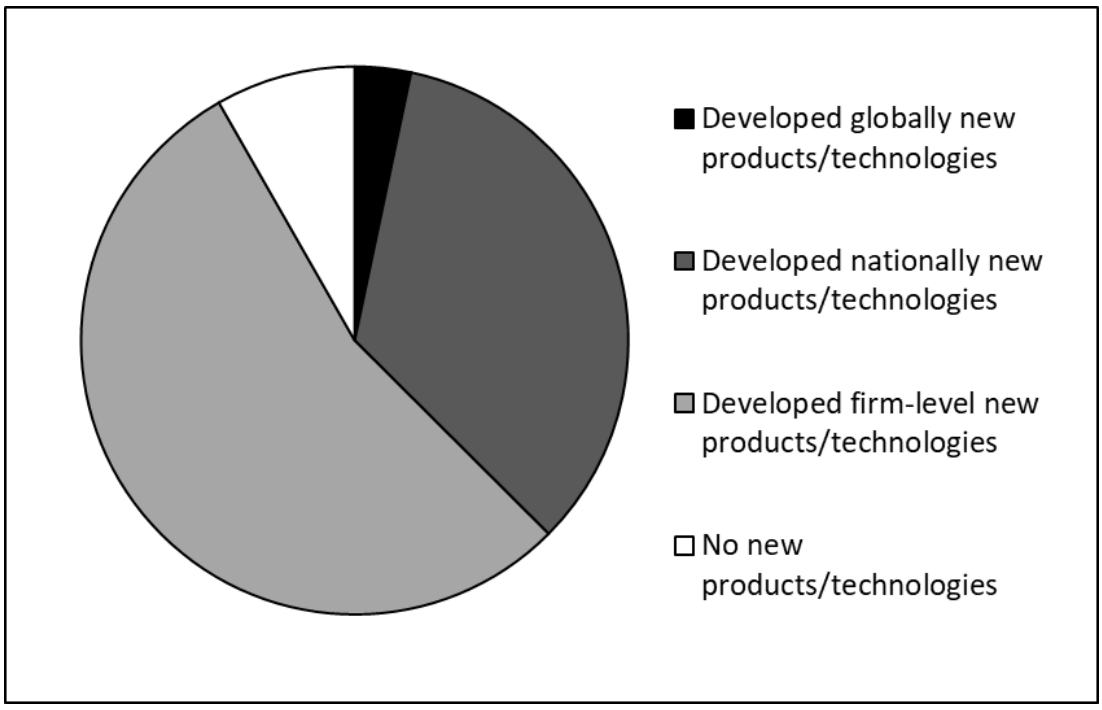


Figure 5.31 Development of different levels of new products/technologies

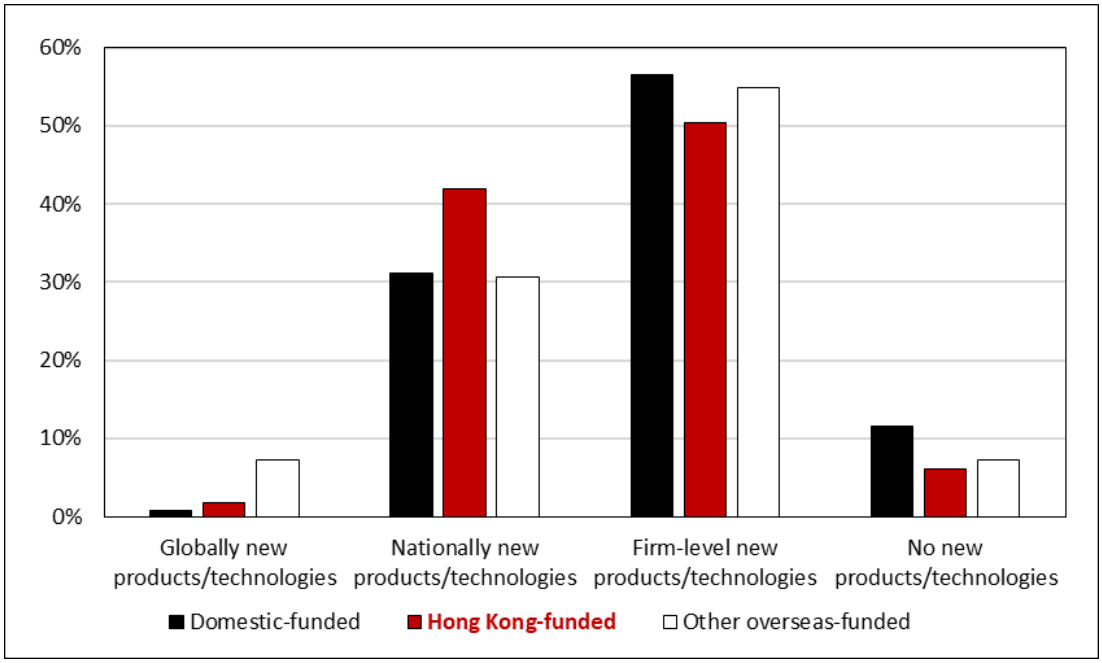


Figure 5.32 Development of different levels of new products/technologies of sample firms with different capital sources

5.4.2.5.d Patent Application

Nearly half of the sample firms applied patents during the years 2017-2019. This number is quite consistent with that of firms with independent research institutes (Figure 5.33). The different types of firms with patent application reveals that Hong Kong- and domestic-funded firms are more active than other overseas-funded firms in innovation-related activities. In particular, compared with the other two types of firms, Hong Kong-funded firms have a larger number of invention patent applications, and high-tech ones are evidenced to be technologically dynamic and advantageous in the PRD.

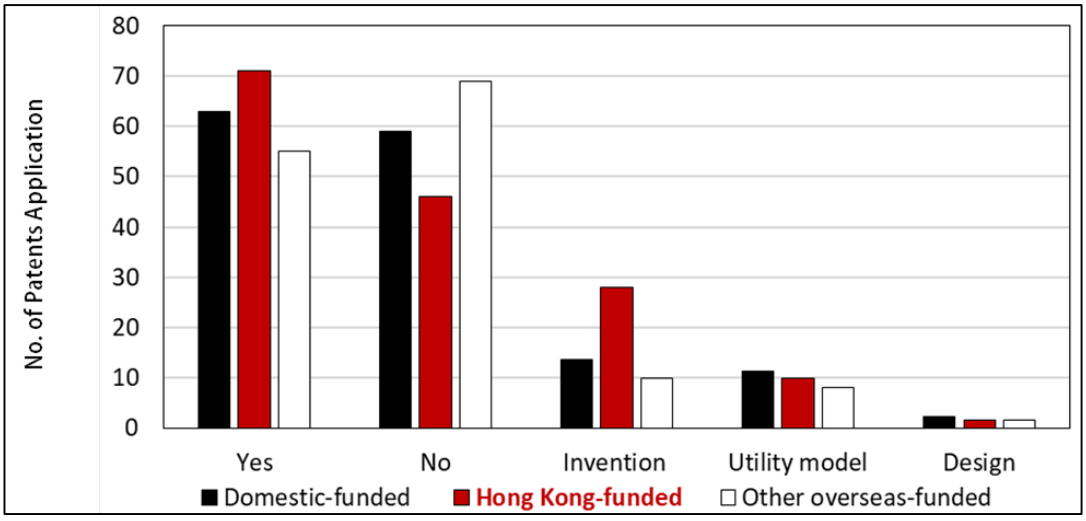


Figure 5.33 Patent application of sample firms with different capital sources

5.4.3 Hong Kong Participation in High-technology Industrial Development in the PRD

5.4.3.1 Geographical origin

5.4.3.1.a Founding of the Firm

Domestic-funded firms are mostly founded from entrepreneurship, and those invested in by parent firms only account for less than 30% (Figure 5.34). However, by contrast, Hong Kong- and other overseas-funded firms are much more relying on investment from parent firms. Over 90% of Hong Kong-funded firms are founded by foreign direct investment (FDI) instead of entrepreneurship, implying that indigenous entrepreneurship is more dynamic than exogenous forces. Cross-border entrepreneurship in the GBA remains insignificant. Due to the cross-border issues, Hong Kong capital engages in the high-tech industrial development in the PRD mainly through FDI.

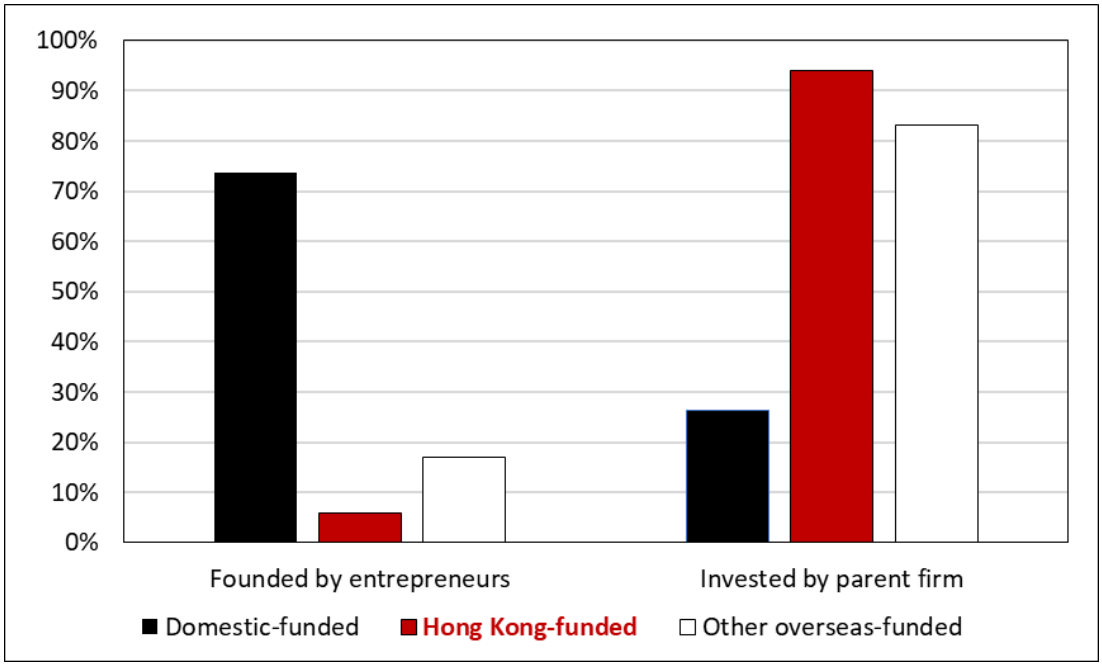


Figure 5.34 Founding of the Firm

5.4.3.1.b Geographical Origin

Geographical origin is defined as where the firm originated, such as the geographical source of spin-offs and firm diversification, or the source of FDI. Figure 5.35 shows that domestic-funded firms are primarily developed from Shenzhen and Dongguan, which indicates emerging endogenous industrial dynamics in this mega-ctiy

region. However, Guangzhou, Shanghai and Beijing do not contribute much to high-tech industrial development in Shenzhen and Dongguan. By stark contrast, Hong Kong-funded firms are almost exclusively invested from their parent firms, which means that the ways in which Hong Kong participates in high-technology industrial development in the PRD mainly zooms in on its own outward direct investments, with the parent firms headquartered in Hong Kong for decision making and strategic management. Other overseas-funded firms are more geographically evenly distributed among different countries, with the United States, Japan, Shenzhen, Taiwan and Germany as the leading sources. Apart from Shenzhen as the place where various branches of multinational corporations are located, most of other regions/countries of overseas-funded firms are leading high-tech industrial development in the world.

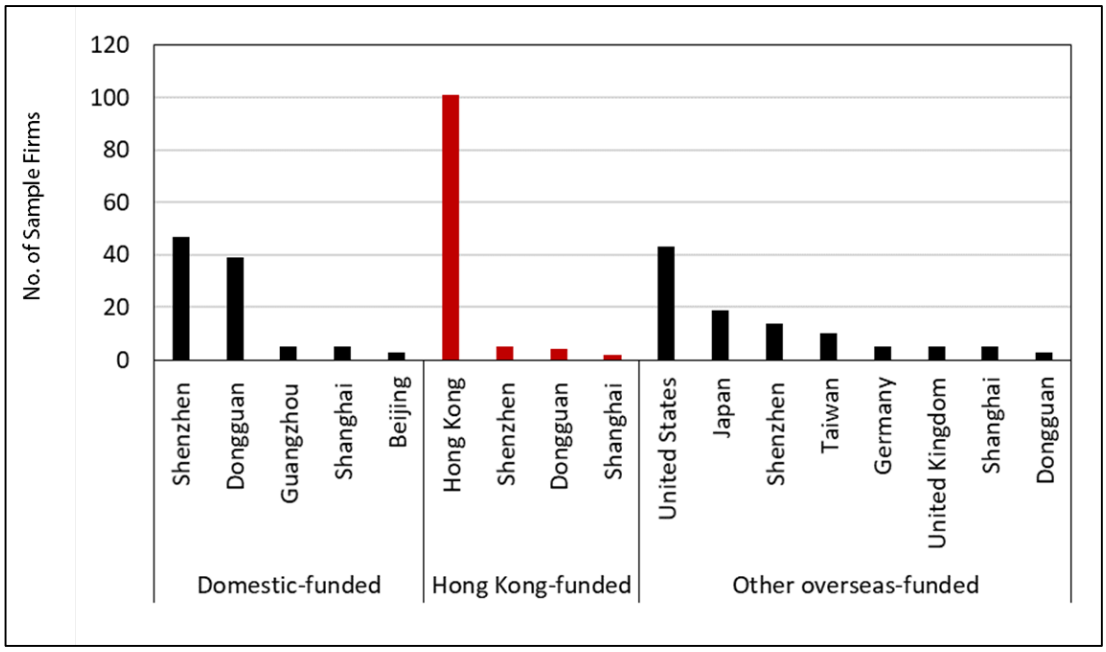


Figure 5.35 Geographical origin of sample firms with different capital sources

5.4.3.2 High-Skilled Labour Force

The geographical sources of R&D workforce and senior managers suggest that the PRD has become the dominant source of high-skilled labour force for high-technology industrial firms (Figure 5.36). High-skilled labour force circulation in the PRD is pervasive, leading to a booming market. Other regions in Mainland China perform as the second most important geographical source for supplying high-skilled labour force, possessing approximately 20% of all high-skilled labour force. Notably, compared with

domestic-funded firms, Hong Kong- and other overseas-funded firms rely more on other regions in Mainland China as sources of high-skilled labour force. However, apparently, only Hong Kong-funded firms source high-skilled Hong Kong labour force, whom domestic- and other overseas-funded firms rarely hire. Other overseas-funded firms hire more high-skilled labour force from other overseas countries/regions than Hong Kong- and domestic-funded firms (Figure 5.37).

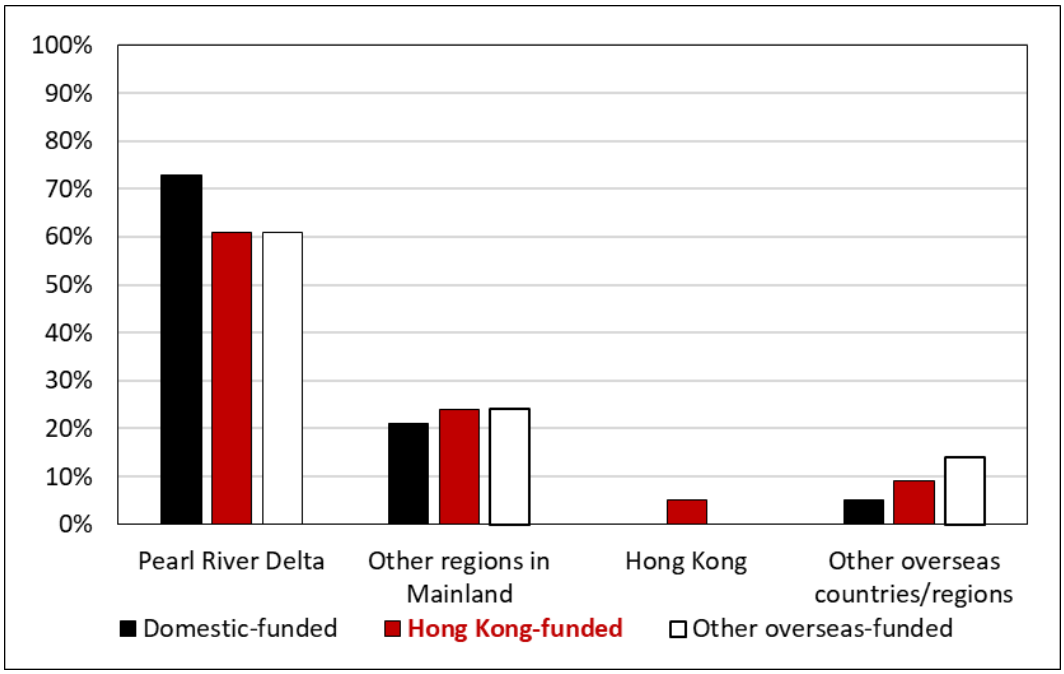


Figure 5.36 Geographical origin of R&D workforce

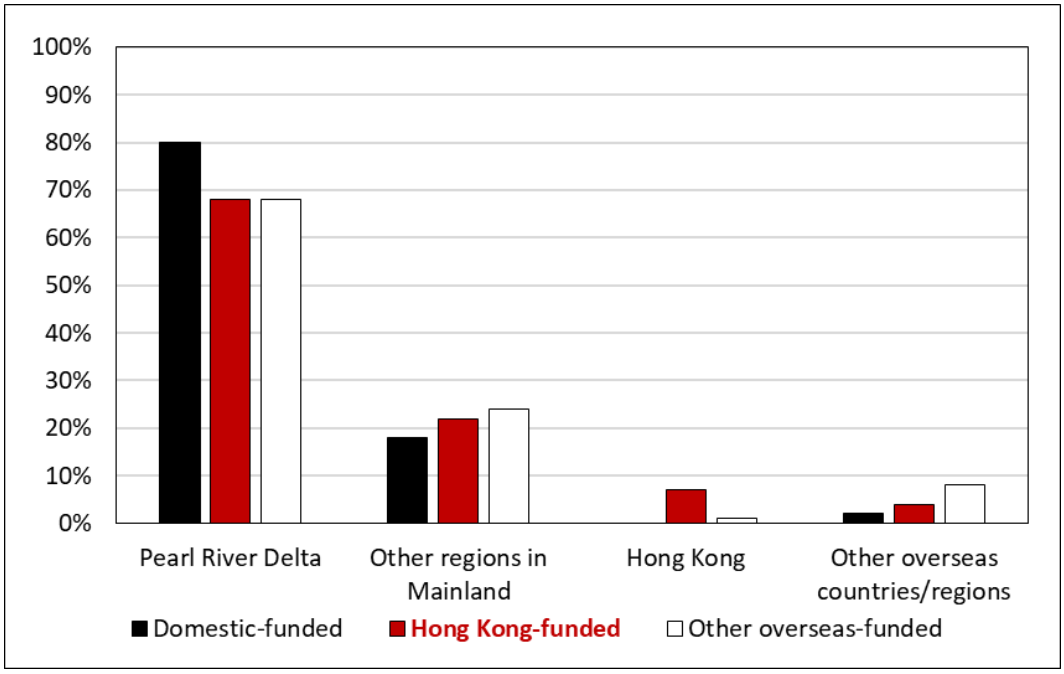


Figure 5.37 Geographical origin of senior managers

5.4.3.3 Major Technological Source

The PRD has become the most important technological source for the sample firms. Over 80% of domestic-funded firms, 60% of Hong Kong-funded firms and 50% of other overseas-funded firms use the PRD as their major technological source (Figure 5.38). This finding demonstrates that the PRD is transforming from the well-known 'World Factory' towards a high-tech R&D centre. Domestic-funded firms exclusively ride on the PRD as their technological centre, while Hong Kong- and other overseas-funded firms source major technologies both from the PRD and their home region/country. More domestic-funded firms tend to source technologies from other regions in Mainland China than Hong Kong- and other overseas-funded firms. However, the decoupling of Hong Kong technologies with domestic- and other overseas-funded firms in the PRD are apparent and alarming.

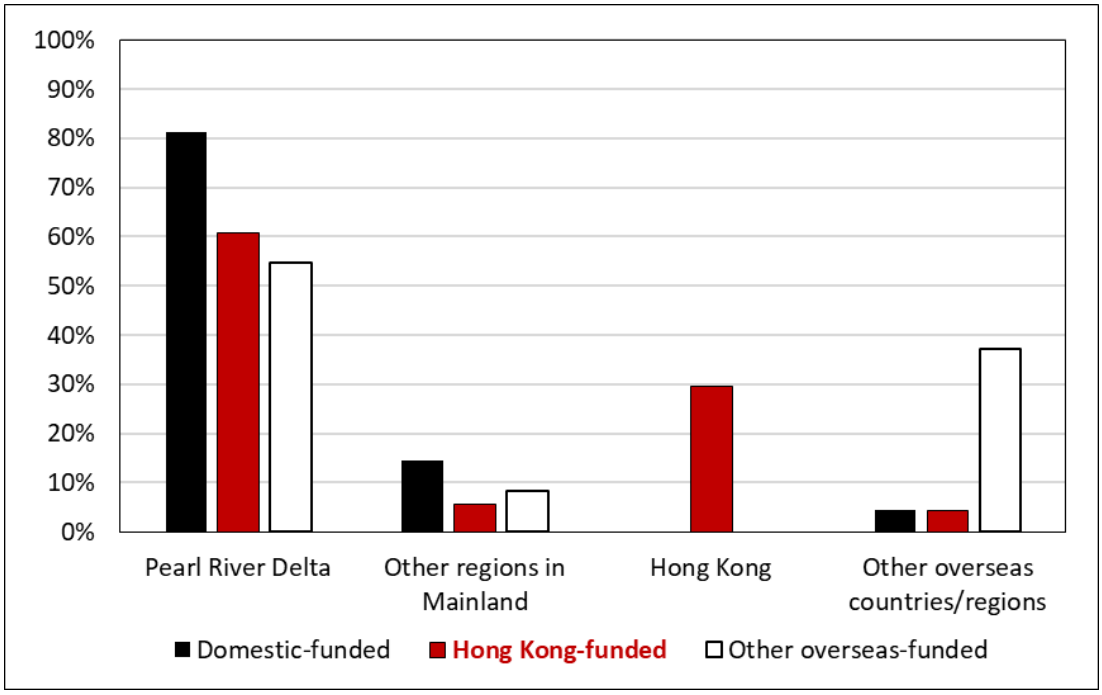


Figure 5.38 Geographical distribution of major technological sources

5.4.3.4 External Technological Support

More domestic-funded firms are using the PRD as a source to pursue external technological support, particularly from technology service firms, suppliers and clients (Figure 5.39). Parent firms, universities and research institutes, technology transfer centres and competitors are not effective. In terms of the importance of sources as

external technological support (Figure 5.40), suppliers and clients in the PRD are more significant than other regions. Technology transfer centres in Mainland China are also more important.

Hong Kong-funded firms rely the most on their parent firms in Hong Kong (Figure 5.41), followed by technology service firms and suppliers in the PRD. Universities and research institutes, technology transfer centres, clients and competitors are thus far ineffective. In terms of the importance of sources for external technological support, actors in the PRD (Figure 5.42), other regions in the Mainland, Hong Kong, and other overseas countries/regions perform more or less the same, except the clients in the PRD that function more remarkably than those from other places.

Other overseas-funded firms rely on their parent companies, technology service firms from other overseas countries/regions (Figure 5.43) and suppliers in the PRD as key sources for external technological support. In terms of the importance of external technological sources (Figure 5.44), variations are not significant among different places. Overall, the PRD and other overseas countries/regions serve as important sources for external technological support.

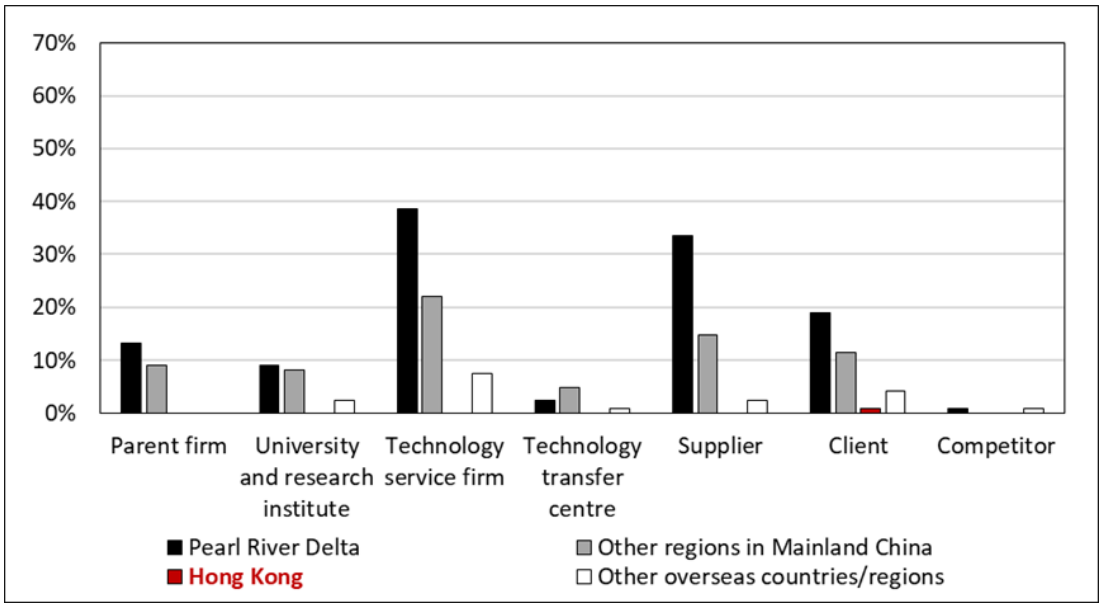


Figure 5.39 Geographical distribution of actors for domestic sample firms' external technological support

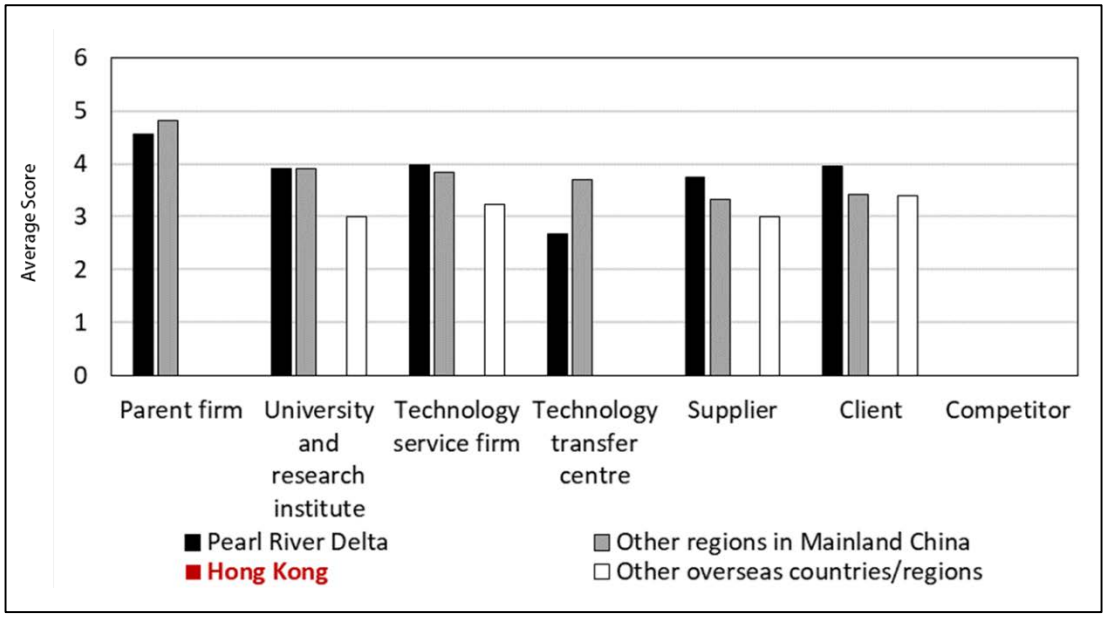


Figure 5.40 Importance of actors in different regions for domestic sample firms' external technological support

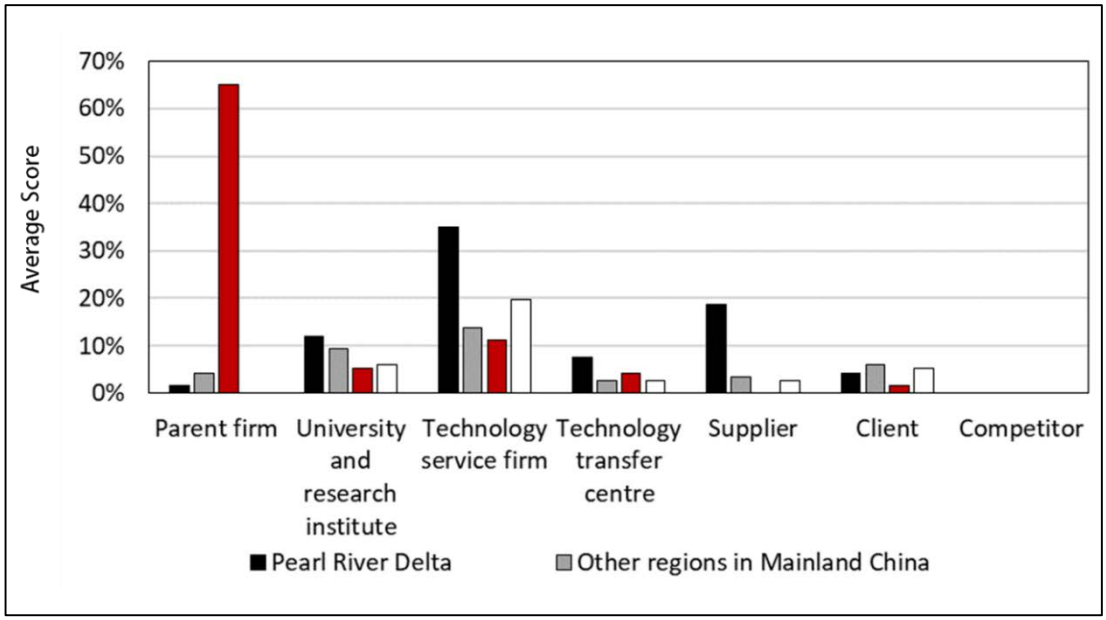


Figure 5.41 Geographical distribution of actors for Hong Kong-funded sample firms' external technological support

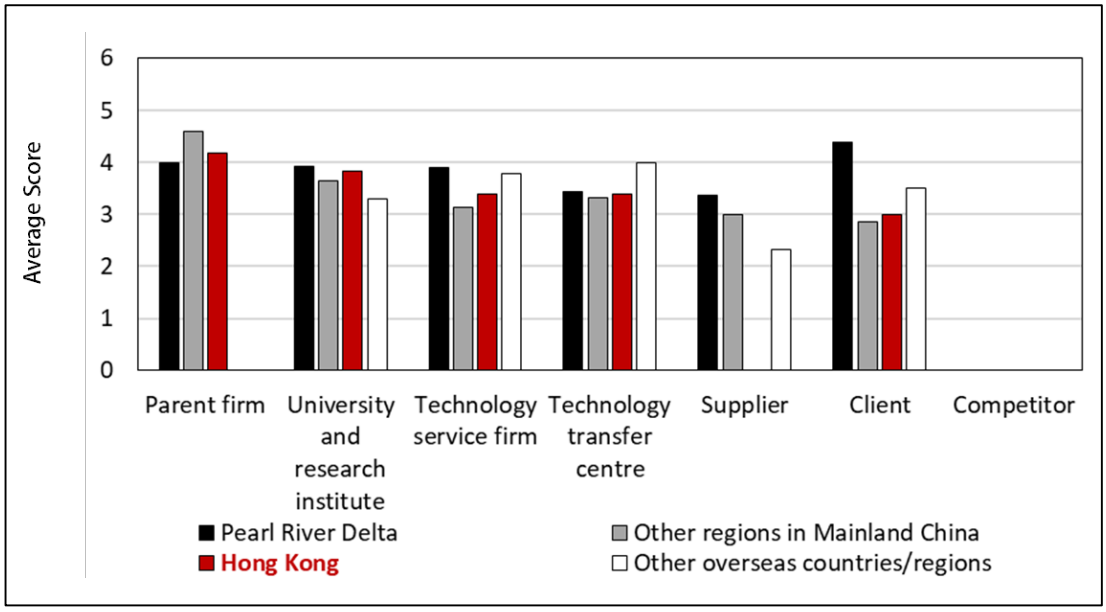


Figure 5.42 Importance of actors in different regions for Hong Kong-funded sample firms' external technological support

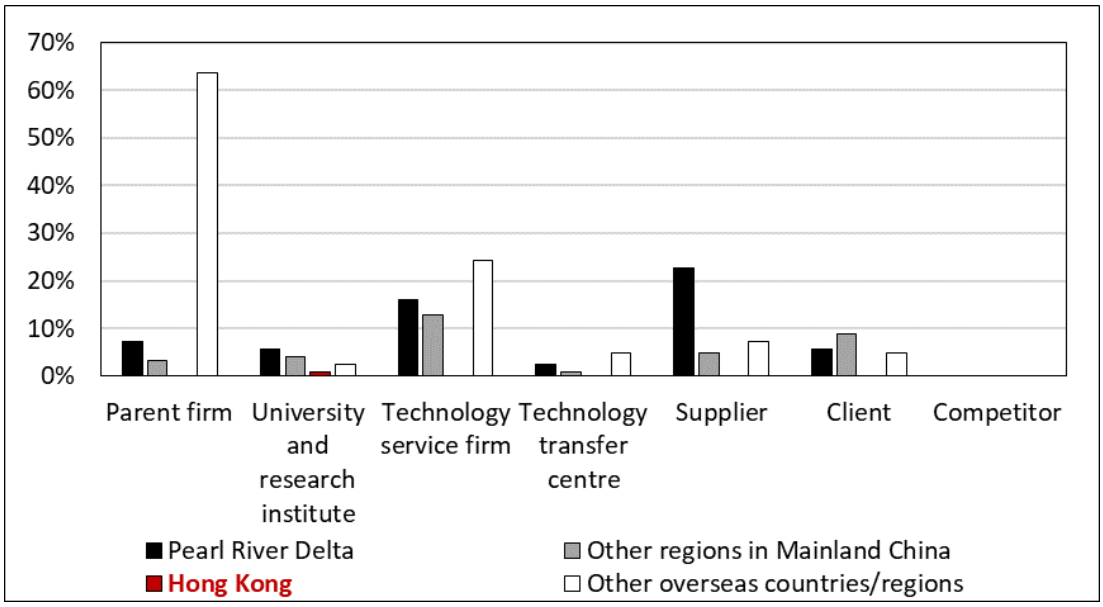


Figure 5.43 Geographical distribution of actors for other overseas-funded sample firms' external technological support

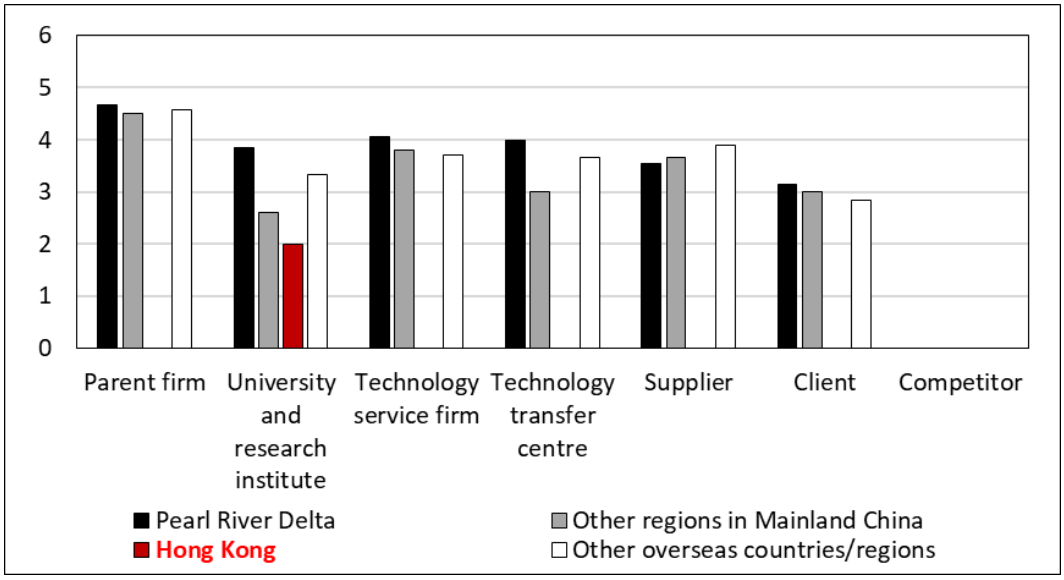


Figure 5.44 Importance of actors in different regions for other overseas-funded sample firms' external technological support

5.4.3.5 Producer Service Linkages

5.4.3.5.a Buying Different Types of Producer Services from Different Sources

Generally, sample firms tend to buy legal, logistics, product testing and trade services while R&D- and market-related and financial services are less externally acquired. This case indicates that high-tech industrial firms at the present stage prefer to internalise R&D, market and financial activities while subcontracting more routinised activities to producer service providers.

The five producer services bought by the largest number of domestic-funded high-tech industrial firms are as follows: Legal; Cargo transport; Testing, inspection and certification; Accounting, auditing and tax; and Intellectual property services (Table 5.7). These firms are becoming active in R&D activities and improving in product quality and market distribution. By contrast, the five producer services bought by the largest number of Hong Kong-funded high-tech industrial firms are as follows: Trade agency; Law; Testing, inspection and certification; Accounting, auditing and tax; and Cargo transport (Table 5.8). These firms focus more on downstream market activities, such as trade and logistics. Finally, the five producer services bought by the largest number of other overseas-funded firms are as follows: Testing, inspection and certification; Cargo transport; Accounting, auditing and tax; Trade agency; and Law (Table 5.9). These firms, apart from concentrating more on product testing, acquire

producer services similar to Hong Kong-funded firms. However, in contrast to Hong Kong-funded high-tech industrial firms that mainly focus on market-oriented trade, other overseas-funded high-tech industrial firms pay more attention to product testing to better compete in the domestic and overseas markets.

The number of firms using producer services from different sources (Figure 5.45) reveals that all of the sample firms use domestic-funded producer services, mirroring strong linkages between high-tech industrial firms from different sources with domestic-funded producer services. For firms using Hong Kong-funded producer services, it is mainly used by Hong Kong-funded high-tech industrial firms (nearly 80%), followed by less than 30% of other overseas-funded high-tech industrial firms and not more than 10% of domestic-funded firms. Apparently, only Hong Kong-funded high-tech industrial firms use Hong Kong-funded producer services, with limited use by other overseas- and domestic-funded firms. Other overseas-funded producer services are used by over 90% of other overseas-funded high-tech industrial firms followed by over 80% of Hong Kong-funded ones and 50% of domestic-funded ones. This finding shows that Hong Kong-funded producer services are not as competitive as their domestic- and other overseas-funded competitors. Only Hong Kong-funded high-tech industrial firms rely on Hong Kong-funded producer services, although they also buy much of domestic- and other overseas-funded service providers.

Table 5.7 Number of domestic-funded firms buying different types of producer services and average percentage of market turnover

Type	Number of firms buying	Average percentage of market turnover
Legal services	50	21%
Cargo transport	49	33%
Testing, inspection and certification	45	31%
Accounting, auditing and tax	43	21%
Intellectual property	41	25%
Trade agency	37	23%
Commercial bank	36	22%
Productive insurance	32	26%
R&D and design	26	33%
Transformation of S&T achievements	22	29%
E-commerce support	22	27%
Information technology	19	27%
Advertisement	19	24%
Business consulting	18	23%
Capital market	8	23%
Market research	3	21%

Table 5.8 Number of Hong Kong-funded firms buying different types of producer services and average percentage of market turnover

Type	Number of firms buying	Average percentage of market turnover
Trade agency	55	22%
Legal services	52	20%
Testing, inspection and certification	50	27%
Accounting, auditing and tax	45	16%
Cargo transport	40	30%
Intellectual property	39	24%
Productive insurance	34	25%
Information technology	28	26%
E-commerce support	28	26%
Business consulting	25	22%
Commercial bank	22	26%
Transformation of S&T achievements	19	25%
Capital market	18	22%
R&D and design	16	27%
Advertisement	10	30%

Market research	8	18%

Table 5.9 Number of other overseas-funded firms buying different types of producer services and average percentage of market turnover

Type	Number of firms buying	Average Percentage of market turnover
Testing, inspection and certification	73	23%
Cargo transport	57	26%
Accounting, auditing and tax	53	14%
Trade agency	51	20%
Law	50	16%
Commercial bank	47	23%
Intellectual property	44	21%
Productive insurance	38	18%
E-commerce support	34	25%
Information technology	29	22%
Business consulting	27	17%
R&D and design	24	24%
Capital market	22	22%
Transformation of S&T achievements	19	22%
Advertisement	16	23%
Market research	12	16%

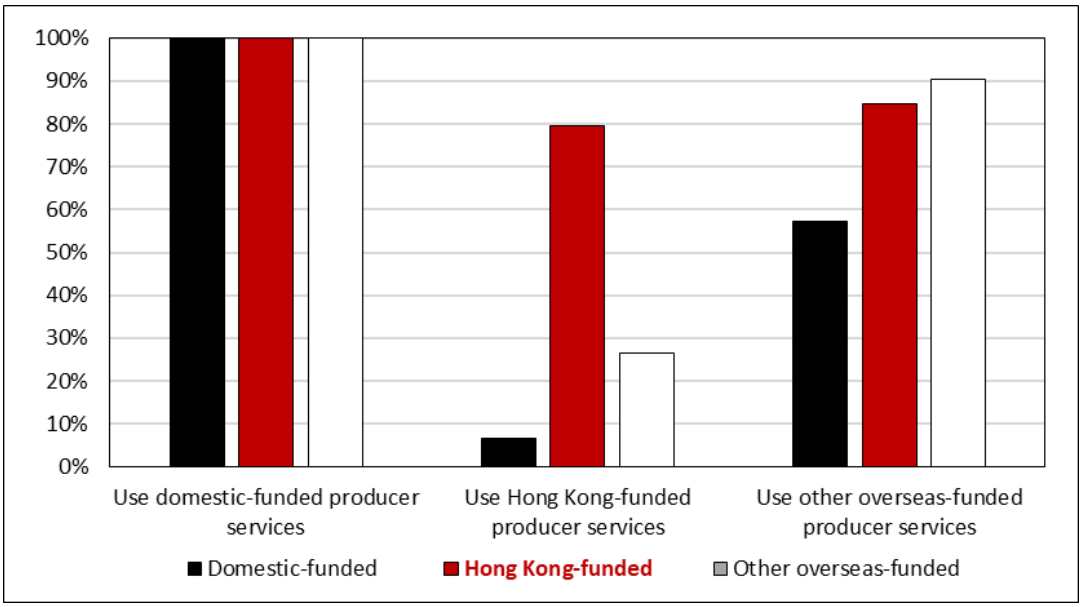


Figure 5.45 Number of sample firms using producer services from each capital source

A further subdivision of using producer services from different sources into 'Only use domestic-funded producer services', 'Use domestic- and Hong Kong-funded producer services', 'Use domestic- and other overseas-funded producer services', and 'Use all sources of the producer services from three sources' shows that the categories of using producer services from different sources can be mutually exclusive and collectively exhaustive (Figure 5.46). Apparently, domestic-funded firms mainly use domestic-funded producer services. Very limited number of high-tech industrial firms use both domestic- and Hong Kong-funded producer services, except approximately 10% of Hong Kong-funded high-tech industrial firms. However, over 50% of domestic- and other overseas-funded high-tech industrial firms use both domestic- and other overseas-funded producer services. Nearly 70% of Hong Kong-funded high-tech industrial firms use producer services from all three sources instead of using only Hong Kong- or domestic-funded ones. Hong Kong high-tech industrial firms perform well in collaborating with all types of producer services, implying that Hong Kong indeed has the advantage of being a 'super connector' to mobilise and synthesise resources and knowledge from different geographical sources.

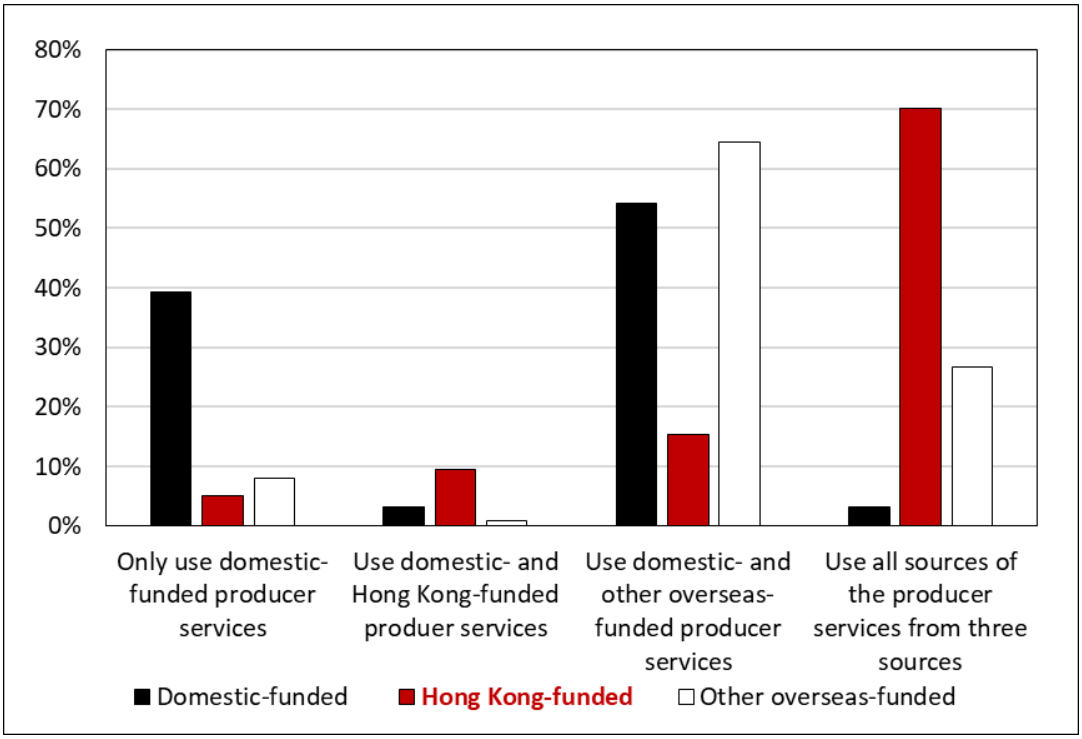


Figure 5.46 Number of sample firms using producer services from different capital sources

5.4.3.5.b Market turnover of buying producer services from different sources

Analysis of the market turnover of buying producer services from different sources by high-tech industrial firms (Figure 5.47) shows the dominant market status of domestic-funded producer services, which reach an average 80%, 70% and 70% of the market shares in serving domestic-, Hong Kong-, and other overseas-funded high-tech industrial firms, respectively. Hong Kong-funded producer services account for only 1% of market turnover in offering domestic- and other overseas-funded high-tech industrial firms. In serving Hong Kong-funded industrial firms, Hong Kong-funded producer services occupy no more than 15% market share, suggesting that Hong Kong producer services as a whole are rapidly diminishing in the PRD service market, with domestic-funded producer services shockingly rising up in the PRD. Other overseas-funded producer services share a medium market position, accounting for approximately 35% of service market of other overseas-funded high-tech industrial firms, followed by 13% of domestic-funded high-tech industrial firms and 10% from Hong Kong-funded high-tech industrial firms. In summary, domestic-, other overseas-, and Hong Kong-funded producer services occupy approximately 75%, 20%, and 5% of the total service market. Meanwhile, nearly all markets taken up by Hong Kong-funded producer services are generated by serving Hong Kong-funded high-tech industrial firms. For trade-related services, the structure of market turnover of producer services from different sources is also rather balanced. The key difference from financial services lies in a better performance of Hong Kong-funded trade services, which account for over 25%, over 10% and approximately 5% of market turnover in serving Hong Kong-, other overseas-, and domestic-funded high-tech industrial firms, respectively. Therefore, Hong Kong trade and financial services still enjoy a competitive edge over R&D services for the moment.

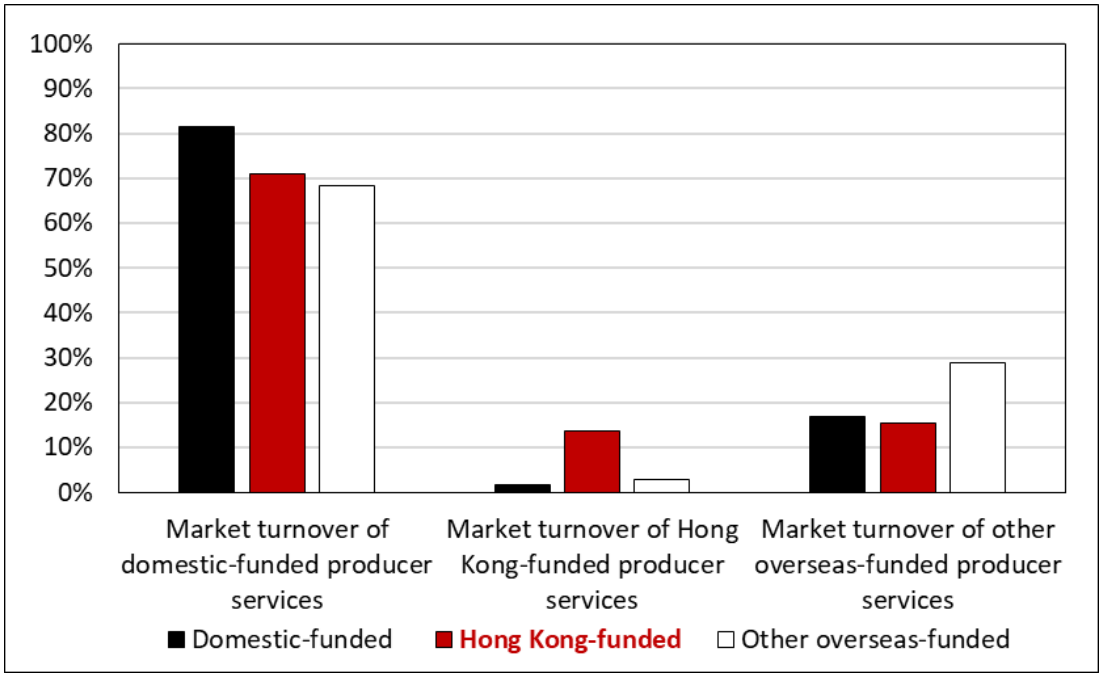


Figure 5.47 Market turnover of buying different types of producer services by sample firms from different capital sources

Dissimilarities are observed among the usage of different types of producer services, which may better unveil the advantages and disadvantages of producer services from different sources in serving different fields of services to high-tech industrial firms in the PRD. For R&D services, domestic-funded producer services dominate in serving domestic- and Hong Kong-funded high-tech industrial firms, but not for other overseas-funded high-tech industrial firms (Figure 5.48). Other overseas-funded producer services perform better in serving other overseas-funded high-tech industrial firms, with a market share above 35%. This finding points out that domestic-funded R&D services in the PRD are growing fast and enjoying competitive advantages over Hong Kong-funded ones, while other overseas-funded producer services retain a not-so-small client base of other overseas-funded high-tech industrial firms. For financial services, the market turnovers of producer services from different sources are more balanced (Figure 5.49). Domestic-funded producer services perform slightly better than, if not equally with, other overseas-funded producer services. Hong Kong-funded producer services occupy approximately 30% of market turnover in serving Hong Kong-funded high-tech industrial firms. Thus, Hong Kong-funded producer services perform well in channelling financial resources to Hong Kong-funded high-tech industrial firms in the PRD. The trade services perform similarly as the financial services (Figure 5.50).

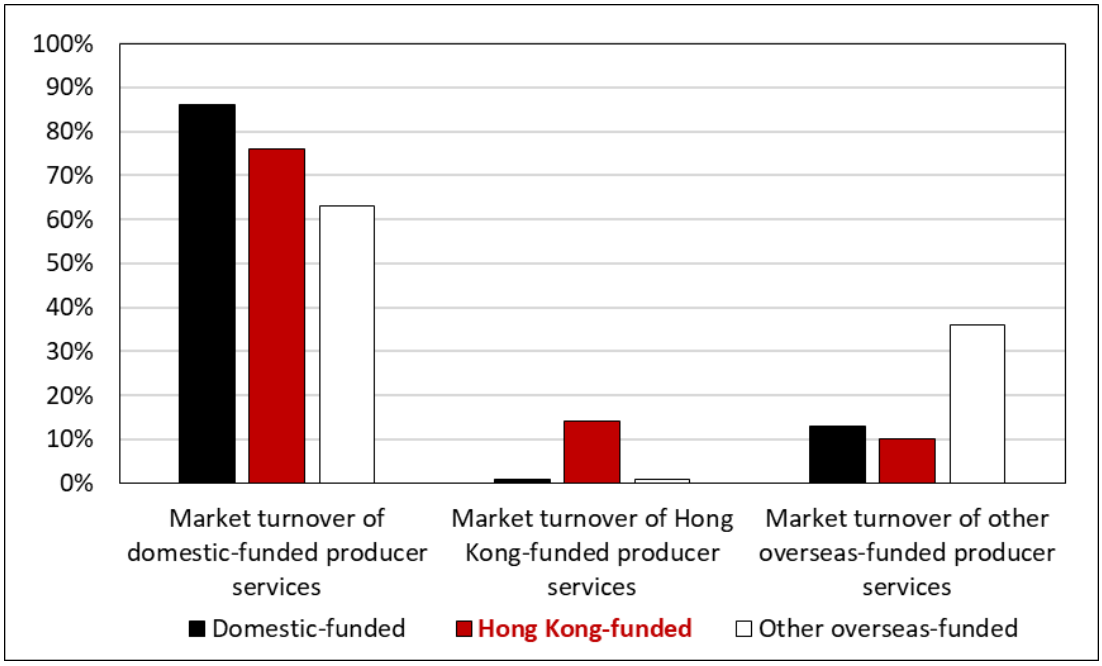


Figure 5.48 Market turnover of buying R&D services by sample firms from different capital sources

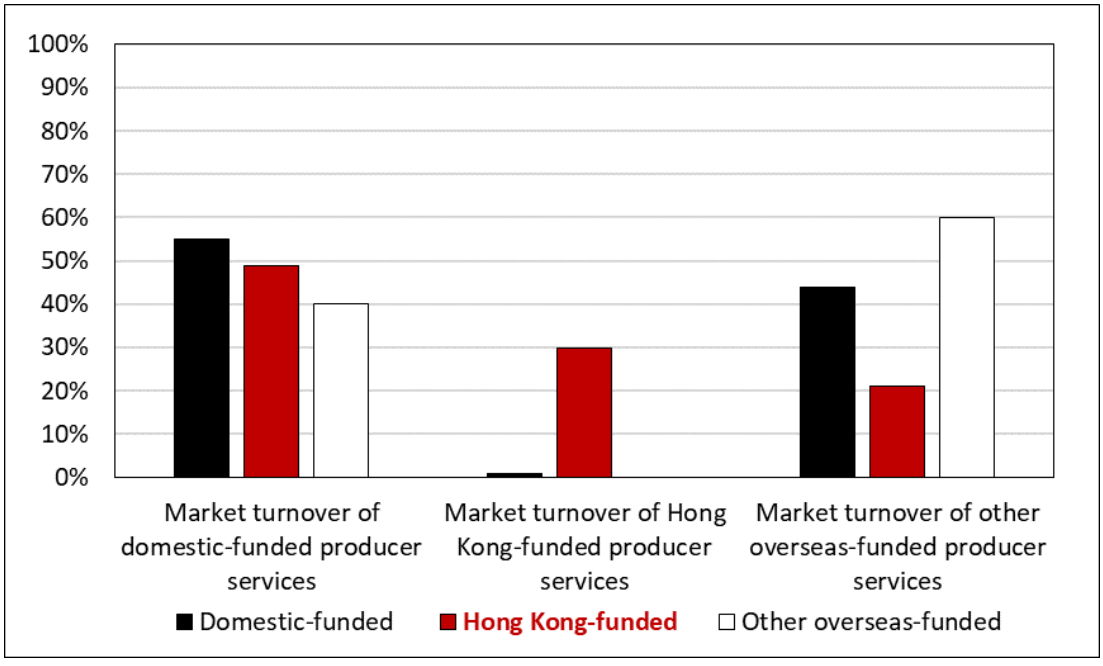


Figure 5.49 Market turnover of buying financial services by sample firms from different capital sources

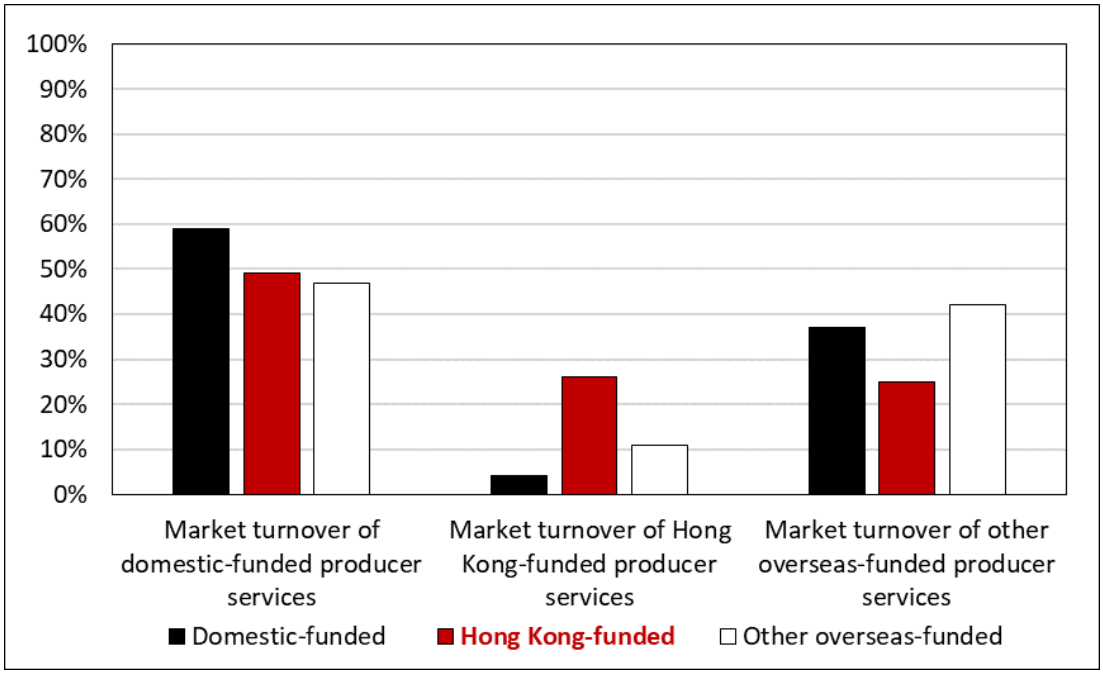


Figure 5.50 Market turnover of buying trade services by sample firms from different capital sources

5.4.3.5.c Market Turnover of Buying Producer Services from Different Regions

The market turnover of buying all producer services as a whole from different regions (Figure 5.51) show that the PRD shares a more primary status than the previous results divided by different capital sources. This result indicates that high-technology industries and producer services have formed strong alliances within the PRD, which generates endogenous growth dynamics, though driven by various sources of capital. Other regions in Mainland China, Hong Kong, and other overseas countries/regions have similar market shares. Domestic-funded high-tech industrial firms tend to use more producer services located in other regions in Mainland China. Hong Kong-funded high-tech industrial firms tend to use more producer services located in Hong Kong, with a highly polarised structure in market source distribution. Other overseas-funded high-tech industrial firms use more producer services outside Mainland China and Hong Kong, demonstrating that high-tech industrial firms in the PRD prefer to use producer services that originate from their home region/country, while the PRD moves towards a self-sustaining regional industrial-service complex comprising heterogeneous firms.

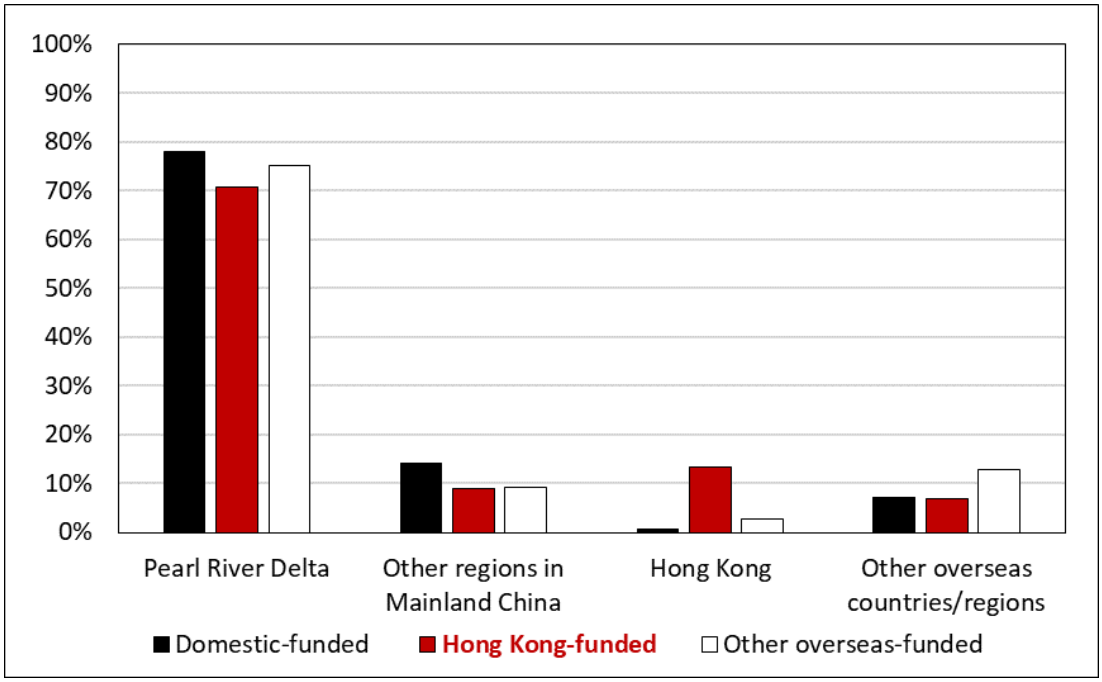


Figure 5.51 Market turnover of buying producer services from different regions by sample firms from different capital sources

Dissimilarities are also observed among using producer services in different regions by sample firms, which may better unveil the advantages and disadvantages of producer services from different regions that cater to different fields of high-tech industrial firms in the PRD. In this case, the PRD enjoys an overwhelming status in offering R&D services to its high-tech industrial firms (Figure 5.52), showing the regionally endogenous nature of R&D activities internally and externally to firms. For financial services, a more balanced market structure stands out (Figure 5.53). Producer services from other regions in Mainland China, Hong Kong and other overseas countries/regions all share approximately 20% of the market. This finding indicates that high-tech industrial firms in the PRD acquire financial resources around the world. For trade services, the PRD, Hong Kong, and other overseas countries/regions are the main sources. Domestic-funded high-tech industrial firms clearly seek more trade services in the PRD than Hong Kong- and other overseas-funded firms. A limited number of firms use producer services in other regions in Mainland China. Approximately 30% and over 10% of the market turnover for buying trade services from Hong Kong-funded high-tech industrial firms and other overseas-funded industrial firms, respectively, are from Hong Kong (Figure 5.54). All three types of high-tech industrial firms source trade services from other overseas countries/regions. Hong Kong trade and financial services are more competitive in capturing the domestic market than R&D services. Furthermore,

by comparing the usage of producer services with different sources and from different regions by high-tech industrial firms in the PRD, Hong Kong-funded producer services that capture the PRD market are mostly based in Hong Kong, while their offices and branches in the PRD are still limited in tapping into the rising regional and domestic market for producer services.

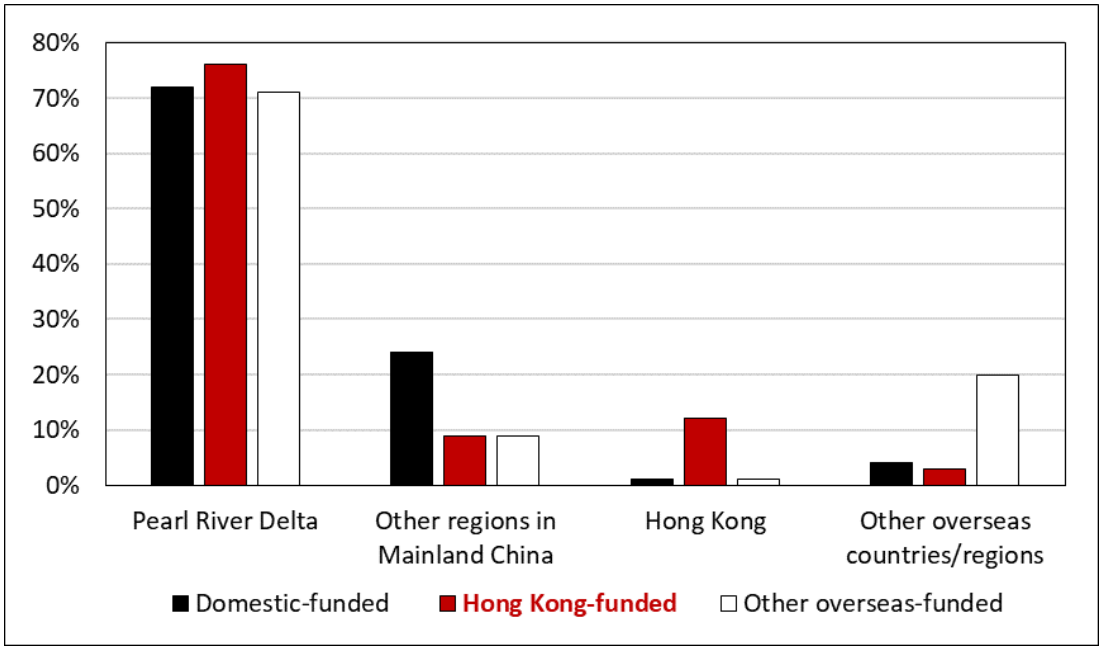


Figure 5.52 Market turnover of buying R&D services from different regions by sample firms from different capital sources

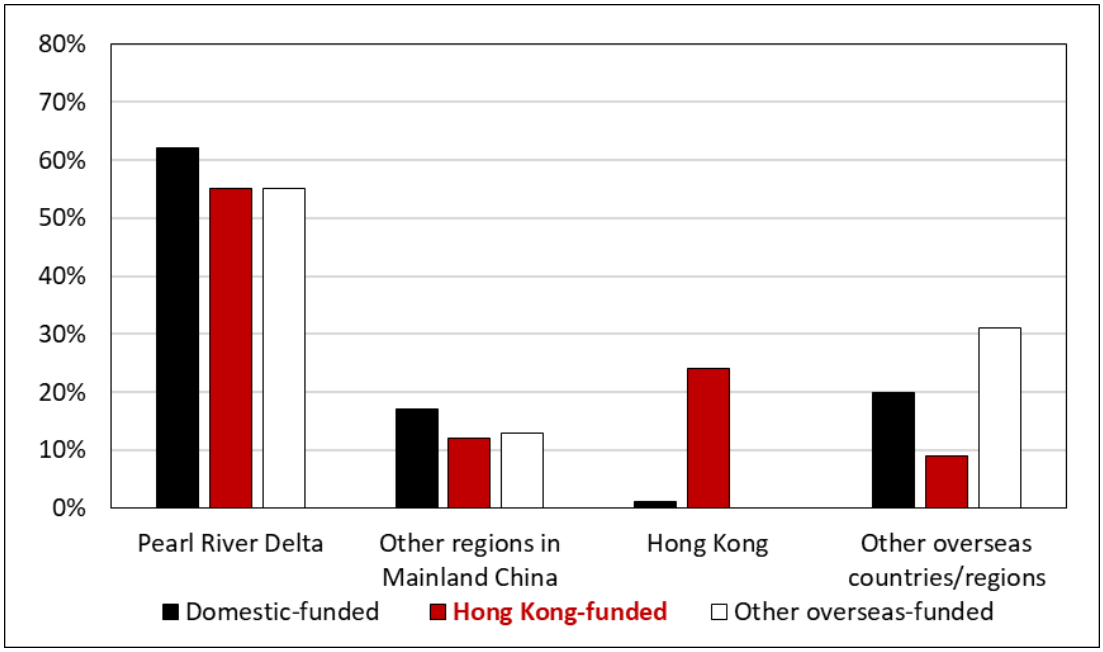


Figure 5.53 Market turnover of buying financial services from different regions by sample firms from different capital sources

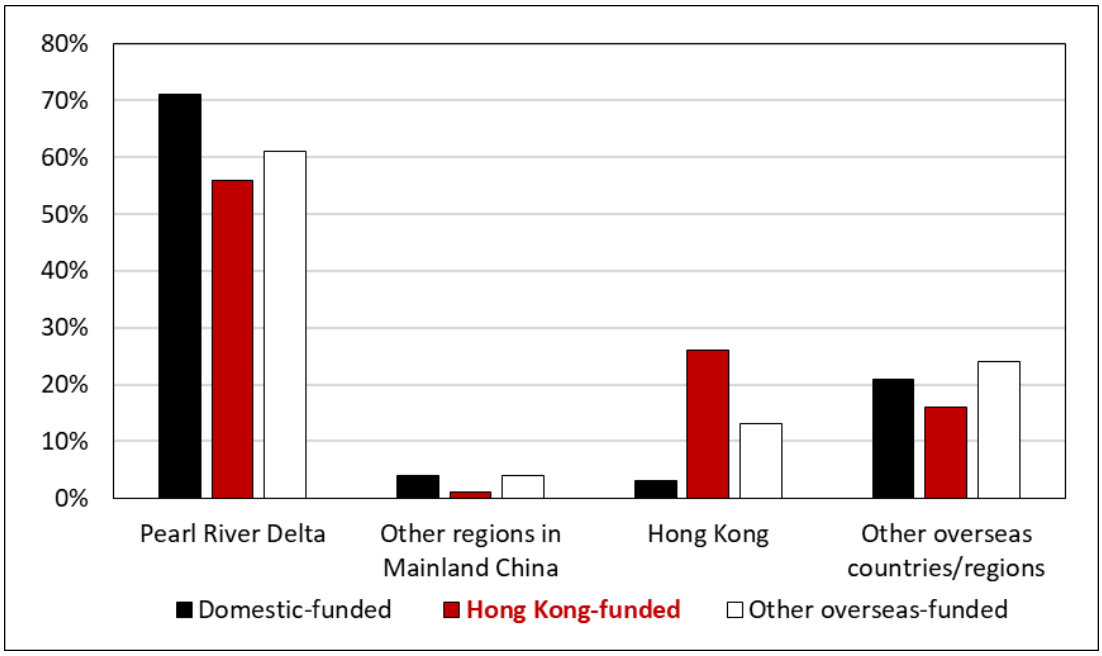


Figure 5.54 Market turnover of buying trade services from different regions by sample firms from different capital sources

5.4.4 Views of High-technology Industrial firms in the PRD on Hong Kong's roles

5.4.4.1 Overall Role

Considering the role of Hong Kong in developing high-tech industrial firms in the PRD, the high-tech industrial firms in the PRD highly evaluate market expansion, international cooperation, corporate financing and producer services (Figure 5.55). Compared with these four supporting forces, investment and entrepreneurship, technological support and labour supply are less important. An interesting finding is that the roles of market expansion, international operation, corporate financing and producer services more or less hinge upon Hong Kong's special status as an international window to the PRD and Mainland China, as well as Hong Kong's special institutional context under the 'One Country, Two Systems'. However, alarmingly, the previously key role of investment and entrepreneurship in the 'Front Shop, Back Factory' regional division of labour has declined in importance. New potential regional collaboration relying on technological support and labour supply has not yet taken effect.

Unlike domestic- and other overseas-funded firms, however, Hong Kong-funded firms view technological support and labour supply as highly important (Figure 5.56). Domestic-funded firms rely more on Hong Kong in terms of investment, corporate financing and market expansion. Other overseas-funded firms deem producer services, market expansion and international operation as Hong Kong's key roles. These findings reflect the varying roles of Hong Kong in fuelling high-tech industrial firms with different capital sources.

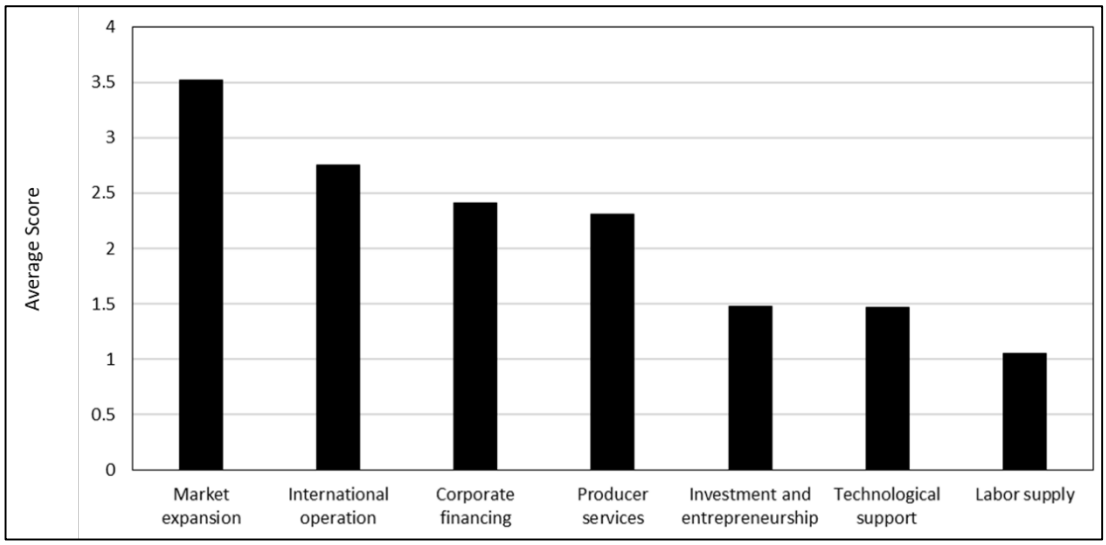


Figure 5.55 Importance of the role of Hong Kong in high-tech industrial development in the PRD rated by sample firms

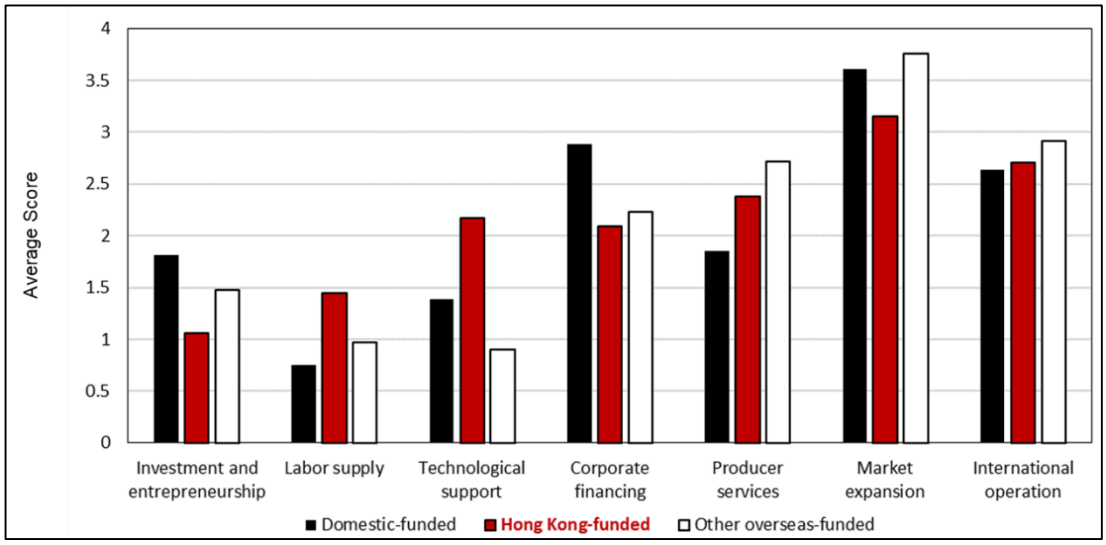


Figure 5.56 Importance of the role of Hong Kong in high-tech industrial development in the PRD rated by sample firms with different capital sources

5.4.4.2 Labour Force Provision

The previous sections clearly show that Hong Kong labour force is unattractive to high-tech firms in the PRD (Figure 5.57). Nearly 90% of the domestic-funded firms, 60% of other overseas-funded firms and less than 10% of Hong Kong-funded firms do not hire Hong Kong labour force. Over 80% of the latter hire Hong Kong labour force, serving as their main employers. Between domestic- and Hong Kong-funded firms, other overseas-funded firms previously hired Hong Kong labour but nearly 20% no longer do so at present. Hong Kong labour force is clearly becoming less attractive to other overseas-funded firms.

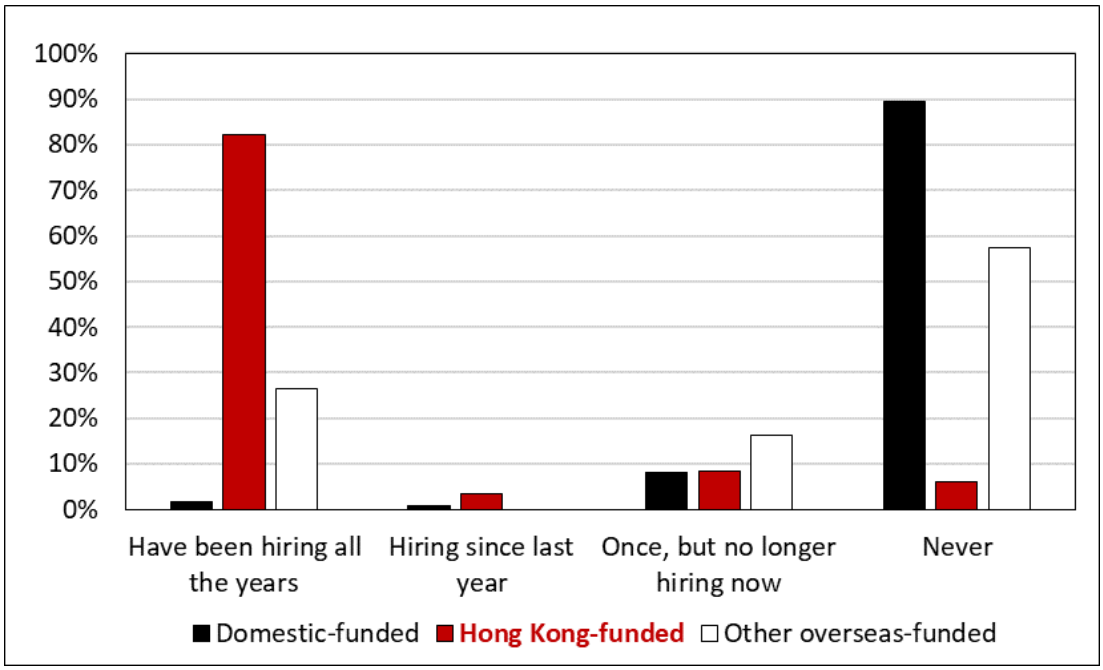


Figure 5.57 Hiring of sample firms in HK labour force

As for reasons for hiring Hong Kong labour force (Figure 5.58), domestic- and overseas-funded firms have apparent differences between them. Domestic-funded firms hire Hong Kong labour force mainly due to their international vision and good understanding of international rules. Hong Kong- and other overseas-funded firms hire Hong Kong labour force because of their advantageous human resources, such as high comprehensive quality, responsible and dedicated working attitude and strong capability to solve problems. However, all types of firms agree that Hong Kong labour force lacks a strong willingness to work in Mainland China.

As for the reasons of not hiring Hong Kong labour force (Figure 5.59), high costs appear to discourage all the types of firms. In addition, domestic-funded firms are impeded mainly by the unfamiliarity of the Hong Kong labour force with Mainland China and the difficulties in mutual communications. Hong Kong- and other overseas-funded firms are warier of the ineffectiveness of Hong Kong labour force in solving problems. Notably, all firms reach high consensus on the difficulties in hiring Hong Kong residents, as they at present still have low intention to work in Mainland China.

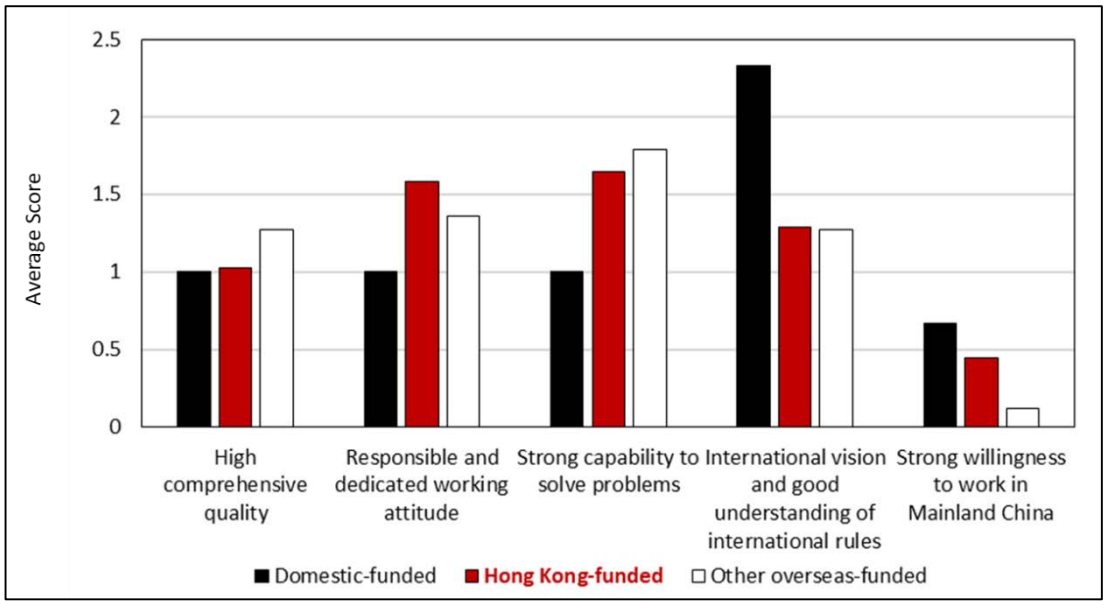


Figure 5.58 Reasons of sample firms to hire Hong Kong labour force

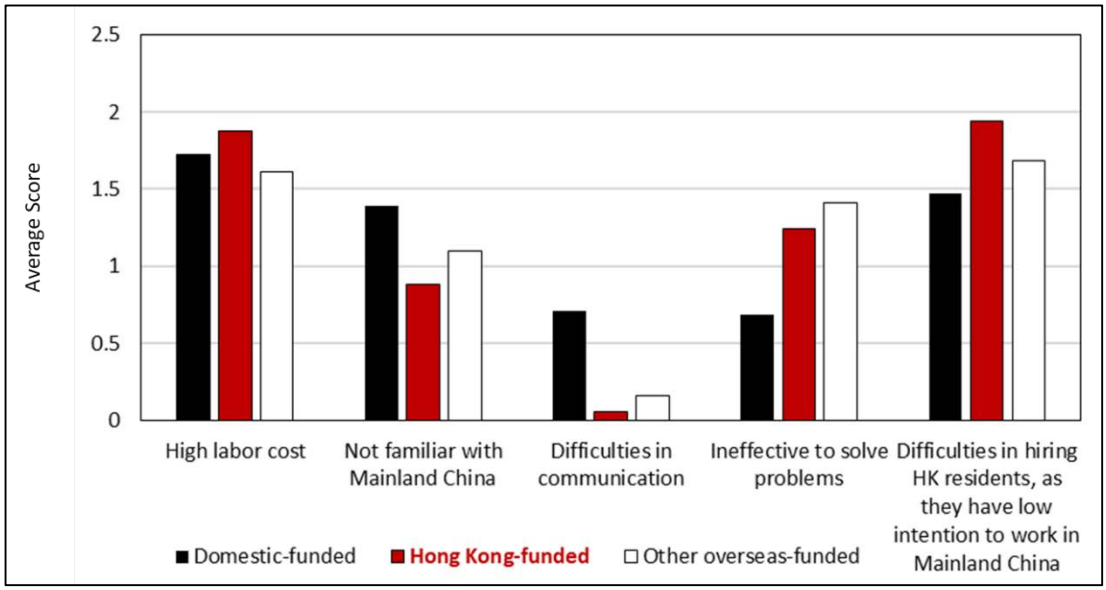


Figure 5.59 Reasons of sample firms to not hire Hong Kong labour force

5.4.4.3 Technological Support

The need of sample firms for technological support from Hong Kong has a highly polarised structure (Figure 5.60). Hong Kong-funded firms have always used Hong Kong technology, which however is not used by over 90% of the domestic- and other overseas-funded firms. Approximately 10% of all firms that once used Hong Kong technological support no longer uses it at present. Primarily, Hong Kong technologies flow into the PRD through the direct investments of Hong Kong-funded firms.

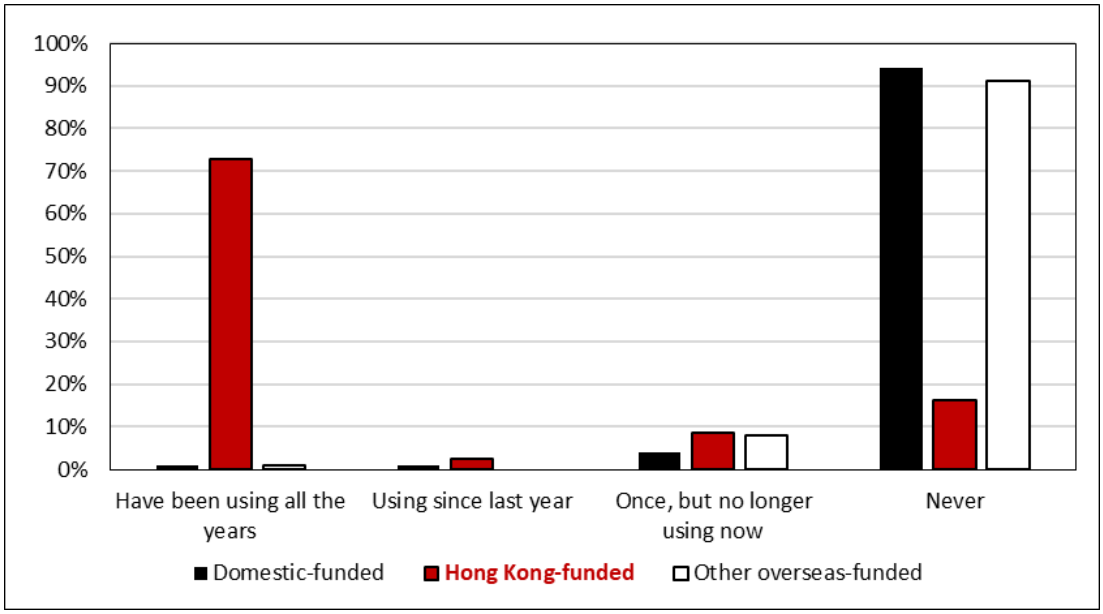


Figure 5.60 Usage of Hong Kong technological support among sample firms

The reasons for the above polarisation are also evident (Figure 5.61). Hong Kong-funded firms are more familiar with Hong Kong technologies, which they believe to effectively meet current demands and are in line with international standards, more competitive than those derived from Mainland China and cutting-edge and advanced.

By contrast, the lack of understanding of Hong Kong scientific and technological achievements is the most significant reason why domestic- and other overseas-funded firms do not use Hong Kong technologies (Figure 5.62). High technological cooperation cost is another critical reason for these firms. In addition, other overseas-funded firms are also discouraged by unsound technological cooperation mechanism, low efficiency to meet technological demands of Hong Kong technologies and weak cooperation intention from Hong Kong technological companies.

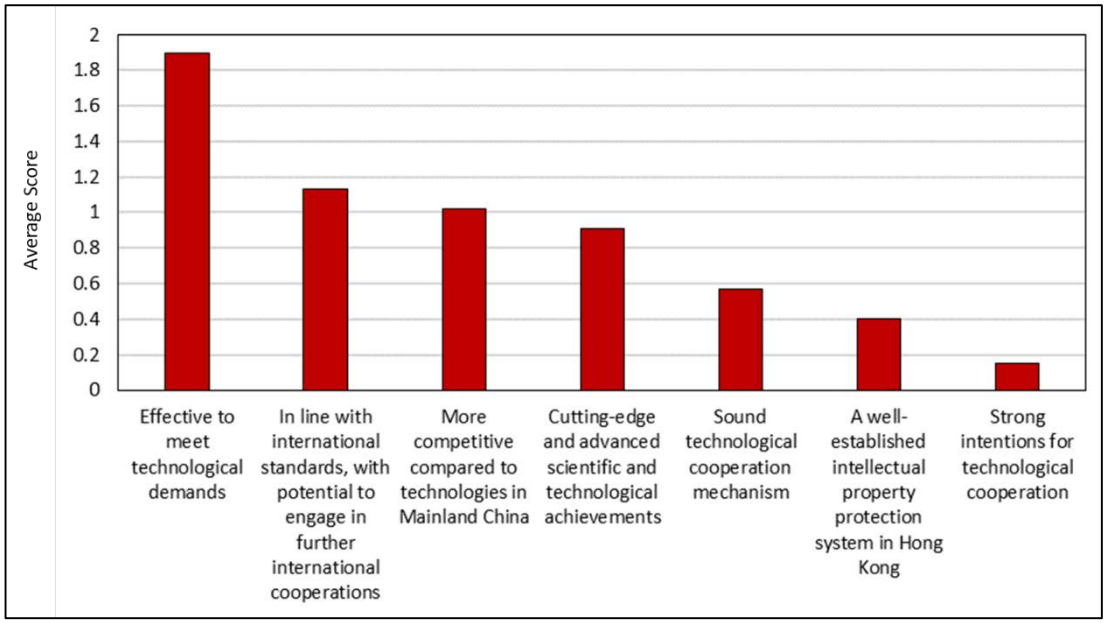


Figure 5.61 Reasons for sample firms to use Hong Kong technological support

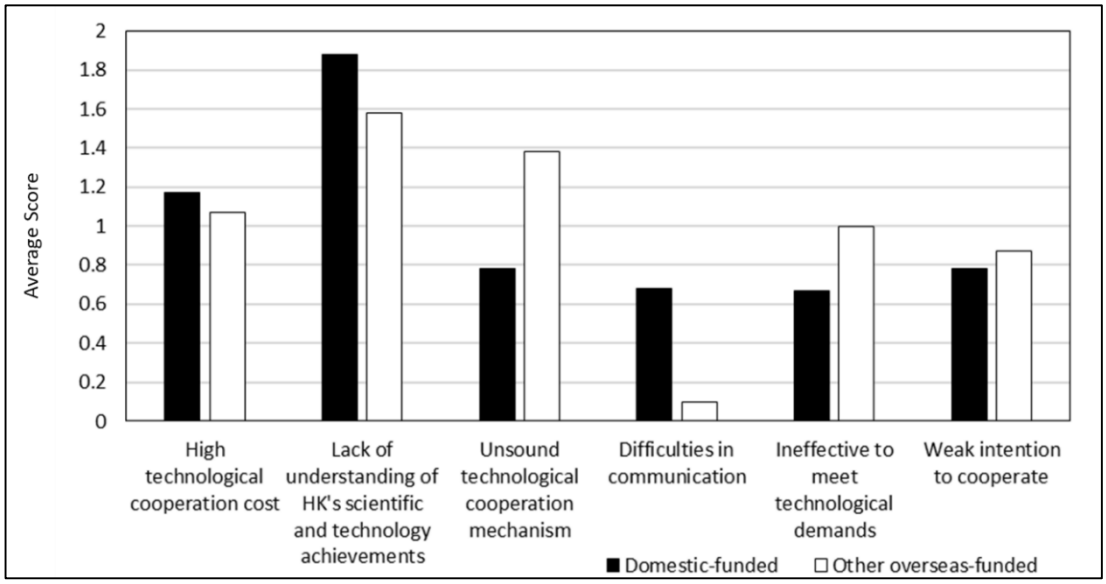


Figure 5.62 Reasons for sample firms not to use Hong Kong technological support

5.4.4.4 Producer Service Linkages

Producer services have long been a brand business of Hong Kong, particularly for the export-oriented development in the PRD over the past several decades. Approximately 75% of Hong Kong-funded firms and 30% of other overseas-funded firms have been buying Hong Kong-funded producer services all these years (Figure 5.63). Nevertheless, over 80% of domestic-funded high-tech industrial firms and 50% of other overseas-funded high-tech industrial firms have not used Hong Kong-funded

producer services. This decoupling is alarming, which almost applies to each type of high-tech industrial firms. Over 20% of other overseas-funded high-tech industrial firms and over 10% of Hong Kong- and domestic-funded high-tech industrial firms that once purchased Hong Kong-funded producer services are no longer buying now. The decoupling of Hong Kong producer services with the PRD high-tech industries merits serious attention.

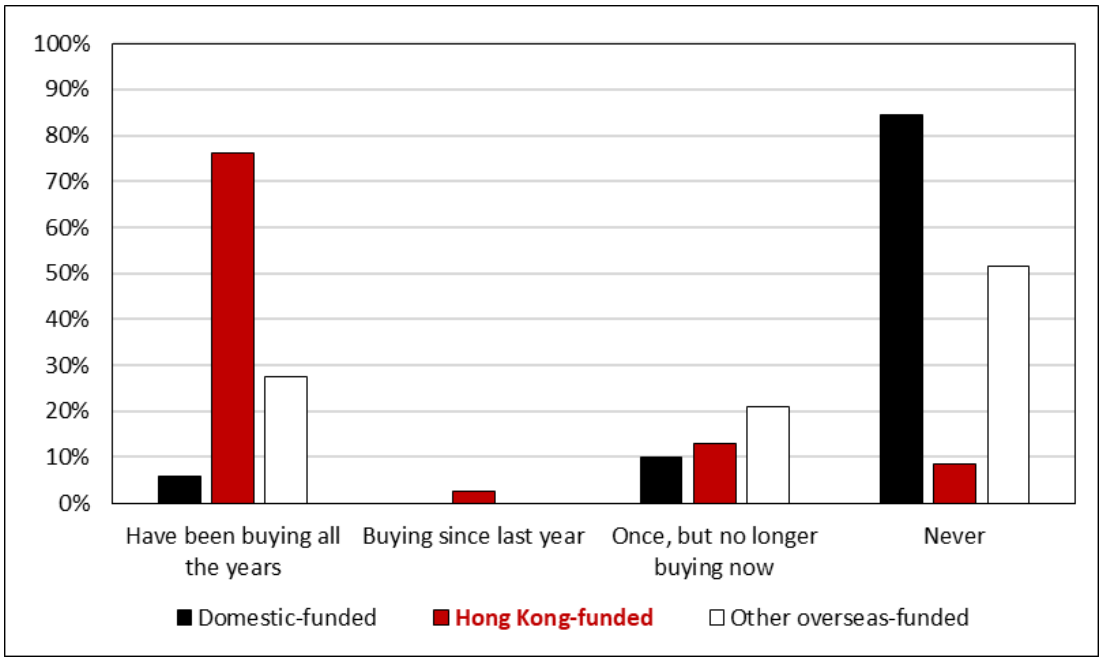


Figure 5.63 Purchase of sample firms for Hong Kong producer services

All types of firms report a good understanding of international markets and rules as a reason to purchase Hong Kong-funded producer services (Figure 5.64). Other reasons differ across the types of firms. For example, domestic-funded high-tech industrial firms highlight high professional level and lower price than other overseas-funded suppliers, effective to meet service demands, and client requirements. Having business in Hong Kong and abroad and sound cross-border service mechanism work better for other overseas- and Hong Kong-funded firms.

The reasons discouraging domestic- and other overseas-funded firms in using Hong Kong-funded producer services are high cost, unfamiliarity with Mainland China, incapability to meet demands and unsound cross-border service mechanism (Figure 5.65). The reasons that prevent the three types of firms from using Hong Kong-funded producer services do not considerably vary across high-tech industrial firms with different capital sources.

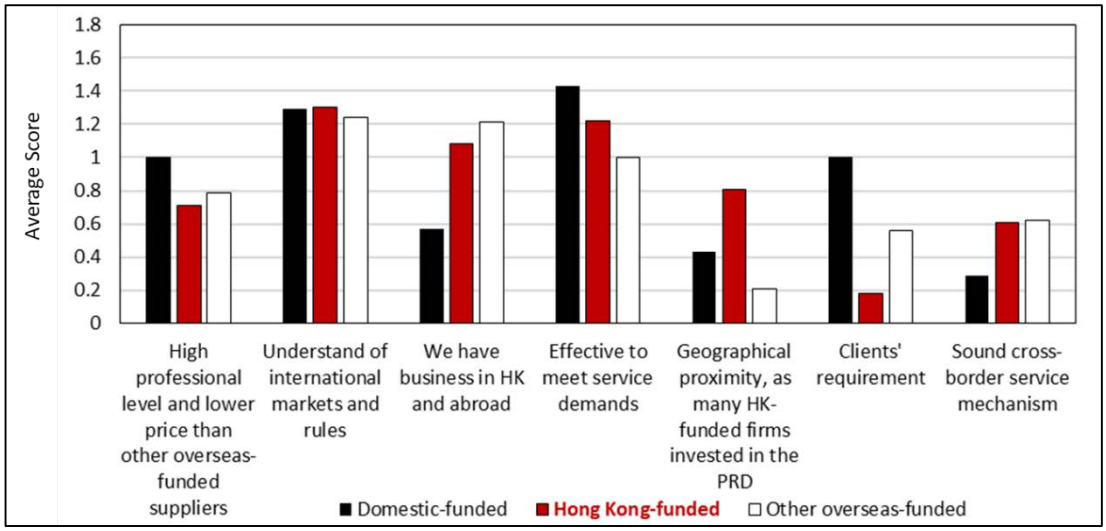


Figure 5.64 Reasons of sample firms to buy Hong Kong producer services

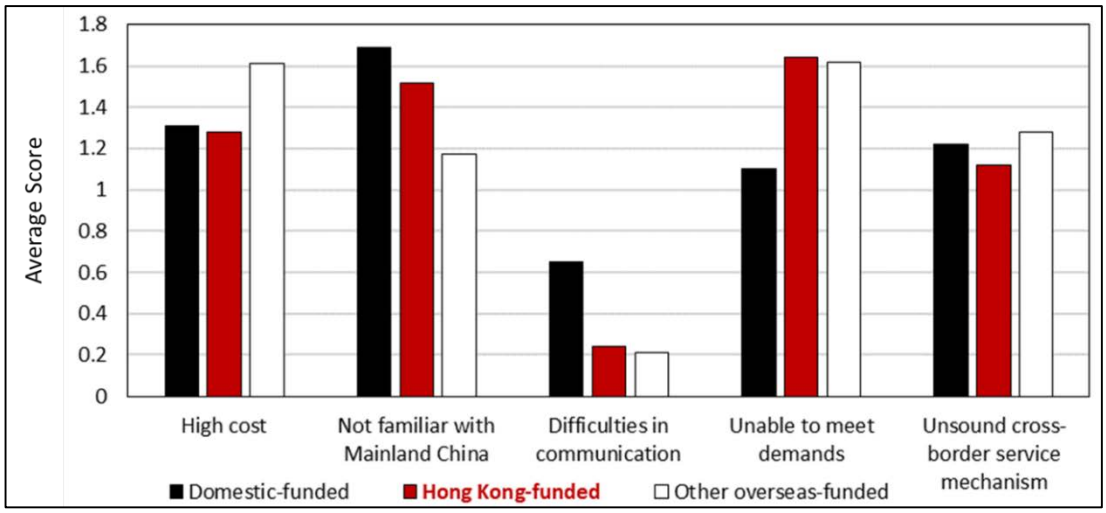


Figure 5.65 Reasons of sample firms not to buy Hong Kong producer services

5.5 Summary

5.5.1 Economic Restructuring in the PRD and Explanations

PRD has been experiencing economic restructuring since the mid-2000s, if not earlier, which is a remarkable deviation from the classical export-oriented labour-intensive development model during the two decades since the reform and opening-up in 1978. Despite its continuing economic growth, PRD has a slowing annual growth rate from approximately 20%-30% in the 1990s to approximately 10% in the mid-2000s. Behind the scenes of economic dynamics conversion, industrial structural changes

move onto the track. Export/GDP ratio has decreased steadily and consistently since the mid-2000s, coupled with a rising proportion of high-technology industries (e.g. electronics, electrical machinery and equipment), strategic emerging industries (e.g. biotechnology, new energy, robotics) and tertiary industries, particularly higher value-added producer services.

Four aspects of explanations are discussed. Firstly, PRD's industrial restructuring can be attributed to changing production resources (e.g. rising production cost and shortage of low-salary migrant labour) and market structure (e.g. shrinkage of overseas market and enlarging domestic market). Secondly, PRD's institutional context has departed from an earlier flexible and *laissez-faire* environment towards a more stringent and regulatory context, such as the restructuring of processing and trade regime in a top-down manner and the 'double-relocation' policy promulgated by the provincial government. Thirdly, the GPNs led by Western flagship corporations have been reconfigured under the trends of domestic high-technology giants (e.g. Huawei, ZTE) growing as leading players. Last but not the least, through the lens of Evolutionary Economic Geography, PRD's historical industrial accumulation paves the path for regional endogenous economic evolution from the former classical model to new waves of economic transition.

5.5.2 High-technology Industrial Development Characteristics in the PRD

Notably, high-technology industrial development in the PRD has been remarkably restructured from the cost-driven export-oriented industrial development in the early post-reform period when the region was 'one step ahead' of Mainland China. Still at its early stages of development and driven mainly by innovation, this new wave of high-tech industrial development in the PRD has risen since the mid-2000s, particularly in the aftermath of the 2008 global financial crisis. The remarkable changes can be considered in two key levels, namely, firm and regional levels.

Firstly, the characteristics of high-tech industrial firms differ from those of export-oriented labour-intensive ones, which are mainly small-scale, raw material-based processing and trade, OEM in the forms of *sanlai yibu* and collectively-owned rural industrial companies. By stark contrast, current high-tech industrial firms in the PRD tend to have medium size and are final product-based ODMs and OBMs in the forms of Wholly Foreign-Owned and Private or Private Holding Enterprises. Secondly, for

regional changes, locational factors for firms to choose PRD as their base have shifted from low production cost, abundant supply of low-skilled labour force and low-cost land, half-hearted regulatory business environment and geographical proximity to overseas market towards a relatively complete regional self-contained supply system, an enlarging regional and domestic market, high-quality infrastructure provision, abundant high-skilled labour force and emerging endogenous agglomeration economies. In summary, the PRD has undergone dramatic changes, as reflected in both its firms and the regional context shaped by various stakeholders.

5.5.3 Hong Kong Participation in High-Tech Industrial Development in the PRD

Analysis of the ways in which the sample firms were founded, geographical origins, high-skilled labour force sources, major technological sources, external technological support and producer services reveals how and to which extent Hong Kong engages with the high-tech industrial development in the PRD. Apparently, Hong Kong participates in the new wave of high-tech industrial development in the PRD primarily via its own direct investments and its special institutional role as an international window to the region and the entire Mainland China.

Firstly, entrepreneurship of Hong Kong residents in the PRD has not become a mainstream trend. Hong Kong still relies on business entities to invest in the PRD with their headquarters located in Hong Kong. Secondly, Hong Kong- and other overseas-funded firms to a greater degree use Hong Kong production resources, which are less used by domestic-funded firms. Hong Kong, as an international window, is clearly more outward- than inward-oriented. Organisational linkages between parent firms in Hong Kong and branch offices in the PRD form critical channels for labour mobility, technology transfer and producer service provision. Organisational proximity plays a vital role in supplementing the lack of institutional proximity between Hong Kong and the PRD under the ‘One Country, Two Systems’ framework. However, the emerging decoupling effects between Hong Kong producer services and PRD high-technology industries are alarming. The rapid development of indigenous producer services and increasing international linkages of the PRD results in Hong Kong facing the danger of losing its role as an intermediary window bridging the PRD to the rest of the world.

5.5.4 Views of High-technology Industrial Firms in the PRD on Hong Kong Roles

In general, the key roles of Hong Kong in facilitating high-tech industrial development in the PRD mainly dwell in market expansion, international operation, corporate financing and producer service provision, while the positive effects of investment and entrepreneurship, technological support and labour supply remain underdeveloped. Although technological support and labour supply function better for Hong Kong-funded high-tech industrial firms due to organisational linkages, these roles have very limited importance for other types of firms.

Domestic-funded firms in the PRD mainly use services and technology in the PRD and other parts of Mainland China. They rarely use Hong Kong's services and technology primarily because of Hong Kong's higher cost in comparison to that in Mainland China, insufficient alignment with Mainland China's market demands, limited penetration into the awareness of business communities in the PRD, unawareness of Hong Kong's producer services and technology achievements, and Hong Kong residents' low intension to work in the PRD. Although Hong Kong- and other overseas-funded high-tech firms also mainly use services and technology in the PRD, they still use some Hong Kong services and technology, but not a lot. Reasons for using Hong Kong services include effectiveness to meet service demands and good understanding of international markets and rules. Reasons for using Hong Kong technology include effectiveness to meet technological demands, alignment with international standards, and competitive advantages compared to technologies in Mainland China. Hong Kong labour forces have been hired mainly by Hong Kong-funded firms ◦ Most domestic-funded firms never hire Hong Kong labour forces. Reasons for hiring Hong Kong labour force include international vision and good understanding of international rules, strong capability to solve problems, and responsible and dedicated working attitude. Reasons for not hiring Hong Kong labour force include high labour cost, low intention to work in Mainland China, and unfamilarity with Mainland China.

Chapter 6
Hong Kong Riding on Increasing Affluence of Mainland Consumers and Investors

6.1 Introduction

After the reform and opening up in 1978, as the Chinese economy continues to flourish for 40 years and its investors increase in wealth, the importance of the Mainland as a market for wealth management and consumption has risen.

As a financial market pilot in China's reform and opening up and as the world's largest centre of offshore RMB transactions, Hong Kong has a well-established and internationally integrated insurance industry and security market. Hong Kong must continue to be developed as a world top-class investment and wealth management centre, bring the top-class financial services to the Mainland and provide its investors with one-stop services for wealth management. The Shanghai-Hong Kong Stock Connect and Shenzhen-Hong Kong Stock Connect, which have been launched for several years, provide excellent opportunities for Mainland investors to invest in Hong Kong. In future, Hong Kong can strengthen the depth of the connectivity mechanism in the security market and broaden the scope of the mechanism into areas such as insurance and funds, to further endorse development coordination of the Mainland and Hong Kong capital markets.

Hong Kong must gradually turn its attention from the European and American markets onto the Mainland market, refocus from serving production-oriented enterprises to customers, and help Mainland consumers to procure goods globally. In 2018, the tertiary sector accounted for over 50% of China's GDP, and consumption has gradually become the main driving force of economic growth. Hong Kong can take advantage of its internationally duty-free port and adjacency to the GBA to gain a more explicit understanding of and respond to the expanding import demand of the Mainland market and the increasing consumption of its middle- and upper-class families. Hence, Hong Kong can diversify its import product types and countries of origin, aid Mainland consumers in searching global suppliers and form competitive advantages that differentiate itself to the major Chinese ports.

6.2 Research Methodology

The recent decade sees increasing importance of Mainland China as a market and a source of capital. Hong Kong can offer one-stop wealth management service for Mainland investors and help the Mainland import diversified consumption goods from the world market. First, we use datasets from the Insurance Authority of Hong Kong to analyse the advantages and disadvantages of insurance products sold in Hong Kong. Second, we use the data from HKEX to study the stock investment options for the Mainland investors. Third, we use custom data to examine how well Hong Kong serves as an importer for the Mainland consumers.

6.3 Stock Investment

6.3.1 Mainland Investors Have Product and Channel Restrictions on Hong Kong Stocks, and thus Trade Mostly Under the Connectivity Mechanism

As a global financial centre, Hong Kong has a well-established stock market. In 2018, the number of Initial Public Offering (IPO) in the Hong Kong Stock Exchange and the value of fundraising ranked first in the world. The Hang Seng Index also reached an all-time high of 33,154 points in 2018. However, at present, Mainland investors can only invest in the Hong Kong stock market in four ways (listed below), and the products and channels for investments have restrictions. QFII, QDII, RQFII, SH-HK-SC and SZ-HK-SC are a series of programmes that aim to intensify the liberalisation of the capital market. In examining the price co-movements of Hong Kong between the Mainland Chinese and US stock markets before and after the SH-HK-SC and SZ-HK-SC programmes, Li and Chen (2021) find no significant daily price co-movement between the first two stock markets. The reason may be that the mutual trading volume is low due to trading quota restrictions on both markets and the number of stocks available for the investors in their counterparts. The four ways in which Mainland investors can invest in the Hong Kong stock market are as follows.

- Directly open Hong Kong stock accounts to invest overseas assets. This type of investment presents various problems such as cumbersome procedures at account opening, high transaction costs and commissions (usually much greater than A-shares).

- Invest through Mutual Recognition of Funds (MRF). Mainland investors can attend to fund companies and purchase Hong Kong MRF products through a detailed procedure that is the same for Mainland products. However, only a limited number of fund products are authorised for sales between the two places—a total of 42 products, including 36 Southbound Funds and six Northbound Funds. Moreover, the products are complex. For funds in different currencies, investors themselves are responsible for exchanging currencies and bearing the corresponding expenses.

- Invest through Mainland Qualified Domestic Institutional Investor (QDII) funds (including Exchange Traded Fund and structured funds). These funds can be purchased at banks, fund company websites or third-party agencies, but most have minimum investment limits.

- Open Southbound trading account in Shanghai-Hong Kong Stock Connect or, inclusively, Shenzhen-Hong Kong Stock Connect, under the market from the Connectivity Mechanism. Shanghai-Hong Kong Stock Connect opened in November 2014 and Shenzhen-Hong Kong Stock Connect opened in December 2016. These two exchanges enabled Mainland and Hong Kong investors to both buy and sell securities listed on the other side of the exchanges within the prescribed range through local security companies or brokers. Table 6.1 shows that the Shanghai-Hong Kong Stock Connect includes Northbound Shanghai (Northbound trading) and Southbound Shanghai (Southbound trading); the former refers to foreign investors investing in stocks listed on the Shanghai Stock Exchange and the latter refers to Mainland investors investing stocks in the Hong Kong Stock Exchange under Shanghai-Hong Kong Stock Connect. Table 6.1 shows the connectivity mechanism. Table 6.2 shows the IPO funds raised by the stock exchange in the world and HKEX ranks the top.

Table 6.1 Stock market under the Connectivity Mechanism

	Shanghai-Hong Kong Stock Connect	**Shenzhen-Hong Kong Stock Connect**
Northbound	Northbound Shanghai	Northbound Shenzhen
Southbound	Southbound Shanghai	Southbound Shenzhen

Source: Hong Kong Stock Exchange

Table 6.2 Top 10 Stock Exchange of Total IPO Funds Raised in 2018

Rank	Exchange	IPO Funds Raised (US $ million)
1	HKEX	36,563
2	NYSE	25,574
3	Japan Exchange Group	25,567
4	NASDAQ	24,501
5	Deutsche Boerse Group	13,916
6	Shanghai Stock Exchange	13,472
7	London Stock Exchange	10,564
8	Mexican Stock Exchange	9,238
9	Shenzhen Stock Exchange	7,787
10	BSE and NSE-India	5,069

Source: Recreated from HKEX (2018, p.5)

The convenience of trading under the Connectivity Mechanism is that investors with A-shares accounts that have opened in the Shanghai Stock Exchange or the Shenzhen Stock Exchange do not need to reopen another at the Hong Kong Stock Exchange. Only before trading stocks in any connects, they need to sign Southbound Securities Trading Proxy Agreements, Risk Disclosure Statements, and other forms with the securities company. Then, investors can directly use their A-shares accounts to buy and sell Southbound stocks. At the same time, all tradable stocks for investors in Southbound trading are directly transacted in RMB, and the investors do not need to exchange currencies themselves; instead, the Shanghai Stock Exchange and China Securities Depository and Clearing Co., Ltd. execute all intermediate currency exchange processes. However, individual Mainland investors have investment qualification restrictions and an investment quota. The Hong Kong stock market requires that the total assets of securities accounts and capital account should be no less than RMB 500 thousand. At the same time, the transaction has an upper limit. The daily transaction amount of the Northbound trading is RMB 13 billion and that of the southbound direction is RMB 10.5 billion.

6.3.2 Financial Industry stands ahead among Hong Kong Stocks

According to the stock turnover of the Hang Seng Industry Classification System in Figure 6.1 prominent among Hong Kong stocks is the market turnover of financials, followed by information technology, while other industries have fewer turnover. In 2018, financials accounted for over 30% of Hong Kong's stock market turnover; the information technology, consumer discretionary and properties and construction

accounted for 18.45%, 14.06% and 12.44%, respectively. Other industries had less than 6% each. Among the top 10 stocks in the Hong Kong Stock Exchange in 2018, financials accounted for seven. This finding is consistent with the industry overall turnover distribution of the Hong Kong stock market, with a prominent share on financials.

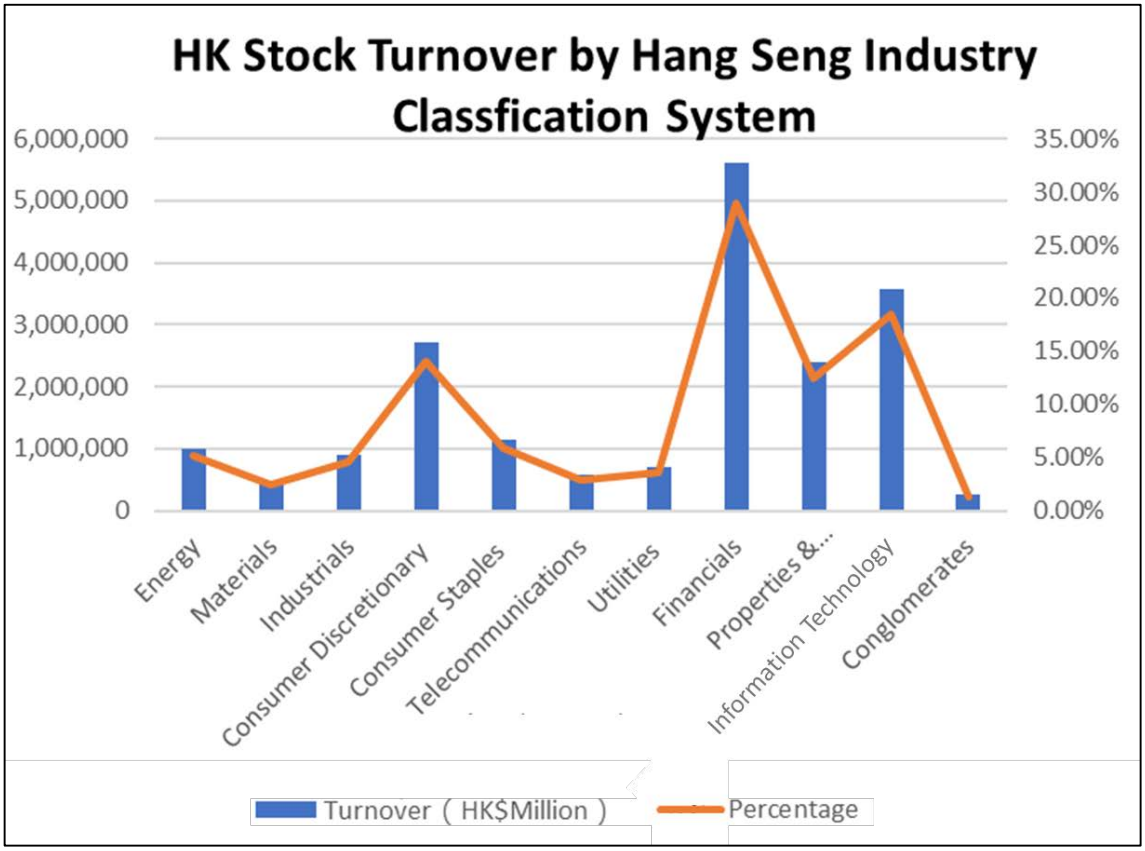


Figure 6.1 Hong Kong stock market turnover in 2018 according to the Hang Seng Industry Classification System

Source: Hong Kong Stock Exchange

6.3.3 Mainland A-shares Market is Better Diversified

Figure 6.2 shows that the turnover is more diversified in the A-shares market in the Mainland, mainly based on Information Technology, Materials and Industrials. In 2018, the three major industries accounted for 22.30%, 17.63%, and 16.06% of the A-shares market turnover. The financial industry that is beneficial for investors in the Hong Kong stock market is not prominent in A-shares, with a turnover of only 9.43%. This finding is consistent, as shown in Table 6.3, with the top 10 stocks in A-shares turnover mainly in the consumer staples, electrical appliance manufacturing and a few other industries.

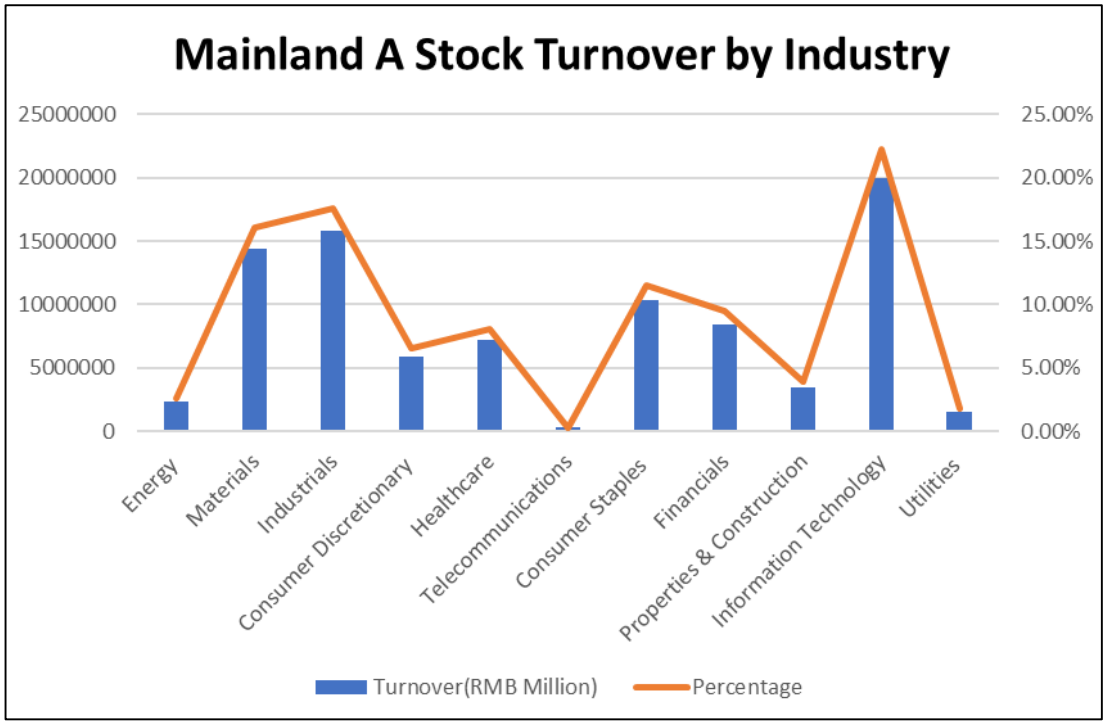


Figure 6.2 China's A-shares in 2018 according to the industry classification by China Securities Regulatory Commission

Source: Wind Database

Table 6.3 Top 10 stocks in Hong Kong stock market in 2018 (in HK$ Million)

Security Code HK	Security Name	Entity Type	Industry Classification by Hong Kong Stock Exchange	Turnover	Listing both in A-share and Hong Kong
00700	TENCENT	Private	Information Technology	2,468,256.22	No
02318	PING AN	Private	Financials	987,535.61	Yes
00939	CCB	State-owned	Financials	653,934.42	Yes
01398	ICBC	State-owned	Financials	439,575.26	Yes
00388	HKEX	Other	Financials	421,944.95	No
00005	HSBC HOLDINGS	Other	Financials	410,485.73	No
01299	AIA	Other	Financials	371,185.50	No
03988	BANK OF CHINA	State-owned	Financials	340,440.72	Yes
00941	CHINA MOBILE	State-owned	Telecommunications	318,851.82	No
00175	GEELY AUTO	Private	Consumer Discretionary	266,870.78	No

Source: Hong Kong Stock Exchange

6.3.4 Mainland Investors Mainly Invest in Private Enterprises for Hong Kong Stocks Centred at Information Technology and Biotechnology

As shown in Table 6.4 and Table 6.5, among the top 10 on Hong Kong stocks turnover favoured by Mainland investors through the Connectivity Mechanism, 80% are private enterprises and 20% are state-owned enterprises. According to the industry classification, the information technology industry accounts for 35%, biotechnology accounts for 20%, and financial services account for 25%. These figures show that Mainland investors are more inclined to the emerging biotechnology and information technology industries, and particularly in favour of large private enterprises that did not list on Mainland stock markets. This inclination may be due to higher stock returns and fewer fluctuations in information technology and biotechnology private enterprises, and better industry investment prospects.

Table 6.4 Top 10 stocks in the Mainland A-shares market in 2018 (in RMB Million)

Security Code for A Shares	Security Name	Entity Type	Industry Classification by Global Industry Classification Standards(GICS)	Industry Classification by Mainland	Turnover
601318.SH	PING AN OF CHINA	Private	Financials	Finance	1,202,486
600519.SH	KWEICHOW MOUTAI	State-owned	Consumer Staples	Manufacturing	743,312
000651.SZ	GREE	State-owned	Consumer Discretionary	Manufacturing	633,621
000725.SZ	BOE	Private	Information Technology	Manufacturing	582,128
000063.SH	ZTE	Private	Information Technology	Manufacturing	549,102
600030.SH	CITIC Securities Co., Ltd	State-owned	Financials	Finance	508,363
000858.SZ	WLY	State-owned	Consumer Staples	Manufacturing	491,837
300059.SZ	Eastmoney	Private	Financials	Finance	442,134
600887.SH	YILI	Private	Consumer Staples	Manufacturing	412,526
000333.SZ	Midea Group	Private	Consumer Discretionary	Manufacturing	367,012

Source: Wind Database

Table 6.5 Top 10 stocks in Southbound trading Shanghai in 2018 (in HK$ Billion)

Security Code HK	Security Name	Entity Type	Industry Classification by HKEX	Buy Trades	Sell Trades	Turnover
00700	TENCENT	Private	Telecommunications	304.00	303.62	607.62
01918	SUNAC	Private	Properties and Construction	107.72	170.07	277.79
01398	ICBC	State-owned	Financials	195.21	67.57	262.77
00939	CCB	State-owned	Financials	164.79	66.79	231.58
01093	CSPC PHARMA	Private	Healthcare	89.12	140.48	229.60
00586	CONCH VENTURE	Private	Industrials	108.27	110.64	218.90
01177	SINO BIOPHARM	Private	Healthcare	71.65	94.11	165.76
03988	BANK OF CHINA	State-owned	Financials	108.65	25.06	133.71
02196	FOSUN PHARMA	Private	Healthcare	80.20	40.04	120.24
00981	SMIC	Private	Information Technology	80.91	29.62	110.53

Source: Hong Kong Stock Exchange

6.3.5 Hong Kong Foreign Investors Mainly Invest in Mainland Stocks in the Manufacturing Industry

Table 6.6 and Table 6.7 show that from the turnover of Northbound Shanghai and Shenzhen, Hong Kong and overseas investors prefer to invest in stocks of the Mainland large and medium-sized enterprises in the manufacturing industry. Among the top 10 stocks in Northbound Shanghai in 2018, five were in the pharmaceutical, food and automotive manufacture industries. Private enterprises and state-owned enterprises account for 50% each. In the same year, among the top 10 stocks in Northbound Shenzhen, seven are from electrical appliance, food and electronic components manufacturing industries, and 60% are private enterprises.

Table 6.6 Top 10 stocks in Southbound trading Shenzhen in 2018 (in HK$ Billion)

Security Code HK	Security Name	Entity Type	Industry Classification by HKEX	Buy Trades	Sell Trades	Turnover
00700	TENCENT	Private	Information Technology	412.27	399.34	811.61
01918	SUNAC	Private	Properties and Construction	154.52	146.24	300.76
01548	GENSCRIPT BIO	Private	Healthcare	143.43	136.57	280.00
00763	ZTE	Private	Information Technology	136.53	129.73	266.27
00268	KINGDEE INT'L	Private	Information Technology	94.78	91.39	186.17
00939	CCB	State-owned	Financials	80.06	71.14	151.20
00175	GEELY AUTO	Private	Consumer Discretionary	76.12	70.19	146.31
00981	SMIC	Private	Information Technology	71.22	69.32	140.54
02382	SUNNY OPTICAL	Private	Information Technology	68.84	69.80	138.64
02318	PING AN	Private	Financials	64.82	71.99	136.81

Source: Hong Kong Stock Exchange

Table 6.7 Top 10 stocks in Northbound trading Shanghai in 2018 (in HK$ Billion)

Security Code SH	Security Name	Entity Type	Industry Classification by Mainland	Buy Trades	Sell Trades	Turnover
601318	PING AN OF CHINA	Private	Finance	849.16	663.08	1,512.24
600519	KWEICHOW MOUTAI	State-owned	Manufacturing	631.90	540.01	1,171.92
600276	HR	State-owned	Manufacturing	420.17	611.96	1,032.12
600036	CMB	Private	Finance	503.86	251.38	755.24
600887	YILI	Private	Manufacturing	221.65	205.61	427.26
600104	SAIC MOTOR	State-owned	Manufacturing	133.24	219.28	352.52
600585	ACC	State-owned	Manufacturing	129.91	210.76	340.67
601668	CSCEC	State-owned	Real Estate	174.91	156.26	331.17
601166	INDUSTRIAL BANK	Private	Finance	194.31	120.79	315.10
600030	CITIC Securities	Private	Finance	149.81	150.66	300.47

Source: Hong Kong Stock Exchange

6.4 Insurance Financing

6.4.1 Hong Kong Insurance Industry Ranks First in Asia and is in the Upper Echelon Worldwide

Hong Kong being a world first-class comprehensive financial centre, its insurance industry is critical in financial services and an essential component integrating Hong Kong into the development of the GBA. According to the Global Financial Centres Index (GFCI) from March 2019, Hong Kong ranked third after New York and London, the same as the insurance industry sub-index in GFCI. In 2018, the insurance density (per capita premium expenditure) in Hong Kong was US$ 8,320, ranking first in Asia and third in the world; insurance penetration rate (ratio of premiums to GDP) was 18%, ranking first in Asia and second in the world. In the same year, Singapore's insurance density and the penetration rate are US$ 6,440 and 7%, respectively, ranking fourth in the world.

As shown in Figure 6.3, Hong Kong insurers have two main businesses: the first is general insurance business (including accidents and health, general legal liability, property damage and automobile, employee compensation). This business aims to meet local needs mainly. The premium scale and growth are small. The second is the long-term insurance business (including individual life, annuity, permanent health). The premium scale and growth are both sizable. Among long-term insurance business, personal life insurance accounts for over 90%, clearly the most substantial insurance within the category. During the financial crisis of 2007-2008, the insurance market in Hong Kong once contracted, but proliferated after 2009. By 2018, the general insurance business reached approximately HK$ 53.6 billion and the long-term premiums reached approximately HK$ 478.2 billion, in which individual life premiums were approximately HK$ 431.3 billion.

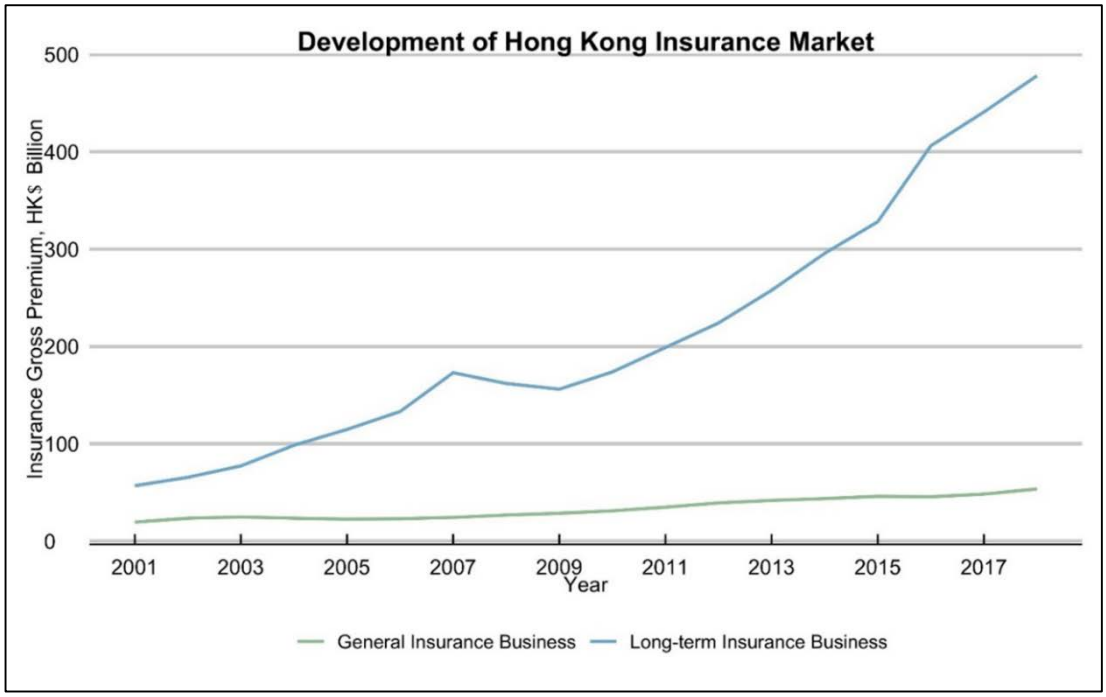


Figure 6.3 Overview of the Hong Kong insurance industry gross premiums in 2001-2018

Source: Hong Kong Insurance Authority

6.4.2 Annualised Non-Linked Insurance Is the Prominent Type of Individual Life Insurance on Demand

Based on the payment method and investment purpose, individual life insurance has seven categories: Single Investment-Linked [1], Single Non-Linked [2], Single Supplementary Contracts, Annualised [3] Investment-Linked, Annualised Non-Linked, Annualised Permanent Health and Annualised Supplementary Contracts. Since 2002, the share of non-linked insurance has gradually increased. In 2018, non-linked insurance accounted for 88% of the new individual life insurance premiums, of which annual payment accounted for 61% and single payment accounted for 27%. For investment-linked business, the single payment and annual payment together is approximately 11% (Figure 6.4).

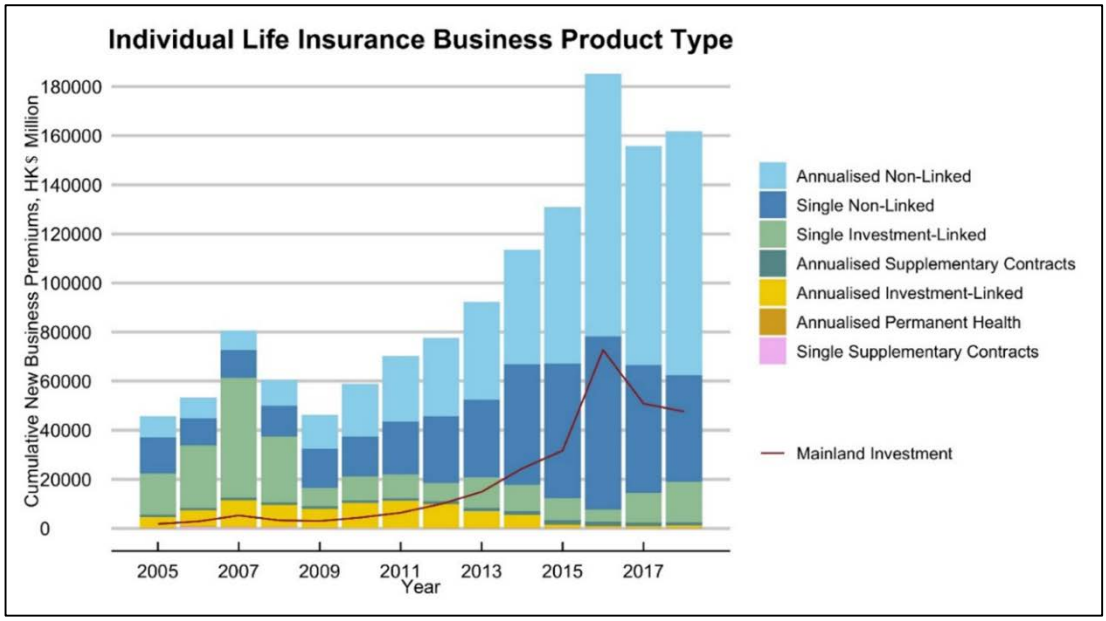


Figure 6.4 Development of Hong Kong individual life insurance business in 2005-2018

Source: Hong Kong Insurance Authority and Bloomberg

6.4.3 Hong Kong's Individual Life Insurance Business Concentrates on US Dollars (US$) and HK Dollars (HK$)

Figure 6.5 shows the unusually rapid development of US$ insurance, which increased year by year after 2013 and reached a peak in 2016. Its new US$ business premiums are HK$ 136.1 billion (74% of the market share). Subsequently, due to policy changes, office premiums sharply declined to HK$ 113.3 billion in 2018 (70% of market share). HK$ insurance has also been growing steadily; in 2018, the new premiums is HK$ 43.5 billion (27% of market share). From 2013 through 2017, the level of RMB premiums continued to decline (market share decreased sharply from 13% to 1%), but the new premiums in 2018 achieved a substantial increase for the first time, reaching HK$ 4.4 billion (market share rebounded to 3%).

Two reasons are assumed for the deficient development of RMB insurance products in Hong Kong. Firstly, Mainland customers have less demand for RMB insurance products. To diversify investment risks, these customers prefer US$ and HK$ insurance products, and mainly US$ insurance. Although the Hong Kong dollar is pegged to the US dollar, investors still lack confidence in the former. Secondly, British and American multinational insurers dominate the Hong Kong insurance market,

notably AIA Insurance Holdings Co., Ltd., the HSBC Group — including HSBC Insurance (Asia-Pacific) Holding Ltd. and Hang Seng Insurance Co., Ltd. — and Prudential Insurance Co., Ltd., which all specialise in overseas market investments. These firms are not as well-versed in RMB asset investment.

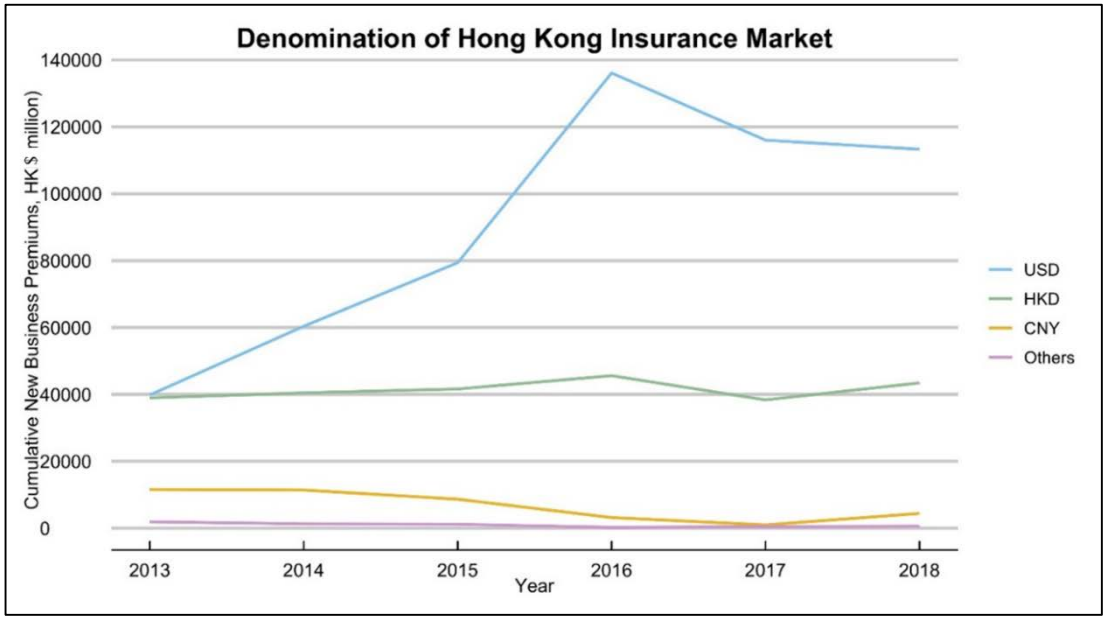


Figure 6.5 Development of Hong Kong insurance in various currencies for 2013-2018

Source: Hong Kong Insurance Authority

6.4.4 Three Giants Standing in the Hong Kong Insurance Market

As of 30 June 2019, Hong Kong had 162 authorised insurers, of which 92, 50, and 20 were pure general, pure long-term, and composite insurers, respectively. For individual life insurance, three giants stand in the market. Figure 6.6 shows that among these three insurance giants, AIA International accounted for 22% of the individual life new premiums; HSBC Group insurers accounted for 16% with the channel advantage from its banks; and Prudential (HK) Life accounted for 13%. These three are followed by Manulife (International), China Life Insurance (Overseas) and BOC Life. According to the comparison of insurance product types, AIA has absolute advantages in both single premium and annualised premium insurance. Manulife's single premium business volume is second only to AIA, while Prudential, HSBC and BOC Life are advantageous in annualised premium insurance （Table 6.8）. According to the

currency comparison (Table 6.9), AIA, Prudential and HSBC have apparent advantages in US$ insurance while Manulife (International) and China Life Insurance (Overseas) have a large amount of HK$ insurance business. By contrast, BOC Life has a balanced business in three currencies and its RMB insurance business is dominant.

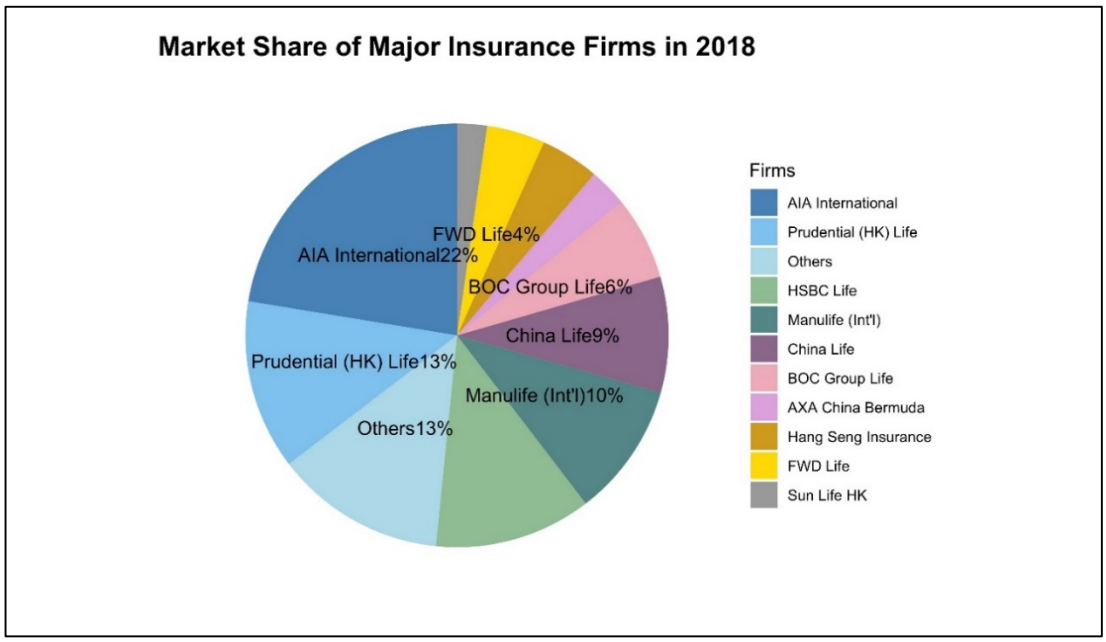


Figure 6.6 Hong Kong insurance market shares in 2018 (annual cumulative new policy premiums)

Source: Hong Kong Insurance Authority

Table 6.8 Cumulative new insurance premiums for products in different types from major insurers in 2018 (in HK$ Million)

Major insurers	**New single business premiums**	**New annualised business premiums**
AIA (International)	18,815	17,453
Prudential (HK) Life	3,577	17,402
HSBC Insurance	3,166	16,247
Hang Seng Insurance	2,749	4,498
BOC Life	1,160	9,124
China Life (Overseas)	0	14,281
Manulife (International)	13,101	3,474

Source: Hong Kong Insurance Authority

Table 6.9 Cumulative new insurance premiums for products in different currencies from major insurers in 2018 (in HK$ Million)

Major insurers	US$ insurance	HK$ insurance	RMB insurance
AIA (International)	34,422	1,840	5
Prudential (HK) Life	19,078	1,847	55
HSBC Insurance	15,269	1,047	27
Hang Seng Insurance	2,593	4,324	330
BOC Life	3,051	3,331	3,906
China Life (Overseas)	2,039	12,221	22
Manulife (International)	1,293	15,282	0.1

Source: Hong Kong Insurance Authority

6.4.5 Business is the Main Driving Force of the Growth of the Hong Kong Insurance Industry

Expansion into the Mainland market is not only due to choices made by Hong Kong insurers, but also driven and motivated by Mainland and Hong Kong policies. In recent years, the issue of the social security fund gap in the Mainland has drawn attention. The central government has encouraged commercial insurance purchases as a supplement. The policies introduced by the GBA has warranted many opportunities for the Hong Kong insurance industry to expand themselves further to the Mainland market. The *Outline development plan for the Guangdong-Hong Kong-Macao Greater Bay Area* explicitly mentions 'to support Mainland, Hong Kong and Macao insurance institutions in developing cross-boundary RMB reinsurance business', 'to progressively promote cross-boundary transactions of financial products such as funds and insurance within the Greater Bay Area', 'to support eligible Hong Kong and Macao banks and insurance institutions in setting up operations in Qianhai of Shenzhen, Nansha of Guangzhou, and Hengqin of Zhuhai' and other related content to support the development of the Hong Kong insurance industry in the GBA. The Hong Kong Insurance Authority suggested that building Hong Kong as the insurance service centre in the GBA is a feasible short-term plan. For Insurance Connect, this strategy is a medium- and long-term goal. Insurance Connect means that Mainland applicants can directly purchase Hong Kong insurance locally, pay new and renewal premiums and handle claims and dividends. In the early stage, Hong Kong implemented trials on protective products, such as medical

and critical illness insurances, which have relatively straightforward marketing features. After expiration of the policy, extra dividends and claim settlements return to the Mainland to prevent illegal activities and capital outflows.

From 2001 to 2016, Hong Kong's individual life insurance premiums grew 11 times, from which the industry greatly benefited owing to the economic integration of Hong Kong and the Mainland in the past 20 years. Individual life insurance premiums from the Mainland increased by nearly 40 times. In 2005, the new business premiums of individual life insurance from the Mainland was only HK$ 1.8 billion, accounting for less than 5% of the annual new business premiums. By 2016, the new business premiums reached HK$ 72.7 billion, accounting for 40%. Similar to Hong Kong, Singapore's insurance business has also proliferated in the past 20 years. The new business premiums have increased from approximately HK$ 4 billion in 2000 to approximately HK$ 20 billion[4] in 2018, growing five-fold. The difference between the two cities upon the insurance industry is that Singapore has less proportion of offshore insurance[5] (i.e. the applicant does not have a local identity card; excluding reinsurance) than Hong Kong. Singapore offshore insurance reached 11% in 2010 and then continuously declined, until reaching only 3% in 2018, signifying a low dependence on foreign investment.

After 2016, Hong Kong's individual life insurance business suffered a cold winter. The business volume in the Mainland declined year by year. New premiums in 2018 were HK$ 47.6 billion, accounting for 30% of the total new business. This decline was caused by the strengthening of foreign exchange control in the Mainland (US$ 50,000 per person per year), UnionPay restriction on the purchase of Hong Kong insurance with natures of saving and dividend, and the sharp fall of Mainland customer purchase of insurance in USD. In the first half of 2019, the number of new premiums from the Mainland was approximately HK$ 26.3 billion, accounting for 26% of the total. The recent social unrest and the COVID-19 pandemic caused a decrease in Mainland tourists, which is expected to further affect the insurance sales and delay the opening of insurance connect.

Figure 6.7 shows the specific insurance types of Mainland customer purchasing Hong Kong insurance. Annualised non-linked insurance accounted for most shares from recent Mainland investments, which burgeoned after 2014, up to 79% in 2018. The single non-linked insurance fell substantially after 2016, accounting for nearly 17%

in 2018. The single linked insurance declined after 2013, to 3% in 2018. After 2011, the annualised investment-linked insurance declined year by year, and hit only 0.4% in 2018. Hence, investment-linked insurance (including single and annualised) has gradually diminished in recent years, and the policies after 2016 have fallen sharply. In 2018, for Hong Kong, the cumulative new business premiums from the Mainland totalled HK$ 47.6 billion, which is 29.4% of the annual total. Of these products, approximately 96% are medical and protective insurance products, such as insurance of critical illness, medical, permanent life, term life and annuity; approximately 99% are annualised policies.

Meanwhile, as shown in Figure 6.8, from 2005 to 2018, the number of policies from Mainland investors on individual life insurance in Hong Kong shows an upwards trend, with non-investment policies account for over 8% of the total. After 2016, the growth of policy numbers has slowed down, and the number of voluntarily terminations of various insurance policies sharply increased to a total of 36,956 in 2018.

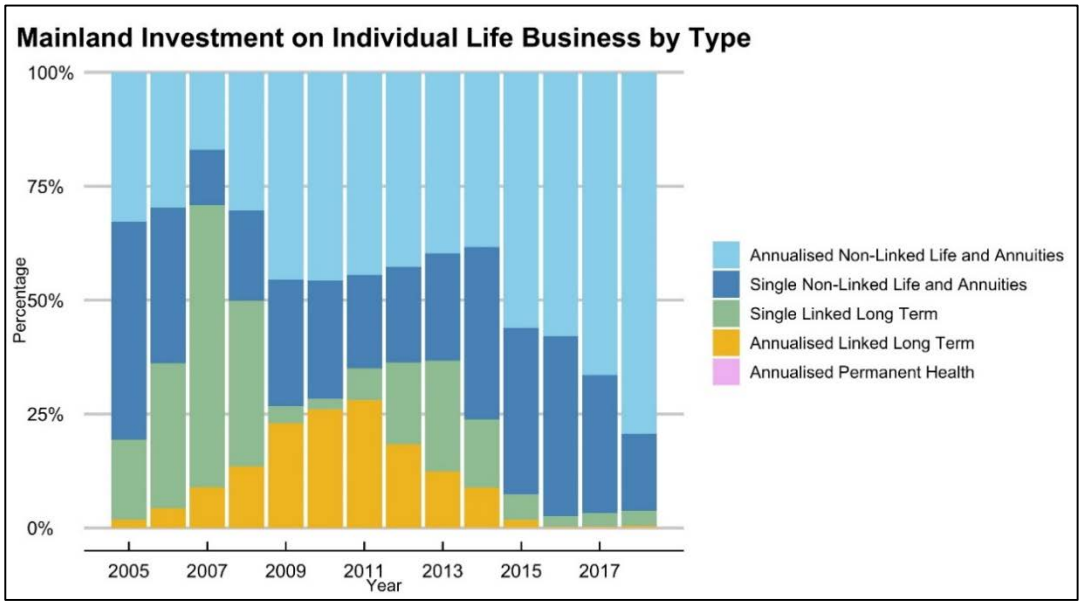


Figure 6.7 Proportion of Mainland investment in different types of individual life insurance in Hong Kong in 2005-2018

Source: Hong Kong Insurance Authority and Bloomberg

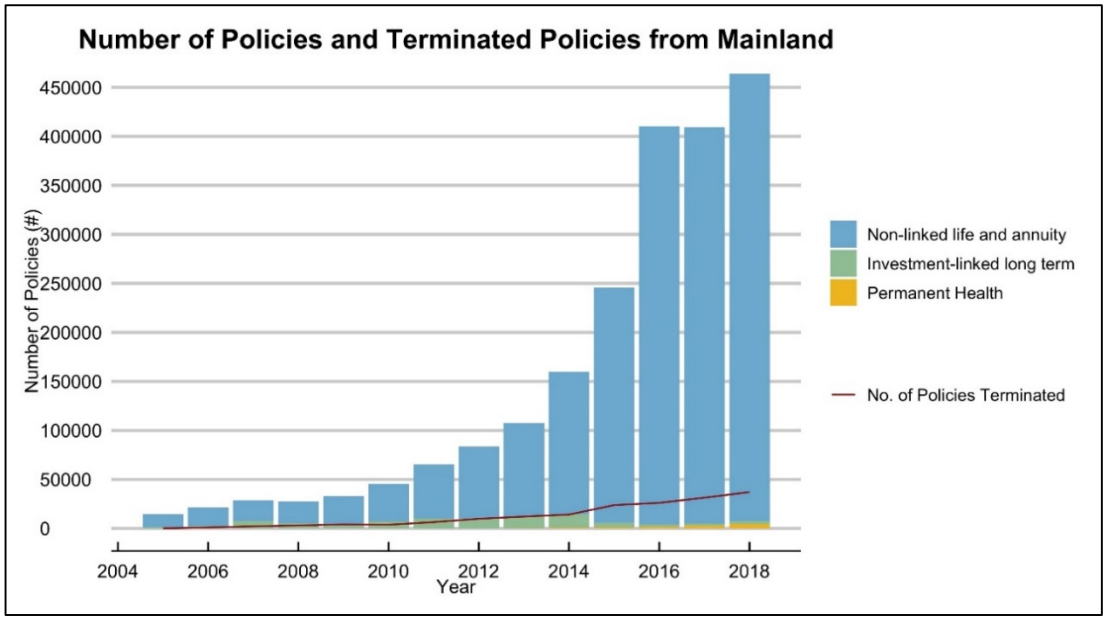


Figure 6.8 Number of policies from Mainland investing individual life insurance in Hong Kong and the number of terminated policies in 2005-2018

Source: Hong Kong Insurance Authority

6.4.6 Core Competitiveness of the Hong Kong Insurance Industry

- Hong Kong insurers can provide global insurance services with apparent advantages in overseas investment, and have higher investment returns. As mentioned above, Hong Kong insurers are mainly large-scaled and well-developed multinational insurers. These firms have connections in many countries and have advantages in globalised investment and multi-currency asset allocation. For example, a particular savings product from Prudential is capable of allocating multi-currency assets, such as USD, Euros and Japanese Yen in the United States, Europe and Asia-Pacific regions at the same time (Table 6.10). This advantage enhances investment income and reduces both exchange and interest rate risks.

Table 6.10 Diversified Investment Allocation of Prudential Insurance for a savings product in 2018

Currencies Investment Portfolio	**US$ Fund**	**HK$ Fund**	**RMB Fund**
US$	74%	14%	25%
HK$	6%	68%	1%
RMB	9%	3%	43%
EUR	5%	5%	11%
JPY	2%	2%	4%
Others	13%	8%	16%
Regional investment portfolio	**US$ Fund**	**HK$ Fund**	**RMB Fund**
US	63%	44%	26%
Asia-Pacific (excluding Japan)	15%	44%	53%
Europe	9%	5%	11%
Japan	1%	1%	3%
Other regions	12%	6%	7%

Source: Eastspring Investments (An investment company, member of Prudential plc)

- Investment in Hong Kong issuance products yields higher returns. Hong Kong insurers are interested in seeking high-yield investment targets and are subject to fewer restrictions. By contrast, Mainland insurers have restrictions on the scope of investment, and are mostly allocated at fixed-income assets, such as monetary funds, bank savings and other low-risk and low-return products, while the proportion of equity products must not exceed 30%. This restriction directly causes a far less dividend yield of Mainland insurance products compared with Hong Kong products. The average annual expected rate of return of Hong Kong insurance is 5%-9%, while that of Mainland life insurance is generically between 3% and 5%. Critical illness insurance and consumptive insurance in Hong Kong also has dividends, but not those in the Mainland. Take AIA and China Life as examples, in 2018, among AIA policyholders and shareholders, 53% of assets are allocated on bonds while 8% are loans and deposits, properties investment and cash; for China Life, only 42% of assets are allocated on Bonds while 18% are time deposits (Figure 6.9).

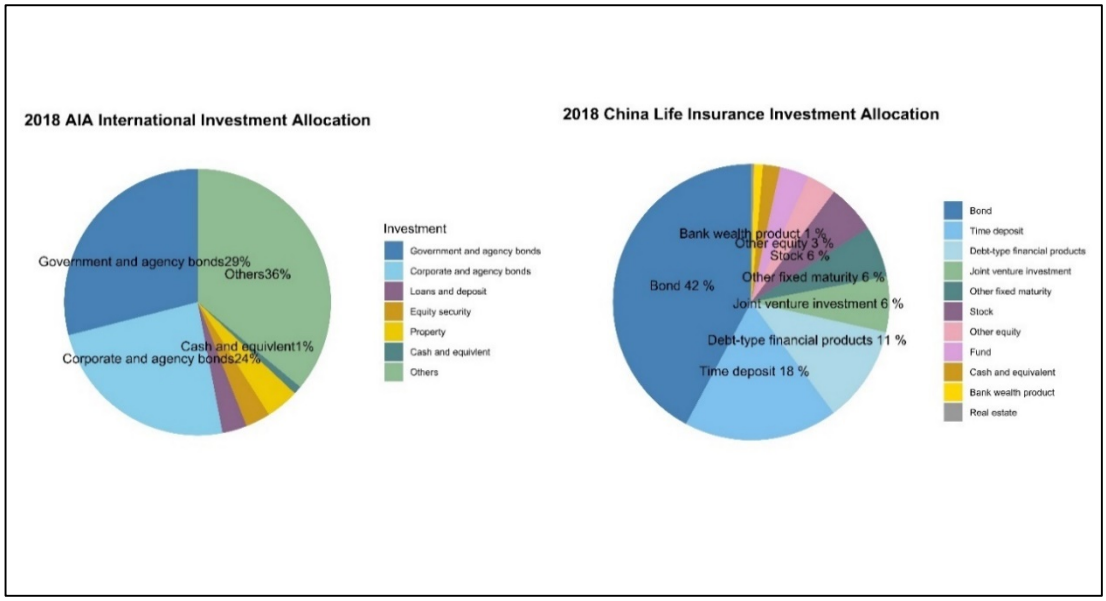


Figure 6.9 AIA (International) and China Life investment allocation distribution in 2018

Source: AIA (International) and China Life Insurance Co., Ltd. 2018 Annual Report

- Hong Kong insurance employees and customers have high quality. Hong Kong insurance licensees must pass a rigorous qualification examination and require continuous studies, completing 12 credits per year. Thus, they have considerable honesty and low mobility. Many Hong Kong insurance brokers and agents have master degrees and have strong personal connections in the Mainland. These highly qualified employees are capable of selecting high-quality Mainland customers, where certainly only middle-class and above families have the investment needs and corresponding economic capabilities to purchase insurance in Hong Kong. High-quality customers can reduce cost on claims and form a virtuous circle.

- Low premiums and high sum assured. Hong Kong premiums are generally 70%-80% of the Mainland ones, whereas their coverage of life insurance is 30% higher. Most of the major critical illness insurances in the Mainland use the 100% compensation method for the first critical illness. By contrast, most of Hong Kong products recognise multiple compensations and set the maximum compensation amount to be several times over the protection policy (300%-700%).

- Hong Kong has a broader scope of protection and claims. (1) Protection of critical illness: normally the Mainland only includes approximately 40 kinds of diseases, and when the insurance comes into effect, usually the life of policyholder is under serious threat. Hong Kong's critical illness insurance commonly includes more than 90 diseases. In addition to protecting more severe diseases, the latter also covers light illnesses and pediatric diseases; (2) Protection on claims: Hong Kong insurers implement strict at application and easy at claims policy. The applicants undergo stringent check underwriting insurance but loose claims processes; (3) Medical protection: Hong Kong policies can file claims worldwide. In particular, Mainland customers do not need to go to Hong Kong to claim for compensation. For example, Prudential Insurance has designated 1765 hospitals in the Mainland, including most of the third-class hospitals, while the Mainland insurers lack experience on claims filed outside of China.

- The function of avoiding debts and avoiding taxes is preeminent, and the supervision is strict: the debt avoidance function of Hong Kong's individual life insurance is protected by law, and has no estate or value-added tax on assets. In addition, the supervision system of the insurance industry in Hong Kong is well-established, and the Hong Kong Insurance Authority and the Hong Kong Federation of Insurers enforces strict supervision.

6.4.7 Risks on Hong Kong Insurance Industry Development

- Great return fluctuation. Similar to the two sides of a coin, the substantial investment ability of Hong Kong insurers also means high risks. The return of insurance products has considerable fluctuations, and the dividends present uncertainty. As shown in Figure 6.10, Prudential's investment share in BBB rating and below assets is 34%, while China Life invests more in credit-worthy products such as AA6 or above corporate bonds. Prudential exposes its funds to higher risks, causing more pronounced fluctuations in returns.

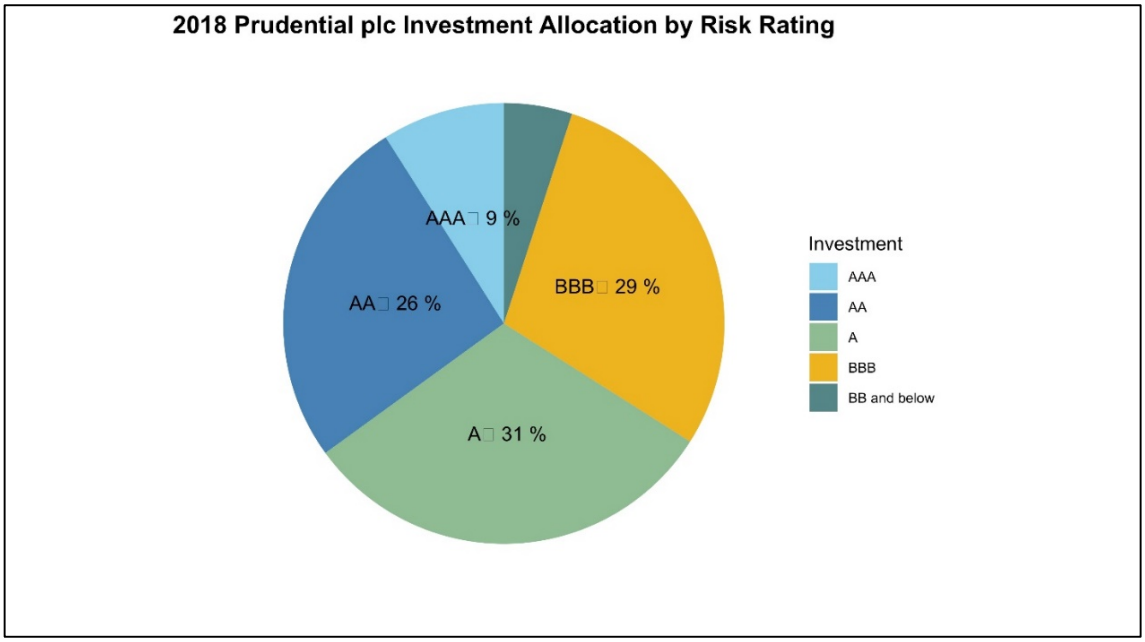


Figure 6.10 Prudential Insurance's investment risk rating in 2018

Source: Prudential Insurance 2018 Annual Report

- Dispute settlement can be troublesome. 'Distance' is the largest problem of Hong Kong insurance. In particular, when a dispute arises, Mainland consumers can only complain or seek resolution through Hong Kong's Insurance Complaints Bureau. The laws of the Mainland cannot protect these consumers and the Mainland regulatory authority cannot help.

- Exchange rate risk is high. Mainland investors who purchase HK$ or US$ insurance in Hong Kong bear the corresponding exchange rate risk themselves. Under the current China-US trade war, the pressure of RMB depreciation is relatively high, and thus the premiums paid by policyholders in the future may increase accordingly. Given factors such as the strengthening of foreign exchange regulations since 2016, the amount and types of purchase of HK$ and US$ by Mainland residents are under restrictions, discouraging Mainland residents from purchasing insurance in Hong Kong.

- Competition from the Mainland is increasingly intensive. In recent years, the insurance markets in the Mainland have become more opened up. Several well-known Mainland insurers have recruited specialised teams. Thus, their services have become better, their insurance products have become more diversified and their overall quality has been guaranteed. Therefore, the risk of reduced Hong Kong insurance market is high, leading to an urgent need for upgrading products and services.

6.5 Import

6.5.1 Growth of Hong Kong Helping the Mainland on Import Downgraded in Recent Years

In addition to helping Mainland investors on wealth management, Hong Kong also helps Mainland consumers to procure goods. However, in recent years, Hong Kong's growth on import for the Mainland has been slower than that of Mainland imports since 2001. Figure 6.11 shows that after 2003, the import performance of Guangdong is becoming inferior to that of the YRD region. Given geographical reasons, Hong Kong helps the Mainland imports mainly through land transportation, and thus mainly serves the PRD region (Figure 6.12). The global economic downturn, escalating international trade frictions, and Coronavirus 2019 pandemic have jointly crashed on the export-oriented industries, which are the main components of the GBA industry structure. These unfavourable factors may be the leading causes of the recent stagnation and declination, directly affecting the growth of Hong Kong imports for the GBA.

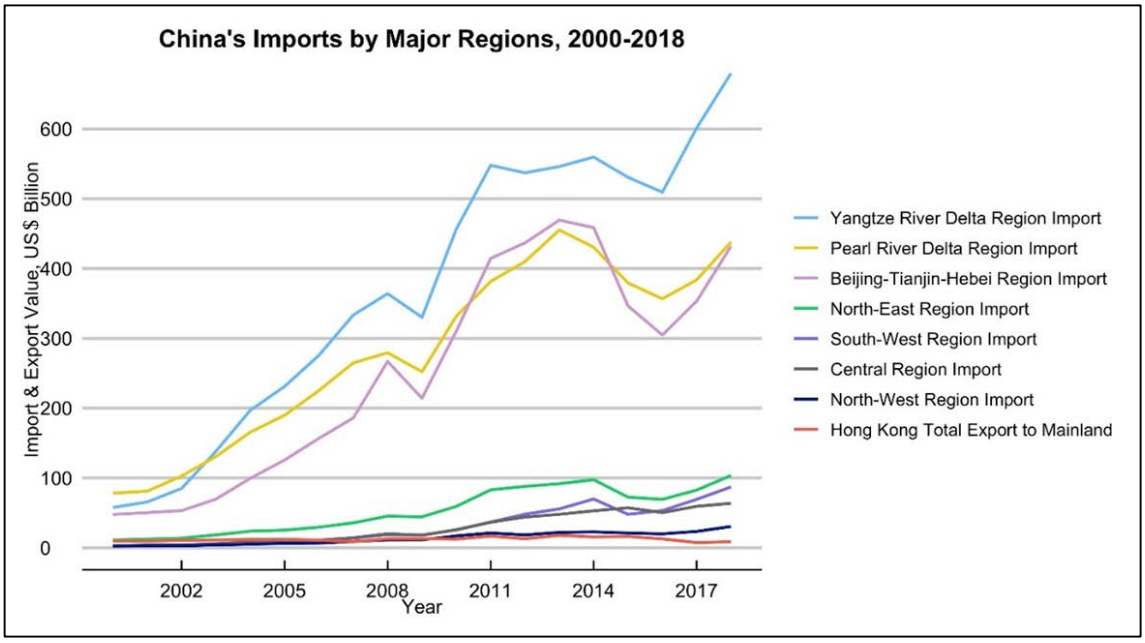


Figure 6.11 Imports by various regions of China and Hong Kong's overall exports to the Mainland in 2000-2018

Source: National Bureau of Statistics of China

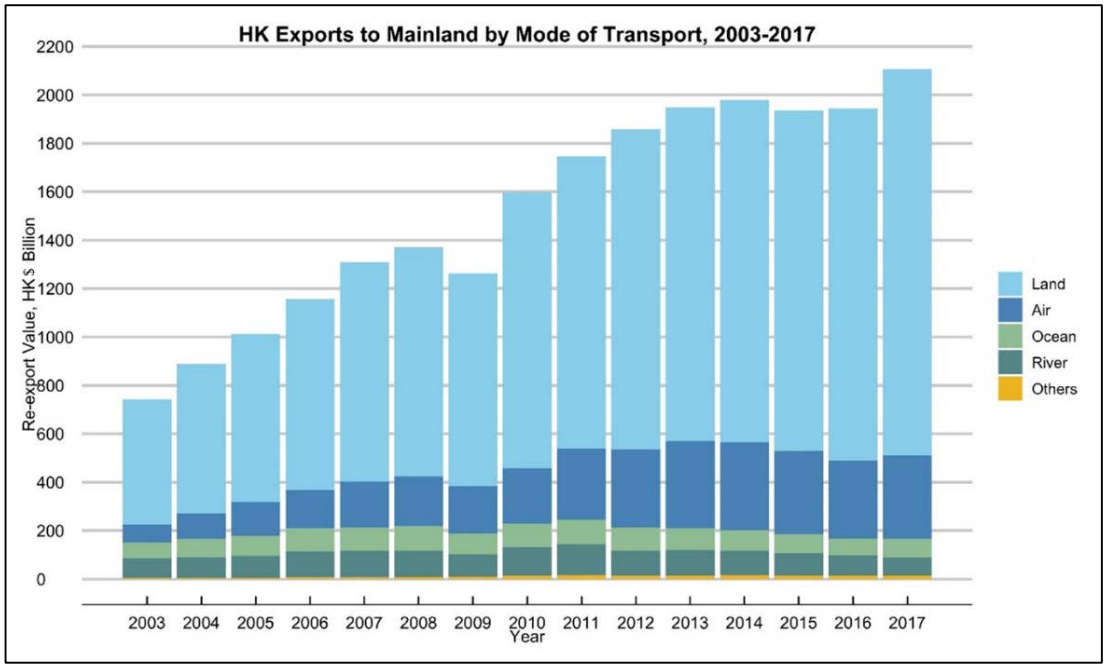


Figure 6.12 Hong Kong's transport mode on assisting Mainland import in 2003-2017

Source: Hong Kong Census and Statistics Department

In recent years, Hong Kong has helped the Mainland importing to Beijing, Shanghai and Guangdong. In 2017, Hong Kong exports to these three provinces accounted for 80% of its total exports to the Mainland (Table 6.1). In addition, the proportion of imports from Hong Kong to the total value of local customs import is lower year after year. In 2010, Hong Kong exports to the Mainland through Beijing customs reached US $633 million, which is 2.1% of local imports. In the same year, Hong Kong exports to Guangdong were US $5.974 billion, which is 1.8% of local imports. In 2014, Hong Kong exports to Shanghai were US $1.488 billion, which is 0.44% of local imports. Then in 2017, the proportion of Hong Kong exports to total local imports of all three provinces descend to less than 1.1%.

Table 6.11 Value and proportion of Hong Kong exports to Beijing, Shanghai and Guangzhou in 2017

Province/ Independent Municipality	Total imports, US$ million	Imports from HK, US$ million	Percentage of local total Imports	Percentage of HK total Exports
Beijing	265,218	243	0.12%	8.73%
Shanghai	364,791	1,698	0.36%	13.69%
Guangdong	383,806	3,397	1.09%	57.26%

Source: National Bureau of Statistics of China, Beijing Customs, Shanghai Customs, Guangdong Statistical Yearbook

6.5.2 Hong Kong Supports the Mainland on Import by Re-Exporting Manufacturing Products

Hong Kong imports for the Mainland mainly includes re-exports and Hong Kong products. Among them, re-export is the mainstay with a proliferating volume in recent years, and its share to Mainland imports has always exceeded 95% (Figure 6.13).

Hong Kong products imported by the Mainland are mainly from manufacturing, accounting for approximately 90% of Hong Kong's total export to the Mainland. Agricultural products, foods, fuel and metals and minerals accounted for 0.4%, 4.8%, 2.3% and 1.2%, respectively. From Figure 6.14, within the industrial products manufactured by Hong Kong, electrical machinery, telecommunication equipment, office machines and professional instruments are the main products for re-export, of which electrical machinery accounted for more than half of the total in 2017.

At the same time, comparing the product types in Hong Kong re-export (Figure 6.15) with those imported by Hong Kong (Figure 6.15), we found that machinery and transport equipment is the primary product type with the largest proportion imported by Hong Kong and then re-exported to the Mainland. Thus, Mainland imports through Hong Kong are mainly capital goods to serve manufacturers in the PRD. Consumption goods account for a relatively small proportion and Mainland consumers gain limited benefits from the trade between Hong Kong and the PRD.

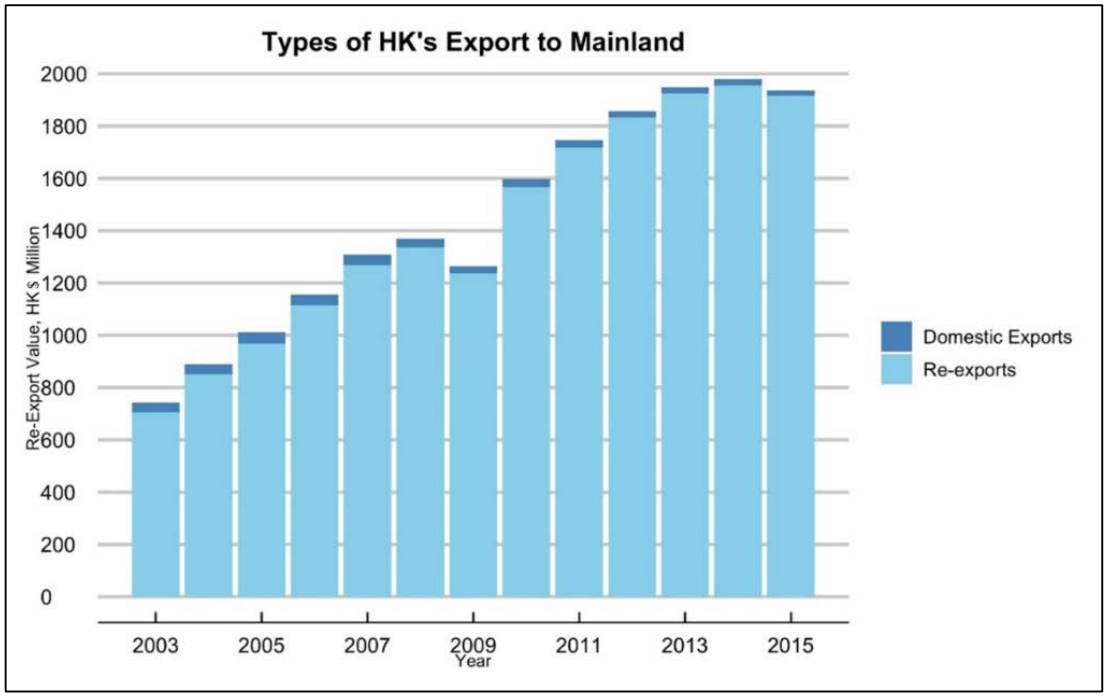


Figure 6.13 Overview of Hong Kong's assistance to Mainland import in 2003-2015

Source: Hong Kong Census and Statistics Department

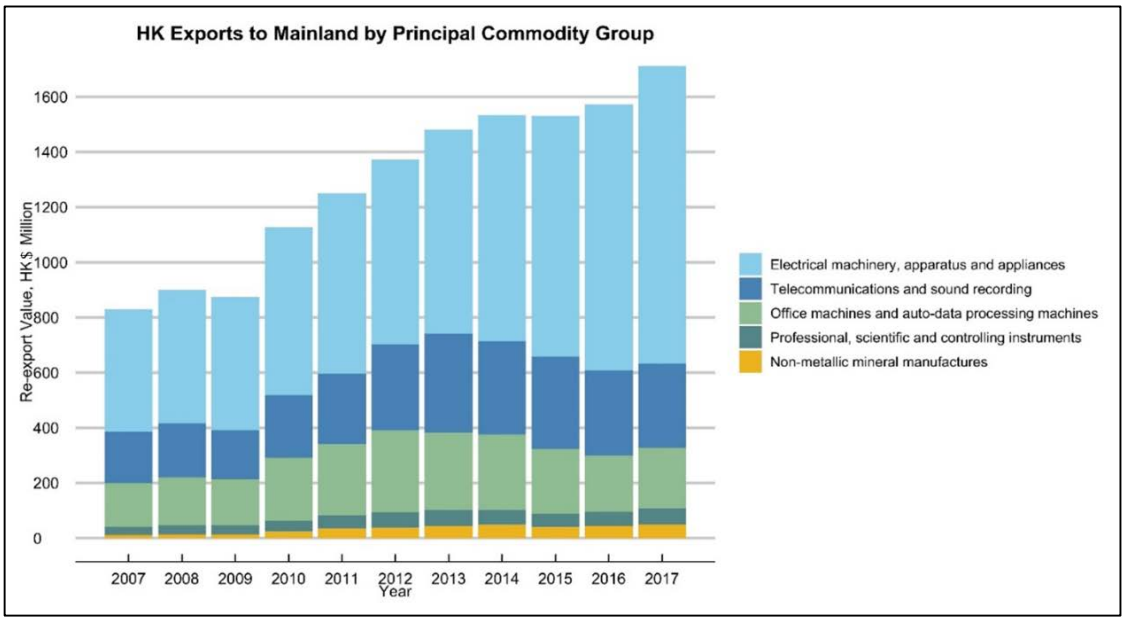


Figure 6.14 Hong Kong commodity exports value by groups to the Mainland | in 2007-2017

Source: Hong Kong Census and Statistics Department

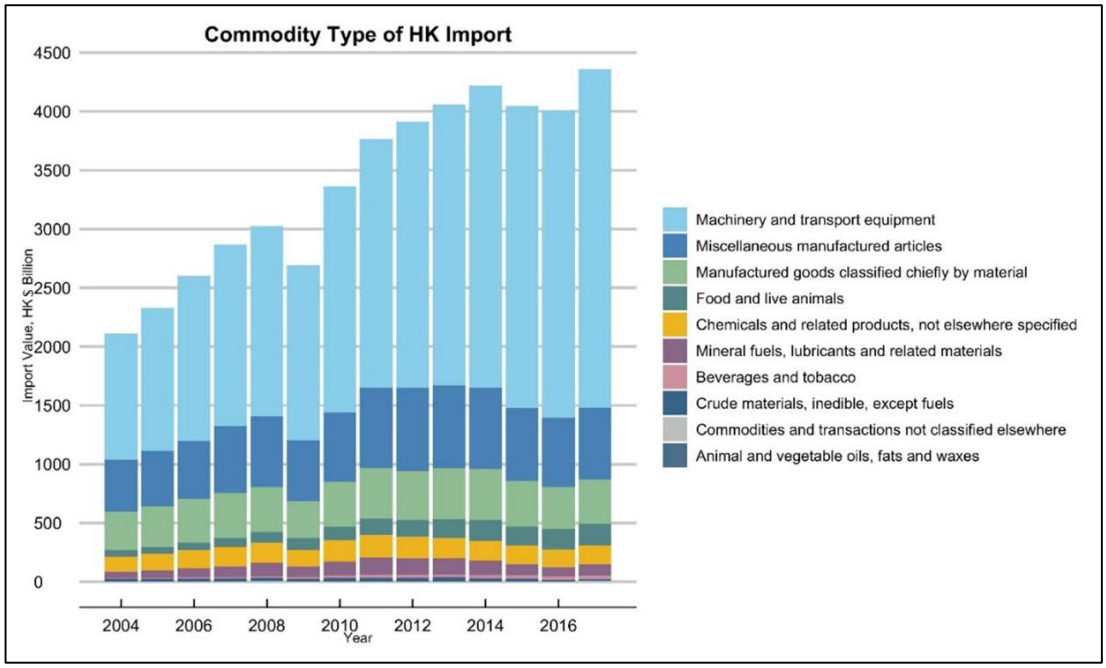


Figure 6.15 Commodity types of Hong Kong import in 2004-2017

Source: Hong Kong Census and Statistics Department

6.5.3 Consumption Goods Re-Exported to the Mainland are Predominantly Food, Luxury Goods and Clothing

The consumption goods that Hong Kong re-exports to the Mainland increased by an average growth rate of 9.8% annually in 2012-2018. In 2012, the re-export value of consumption goods to the Mainland was HK$ 75.5 billion, accounting for 4.6% of the total re-export volume with the year. In 2018, the re-export value of consumption goods to the Mainland was approximately HK$ 130.8 billion, the proportion of total re-export volume increased to 6.8% within the year. Figure 6.16 shows that Hong Kong re-export consumption goods to Mainland mainly consist of non-durable[7] and semi-durable consumption goods[8]. In 2018, these products accounted for 62% of total re-export consumption goods. Figure 6.17 shows that in 2012, the re-exports were: foods at 29% (including 02 meat and edible meat offal, 03 fish and sea foods, 04 dairy product and honey, 08 nuts and fruits, 21 miscellaneous edible preparations); Luxury goods at 27% (including 22 beverages, spirits and vinegar[9], 24 tobacco, 33 perfumery and cosmetics, 42 leather and travel goods, 71 precious metals and jewellery, 91 clocks and watches); and Clothing and footwear at 13% (including 61 clothing, knitted or crocheted, 62 clothing, not knitted or crocheted, 64 footwear). In 2018, the proportion of foods increased to 41%, while luxury goods remained unchanged at 27% and clothing and

footwear declined to 7%. At the same time, the proportion of product composition within each category also changed. In food re-exports, fish and sea foods still dominated and its proportion increased by 5%, while dairy product and honey proliferated from 0.2% in 2012 to 4%. Among luxury goods, the proportion of perfumery and cosmetics boomed from 3% to 10%, showing the fastest growth. The proportion of leather and travel goods decreased from 8% to 5%. In addition, the proportion of 39 plastics and articles also decreased by half, from 6% to 3%.

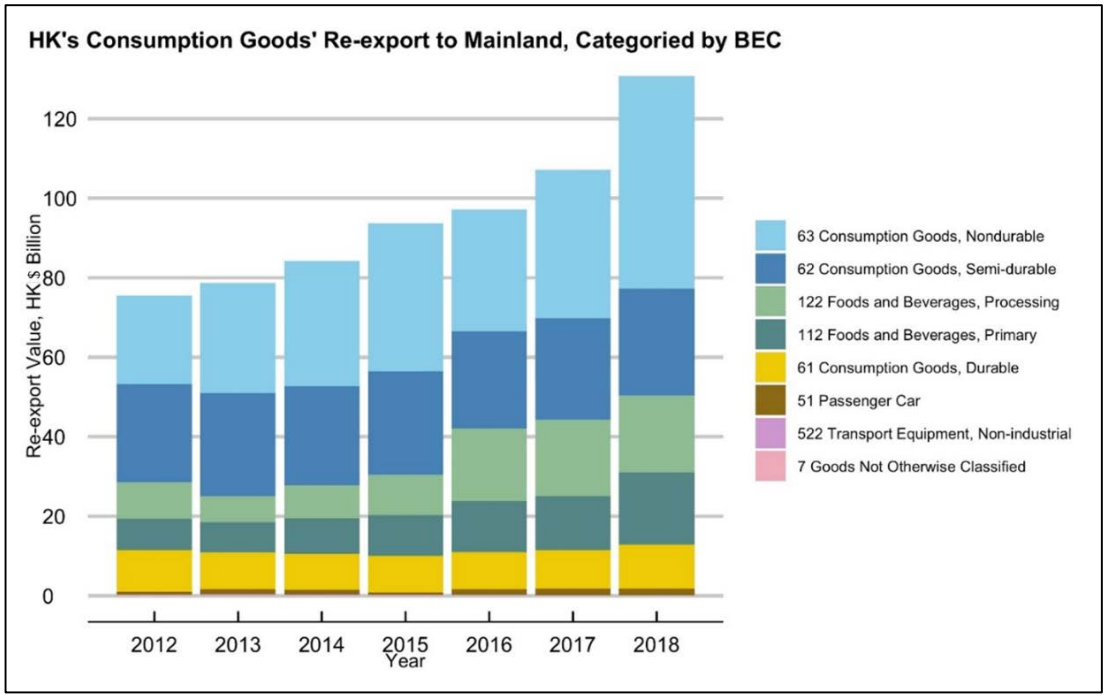


Figure 6.16 Hong Kong consumption goods categories re-exported to the Mainland in 2012-2018, classified by BEC

Source: Hong Kong Census and Statistics Department

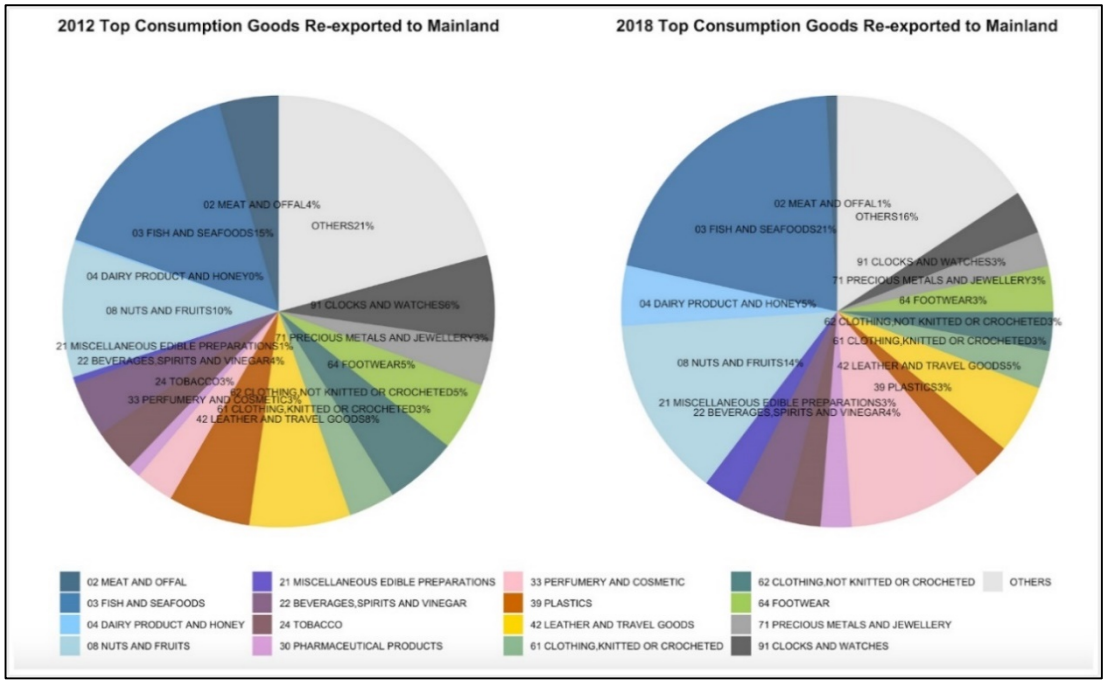


Figure 6.17 Hong Kong major consumption good categories re-exported to the Mainland in 2012-2018, by HKHS

Source: Hong Kong Census and Statistics Department

6.5.4 Hong Kong Is Better Diversified on Categories and Sources of Re-Exported Consumption Goods and is thus Far Advantageous on Intermediate Goods

As shown in Figure 6.18, for Hong Kong re-export, the Mainland is the largest consignment and the largest origin. Other than the Mainland, Hong Kong's re-export consumption goods sourced mainly from the United States (US), Chile, Italy, Japan, France, Thailand and a few other countries. Since 2012, not only the trade volume between Hong Kong and developed countries has proliferated, such as the US, Japan, France and Germany, but also with developing countries, such as Chile, Thailand and South Africa. According to the concentration indicator, which is HHI in this case, the concentration of Hong Kong's trading partners declined from 526 in 2012 to 363 in 2018. This result shows that the diversity of trading partners has improved for Hong Kong re-exporting towards the Mainland, and is better than Shanghai, Guangdong and other Mainland ports.

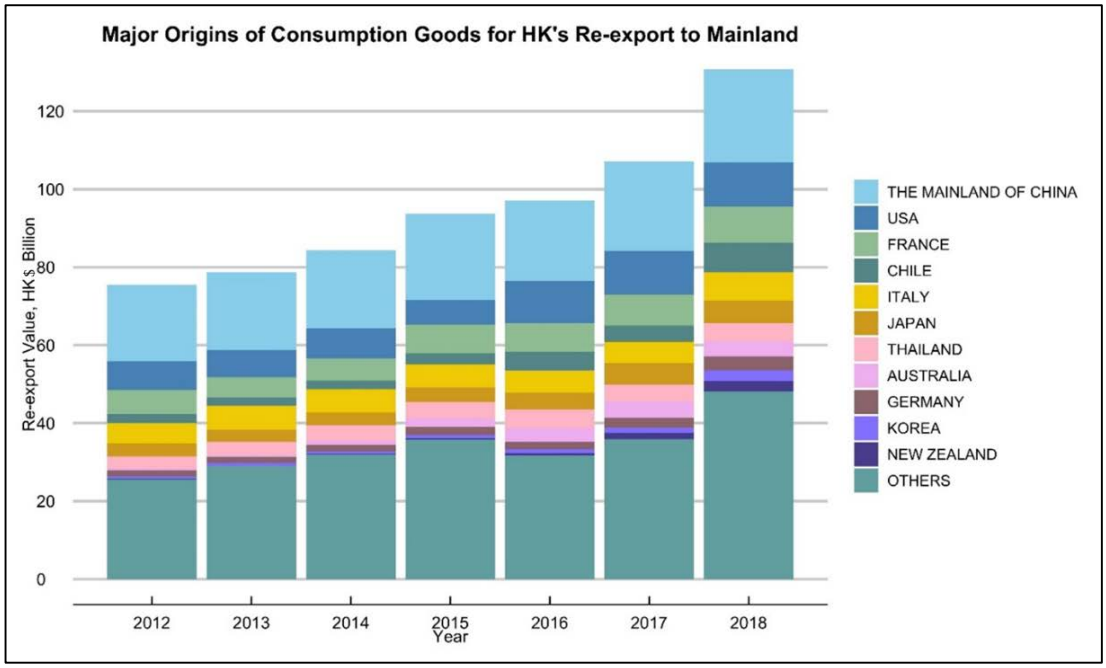


Figure 6.18 Country of origin of consumption goods for Hong Kong Re-export to the Mainland in 2012-2018

Source: Hong Kong Census and Statistics Department

The import origin of consumption goods in the Mainland is more concentrated in a few countries compared with Hong Kong. Take Shanghai and Guangdong as examples. The HHI of the trading partner of Shanghai consumption goods import was 532 in 2018, while that of Guangdong was 500. According to Figure 6.19, Shanghai imports consumption goods mainly from European developed countries such as Germany, Italy and France, while Guangdong imports mainly from both the developed countries, such as the US and Germany, and from developing countries, such as Thailand, Chile and Vietnam. Shanghai imported RMB 329.1 billion of consumption goods in 2018, while Guangdong imported RMB 146.4 billion, accounting for 16% (RMB 2034.3 billion) and 5% (RMB 2890.0 billion) of total local imports for the year, respectively. Thus, Guangdong mainly imports production goods and its demand for consumption goods is less than Shanghai.

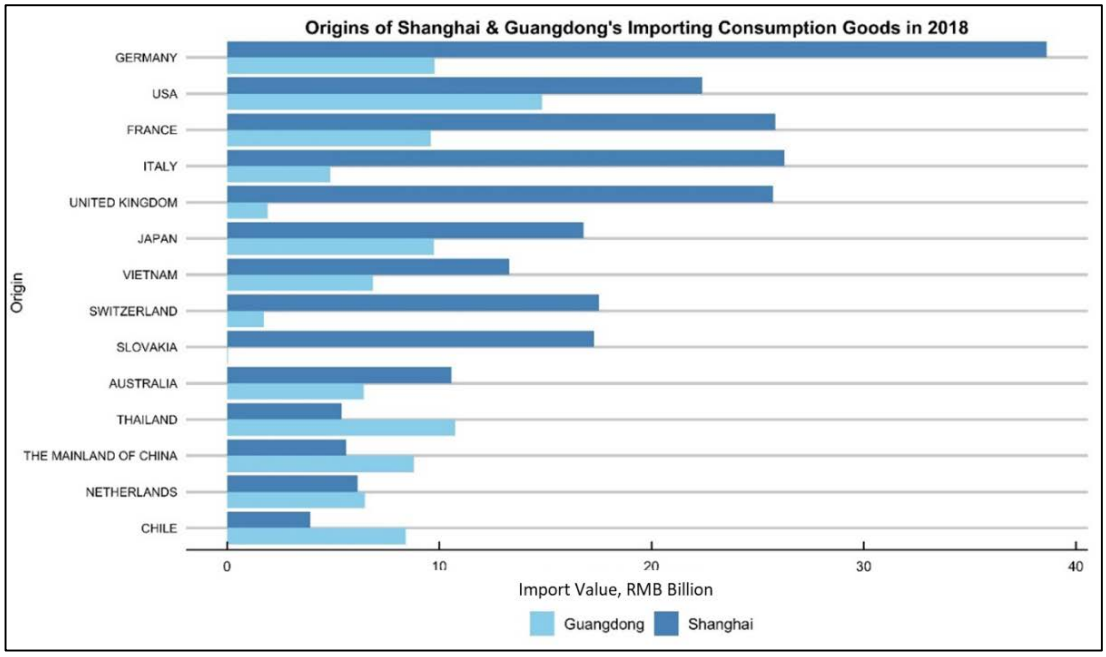


Figure 6.19 Major consumption goods import origin in Shanghai and Guangdong in 2018

Source: General Administration of Customs of the PRC - Customs Statistics Query Platform

Similarly, through HKHS classification, the concentration of Hong Kong re-exports of consumption goods (HHI) in the Mainland is 876, and the concentration of imported goods in Shanghai and Guangdong is 992 and 697, respectively. Hong Kong's re-export concentration is between those of Guangdong and Shanghai, which differ in the categories of consumption good imports (Figure 6.20). The most imported consumption good by Shanghai is 87 Vehicles, followed by 30 Pharmaceutical Products and 33 Perfumery and Cosmetics, accounting for 25%, 10% and 9%, respectively, as a proportion of total consumption goods import of the year. By comparison, the most critical imports of consumption goods in Guangdong are foods, especially 08 Fruits and Nuts, 19 Preparations of Cereals and Flour and 02 Meat and Offal, accounting for 14%, 11% and 9%, respectively. Therefore, we believe that Shanghai's imports concentrate upon Vehicles and a small number of other commodity types, while Guangdong is better diversified. Guangdong not only obtains consumption good imports from Hong Kong but also directly imports from sources such as the US and Thailand. Combining with Figure 6.17, Hong Kong's re-export structure of consumption products is a closer match to that of Guangdong. For 87 Vehicles and 30 Pharmaceutical Products, which Shanghai imports massively and Hong Kong re-exports little.

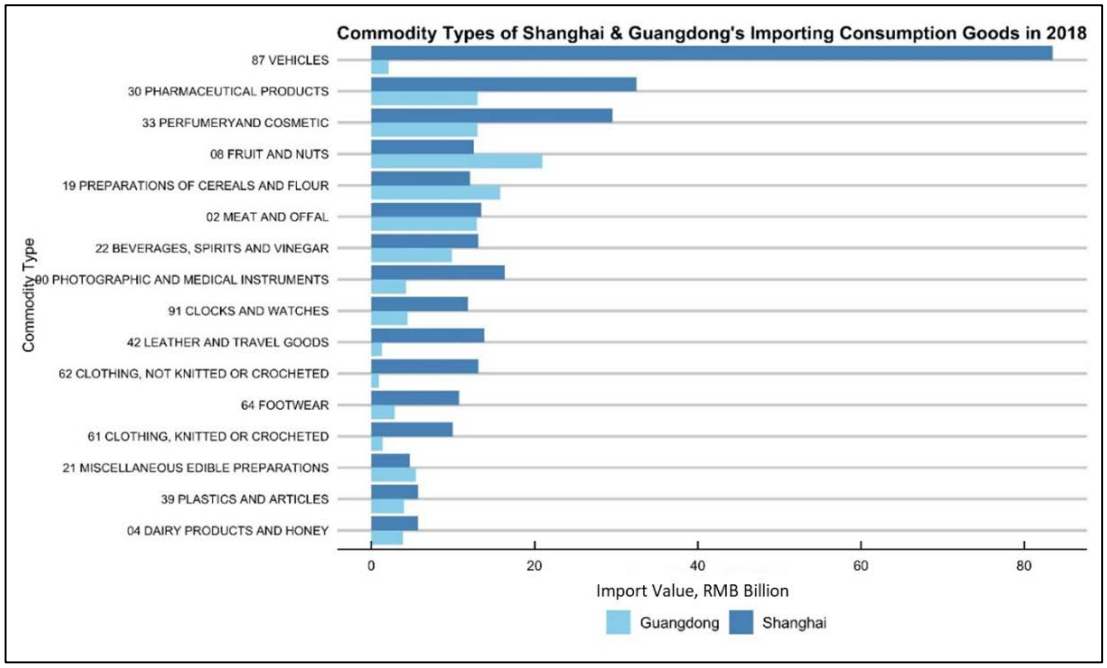


Figure 6.20 Major import consumption goods by type in Shanghai and Guangdong in 2018

Source: General Administration of Customs of the PRC - Customs Statistics Query Platform

The PRD region is the main destination for Hong Kong consumption goods re-exports. Figure 6.21 shows that the products that registered both countries of origin and consignee to be the Mainland are mostly semi-durable consumption goods. These products are mainly plastics, such as tableware, kitchen utensils, sanitary products, decorations and clothing accessories. In 2012-2018, these products consistently account for over 50% of goods from and to the Mainland. Non-durable consumption goods, which are mainly lamps and batteries, follow with 25% in 2018. Products in these two categories are mostly raw materials or processing products under the three-plus-one trading mix [10]. Hong Kong is a globally high-quality duty-free port, and thereby Mainland companies are willing to trade through Hong Kong for tax relief. Hong Kong still obtains advantages on intermediate goods trading.

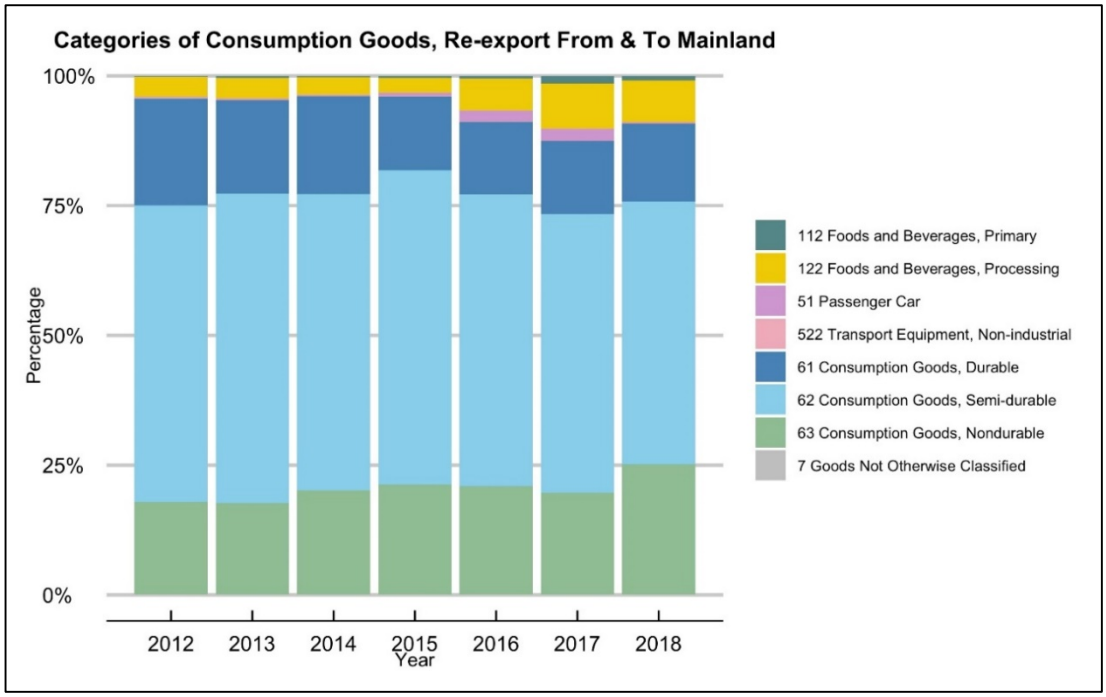


Figure 6.21 Hong Kong re-export of consumption goods in 2012-2018, both from and to the Mainland

Source: Hong Kong Census and Statistics Department

6.5.5 Hong Kong is Advantageous in Luxury Goods Importing and Marketing

In addition to helping the Mainland to re-export consumption goods, tourism accompanying local luxury goods marketing is also an indispensable channel for Hong Kong to conduct import for the Mainland. Figure 6.22 compares the 2018 data of Hong Kong's major luxury goods imports and re-exports to the Mainland. Approximately 25% of imported luxury goods are re-exported to the Mainland, of which 97% of tobacco products. Nonetheless, this value is only 5% for precious metals and jewellery. Thus, we infer that a large percentage of jewellery products are directly sold locally or shipped overseas. In Table 6.12, according to famous jewellery companies such as Chow Sang Sang and Chow Tai Fook, approximately 50% of the profits come from local sales in Hong Kong. Although this part of local sales does not reflect in customs data, we reasonably assume that the majority of their customers are still from the Mainland. Hong Kong thus has clear advantages in luxury goods imports and marketing compared with the Mainland.

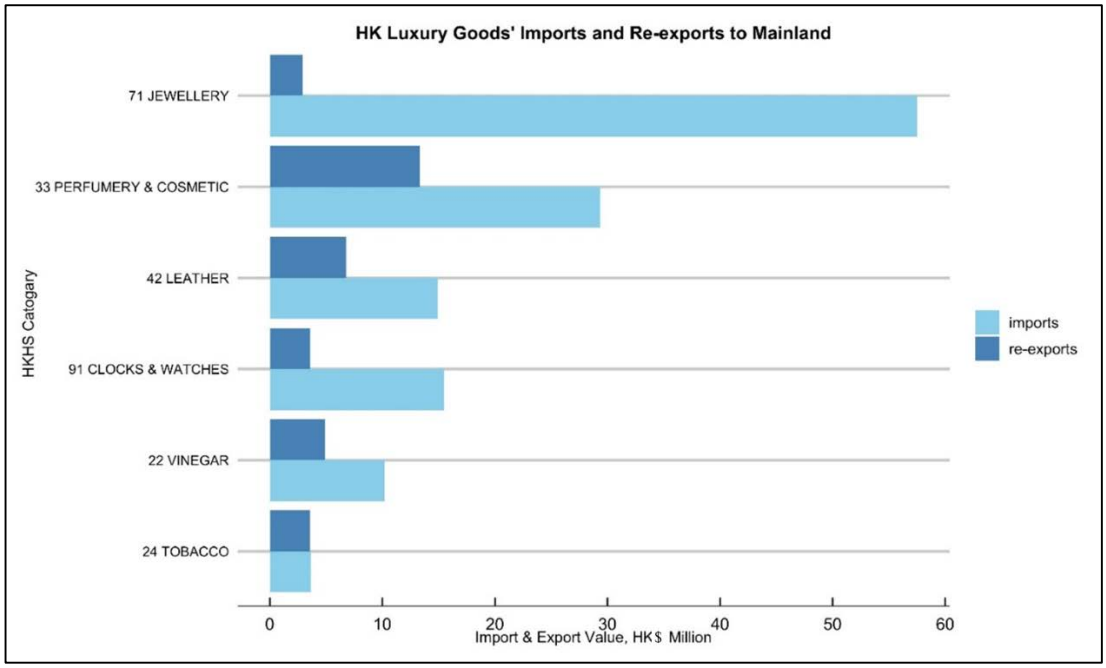


Figure 6.22 Hong Kong major luxury goods imported and re-exported to the Mainland in 2018

Source: Hong Kong Census and Statistics Department

Table 6.12 Sales of Listed Jewellery Enterprises in Hong Kong in 2018

Brands	Overseas sales, including Hong Kong, Macao, and Taiwan (HK $ Million)	Mainland sales (HK $ Million)	Overseas sales, including Hong Kong, Macao, and Taiwan (%)	Mainland sales (%)
Chow Sang Sang	9,364	9,329	49.79	49.60
LukFook Group	7,441	5,256	62.10	37.90
Chow Tai Fook	3,552	4,402	44.66	55.34
Tse Sui Luen	1,492	2,512	36.71	61.79

Source: 2019 Annual Reports of Chow Sang Sang, LukFook Group, Chow Tai Fook, Tse Sui Luen

Chapter 7
Hong Kong Leading the Greater Bay Area to Go Global and Driving New Economic Growth from Enterprise Internationalisation

7.1 Introduction

In terms of the scope of overseas investment and international holding capacity, Hong Kong enterprises deliver impressive performance. As suggested by Tring (2012), being a transaction haven and international platform with banking and legal professionals, the city provides considerable advantages for outbound transactions of Chinese companies. Although merger-and-acquisition (M&A) activities in Hong Kong slightly declined due to the Coronavirus 2019 pandemic, a solid rebound is observed with a focus on Chinese target companies (Cooke et al., 2021). Meanwhile, M&A in Mainland China have been increasing and are expected to expand, with Hong Kong being the top destination of outward investment (Tang and Metawalli, 2012). Through internationalising the scope and adopting a more structured approach as to the resizing of activities, technical and scale efficiency can be constructively affected (Wang and Zhang, 2018). In achieving internalisation, enterprises must be encouraged to expand the scope of investment and strengthen the international holding capacity. The advantageous and disadvantageous positions of Hong Kong enterprises in carrying out M&A are explained below. The SDC Platinum database is used to analyse data from 1998 to 2018 to examine the advantages of Hong Kong enterprises in carrying out M&A. The results are explained as follows.

7.2 Research Methodology

Low-cost manufacturing industries in Mainland China in general and PRD in particular is not sustainable in the long run. Mainland China needs to move up (technological innovation) to high tech industries and move out of some of its low-cost manufacturing industries to low-cost countries and regions (The Belt and Road Initiative). Hong Kong can help Mainland China to acquire technology-intensive firms in the developed countries (dealing with information asymmetry and moral hazard problems) and help Mainland China's firms to operate in economically underdeveloped regions, generally with poorer economic institutions.

In this chapter, we study the acquisition patterns by Mainland Chinese bidders versus Hong Kong bidders, the characteristics of targets and acquirers using SDC dataset, to examine Hong Kong's advantages and disadvantages in helping companies in Mainland China to go global.

We also use Orbis company database, which has information on around 300 million companies from all countries. One special feature of Orbis is that it provides ownership information, so we can examine Hong Kong-invested firms all over the world. In particular, we can understand whether Hong Kong firms have relocated away from the Mainland into other countries and its subsidiaries allocation. We can also examine whether Hong Kong firms participate in Mainland China's drive to the Belt-and-Road countries.

7.3 Comparison on industry distribution of Hong Kong and Mainland Acquirer and Target Companies

In the global M&A, financial enterprises account for over half of Hong Kong acquirer companies. Compared with Mainland acquirers, their industry distribution is relatively simple, but their target companies are more diverse in industry than the Mainland. Firstly, we look at the industry distribution of Hong Kong and Mainland enterprises from the perspective of the acquirer. Sectors are classified according to the top four SIC codes, as follows: Agriculture, Forestry and Fishing; Mining; Construction; Manufacturing; Transportation, Communication, Electric, Gas and Sanitary Service; Wholesale Trade; Retail Trade; Finance, Insurance and Real Estate; Services; Public Administration; and Non-classifiable Establishments.

Next, relevant data from 1998 to 2018 are compared. Figure 7.1 reflects the industry distribution of Hong Kong companies as the acquirer. The rise, fall and rebound in the overall number of M&A activities of Hong Kong companies as acquirers are consistent with the period of the global M&A wave. In the M&A activities, the top three industries in the Hong Kong enterprises as the acquirer are financial insurance and real estate, manufacturing and service industries, among which financial enterprises rank first and account for over half of the total. This result fully reflects Hong Kong's dominant position as an international financial centre and the imbalance in the distribution of Hong Kong companies as acquirers.

Figure 7.2 shows the industry distribution in the Mainland as an acquirer. Although the top three industries of the Mainland companies as acquirers are consistent with Hong Kong, the ratio gap of different industries is small.

To further quantify and compare the balance of Hong Kong companies and Mainland enterprises as acquirers in the industry distribution, we construct the Herfindahl Index (HHI) to indicate the balance of industry distribution (a higher HHI index indicates a more uneven industry distribution). Figure 7.3 shows that compared with Mainland enterprises, Hong Kong enterprises has a significantly higher HHI in 1998-2018, and a higher upward trend in recent years, reflecting their industry distribution as acquirers as unbalanced.

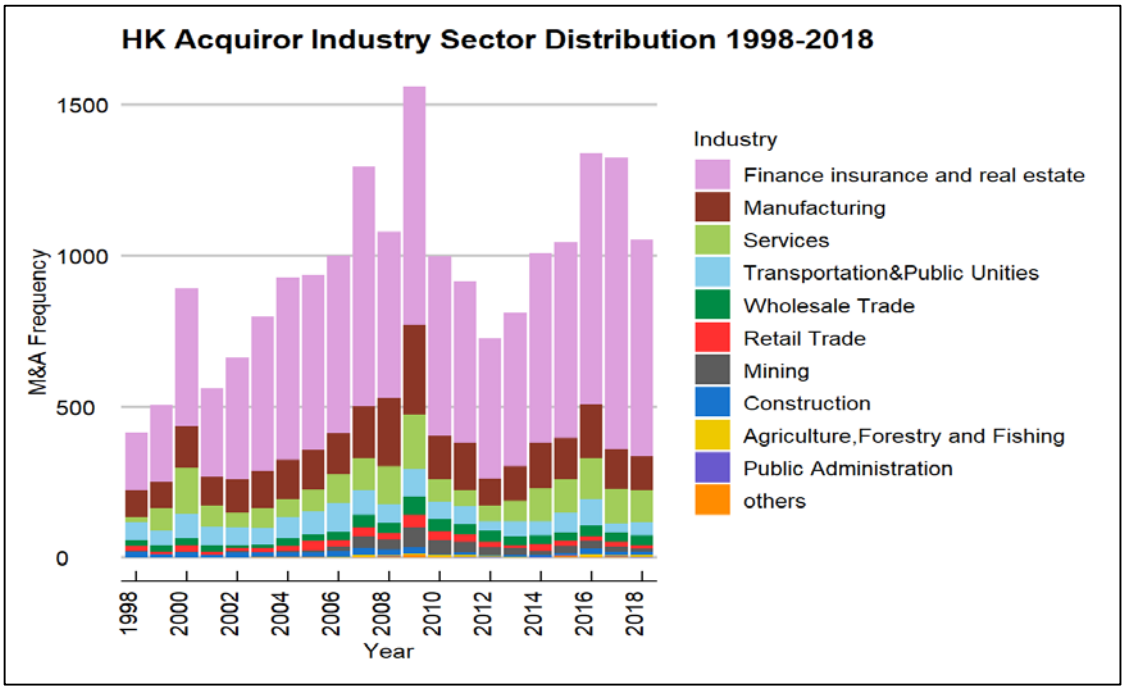


Figure 7.1 Industry distribution of Hong Kong acquirer companies in 1998-2018

Source: SDC Platinum database

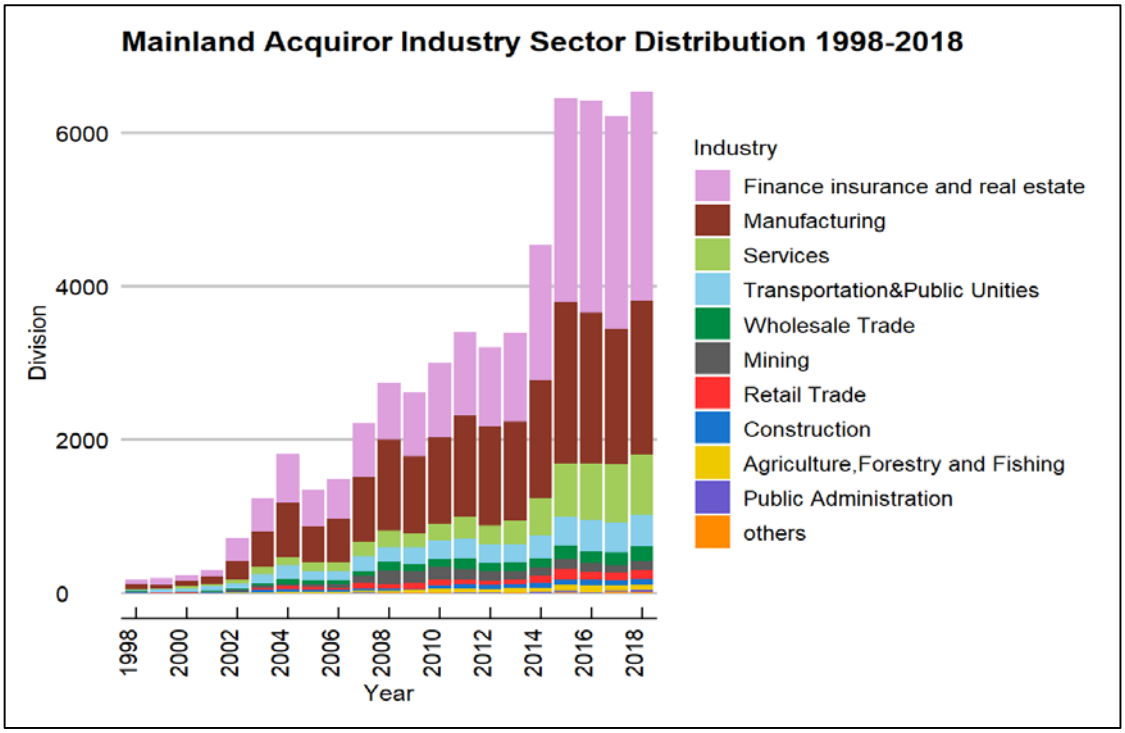


Figure 7.2 Industry distribution of the Mainland acquirer enterprises in 1998-2018

Source: SDC Platinum database

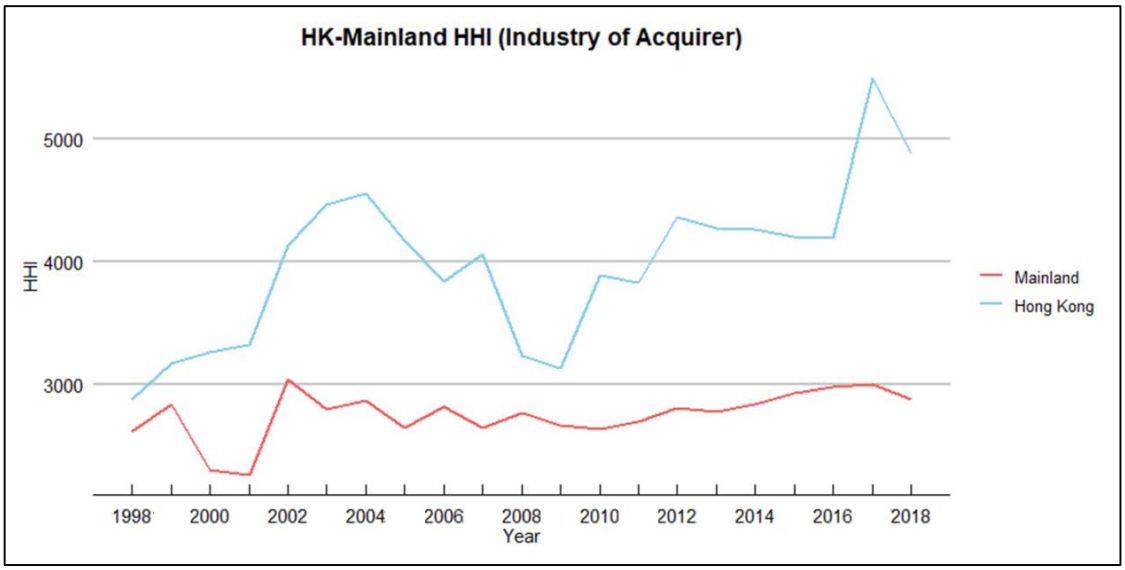


Figure 7.3 HHI of Hong Kong and the Mainland's acquirer industry distribution in 1998-2018

Source: SDC Platinum database

The industry distribution of target companies acquired by Hong Kong and the Mainland are also compared from the perspective of target companies. Figures 7.4 and 7.5 show the industry distribution in which the target companies are acquired in the global M&A business in 1998 and 2018, respectively. Figure 7.4 shows that from 1998 to 2018, the target enterprises acquired by Hong Kong enterprises account for the top three in the industry, namely, finance insurance and real estate, manufacturing and services. Overall, the industry distribution of acquired enterprises is relatively balanced. Figure 7.5 shows the target enterprises acquired by the Mainland firms during the same period. The three industries with top shares are manufacturing, financial insurance and real estate, and service, which the manufacturing has higher proportion. To more accurately reflect the balance of industry distribution of the acquired companies, we consider the HHI of the industry distribution with firms acquired in Hong Kong and the Mainland from 1998 to 2018. Figure 7.6 shows that compared with the Mainland, Hong Kong target company's industry HHI is significantly lower, reflecting the balanced distribution of target companies in Hong Kong.

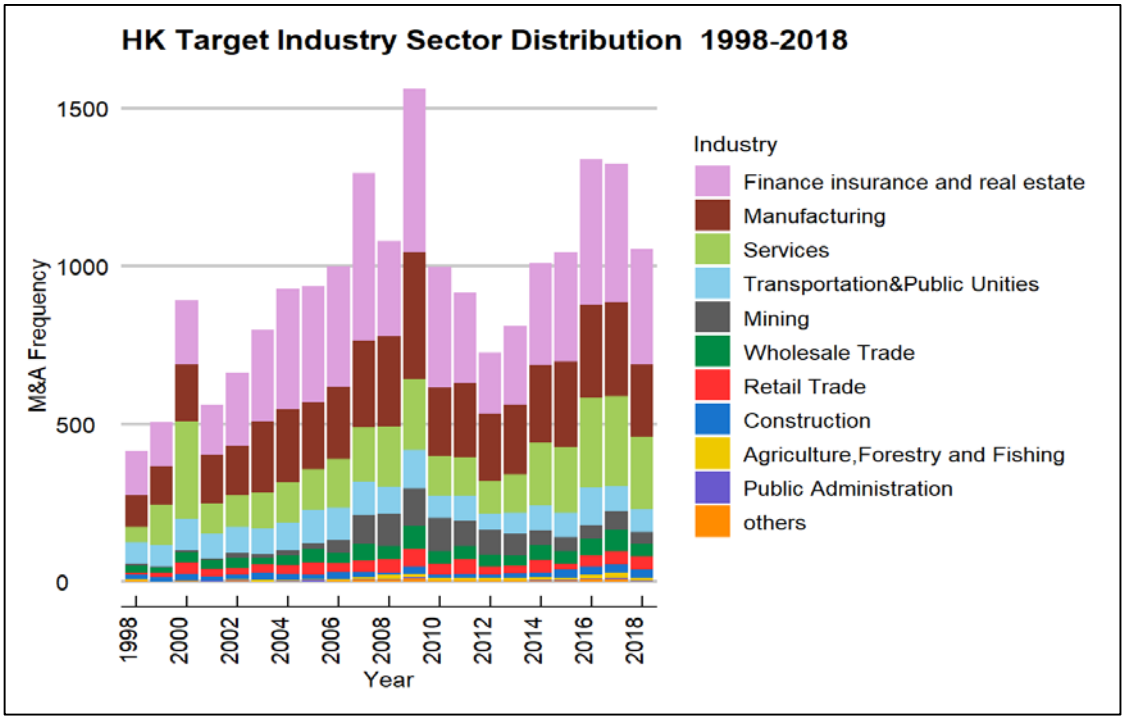


Figure 7.4 Distribution of target industries in Hong Kong 1998-2018

Source: SDC Platinum database

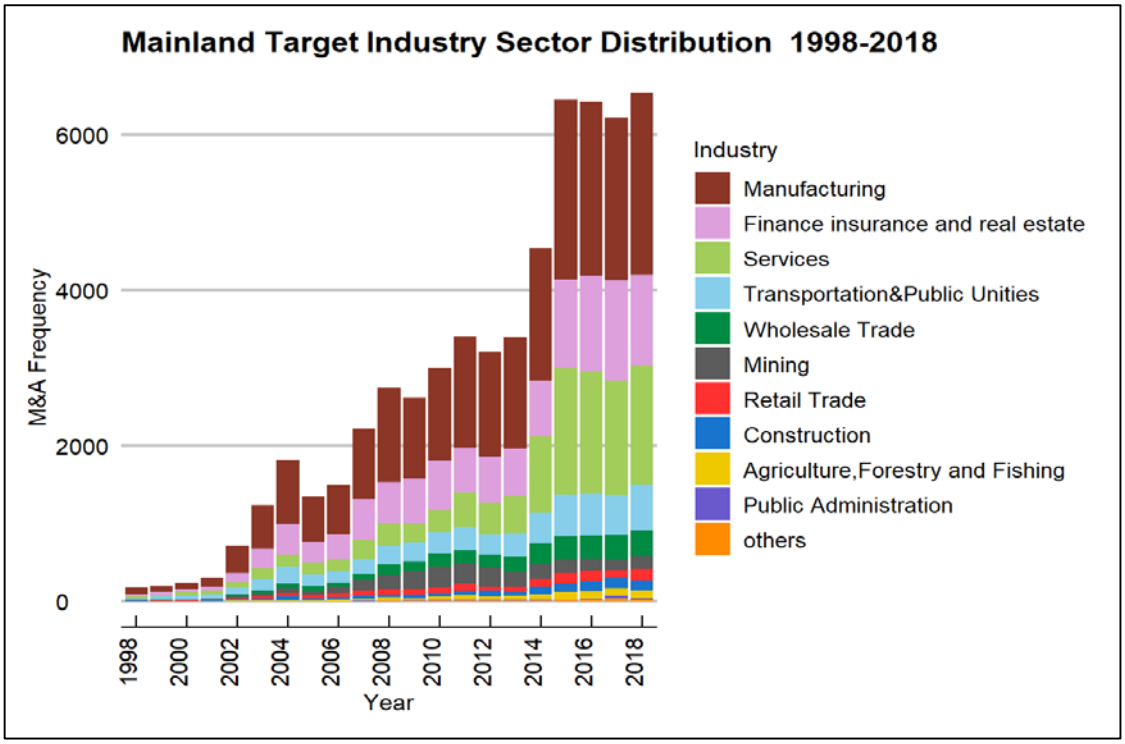


Figure 7.5 Industry distribution of target enterprises in the Mainland 1998-2018

Source: SDC Platinum database

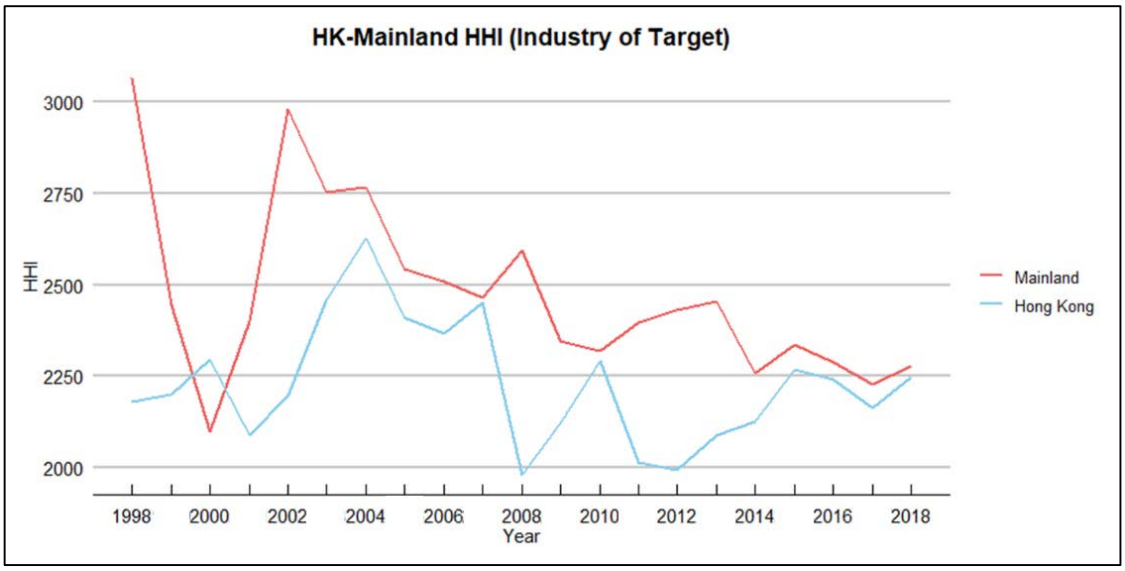


Figure 7.6 HHI on the distribution of target companies in Hong Kong and the Mainland, 1998-2018

Source: SDC Platinum database

7.4 Compared with the Mainland, Hong Kong Enterprises are in a Disadvantaged Position in the Acquisitions of High-Tech Enterprises

Science and technology are developing rapidly, and technological innovation has become a new driving force for global economic development. However, judging from the M&A activities of Hong Kong enterprises in the past 10 years, the proportion of acquired high-tech enterprises is lower than that by the Mainland. The scientific and technological strengths of Hong Kong and the Mainland high-tech enterprises from both parties are analysed. Figure 7.7 shows the proportion of high-tech acquirers in the past 10 years. From the graph, since the demarcation point of 2001, the proportion of high-tech companies acquired by Hong Kong enterprises are significantly lower than that by Mainland buyers. In recent years, the gap has considerably expanded. Figure 7.8 shows the proportion of high-tech companies in the targets acquired by Hong Kong and the Mainland in the past 10 years. Hong Kong must promote the development of high-tech industries, introduce high-tech talents, and enhance its technological innovation capabilities.

The above data show that Hong Kong global M&A businesses are mainly concentrated in the financial industry, reflecting that Hong Kong's level of financial technology does not develop in accordance with its status as an international financial centre. According to the *2018 Global Fintech Hub Report*, Hong Kong's comprehensive index ranks 10th in the world in terms of comprehensive financial technology industry, ecology and supervision, far lower than the ranking of New York (third) and London (fourth). In the rankings of all financial technology development cities in China, Hong Kong also lags behind Beijing, Shanghai and Hangzhou. These findings reflect the lack of technological application and innovation capabilities in Hong Kong's financial industry. Last year, the Hong Kong Securities and Futures Commission revised the Code of Corporate Acquisitions, Mergers and Share Buybacks (two codes) to align them with the new listing system of emerging and innovative industries. The Hong Kong Monetary Authority also issued the 'Acceptance of Virtual Banks' and proposed seven measures to accelerate the development of Hong Kong financial technology. However, to achieve significant short-term results, Hong Kong must speed up management reform of financial industry, guide and support local financial enterprises in terms of technological innovation, encourage local mergers with and acquisitions of overseas or Mainland financial technology enterprises, and hasten to enhance the scientific and technological strength of their financial services.

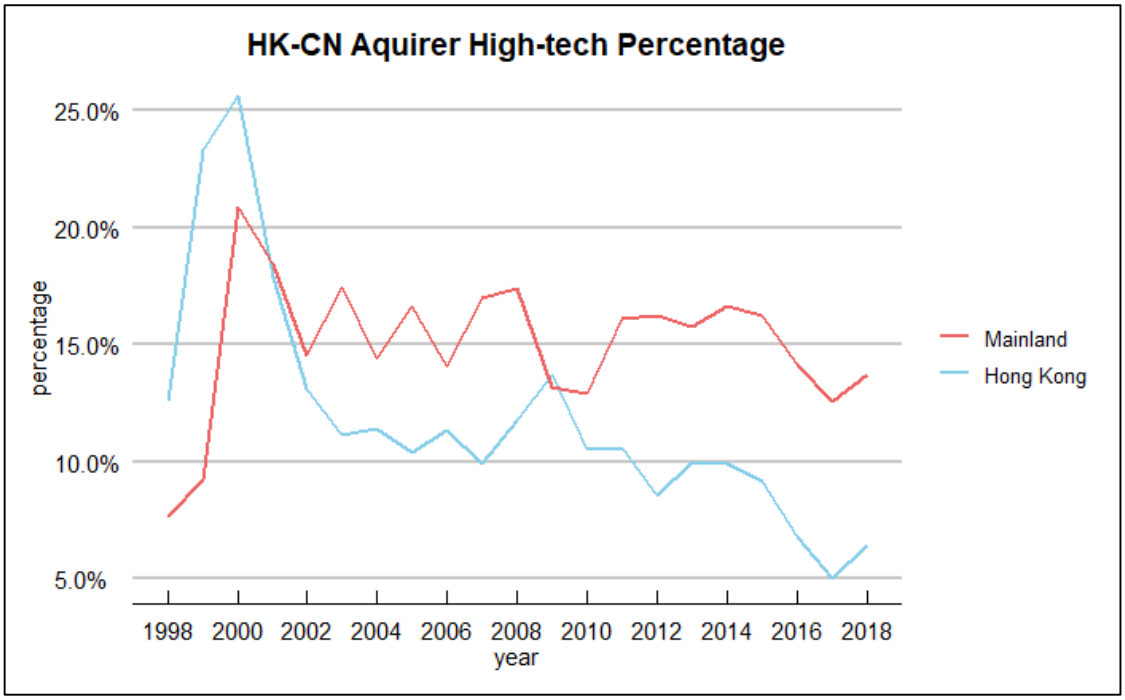


Figure 7.7 Proportion of Hong Kong and Mainland acquirer companies belonging to high-tech industries in 1998-2018

Source: SDC Platinum database

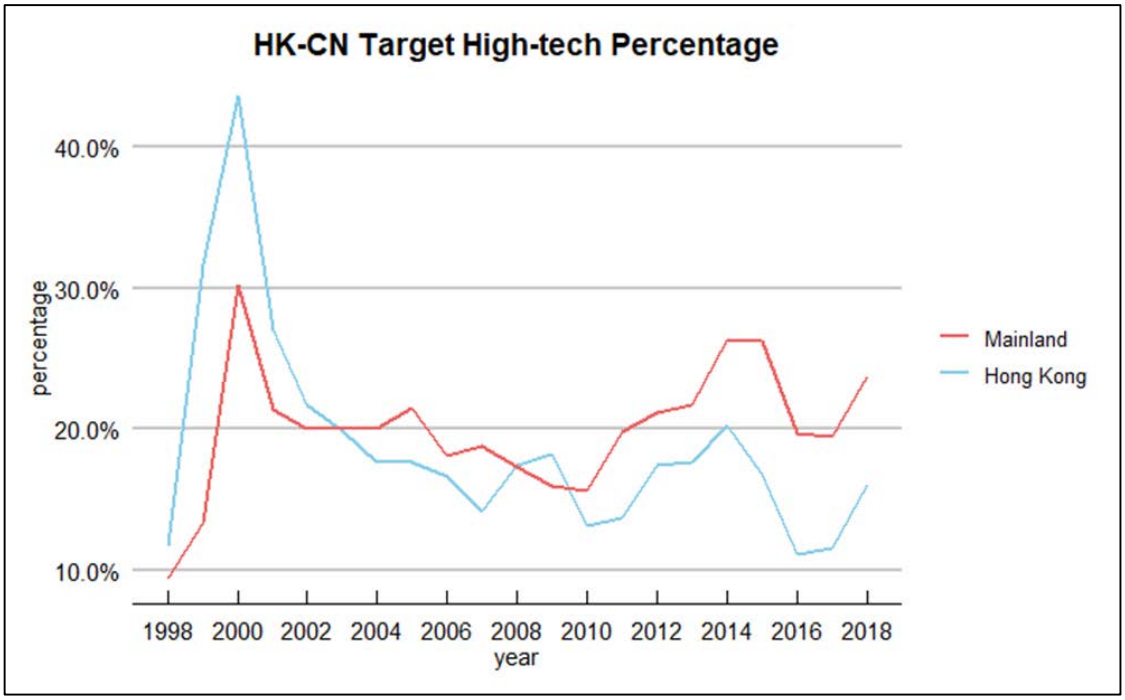


Figure 7.8 Target companies acquired by Hong Kong and the Mainland in high-tech industry in 1998-2018

Source: SDC Platinum database

7.5 Compared with the Mainland, Hong Kong Companies are more Internationalised in M&A Activities

The unique advantage of Hong Kong is its modern and free economy. Hong Kong ranks third in the world and first in Asia in terms of business environment, being a worldwide recognised international financial centre. This internalisation facilitates Hong Kong companies to carry out global M&A. Figure 7.9 below shows the global distribution of target companies acquired in Hong Kong from 1998 to 2018. The graph shows that the target overseas companies for acquisition occupies a large proportion of the total number of acquisitions. Figure 7.10 shows the global distribution of target companies acquired by Mainland buyers during the same period. The two figures show a significant difference between the distribution of its target acquisition companies, of which Chinese buyers are mainly local enterprises in the Mainland. Although overseas companies in the Mainland has increased in recent years, their number is significantly lower than that of Hong Kong companies, indicating the latter's wider geographical distribution of targets.

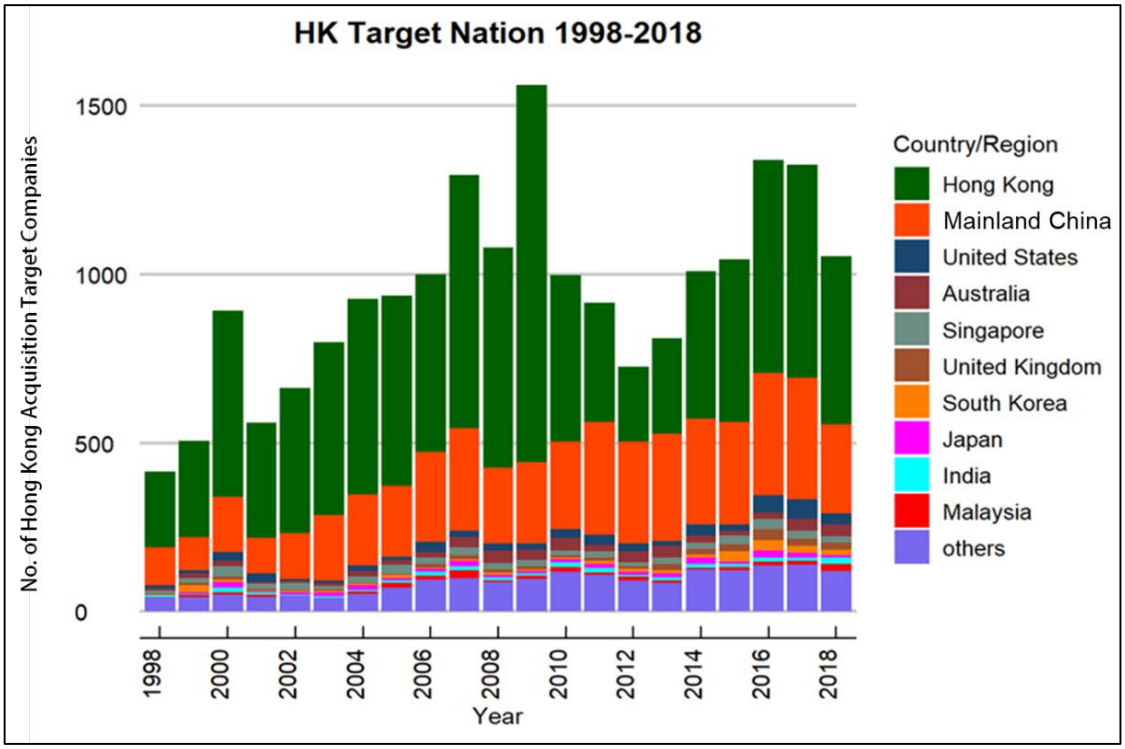


Figure 7.9 Global distribution of Hong Kong acquisition target companies in 1998-2018

Source: SDC Platinum database

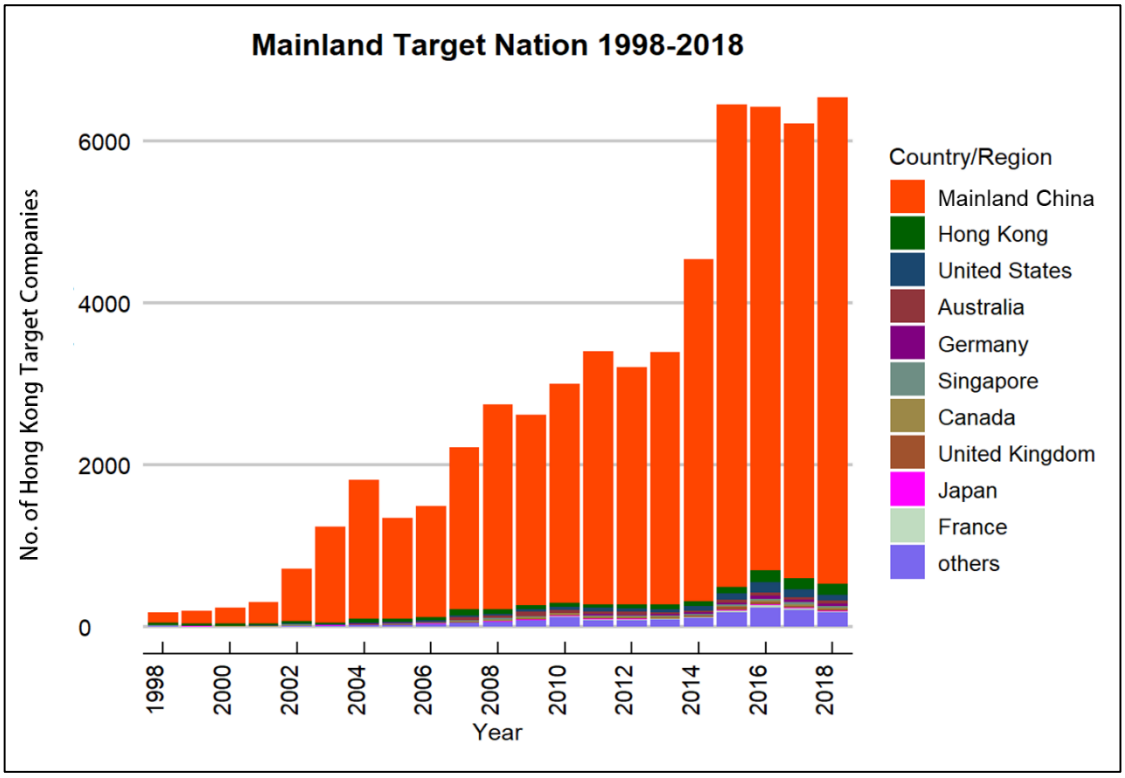


Figure 7.10 Distribution of Hong Kong target companies in the world in 1998-2018

Source: SDC Platinum database

To more accurately reflect the global distribution of target companies acquired by Hong Kong and the Mainland buyers between 1998 and 2018, we use the Herfindahl Index (HHI) below to represent the equilibrium of target companies in various countries. Figure 7.11 shows that compared with that of the Mainland, the distribution of HHI in the target countries of Hong Kong is significantly lower, reflecting a more balanced distribution of Hong Kong target companies in the global region. Figure 7.12 reflects the total number of countries covered by the target companies acquired in Hong Kong and the Mainland. Notably, compared with the Mainland, before 2008, the distribution of countries of target companies acquired by Hong Kong buyers is significantly higher than that of Mainland enterprises, and was surpassed by the latter only after 2008.

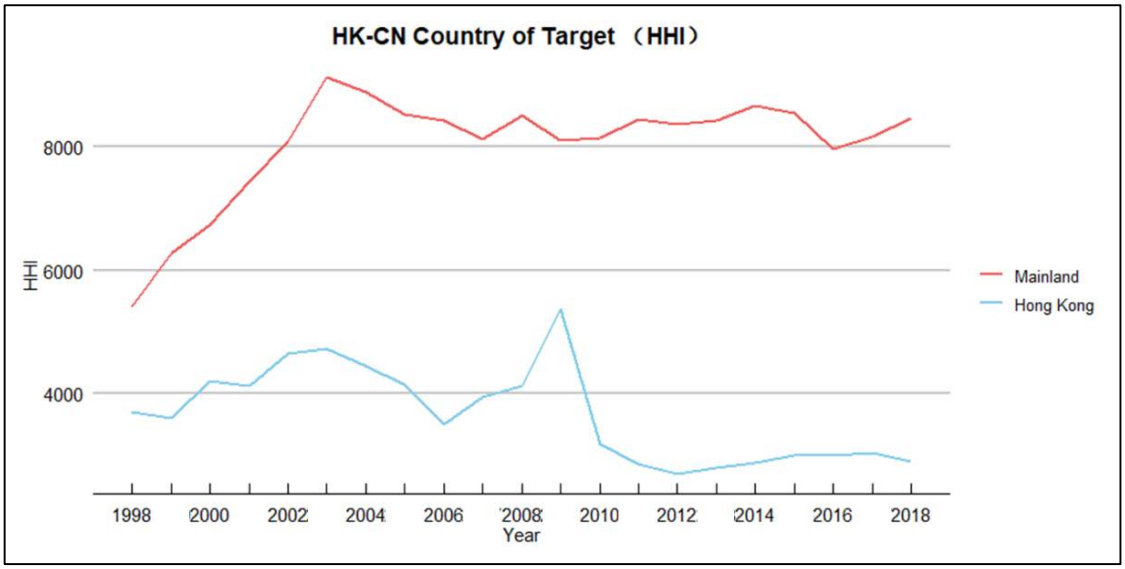


Figure 7.11 HHI for the distribution of target countries in Hong Kong and the Mainland in 1998-2018

Source: SDC Platinum database

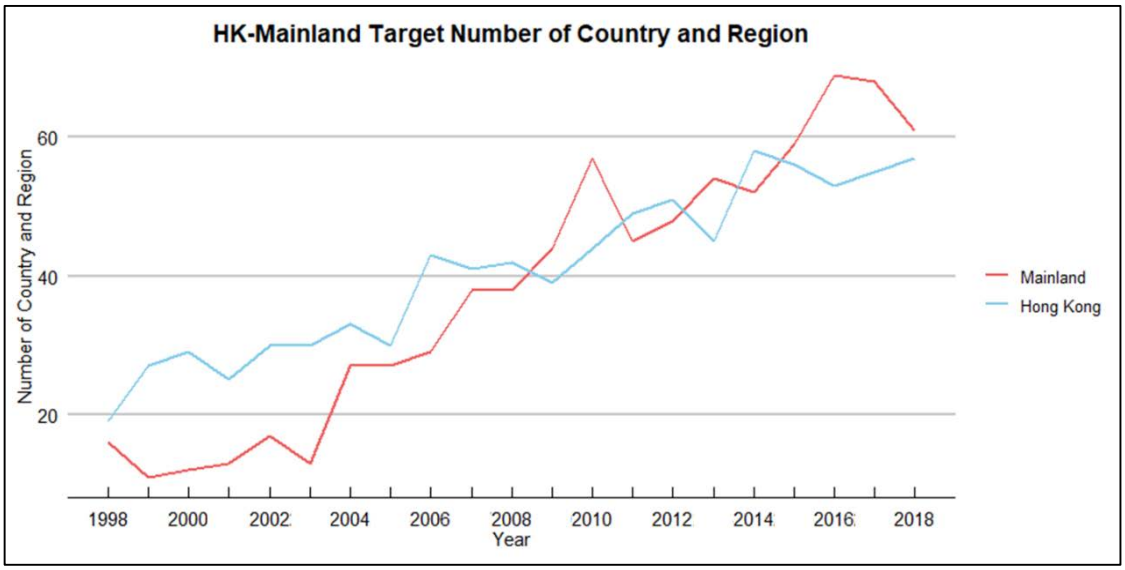


Figure 7.12 Total number of target countries in Hong Kong and the Mainland worldwide in 1998-2018

Source: SDC Platinum database

Subsequently, the specific locations of the target companies are examined. Figure 7.13 and Figure 7.14 below show the top 10 overseas countries where target enterprises are acquired from 1998 to 2018. Hong Kong enterprises have overseas acquisitions mainly from the US, Australia, Singapore, the United Kingdom (UK), South Korea, Japan, India, Malaysia, Taiwan and Canada. Overseas companies acquired by Mainland

companies are mainly located in the US, Australia, Germany, Singapore, Canada, the UK, Japan, France, Italy and South Korea. Hong Kong target companies are mainly distributed in the Commonwealth countries. Among the top 10 countries where the Chinese-acquired firms are Germany, France and Italy. Germany ranks third with their manufacturing industry in a leading position in the world.

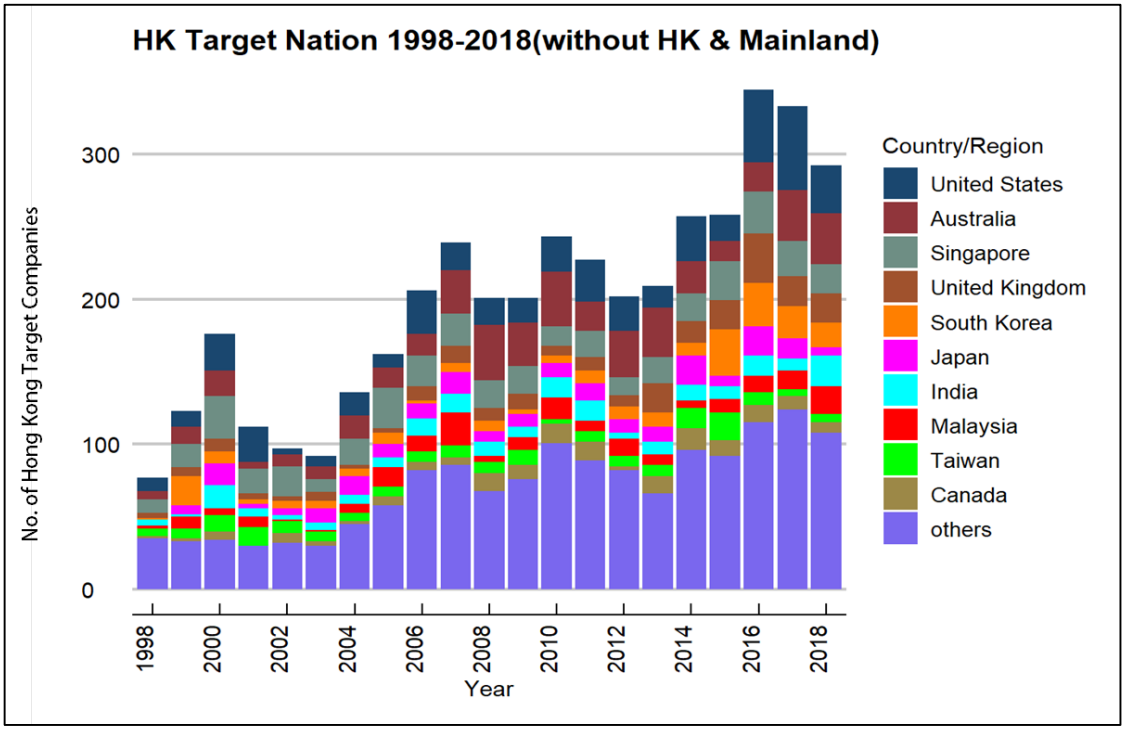


Figure 7.13 Distribution of Hong Kong target companies in overseas countries and regions in 1998-2018

Source: SDC Platinum database

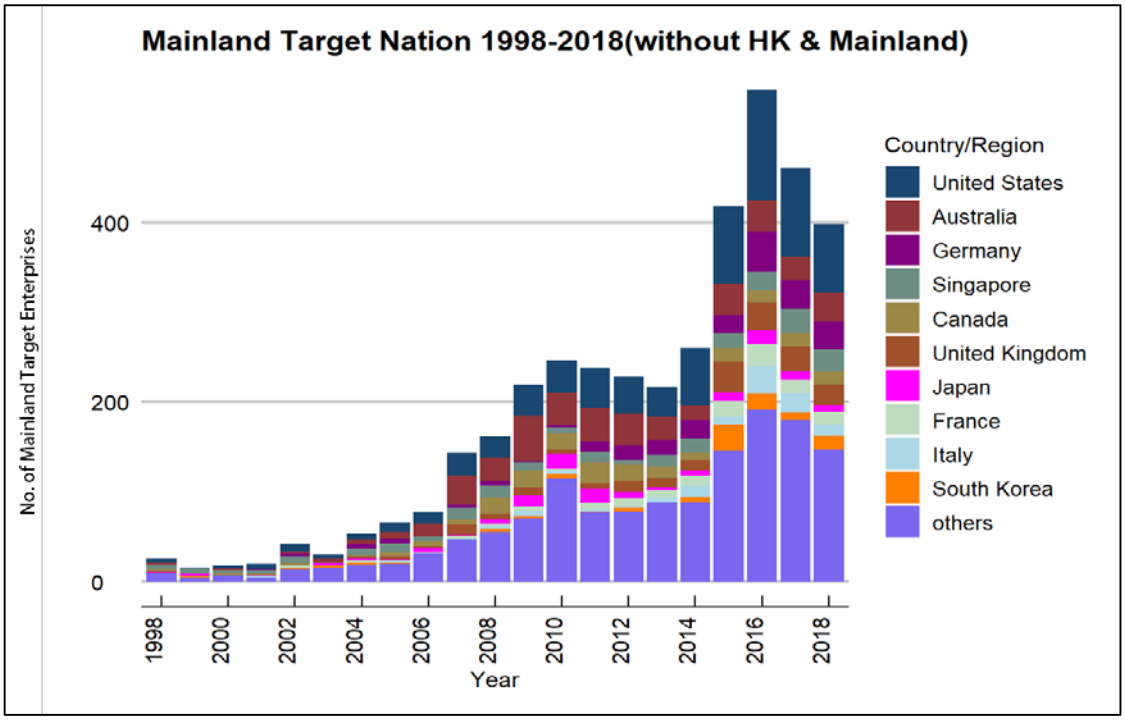


Figure 7.14 Distribution of Mainland target enterprises in overseas countries and regions in 1998-2018

Source: SDC Platinum database

7.6 Hong Kong Companies are More Professional and Efficient in Completing Global M&A than Mainland Companies

In addition, the completion of M&A activities can reflect the degree of specialisation and strength of the acquiring company. Figure 7.15 shows the completion of acquisitions by Hong Kong and Mainland buyers in 1998-2018. The chart shows that before 2004, the Mainland has a small total number of M&A projects, but the completion rate was higher than that in Hong Kong. In 2009, due to the impact of the global M&A wave before the end of the financial crisis, the total number of M&A activities in Hong Kong was higher than any other year and turned to be an outlier. In addition, from 2004 to 2018 (except 2009), the completion of M&A activities in Hong Kong was higher than that in Mainland China. Compared with acquirers in the Mainland, those in HK are more efficient in completing acquisitions, showing a higher standard of business professionalism. Figure 7.16 shows the proportion of M&A projects in HK and the Mainland Chinese enterprises as acquirers from the acquisition year to the end of March 2019, respectively. However, the completion degree of HK and the Mainland significantly decreased in 2016-2018 due to the large number of projects still in progress. During the period of 2002-2018, the proportion of M&A for completion is lower in HK than that in Mainland China.

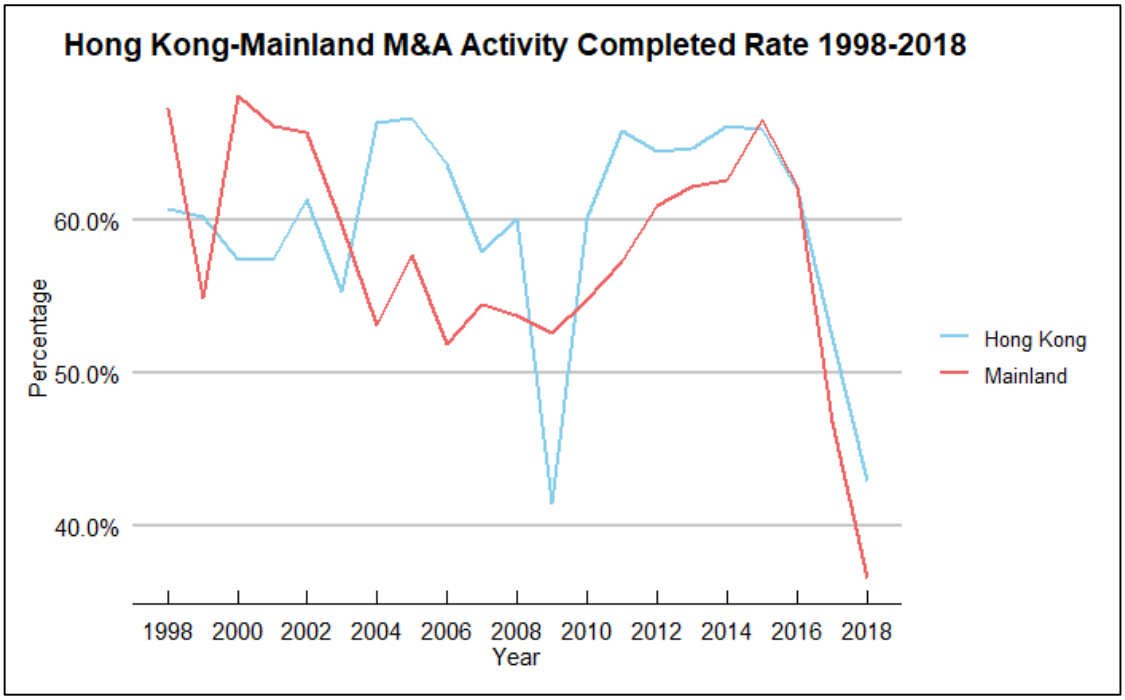


Figure 7.15 Proportion of M&A projects in Hong Kong and Mainland enterprises in 1998-2018

Source: SDC Platinum database

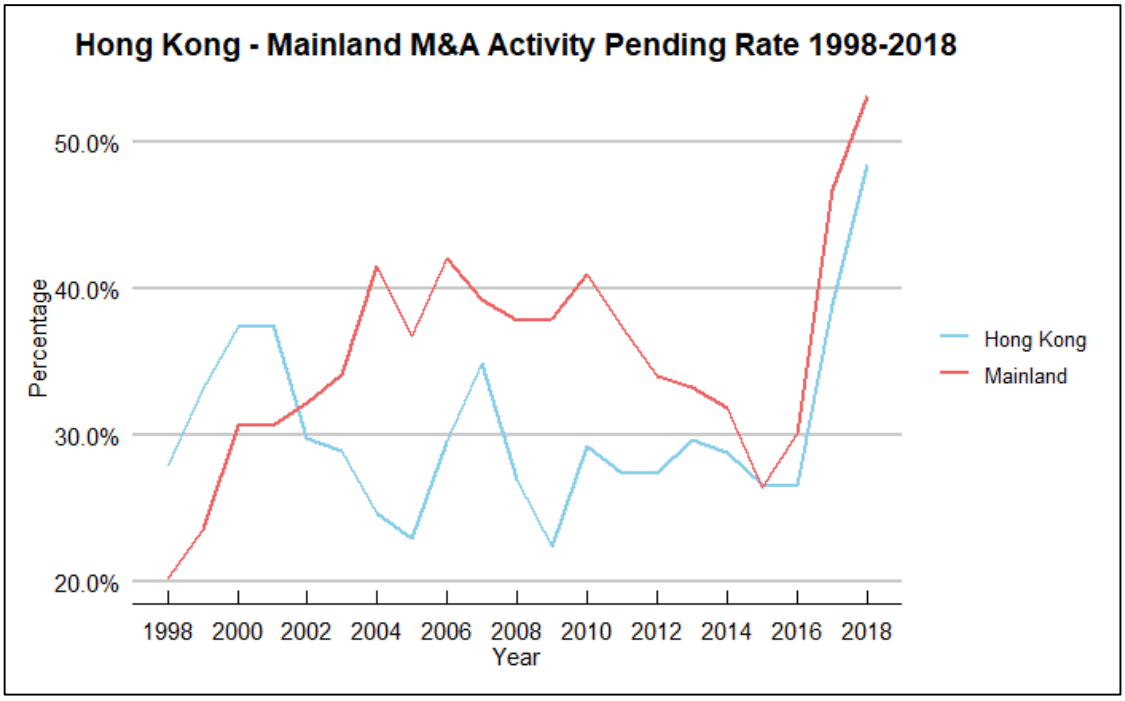


Figure 7.16 Proportion of M&A projects in Hong Kong and Mainland enterprises to be completed in 1998-2018

Source: SDC Platinum database

7.7 Hong Kong Overseas Subsidiaries Account for a Higher Proportion than Mainland Enterprises

The overseas holding ability also demonstrates Hong Kong's dominant position in the international market. In 2016, Hong Kong overseas subsidiaries accounted for 40%, while Mainland overseas subsidiaries accounted for only approximately 2%. Domestic subsidiaries of Mainland enterprises accounted for 96%. Figure 7.17 and Figure 7.18 show the location distribution of Hong Kong and Mainland subsidiaries.

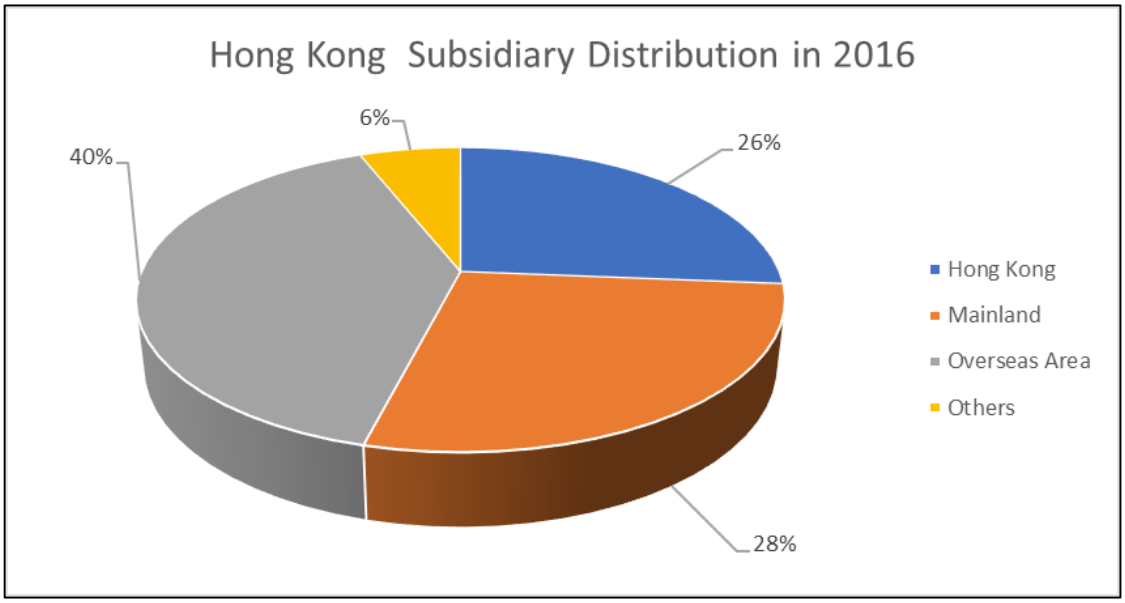


Figure 7.17 Global distribution of Hong Kong enterprise subsidiaries in 2016

Source: Orbis Database

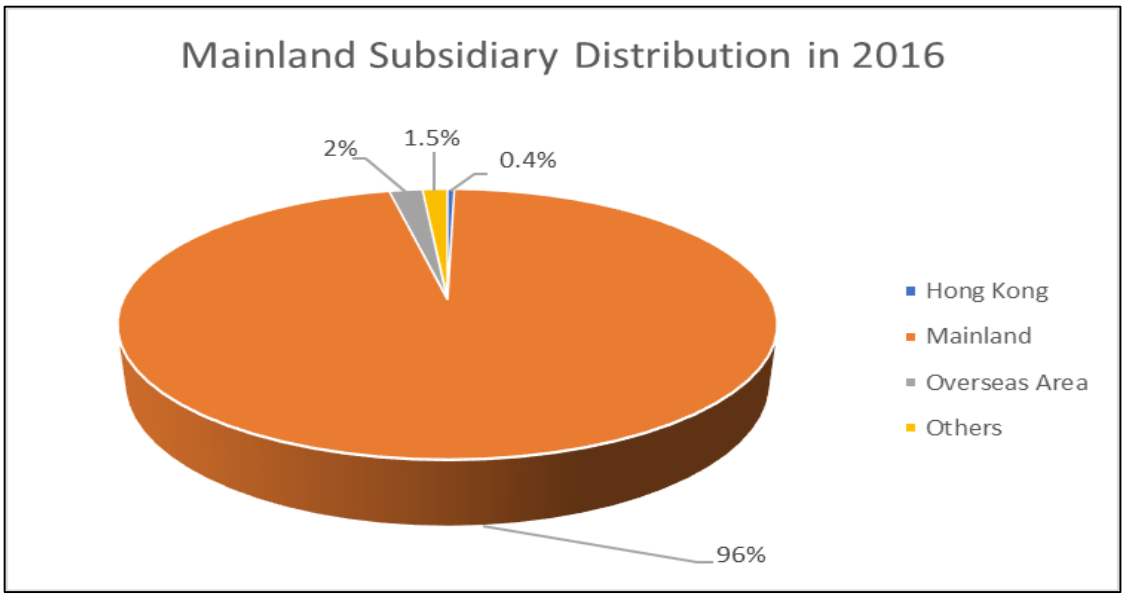


Figure 7.18 Global distribution of Mainland enterprise subsidiaries in 2016

Source: Orbis Database

7.8 Compared with Mainland China, Hong Kong has a Higher Proportion of M&A along the Belt and Road

The strategy of the Belt and Road Initiative brings new opportunities for global economic development. The acquisition of enterprises in countries and regions along the Belt and Road can be more convenient by connecting Hong Kong, the Mainland and overseas capital markets. Figure 7.19 and Figure 7.20 show the countries and regions of the target enterprises acquired by Hong Kong and the Mainland in 1998-2018. Hong Kong enterprises mainly acquired local enterprises, followed by Chinese enterprises. The top 10 are mainly from Commonwealth countries, and Southeast Asian countries and regions, with Germany being the only European country. Similarly, Mainland enterprises mainly acquired Mainland enterprises, and the rest are from only a few other countries and regions. By comparison, among the target companies acquired by Hong Kong, except those from local areas and in the Mainland, the proportion of companies located in countries and regions along the Belt and Road is high.

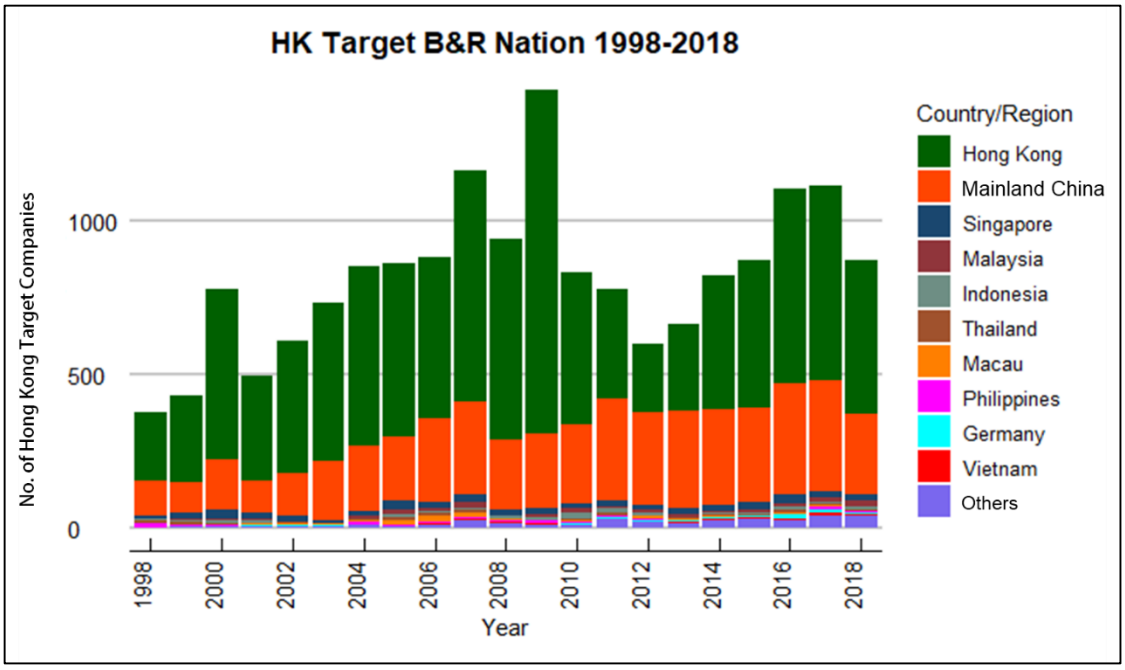


Figure 7.19 Distribution of Hong Kong target companies in the countries along the Belt and Road in 1998-2018

Source: Orbis Database

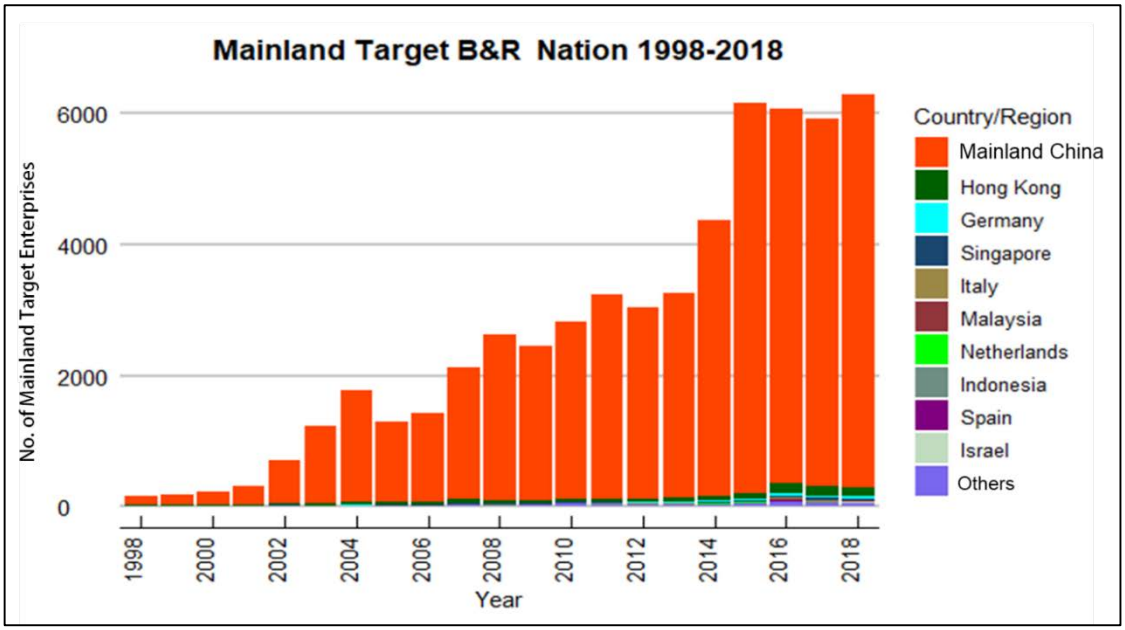


Figure 7.20 Distribution of Mainland Target Enterprises in the countries and regions along the Belt and Road in 1998-2018

Source: Orbis Database

The internalisation of Hong Kong and Mainland enterprises in the countries and regions along the Belt and Road are further examined by excluding local enterprises in Hong Kong and Mainland from all acquired enterprises during 1998-2018. Figure 7.21 and Figure 7.22 show this comparison. In general, the target companies acquired by Hong Kong enterprises in areas along the Belt and Road are mainly in Singapore, Malaysia, Indonesia, Thailand, Macao, the Philippines, Germany, Vietnam, the Netherlands and Italy. By comparison, target companies acquired by Mainland enterprises are mainly in Germany, Singapore, Italy, Malaysia, the Netherlands, Indonesia, Spain, Israel, Thailand and Switzerland. An easy observation is that the target companies acquired by Hong Kong are mostly Southeast Asian countries, while those of Mainland enterprises are European companies.

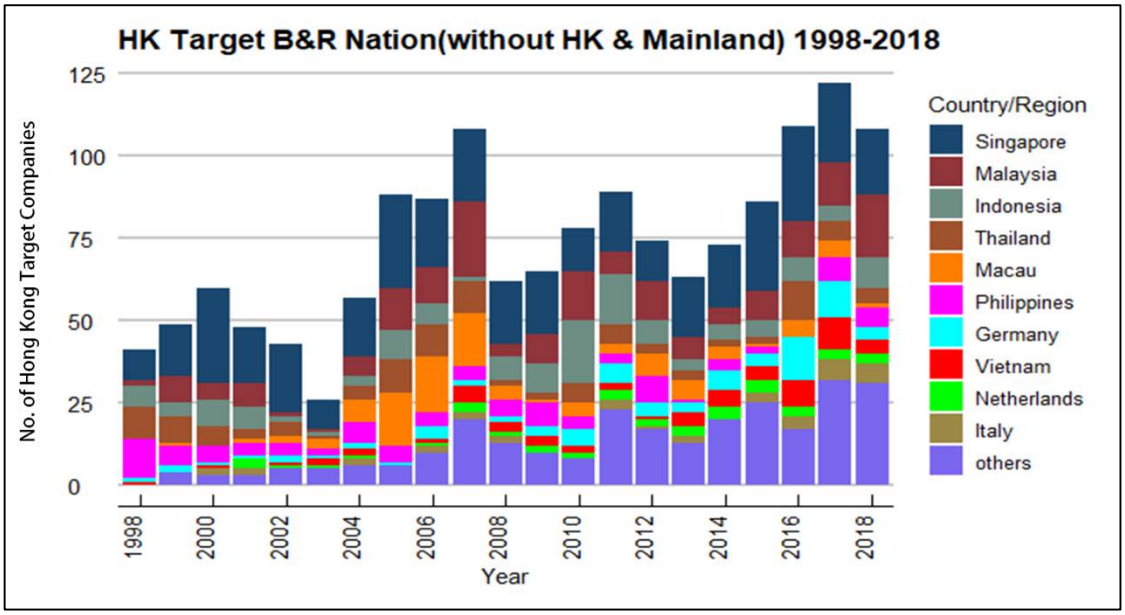


Figure 7.21 Distribution of Hong Kong target companies along the Belt and Road (excluding Hong Kong and the Mainland) in 1998-2018

Source: Orbis Database

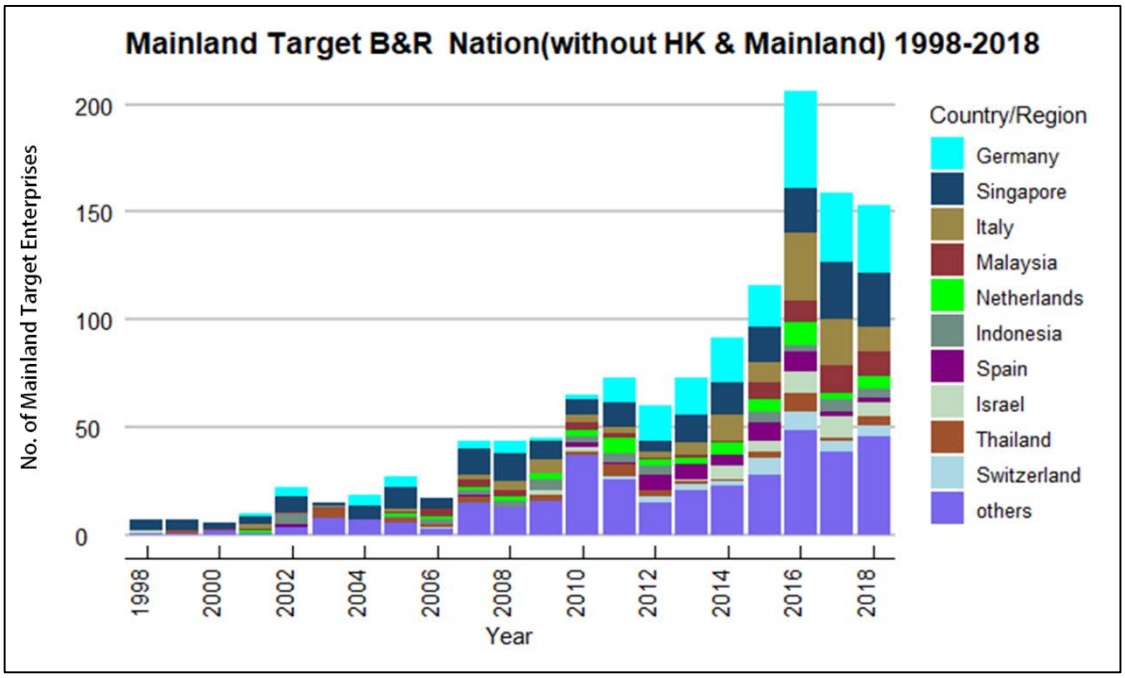


Figure 7.22 Distribution of Mainland Target Enterprises in the countries along the Belt and Road (excluding Hong Kong and the Mainland) in 1998-2018

Source: Orbis Database

Figure 7.23 below shows the total number of target companies in areas along the Belt and Road acquired by Hong Kong and Mainland enterprises from 1998 to 2018. Before 2008, the geographical distribution of target companies in the countries along the Belt and Road acquired by Hong Kong is generally wider than that of the Mainland enterprises. However, after 2008, especially in recent years, the Mainland enterprises gradually acquired more enterprises based near the Belt and Road than Hong Kong.

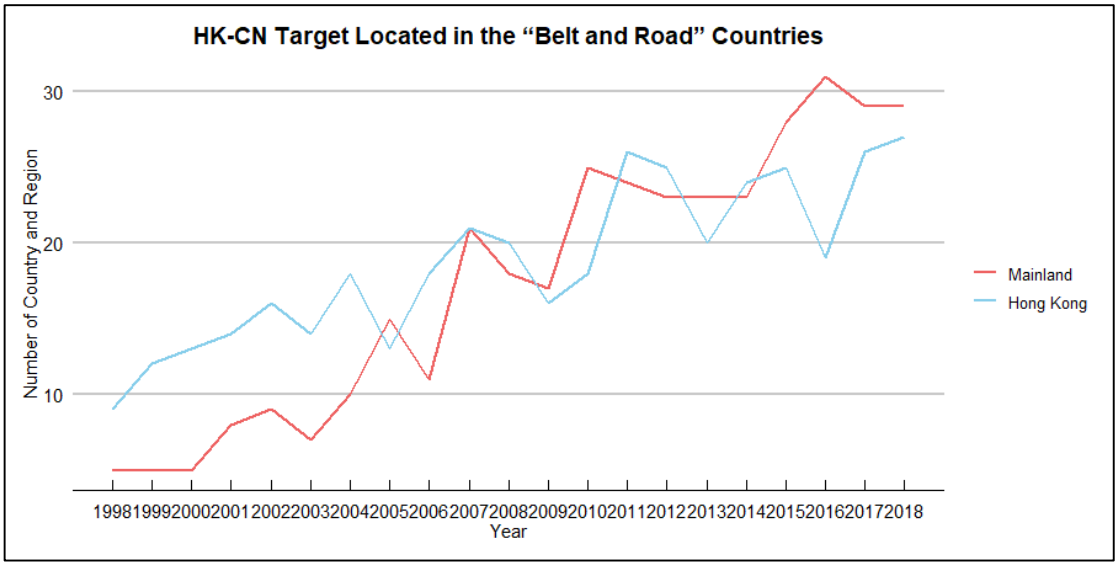


Figure 7.23 Total number of target enterprises in Hong Kong and the Mainland along the Belt and Road in 1998-2018

Source: Orbis Database

7.9 Comparison on Location Distribution of Overseas Subsidiaries between Hong Kong and Mainland Companies

Further analysis of the overseas subsidiary distribution clearly shows that most Hong Kong overseas subsidiaries are in the Commonwealth countries. The top four countries are the British Virgin Islands, the Cayman Islands, the UK and Australia, which show a clear preference for locations. However, the top four countries of subsidiaries of the Mainland enterprises are Russia, Germany, the US and Australia, showing no apparent location preference. Calculations of the HHI of the overseas subsidiaries of the Mainland enterprises yields as 647.18, which is slightly higher than the 605.19 of Hong Kong. Compared with the Mainland, HK's overseas subsidiaries are more evenly distributed rather than concentrated. In terms of the total number of countries, the subsidiaries of Hong Kong enterprises in 2016 are located in 138

countries and regions, while those of the Mainland enterprises are located in 140 countries and regions. Figure 7.24 and Figure 7.25 show the distribution of the top 10 countries of subsidiaries of Hong Kong and Mainland China in 2016.

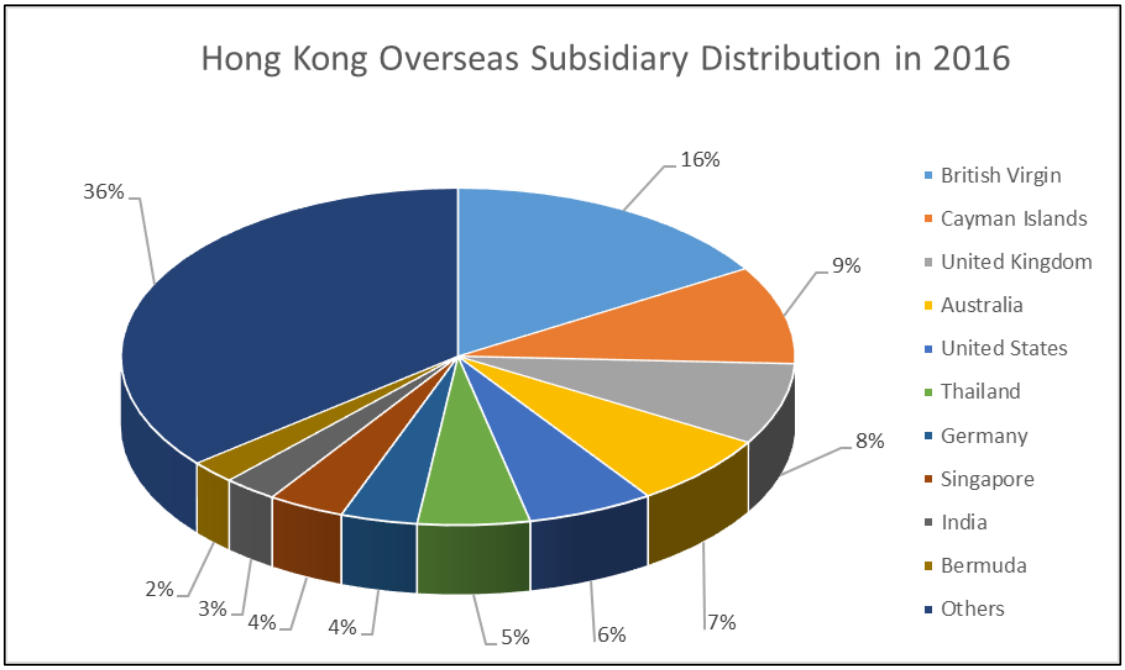


Figure 7.24 Distribution of Hong Kong overseas subsidiaries in 2016

Source: Orbis Database

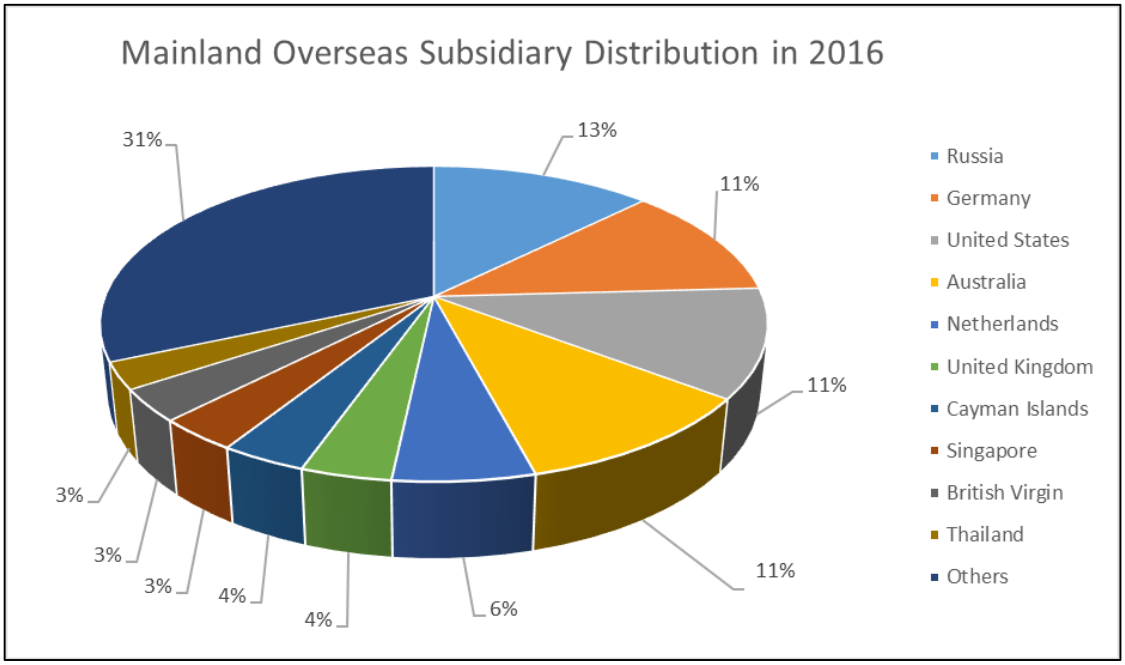


Figure 7.25 Distribution of the Mainland overseas subsidiaries in 2016

Source: Orbis Database

7.10 Comparison on Location Distribution of Overseas Subsidiaries between Hong Kong and Mainland Companies along the Belt and Road

In 2016, corporate subsidiaries along the Belt and Road were spread across 52 countries for Hong Kong enterprises and across 55 countries for the Mainland enterprises. In the countries along the Belt and Road, Figure 7.26 shows that Hong Kong subsidiaries account for less than 50% of their overseas total while Figure 7.27 shows that Mainland subsidiaries exceeds 50% of their overseas total. However, since the proportion of overseas companies in Hong Kong enterprises is much higher than that of Mainland enterprises, the proportion of Hong Kong enterprises distributed in the countries along the Belt and Road is still higher than that of the Mainland. This result shows that Hong Kong subsidiaries have advantages over those in the Mainland, although further improvement can be achieved.

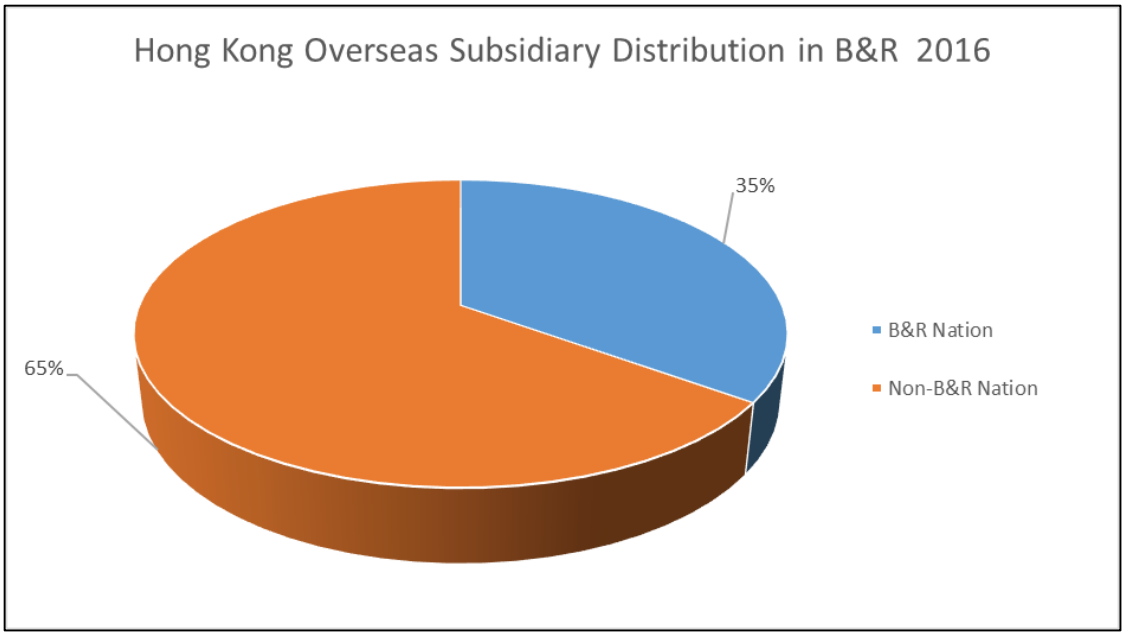


Figure 7.26 Proportion of Hong Kong overseas subsidiaries in countries and regions along the Belt and Road in 2016

Source: Orbis Database

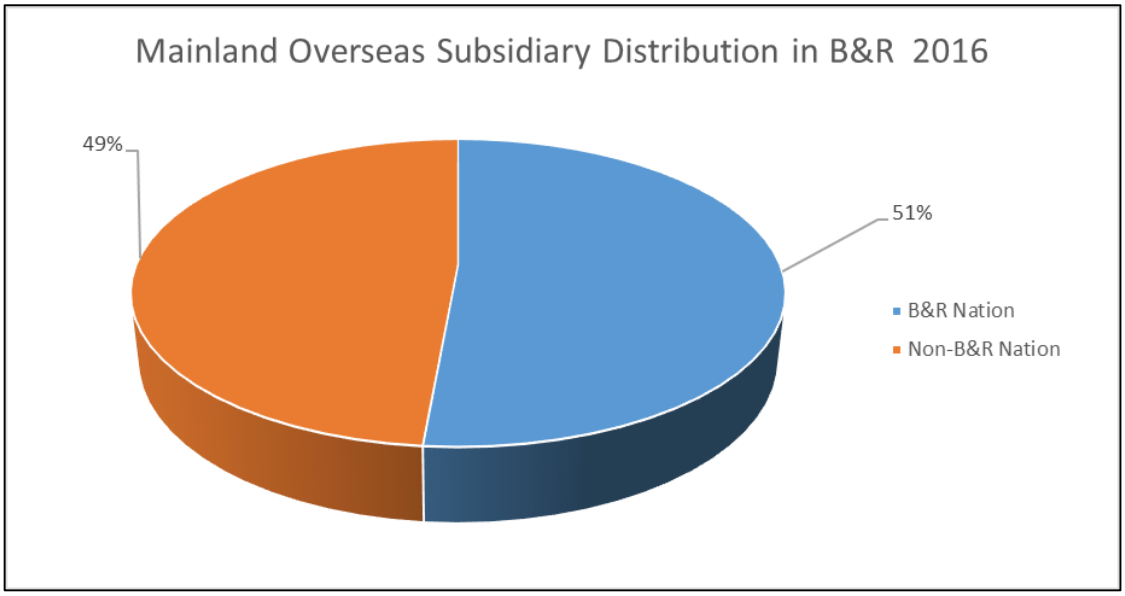


Figure 7.27 Proportion of Hong Kong overseas subsidiaries distributed in countries and regions along the Belt and Road in 2016

Source: Orbis Database

Specifically, in 2016, the top five countries and regions of the Belt and Road with the most extensive distribution of Hong Kong subsidiaries were Thailand, Germany, Singapore, New Zealand and Luxembourg, accounting for over 50%. The top three countries and regions of the Belt and Road with the largest number of Mainland subsidiaries are Russia, Germany and the Netherlands, also accounting for over 50%. Figure 7.28 and Figure 7.29 show the distribution of Hong Kong and the Mainland enterprises in the top 10 countries and regions of the Belt and Road in 2016. Compared with Mainland China, Hong Kong subsidiaries are more balanced in distribution along the Belt and Road, rather than concentrating on a few countries and regions.

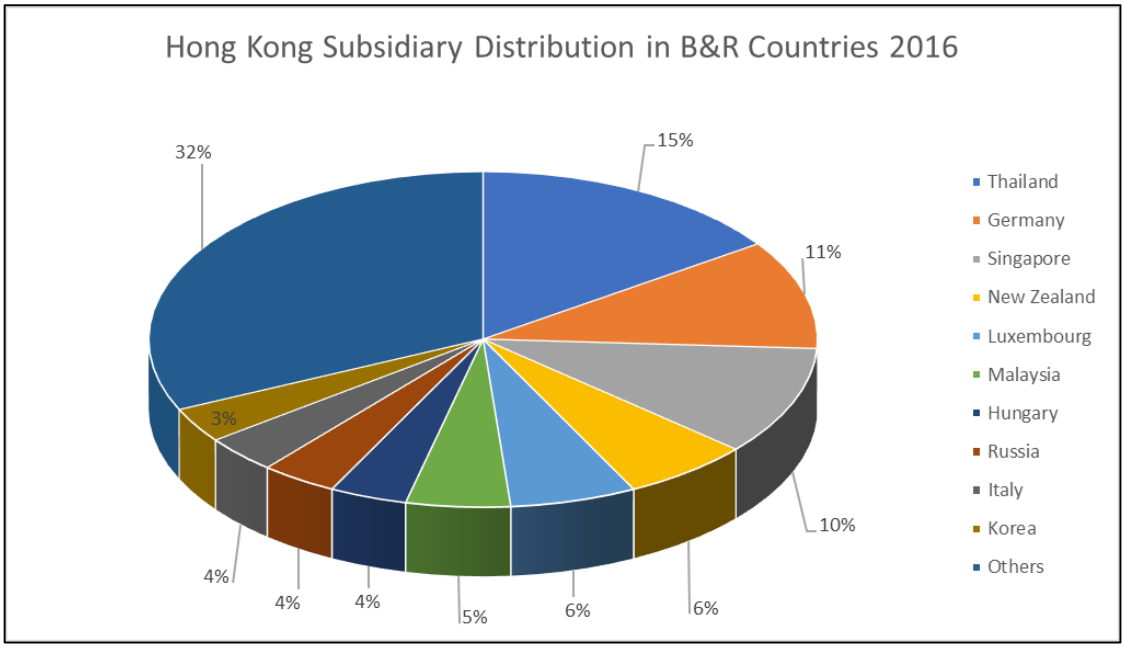


Figure 7.28 Hong Kong Subsidiaries in countries and regions along the Belt and Road in 2016

Source: Orbis Database

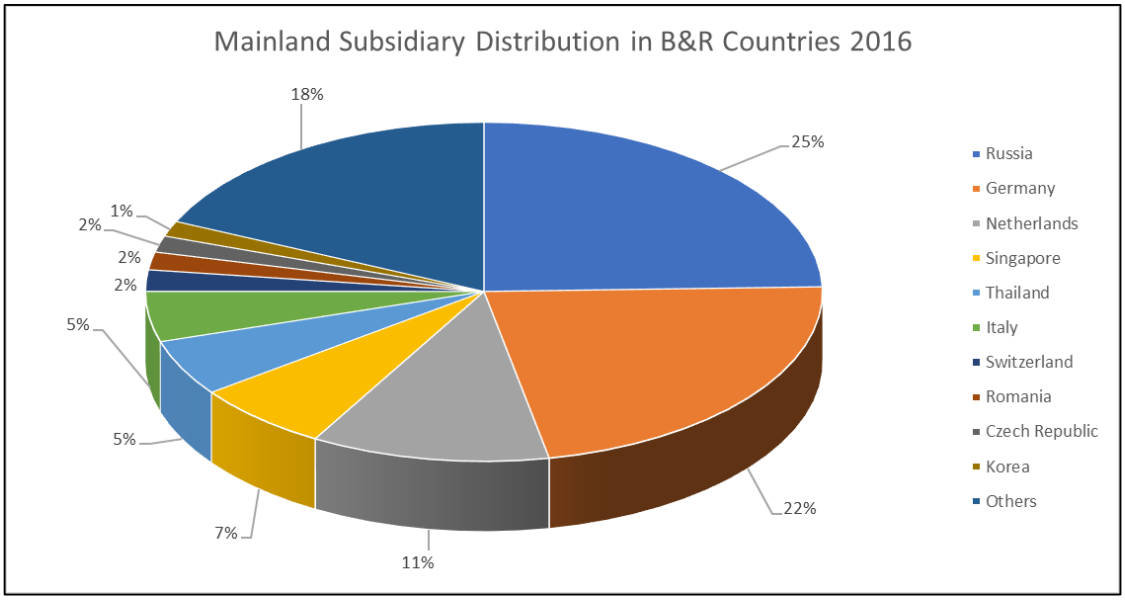


Figure 7.29 Mainland China's subsidiaries in 2016 in countries and regions along the Belt and Road in 2016

Source: Orbis Database

Chapter 8
Continuously Utilising Hong Kong to Provide Producer Services for Mainland Enterprises, Setting up 'Front Store' in the Greater Bay Area and Improving Access to Customer Groups

8.1 Introduction

For many years, the four pillar industries in the Hong Kong economy have been financial services, trading and logistics, professional and other producer services and tourism, which drive the development of other industries and create numerous jobs. These industries serve as the core engines of the Hong Kong economic prosperity. Among them, the financial services and the trading and logistics industries accounted for the largest shares of value addition and employment. From the dataset reported in Hong Kong Census and Statistics Department[11] in 2017, financial services and trading and logistics brought 18.9% and 21.5% on value-added to local gross domestic product (GDP), respectively, and accounted for 6.8% and 19.0% of the total employment, respectively.

In financial services, since the reform and opening up, Hong Kong has served as an international financial centre and as the Mainland's most important overseas capital cluster centre, continuously pumping valuable capital as an essential contribution into the development of the Chinese economy. Even though the current Mainland capital is increasingly high and its hardware has caught up and even exceeds those of Hong Kong, the latter's tangible and intangible advantages in open market, rule of law, internationally aligned supervision system, professional talents in relevant fields, and bilingual environment are prominent and unique. Under the construction of the Greater Bay Area (GBA)[12], Hong Kong is to continuously reinforce its role in its financial services industries, utilise its advantages in equity and bond financing, and attract and fulfil the financing needs of Mainland enterprises.

In the trading and logistics industry, the 'Front Shop, Back Factory'[13] model established between Guangdong province and Hong Kong in the early days of China's reform and opening up now needs an upgrade. With China's entry into the World Trade Organisation (WTO), numerous Mainland enterprises can directly connect with foreign customers without exporting through Hong Kong, and numerous Hong Kong enterprises moved to regions beyond Guangdong province. Thus, Hong Kong has to utilise its rich management and foreign customer communication experience;

continuously provide excellent local services; and set up ‘front shops’ closer to the Mainland market such as the Pearl River Delta (PRD) region, and closer to Mainland producers while providing high-end value-added services.

8.2 Research Methodology

Hong Kong has evolved into a center of producer service in the last forty years, triggered and reinforced by the relocation of Hong Kong’s manufacturing into the Mainland and multinationals moving into Mainland China via Hong Kong. In other words, the demand for Hong Kong’s producer service was mostly from Hong Kong-quartered local and international firms. There is, however, increasing demand for producer services from China’s domestic firms as they experience dramatic growth domestically and also their increasing internationalization drive. In this chapter, we first examine the development and current status of the financial services provided to the Mainland producers, equity and bond financing in particular, using datasets from HKEX. Then we will study the trade and logistics services provided to the Mainland producers. Using China firm-level custom data, we study the allocation and characteristics of the exporters of the Mainland using Hong Kong trade-related services and explore how Hong Kong producer service providers can further enlarge their customer base in the Mainland.

8.3 Equity Financing

8.3.1 Mainland companies account for a large proportion of listed companies in Hong Kong and are increasing every year

Since 1997, the proportion of stocks of Mainland enterprises listed in the Hong Kong stock market has proliferated substantially on their turnovers, market capitalisation and number of issuers. As of 2017, Mainland enterprises had an annual total turnover accounting for over 70%; market capitalisation accounting for over 65%, and number of issuers accounting for over 50% of the total of all stocks listed in Hong Kong (Figure 8.1).

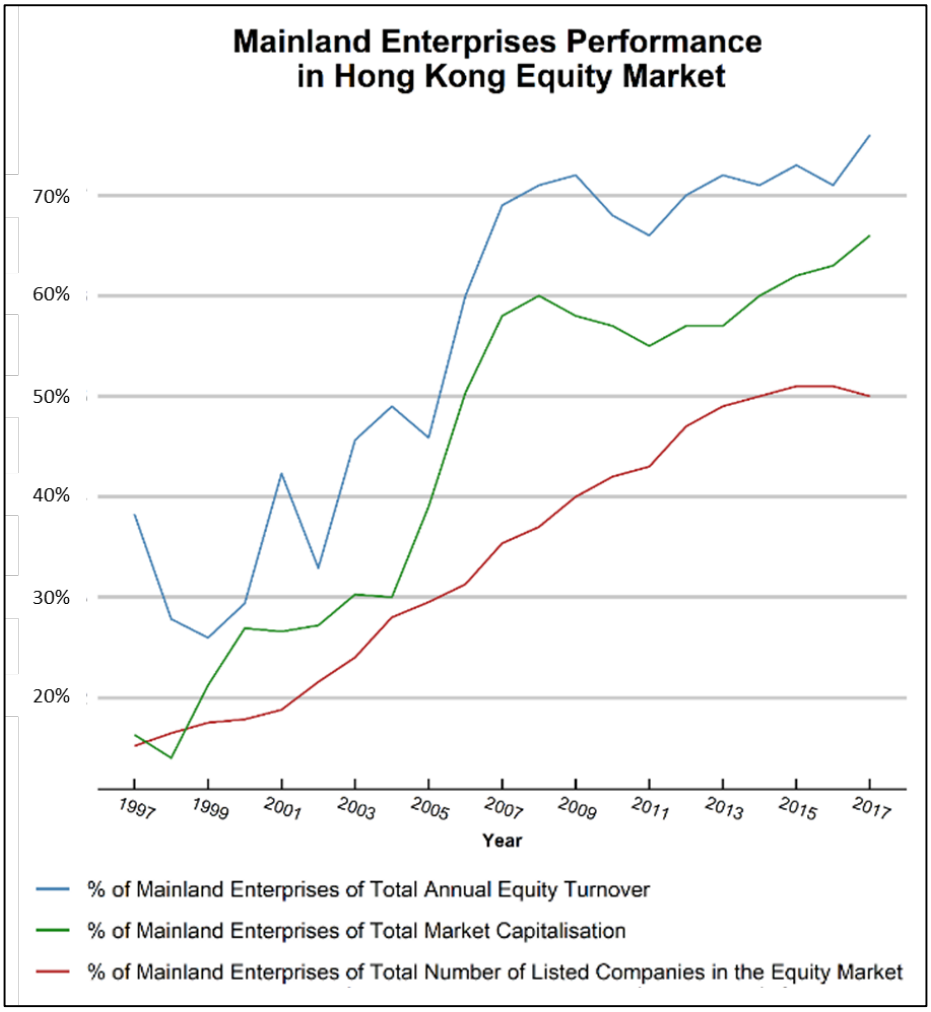


Figure 8.1 1997-2017 Turnover of Mainland companies in the Hong Kong stock market, total market value and issuer ratio

Source: Hong Kong Stock Exchange

8.3.2 Private Enterprises Become the Main Component Among Mainland Companies Listed in Hong Kong

Hong Kong-listed Mainland enterprises are composed of three types: H-share, Red Chip, and Mainland private enterprises. H-share companies are registered in the Mainland and owned by government agencies or Mainland individuals. Red Chip companies are registered outside the Chinese Mainland and controlled by a government agency in the Mainland. Private enterprises in the Mainland refer to those incorporated outside the Mainland but controlled by individuals in the Mainland (Figure 8.2).

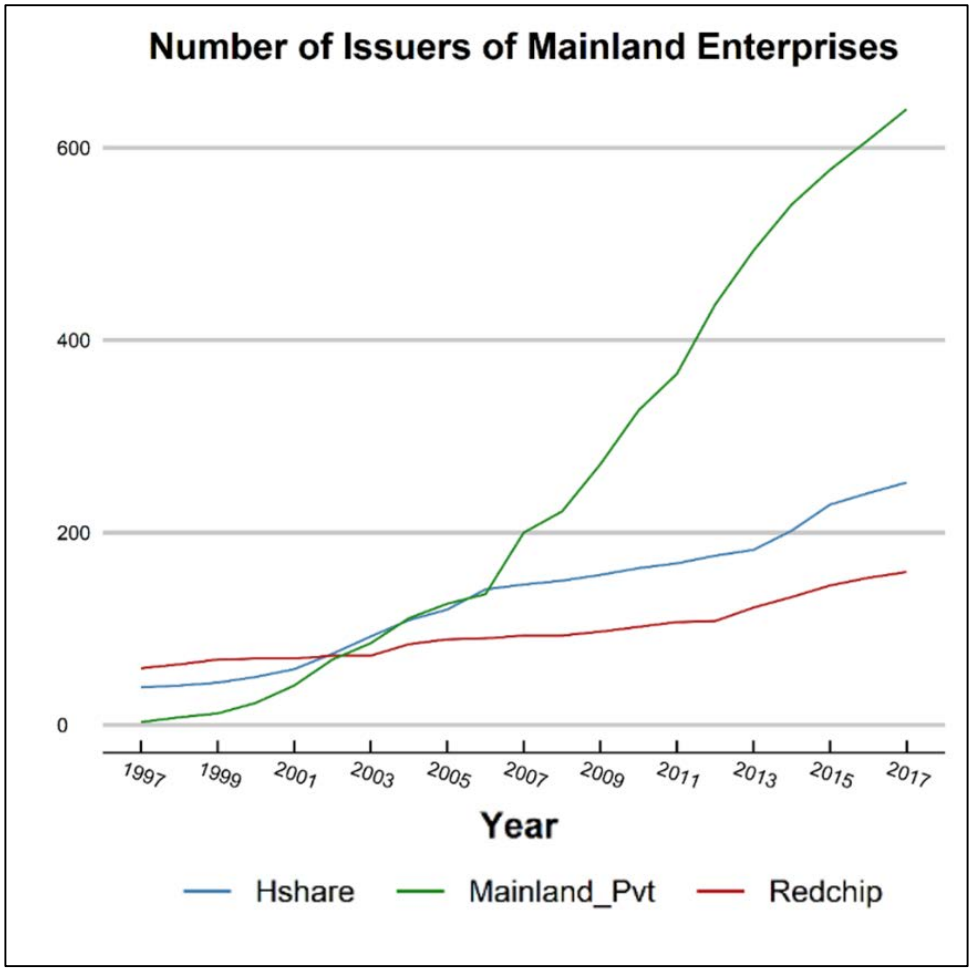


Figure 8.2 Hong Kong Main Board and Growth Enterprise Market (GEM), Issuers of Mainland Enterprises, 1997-2017

Source: Hong Kong Stock Exchange

In 1997, Red Chip companies accounted for the highest number of Mainland enterprises, and private enterprises were limited. Since then, all three types of Mainland enterprises have substantially increased. After 2002, the number of issuers of private enterprises and H-share companies exceeded the Red Chip companies. By 2006, private enterprises surpassed H-shares and has continued proliferating, reaching 640 in 2017 and accounting for 61% of the total number of Mainland issuers (Figure 8.3).

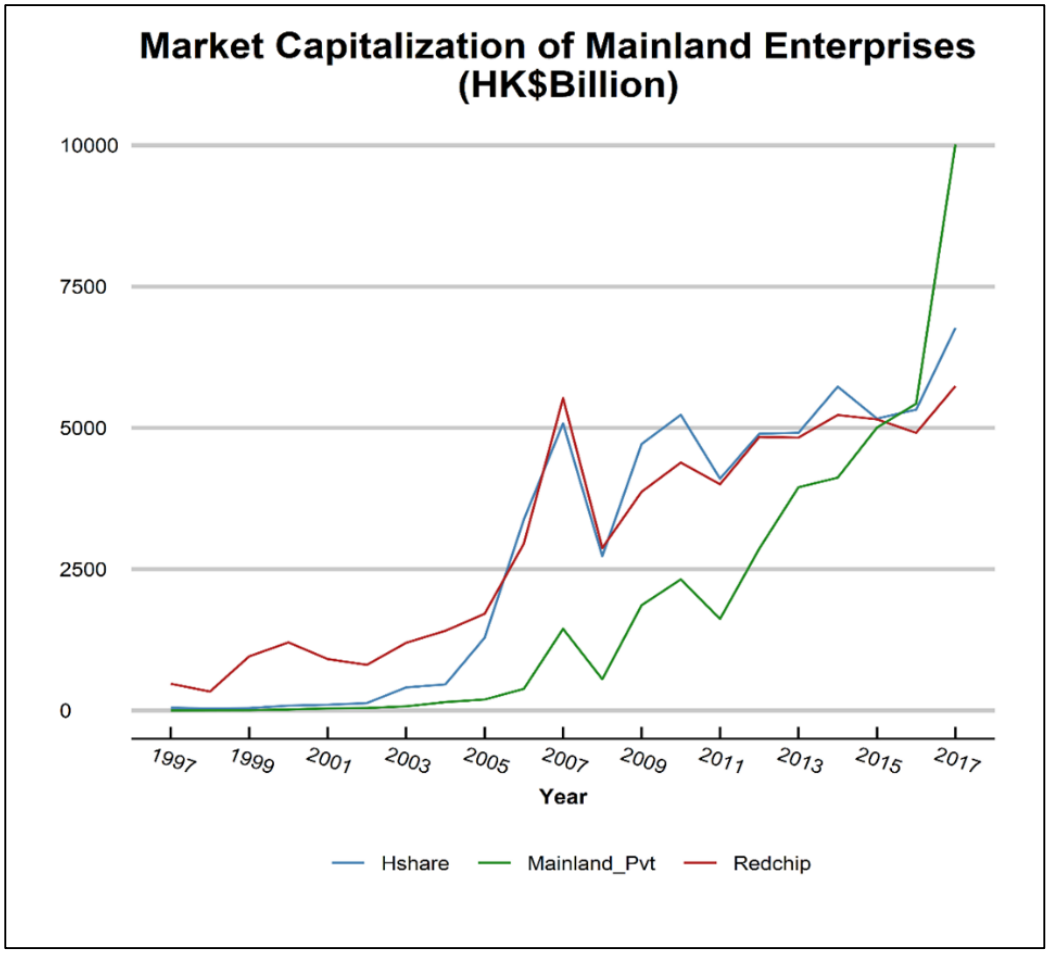


Figure 8.3 Hong Kong Main Board and GEM, 1997-2017 market value of Mainland enterprises

Source: Hong Kong Stock Exchange

The overall market capitalisation of H-shares, Red Chip and Mainland private enterprises in all Hong Kong boards rise with volatility. Despite having a far lower market value than H-shares and Red Chip stocks before 2016, Mainland private enterprises has taken the lead since then.

Figure 8.4 shows that in 1997, the total turnover of Red Chips far exceeded that of H-shares and private enterprises. Red Chips ranks first, followed by H-shares. Mainland private enterprises had almost zero turnovers. Subsequently, the total annual turnover of H-shares continued to grow, exceeding Red Chips and became first in 2003 and hit its peak in 2007 at HK$ 7,772.53 billion. Since then, H-shares rapidly declined, but its turnover is still first place, far ahead of Red Chips and Mainland private enterprises. The annual turnover of Mainland private enterprises was the lowest since 1997, rising to a small peak of approximately HK$ 1,040.43 billion in 2007, but still far below H-shares and Red Chips. Since 2007, Mainland private enterprises have shown significant growth, exceeding Red Chips after 2012 and reaching HK$ 1,485.93 billion. After a short decline, Mainland private enterprises rebounded rapidly in 2016 and continues to catch up with H-shares.

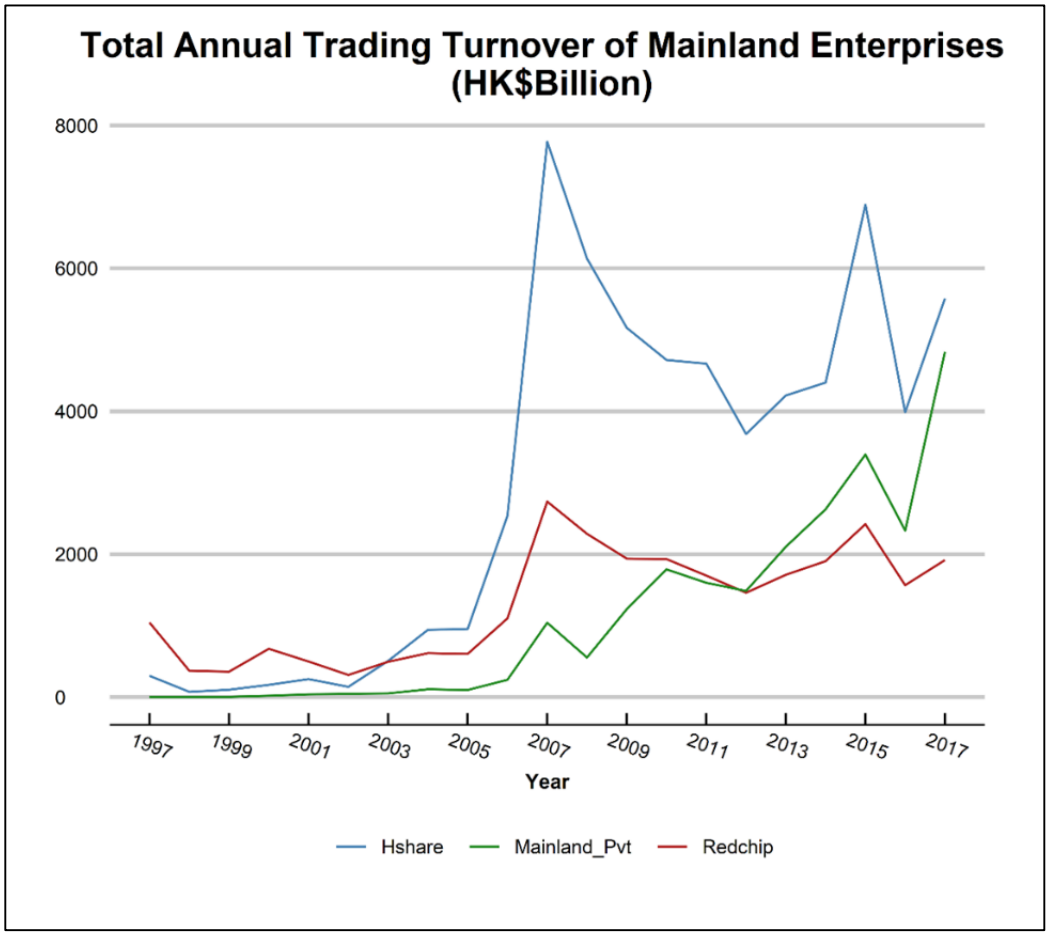


Figure 8.4 Main board and GEM, 1997-2017 total turnover of Mainland enterprises

Source: Hong Kong Stock Exchange

8.3.3 Leveraging the Comparative Advantage of the Hong Kong Stock Market to Attract Mainland Companies' Listing Financing

8.3.3.1 Advantages of Hong Kong Stock Market Financing: Multiple Listing Paths; 'New Listing System' Attracts Emerging Industries and Innovative Companies

Listing in the Hong Kong stock market can be done in various ways in addition to the traditional Initial Public Offering (IPO), Reverse Takeovers[14], or Reverse Merger obtaining listing funds, and the Variable Interest Entity (VIE) model[15] (protocol control). At the same time, the advantages of equity financing in Hong Kong are as follows. Firstly, the low cost of Hong Kong IPO is usually 5%-8% of IPO income, while Mainland IPO expenses commonly account for 5%-30% of IPO income. Secondly, Hong Kong IPO has shorter audit time and processes. The time to market in Hong Kong IPO is high, and the company generally completes the listing process in approximately 6-12 months. A controllable time to listing means that companies can better grasp its opportunities. Thirdly, the threshold for initial issuance is low, without a particularly strict requirement for industry and ownership. Finally, Hong Kong has clear and standardised listing procedures, independent and transparent regulatory authorities, high-quality professional agencies and market-leading IPO value assessment. By comparison, the listing of A-shares in the Mainland has the following disadvantages. Firstly, the cost is high. Costs include taxation, social security, listing preparation expenses, personnel compensation, intermediary expenses and marginal operating and risk costs. Secondly, listings take long review processes that requires authorities to approve securities disclosure issuance in China's market. The time of the IPO depends on the approval of the China Securities Regulatory Commission (CSRC), and subsequent issuance is subject to approval by the relevant authorities. The listing cycle is uncertain, procedures are cumbersome and the transparency is relatively low. Several industries—such as education services, catering, agriculture, highly polluting industries, real estate and new economy companies that use different voting rights—are subject to stricter listing requirements. From the perspective of business operators, market opportunities and development opportunities may pass. If the listing cycle takes too much time, companies can miss out in pursuing business opportunity. Thirdly, the threshold for listing is high. The net profit of the companies listed on the mainboard must be positive in the past three accounting years, accumulated over 30 million or the net cash flow generated from the business activities reach a total of more than 50 million yuan.

The implementation of the new listing regime in Hong Kong in April 2018 established a new concessionary secondary listing route for Greater China and international companies in Hong Kong. This regime can attract qualified global companies with keen listing interest, which helps to push Hong Kong as a financing centre for 'New Economy' companies. By the end of 2018, a number of large Mainland technology companies benefiting from the new regime are preparing to list in Hong Kong. For the first time, Hong Kong ranked first in the world on IPO fundraising (See Table 6.2). In terms of shares of the Hong Kong stock market IPO and secondary offering in 2018, Mainland private enterprises account for the majority. In April 2017, the HKEX revised its listing rules, opening the door for unprofitable biotech and innovative companies for the first time. In a single year, HKEX has welcomed the listing of 40 new economic companies (including nine biotech companies and two different voting rights companies), with a total financing of approximately HK$ 150.4 billion, accounting for more than 50% of Hong Kong funds during this period.

Table 8.1 shows that the top 10 companies with the largest IPO fundraising on the HKEx in 2018 are all Mainland Chinese companies, except for state-owned China Tower Co., Ltd., Gansu Bank Co., Ltd., Shandong Gold Mining Co., Ltd. and Jiangxi Bank. The remaining six are fundraising listings of private enterprises in the Mainland, including five large-scale technology and biological companies (highlighted in blue).

Table 8.1 Hong Kong Stock Exchange 2018 top 10 IPO fundraising companies

Rank	Company Name	IPO Funds Raised (HK$ Billion)	Industry
1	China Tower Corporation Ltd- H shares (788)	58.8	Properties & Construction
2	Xiaomi Corporation - W (1810)	42.61	Information Tech
3	Meituan Dianping - W (3690)	33.14	Information Tech
4	Ping An Healthcare and Technology Co Ltd (1833)	8.77	Biotech/ Medical Devices
5	Jiangxi Bank Co Ltd - H Shares (1916)	8.6	Financials
6	WuXi AppTec Co Ltd - H Shares (2359)	7.92	Pharmaceuticals/ Biotech
7	Haidilao International Holding Ltd (6862)	7.56	Consumer Services
8	BeiGene Ltd - B (6160)	7.08	Biotech/ Medical Devices
9	Bank of Gansu Co Ltd - H Shares (2139)	6.84	Financials
10	Shandong Gold Mining Co Ltd - H Shares (1798)	5.25	Materials

Source: Adapted from HKEX (2018)

8.3.3.2 Disadvantages of Equity Financing in Hong Kong: Uneven Distribution of Industries and the Outstanding Financial Industry; Hong Kong Lacks Understanding of Mainland Listed Companies

In terms of market value under the Hong Kong stock market by industry, financial services gained the highest from 2008 to 2018, accounting for approximately 30% of the total. Financial services slowed down in recent years, from a peak of 34.8% in 2008 to 27.3% in 2018. In the same period, information technology—including telecommunications equipment, computers and hardware equipment, system applications and IT consulting, E-commerce and Internet services, software and semiconductors—increased significantly from 2% to 14.33%. The second is the manufacturing of consumer goods that increased slightly from 6% to 10.64%. This industry includes automobiles, home appliances, textiles, clothing and personal care, food and beverage, medical care and agricultural products). However, compared with the small shares of other industries, financial services occupied the gigantic slice of the Hong Kong stock market.

For many years, listed companies in the Hong Kong capital market have concentrated on financial and real estate companies, while biomedical and technology companies ar*e comparatively much fewer*. The shares are far behind those of the New York Stock Exchange and Nasdaq (Figure 8.5). Alibaba, JD Group and Baidu Group are the best examples. The HKEX group has conducted a comprehensive review of the Hong Kong listing system earlier and found that there are several deficiencies, especially in attracting emerging industries and innovative companies, which has weakened Hong Kong's overall competitiveness relative to other major listing places in the world. The new listing system of the HKEX in 2018 includes the admission of companies with different rights in the same share and non-profitable biotechnology companies, and deregulation on the listing of companies with different rights in the same share that has been listed in markets such as the United Kingdom and the United States (US). Several biotechnology companies are now listed on the Hong Kong stock market. Nevertheless, the Hong Kong stock market still has room for improvement in listing and financing for more companies in different industries.

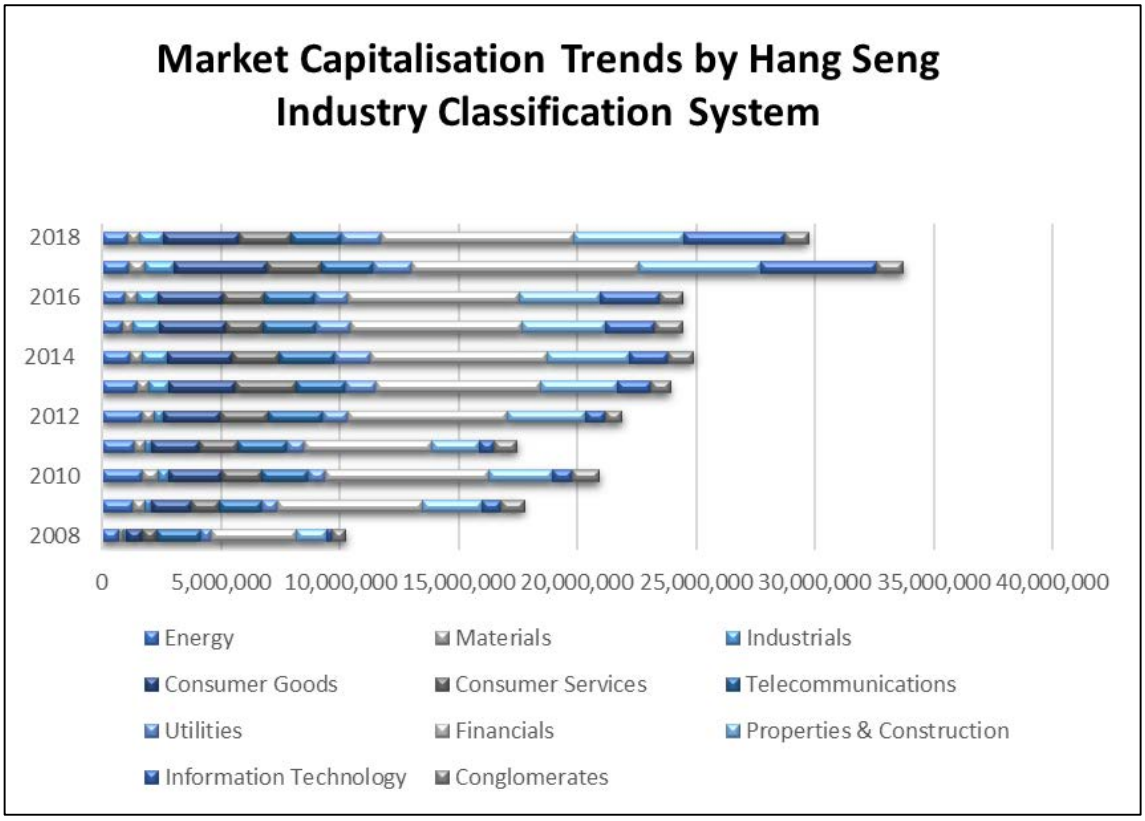


Figure 8.5 2008-2018 Stock value under Hang Seng Industry Classification System16 (in HK$ Million)

Source: Hong Kong Stock Exchange

In theory, the companies listed in both Hong Kong and the Mainland must have the same price in spite of listing in different locations because they are driven by the same long term fundamental value. However, variances are still observed between A- and H-shares of the same company (Figure 8.6). The reasons that cause the segmentation of the stock markets include information asymmetry between domestic and foreign investors, language and cultural differences and other direct or indirect barriers (Wang and Jing, 2004).

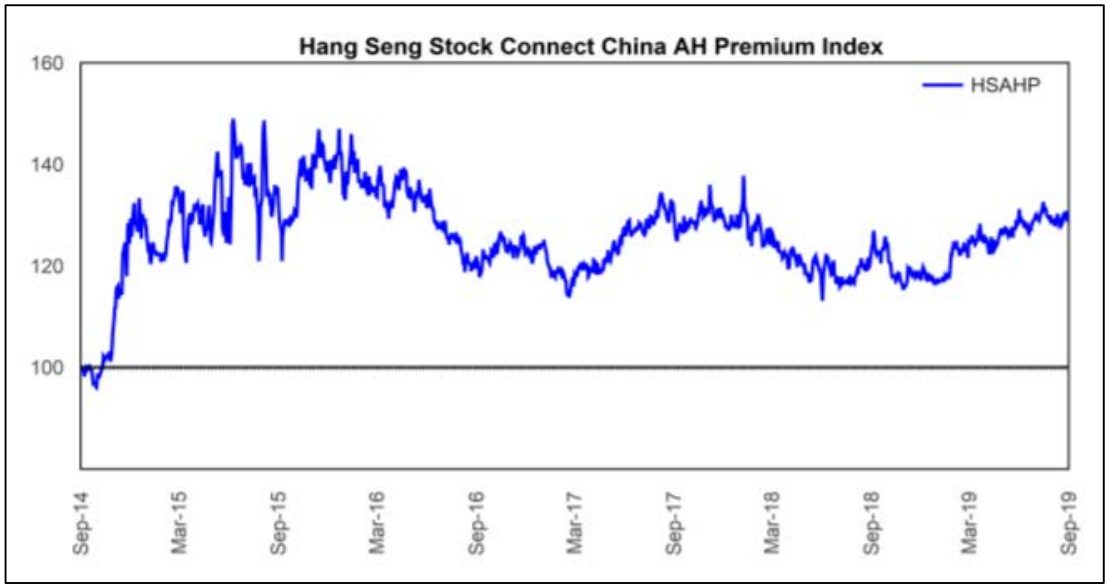


Figure 8.6 2014.6-2019.6 Hang Seng Shanghai-Shenzhen-Hong Kong Stock Connect AH Shares Premium Index

Source: Hang Seng Indexes Company Limited

Since the market activated under connectivity mechanism in 2014, the fluctuation of the A-H share premium index has risen, reflecting that the connection of the two markets has not forced investors to converge on the valuation of stock prices on Mainland companies that are both listed on A- and H-shares. Fan and Wang (2017) and Li and Chui (2014) examine the price premium of pair trading in stock exchanges in Hong Kong and Mainland. A-H share premium remains large but has a decreasing trend. The possible reason is that Hong Kong stockholders lack understanding of Mainland enterprises. Moreover, for a long time, Hong Kong's finance, real estate and consumer goods service have been well-established and employ outstanding professionals in related markets, but lack understanding of new technology companies, such as the information technology industry. In terms of valuation, the Hong Kong market has a high level of understanding of the Mainland's finance, new energy and consumer goods industries, but not of others.

Mis-valuation of stock prices and errors in listing approvals have prevented high-quality companies from smooth development, and poor-quality companies have dragged the Hong Kong stock market down. For a long period, the HKEx has dominated the financial industry. The Hong Kong market lacked analysts specialising in other non-financial industries, especially on emerging technologies and high-tech industries, causing this set of companies to be undervalued in the Hong Kong stock market. Therefore, Hong Kong must attract experts specialising in different industries.

8.4 Bond Financing

8.4.1 Hong Kong is the Largest Offshore Bond Financing Centre for Mainland Enterprises

Mainland authorities encourage private companies to use offshore bonds for offshore financing. The National Development and Reform Commission launched an offshore bond issuance registration system in September 2015, replacing the old preview approval system. In January 2017, the State Administration of Foreign Exchange deregulated its restrictions on repatriating funds from offshore bonds under the 'oversees lending secured by domestic guarantees' framework. The funds raised by the offshore bond issuance can be utilised to support offshore investments on The Belt and Road Initiative and those in the Guangdong—Hong Kong-Macao Greater Bay Area.

Hong Kong's strategic advantage in supporting Mainland corporate bond financing is illustrated by Mainland enterprises issuing dim sum bonds and foreign currency bonds in Hong Kong. The financing costs in this channel are often lower than in other offshore centres. Authorities in Hong Kong have taken various measures to reduce the cost of bond issuing, including the launch of the Pilot Bond Grant Scheme. First-time issuers are sponsored to issue and list offshore bonds in Hong Kong. In response to the steady increase of the issuance and listing of green bonds both in the Mainland and in Hong Kong, authorities have launched a Green Bond Grant Scheme to fund the certification costs of such bonds in Hong Kong.

8.4.2 Hong Kong as an Offshore Bond Centre has Advantages of High Total Issuance and Low Financing Cost

Figure 8.7 shows that Hong Kong is dominant in issuing dim sum bonds for non-financial companies compared with Singapore and Taiwan. Figure 8.8 shows that in terms of US$ bond issuance, compared with other offshore centres, Hong Kong has always ranked first among Mainland non-financial corporate issuance quotas. When the issuance of credit bonds by Mainland non-financial corporations declined in 2017, Mainland non-financial corporations in Hong Kong issued bonds with an all-time high record of US $99.3 billion. Hong Kong's issuance in 2018 was approximately US $72.3 billion, which was approximately 80% more than other offshore centres during the same period. Figure 8.9 shows that the average coupon rate of US$ bonds issued by Mainland non-financial companies in Hong Kong has always been lower than other offshore centres, reflecting an essential advantage in terms of financing cost.

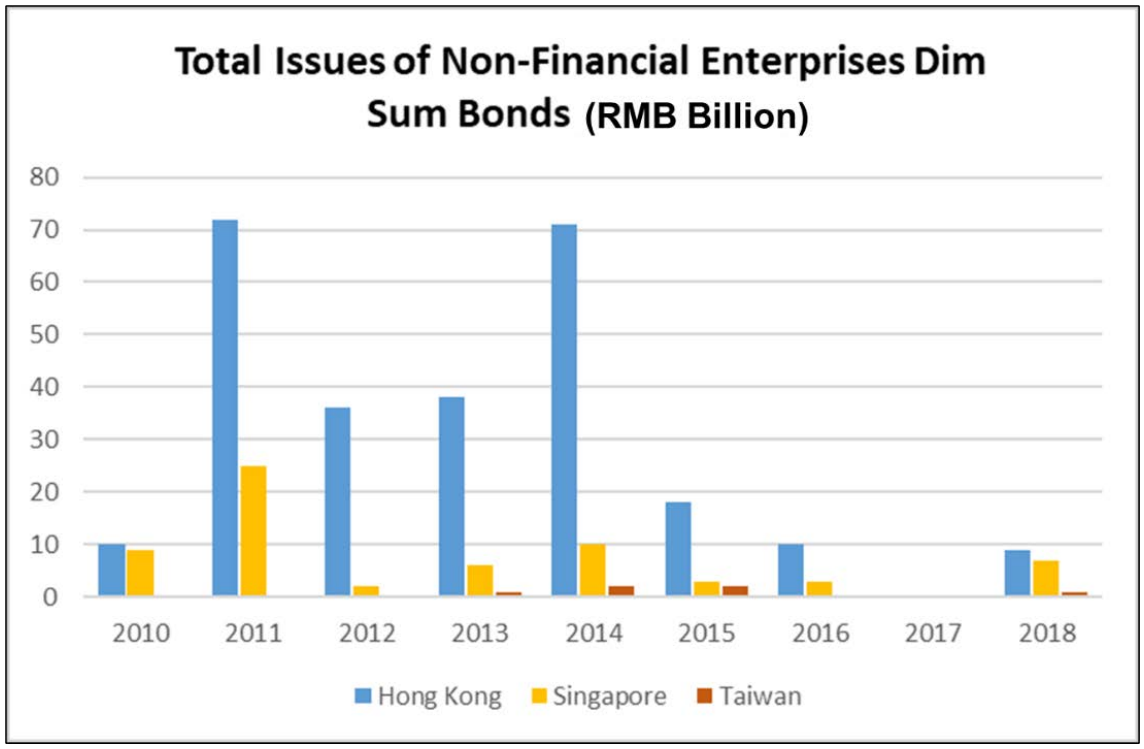


Figure 8.7 Total issues of Dim Sum Bonds by non-financial corporations in Hong Kong, Singapore and Taiwan in 2010-2018

Source: Wind Database

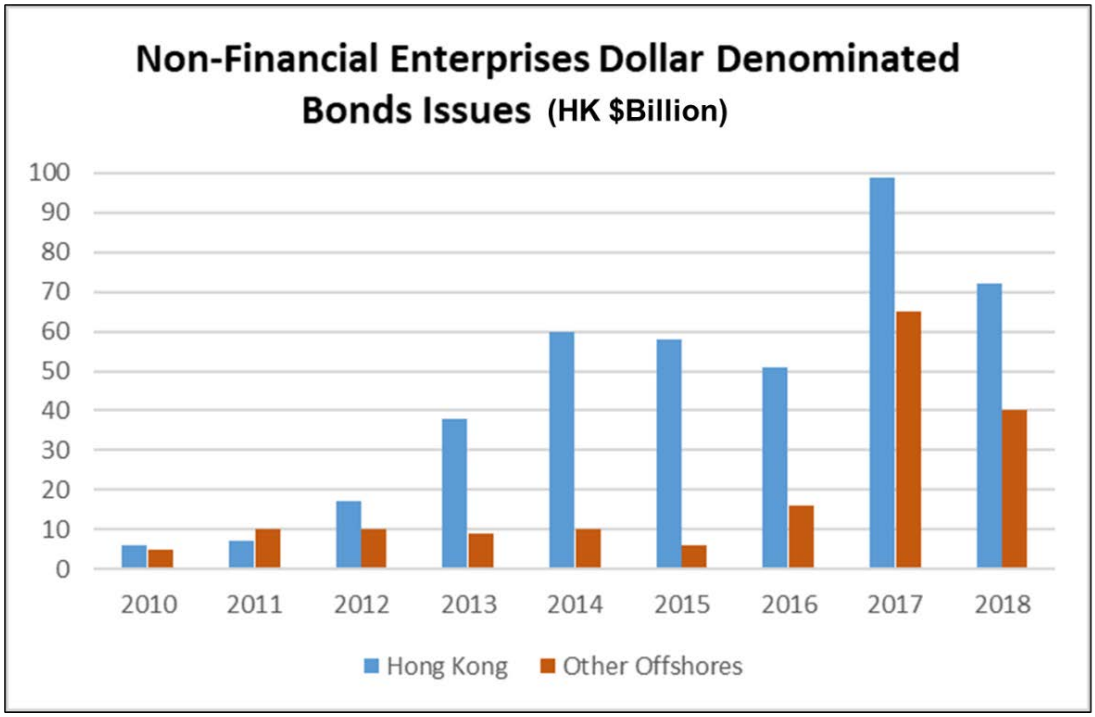


Figure 8.8 Total US$ bonds issued by Mainland companies in Hong Kong and other offshore centres in 2010-2018 [17] (in HK$）

Source: Wind Database

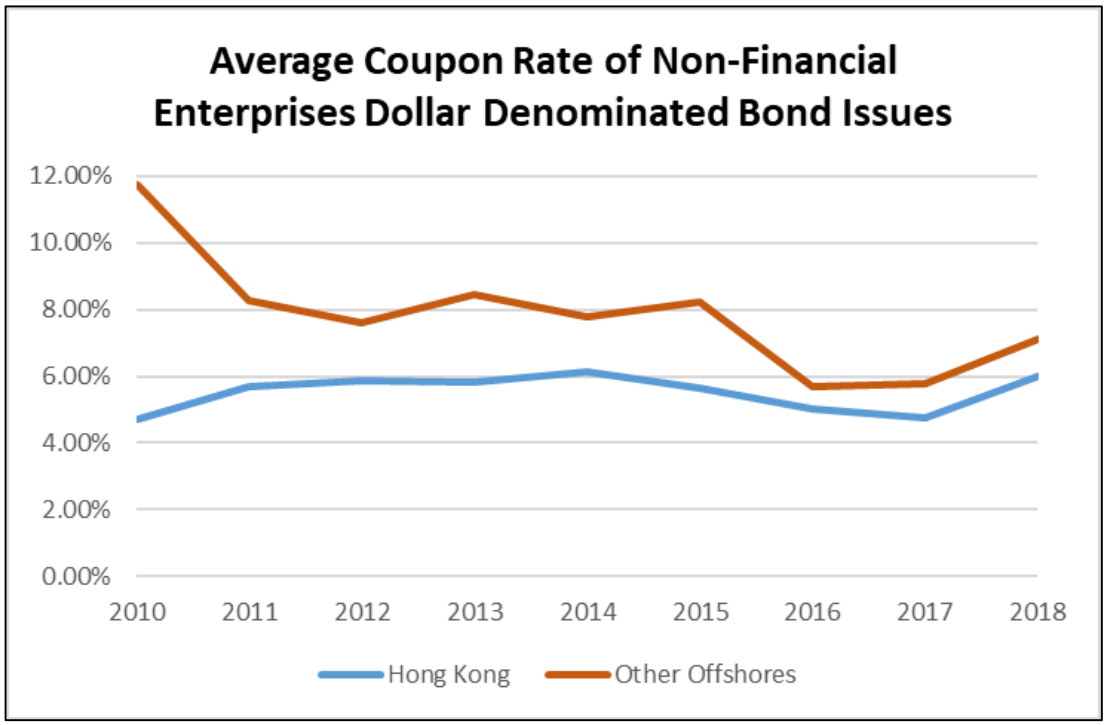


Figure 8.9 Average coupon rate of Mainland non-financial companies issuing US$ bonds in Hong Kong and other offshore centres in 2010-2018

Source: Wind Database

Offshore funds can be a useful supplementary raising fund for the GBA. In the construction of the GBA, infrastructures such as transportation, communications, electrical systems and pipelines are important. The gigantic scale of infrastructure construction has raised an equivalent demand level for financing. Infrastructure financing has characteristics of long construction cycle and slow investment return, which in turn determines the demand for debt financing. The offshore funds market of Hong Kong is a critical support in financing Mainland infrastructure enterprises when the local financing cost is high.

8.4.3 Hong Kong as a Green Financial Centre Endorses the Development of Green Bonds

The green development in the GBA requires Hong Kong assistance as a green financial centre, a strategy that is distinctly supported in the *Outline Development Plan for the Guangdong-Hong Kong-Macao Greater Bay Area*[18] issued by the China State Council Hong Kong is to establish internationally recognised green bond certification institutions. Green finance (Securities and Futures Commission. (2021) is an internal demand for the transformation and upgrading industrial structure at the GBA to achieve sustainable economic development. Enterprises in the GBA are actively seeking extensive projects in green financial cooperation, comprising considerable demand for green bond financing. Hong Kong can provide such important support. The green economy is a global economic development trend. When Hong Kong responds to this trend, the listing and trading of green bonds and the development of related products will also be naturally promoted in the Hong Kong bond market.

Figure 8.10 shows that the proportion of green bonds issued in the offshore market in Mainland China increased from 18% in 2017 to 40% in the first half of 2018, indicating an increase in investment in offshore green projects. The change behind this increase involved the issue of ICBC (Asia) and ICBC (London) on the HKEX and the London Stock Exchange with a total value of US$ 2.3 billion. The previous funds are allocated for offshore green projects in the Guangdong—Hong Kong-Macao Greater Bay Area, while the latter funds are used for countries and regions along the Belt and Road[19].

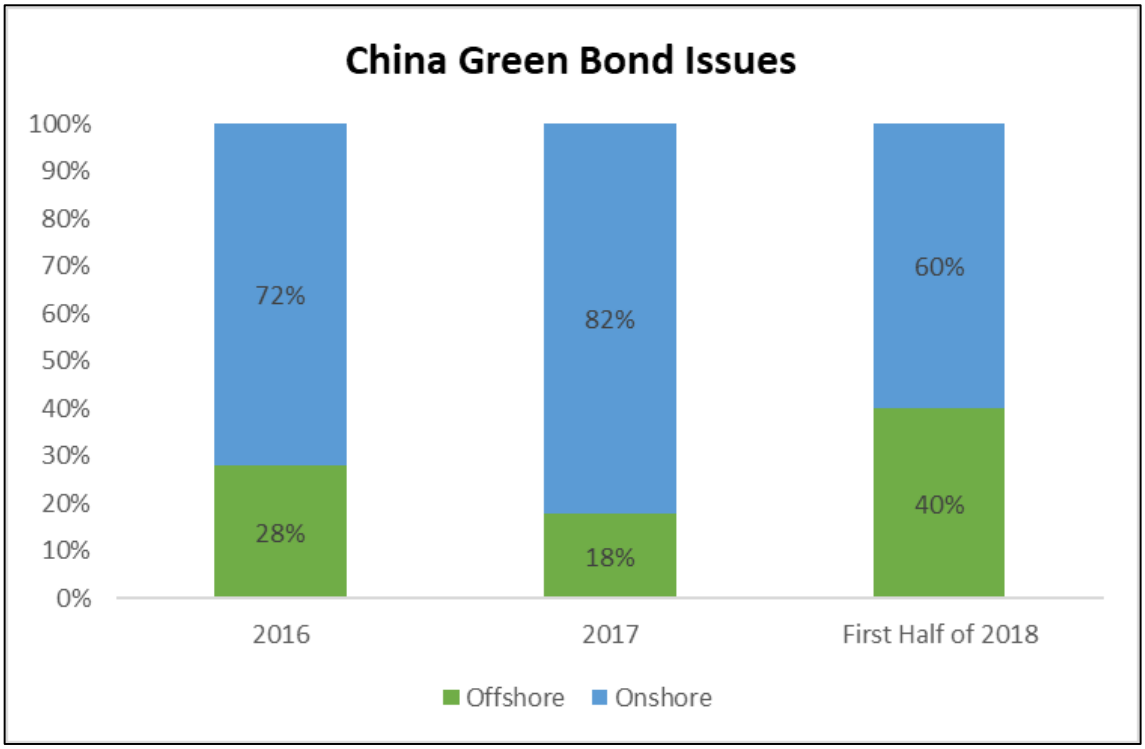


Figure 8.10 2016-2018 Mainland China green bond issuance in the first half of each year

Source: China Green Bond Market Report, CBI

However, the current development of the green bond market has unclear definitions of green standards, and the Guangdong—Hong Kong-Macao Greater Bay Area lacks uniform criteria for green financial cooperation. For example, the Hong Kong and Macao governments have not explicitly promulgated mandatory standards and specifications for green finance. Only documents of reference guide are available, and its content has inconsistencies with the current industry norms in the Mainland. Such details in business often have identification and connection conflicts hampering market cooperation. Moreover, the pursuit of green standards may increase the overall capital cost of enterprises.

Therefore, Hong Kong needs to communicate and cooperate with the Mainland to define uniform green standards, and its government can provide subsidies to green bond issuers to reduce the overall cost burden.

8.5 Trade and Logistics

8.5.1 Mainland China is Hong Kong's Most Important Source/Destination in Trade and the Second-Largest Destination for Offshore Trade for Hong Kong

Figure 8.11 shows that trading and logistics accounts for the highest portion of value-added GDP in Hong Kong, with results in all years above 20%. Clearly, trading and logistics is a pillar industry in the Hong Kong economy. Financial services follow with approximately 15% and 20% at the peak in 2007. With relatively smaller proportions, professional and producer services rank third and tourism ranks fourth, accounting for approximately 10% and 4%, respectively.

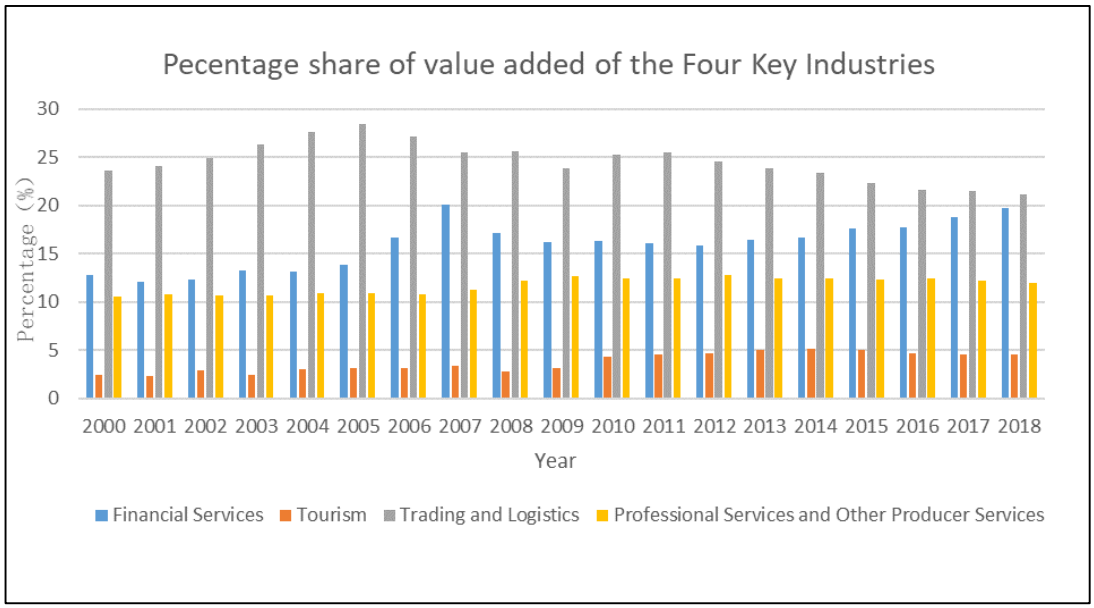


Figure 8.11 Percentage of the four pillar industries in value-added GDP

Source: Hong Kong Census and Statistics Department

Figure 8.12 and Figure 8.13 both show the trade volume between Hong Kong and 10 major countries and regions in terms of level and shares, respectively. In the 1970s, when the Mainland economy was underdeveloped, China had no capability of massive production for export products and the consumption demand via import is low. The trade volume between Hong Kong and the Mainland accounted for less than 10% of all foreign trade in Hong Kong. Later, due to China's reform and opening-up and Hong Kong's geographical adjacency with the Mainland, the trade volume remarkably increased from the 1980s onward and surpassed other countries after 2001, when China joined the WTO. By 2017, trade volume reached a record high at approximately HK$ 4,000 billion and accounted for 50% of Hong Kong's foreign trade. This finding illustrates the importance of the Mainland in Hong Kong trading activities.

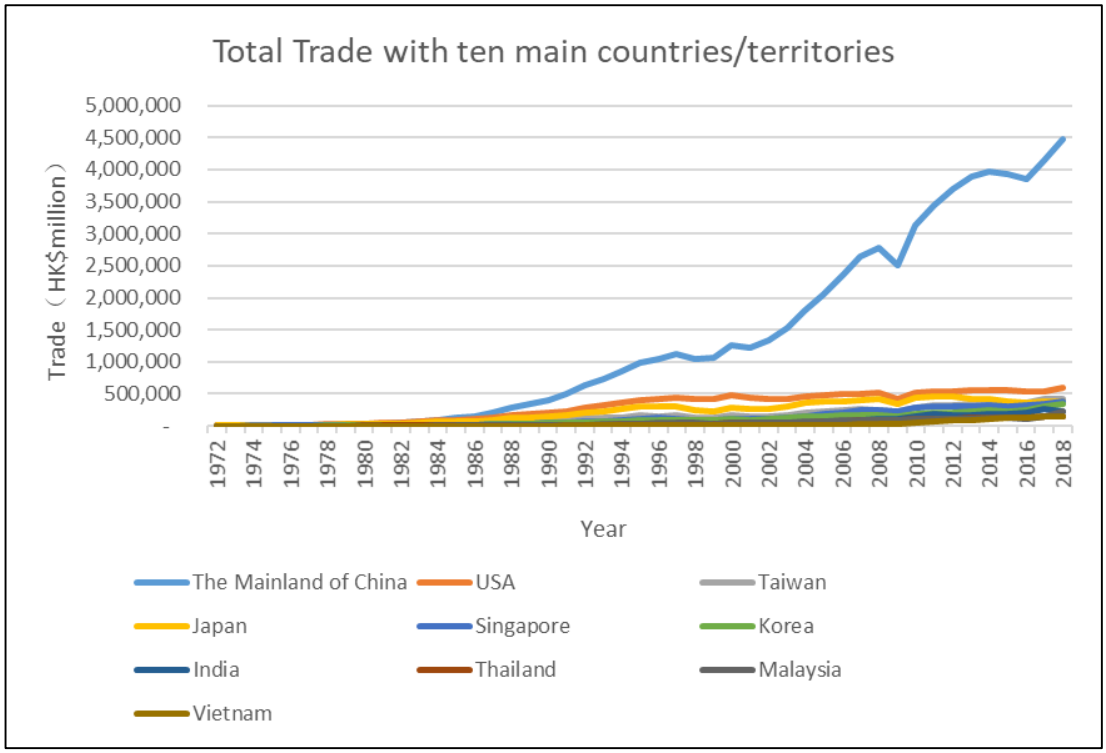


Figure 8.12 Total trade volume between Hong Kong and 10 major countries and regions

Source: Hong Kong Census and Statistics Department

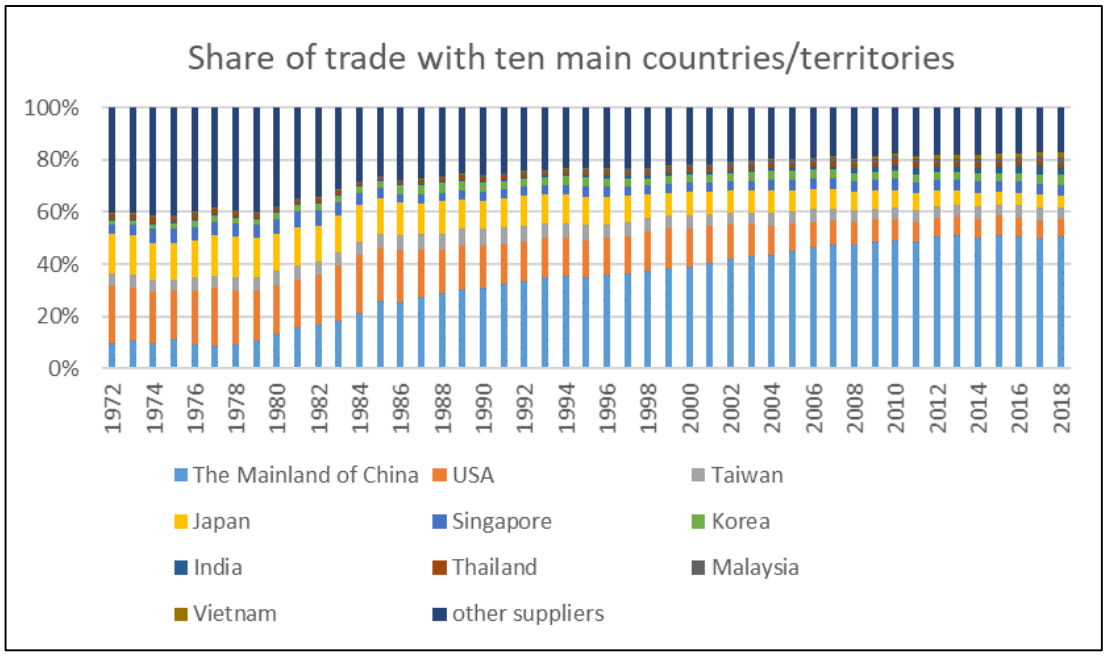


Figure 8.13 100% Stacked column chart of trade volume between Hong Kong and 10 major countries and regions

Source: Hong Kong Census and Statistics Department

As trade-related services increase in the Mainland, the model of trading companies procuring goods offshore [20] then selling in the international market has become standardised. Certain goods are shipped directly through Hong Kong without going through the city itself. These offshore trades are not included in trade statistics via Hong Kong. Data released by the Hong Kong Census and Statistics Department (Figure 8.14) show that, since 2007, Hong Kong offshore trade volume has been growing for most years, except for two exceptions in 2009 and 2015. The negative growth on 2009 was due to massive impact of the financial crisis in December 2008, which caused the growth rate to fall from nearly 30% in 2008 to −10% the following year. Offshore trade volume reached a maximum of HK$5,100 billion in 2014, but the total amount of offshore trade in Hong Kong in 16 years was only HK$4,243.8 billion, a decrease of 20% compared with 2014.

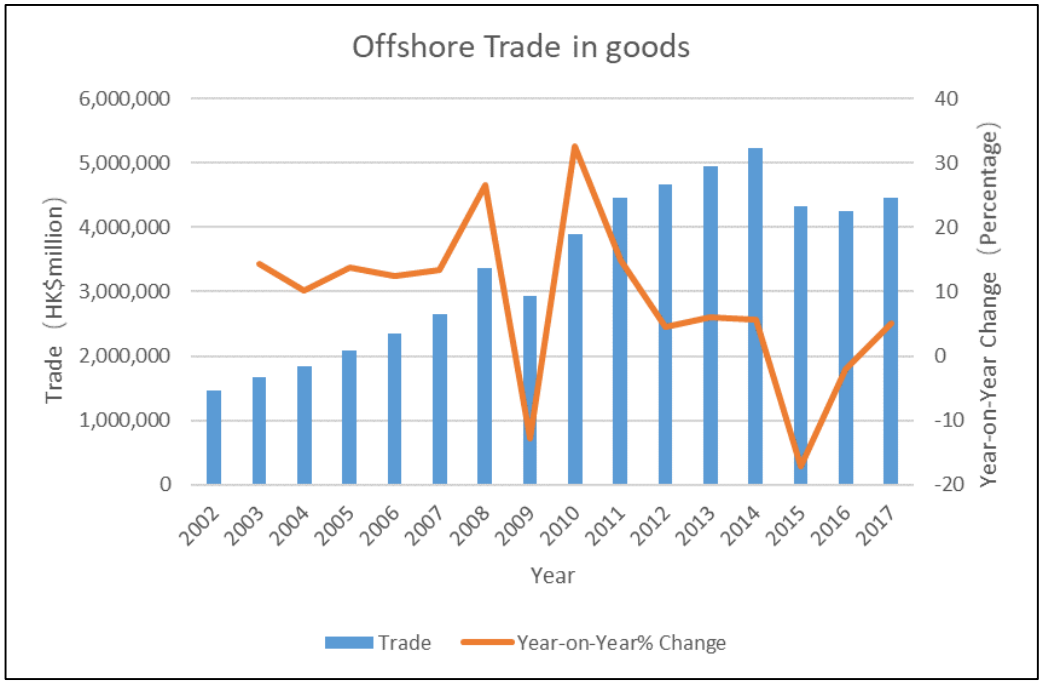


Figure 8.14 Hong Kong offshore trade volume and changes by years (in HK$ million)

Source: Hong Kong Census and Statistics Department

Figure 8.15 displays Hong Kong's gross profit from offshore trades with several major trading nations. From 2012 to 2016, the highest value is from the US, followed by Mainland China, Britain, Germany and Japan.

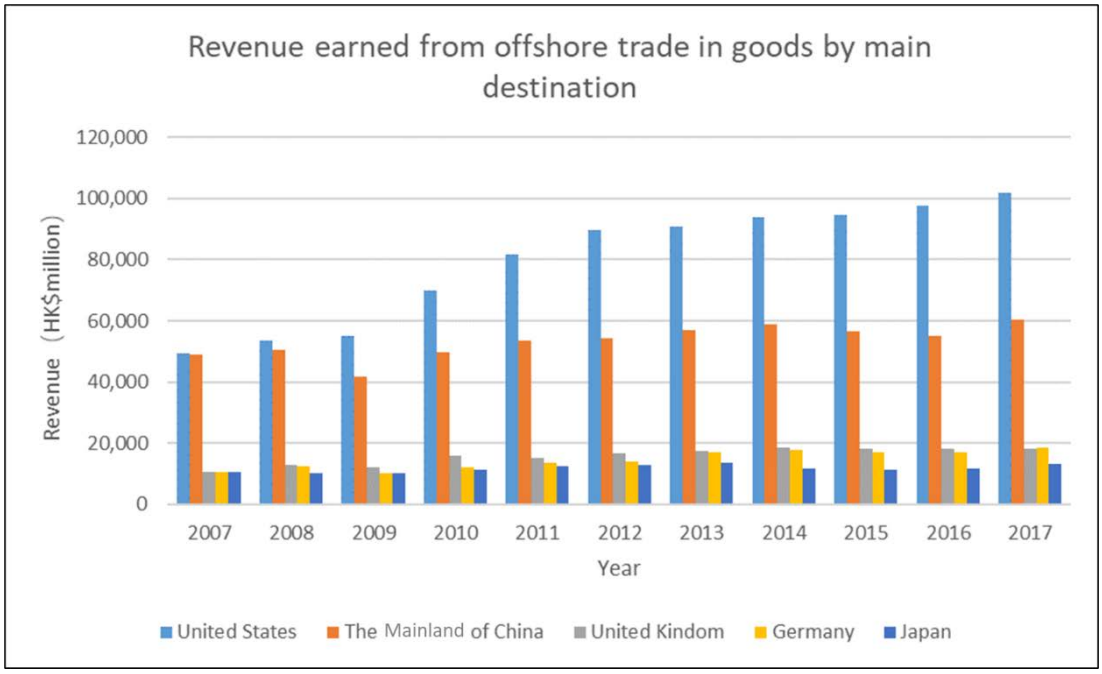


Figure 8.15 Revenue earned from offshore trade in 2007-2017, categorised by destination[21]

Source: Hong Kong Census and Statistics Department

8.5.2 Hong Kong status as a re-export centre for the Mainland has declined gradually

Figure 8.16 shows the export shares of various local Chinese customs in 2014. Shanghai Customs and Shenzhen Customs have the largest export shares and far exceed the rest with 22.32% and 20.05%, respectively. Figure 8.17 shows the export shares of various customs in 2018. Compared with 2014, except for the decline in the proportion of Shenzhen customs, the export share of several major customs outlets has increased. In 2014, Hong Kong exports accounted for 15.50% of the total exports from the Mainland, ranking third among all Mainland gateways. In 2018, Hong Kong exports accounted for 11.93%. Despite maintaining its rank, Hong Kong's share has declined and narrowed its gap from that of Nanjing Customs, which ranked fourth. Figure 8.18 and Figure 8.19 show that, comparing with 2014, Shanghai custom import shares in national total in 2018 has increased over 2%, while according to Figure 8.20, Hong Kong's role to import for China has slightly weaken during the same period.

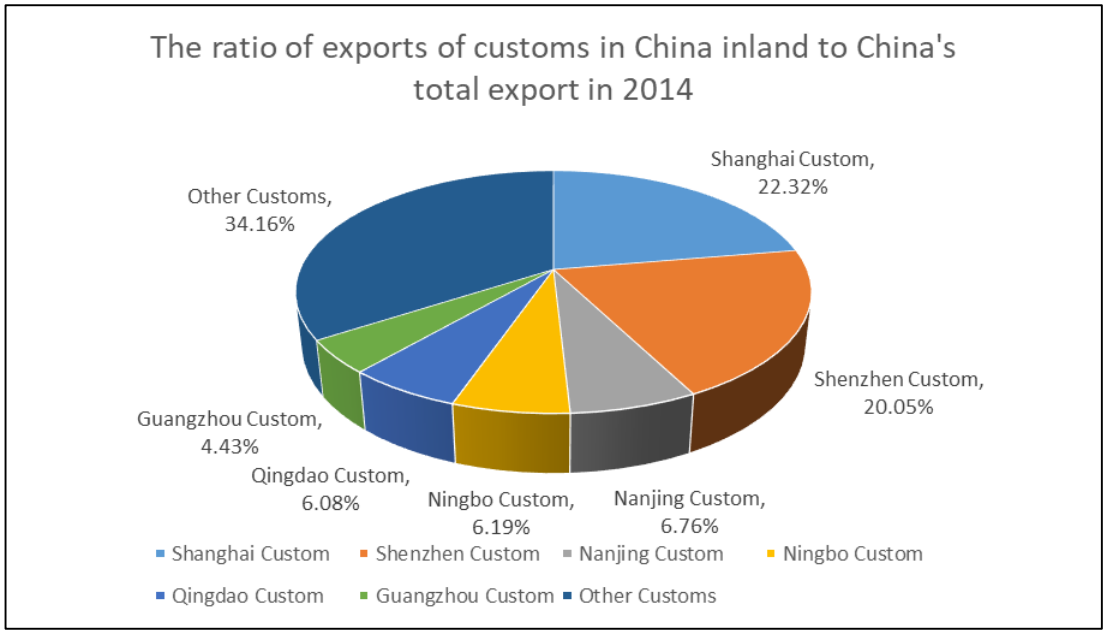


Figure 8.16 Ratio of local Chinese customs export to total exports in 2014

Source: General Administration of Customs of PRC

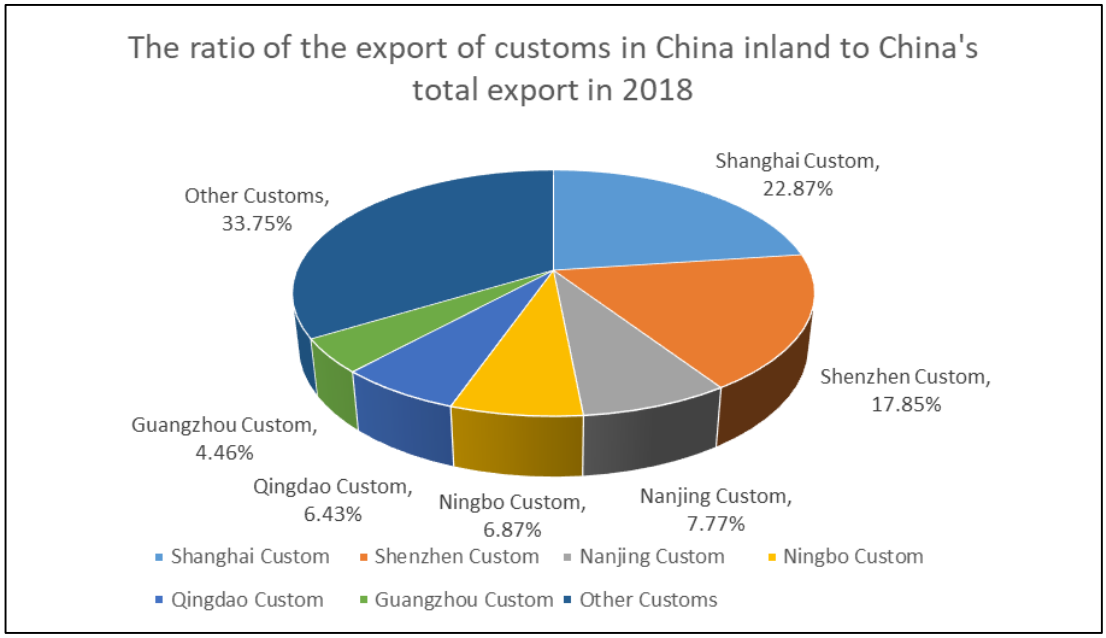


Figure 8.17 Ratio of local Chinese customs export to total exports in 2018

Source: General Administration of Customs of PRC

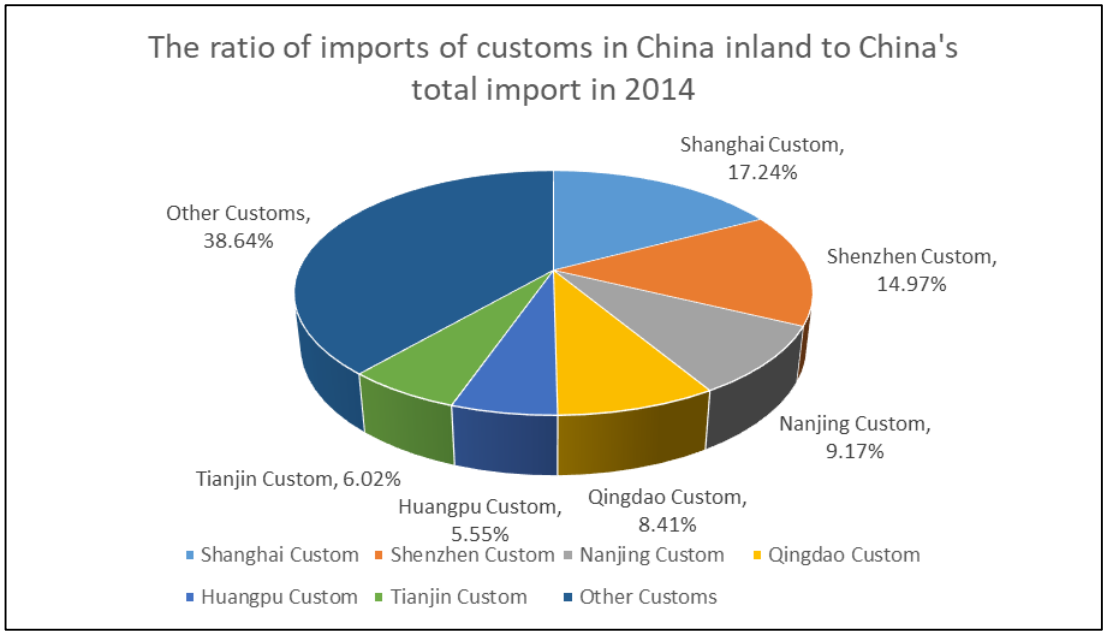


Figure 8.18 Ratio of local Chinese customs import to total imports in 2014

Source: General Administration of Customs of PRC

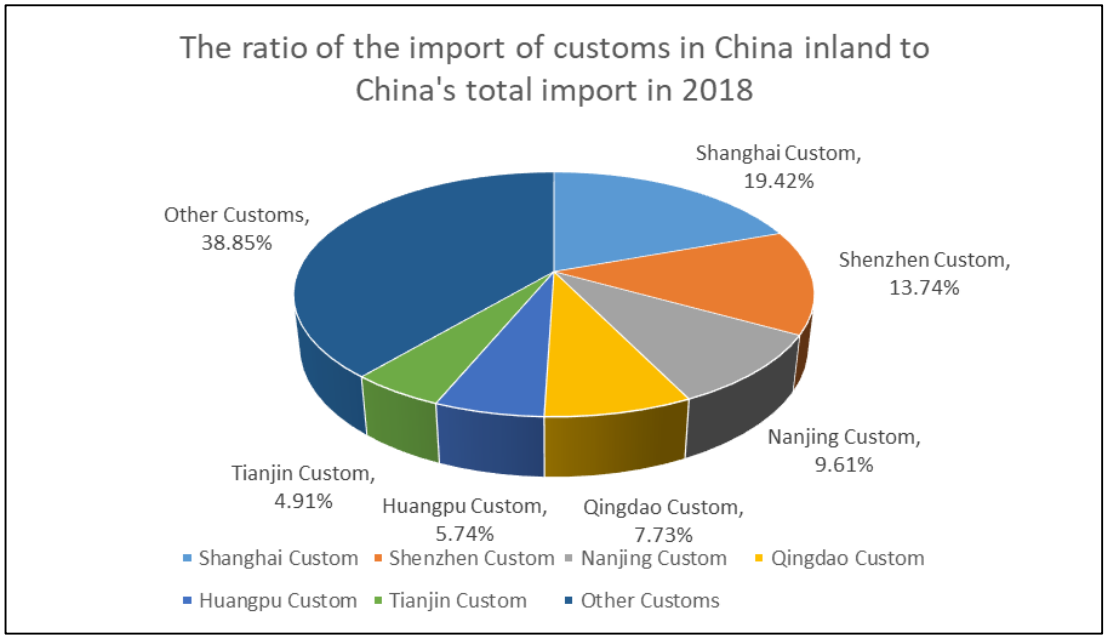


Figure 8.19 Ratio of local Chinese customs import to total imports in 2018

Source: General Administration of Customs of PRC

Figure 8.20 shows the volume of Hong Kong imports and exports as a percentage of those of China. The proportion of Hong Kong imports and exports shows a downward trend. At the same time, if Hong Kong is considered as one port and

compared with other Mainland ports, Hong Kong ranks the third in terms of export share in 2018 and 34th in import share, a decrease from 2014 when Hong Kong ranked third in export share and 25th in import share. These findings show that Hong Kong importance as an importing source has declined during this period.

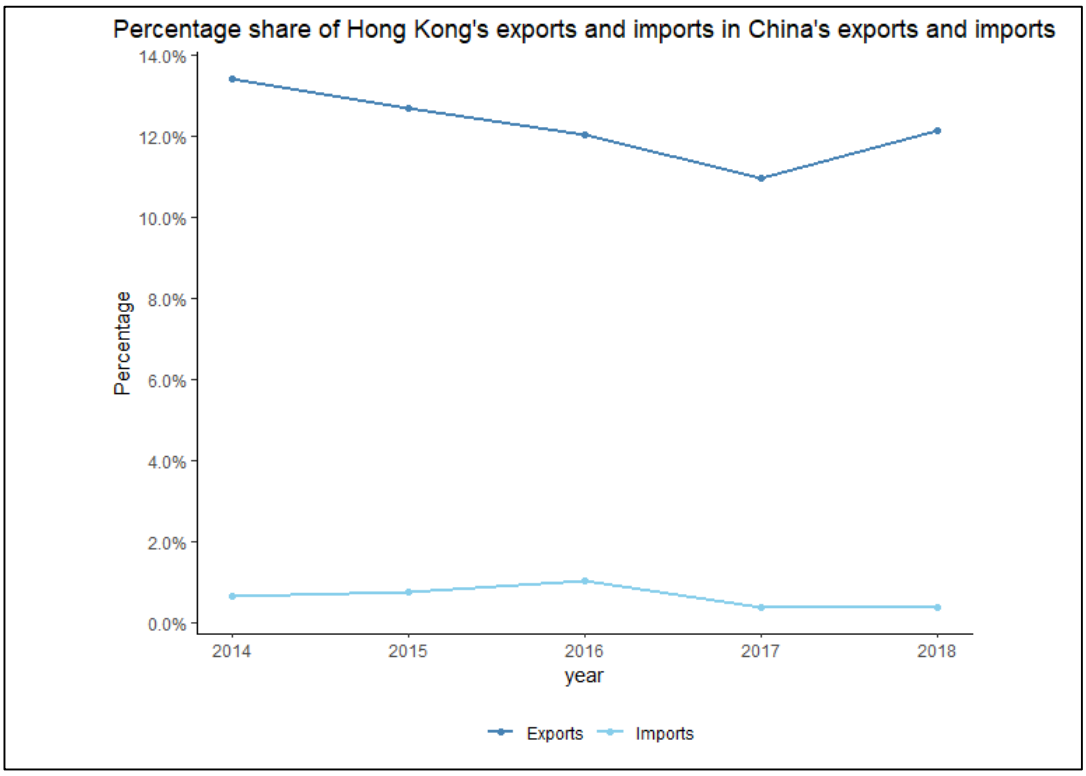


Figure 8.20 Proportion of Hong Kong's total imports and exports in the total of China in 2014-2018

Source: General Administration of Customs of PRC

8.5.3 Share of imports and exports in the YRD region is on the rise, while the share in the PRD region is in decline

Figure 8.21, Figure 8.22 and Figure 8.23 show the ratio of exports and imports to the nation from Shanghai, Ningbo and Nanjing, respectively. Shanghai plays a very important role in national imports and exports. Of China's total between 2014 and 2018, exports accounted for 25% and imports accounted for 20%. The combined ratio of exports and imports was 23%. In Zhejiang Province in the YRD region, Ningbo's export ratio was 6%, and increased year by year. In Jiangsu Province, Nanjing had an export share of approximately 10%, while the import and export ratio was approximately 8% in 2014, rising to 9% in 2018.

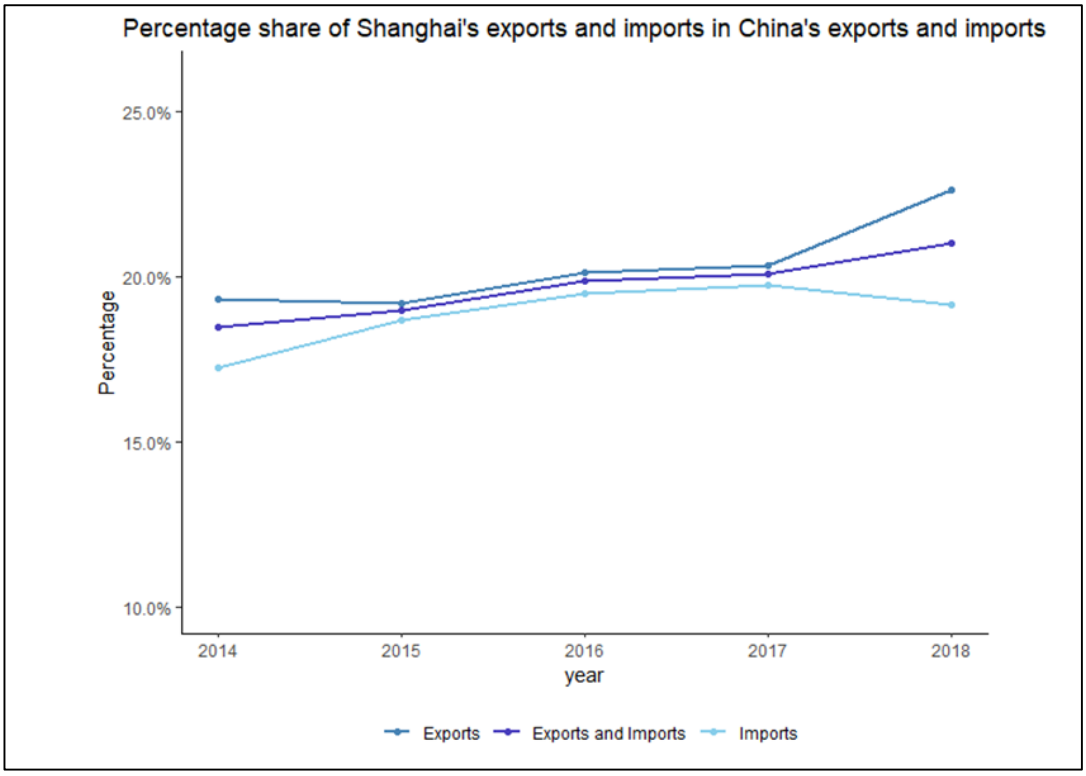


Figure 8.21 Proportion of Shanghai 'customs' total import and export volume in the total import and export volume of China in 2014-2018

Source: General Administration of Customs of PRC

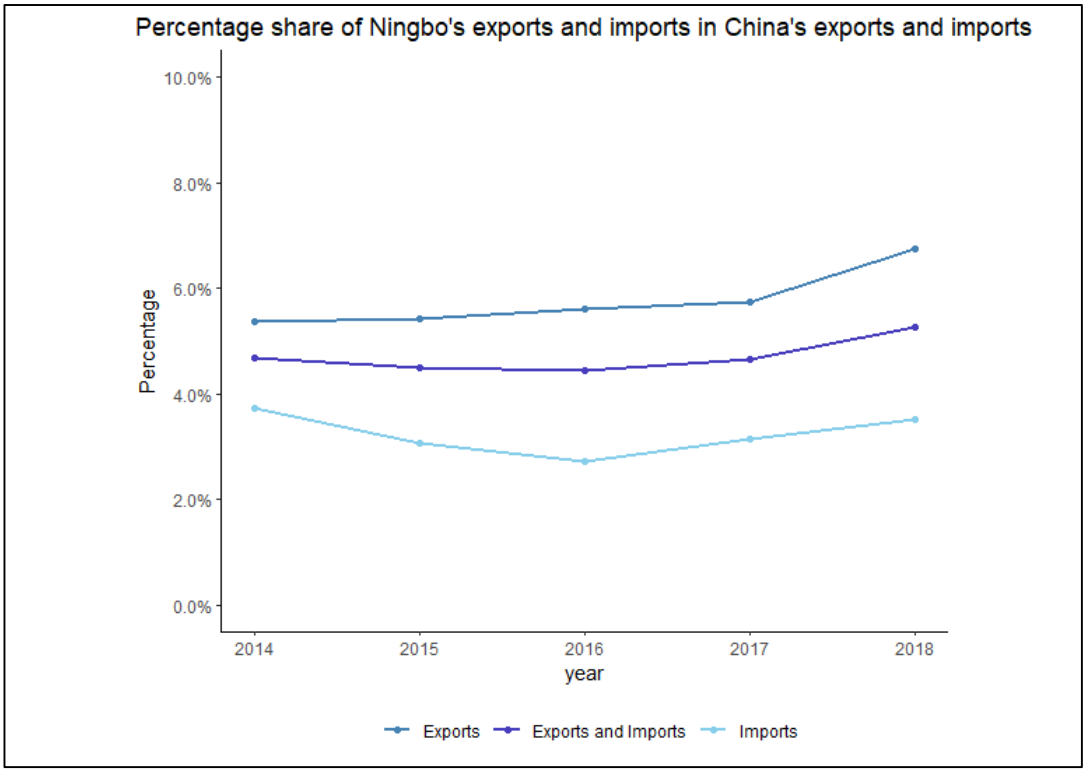


Figure 8.22 Ratio of total import and export of Ningbo Customs to the total import and export volume of China in 2014-2018

Source: China Customs Administration

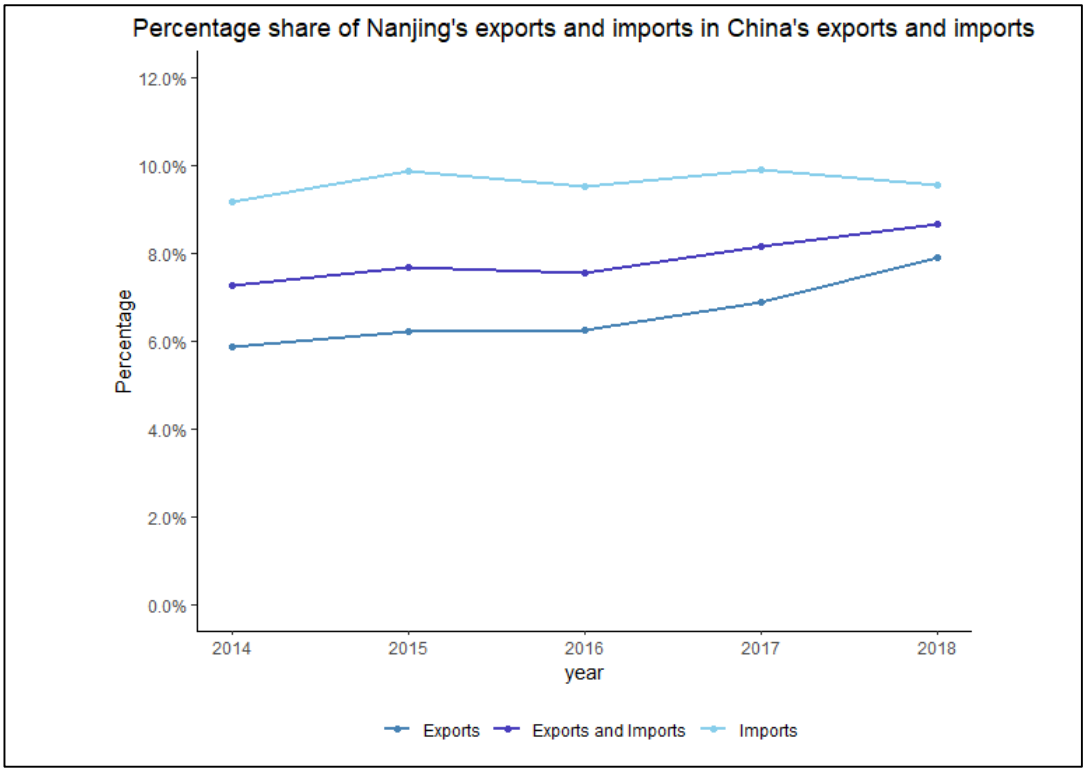


Figure 8.23 Ratio of total import and export of Nanjing Customs to the total import and export volume of China in 2014-2018

Source: China Customs Administration

Figures 8.26 and 8.27 illustrate the proportion of Guangzhou and Shenzhen imports and exports of China. Figure 8.24 shows that the import and export ratio of Guangzhou is only 4%, which is less than that of the abovementioned three cities in the YRD. The ratio initially increased and then declined during 2014-2018. Figure 8.25 shows the proportion of imports and exports in Shenzhen. The proportion accounted for 18% in 2014 and then gradually decreased to 15% in 2018. to the possible reason is the gradual rise of the YRD in occupying the share of exports that originally belonged to the PRD. The figures show that the import and export shares of the YRD has an increasing trend and is greater than that of the PRD, which shows a downward trend.

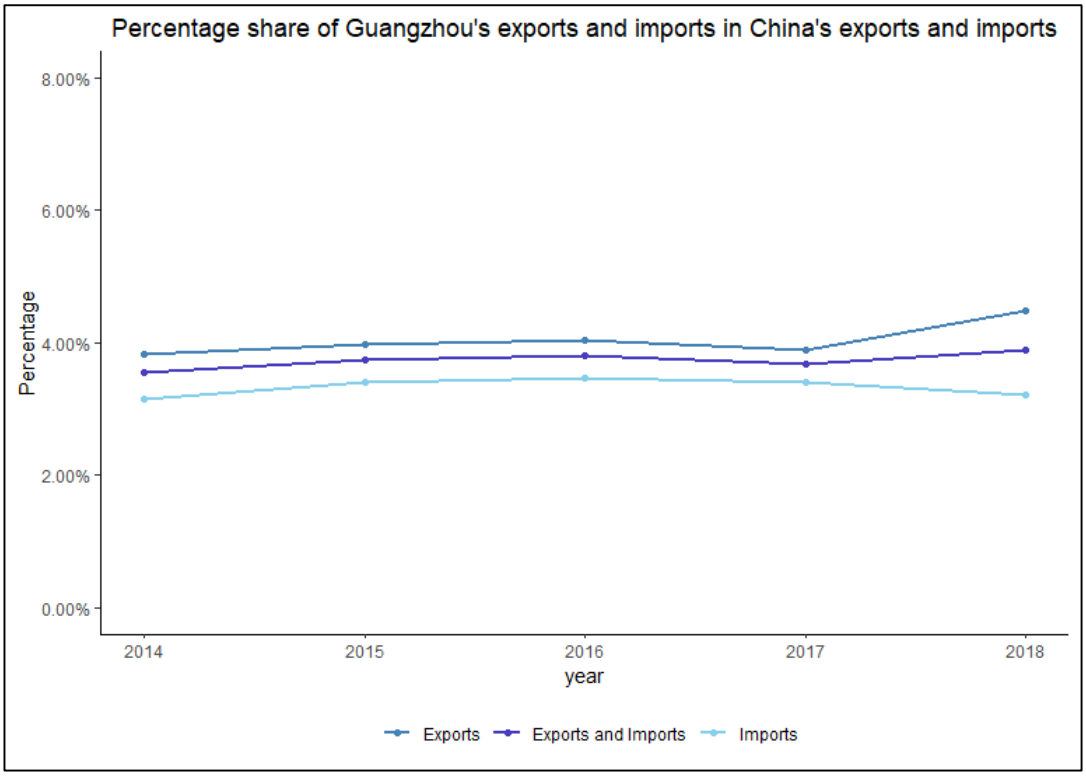


Figure 8.24 Proportion of total import and export of Guangzhou Customs in the total import and export volume of China in 2014-2018

Source: China Customs Administration

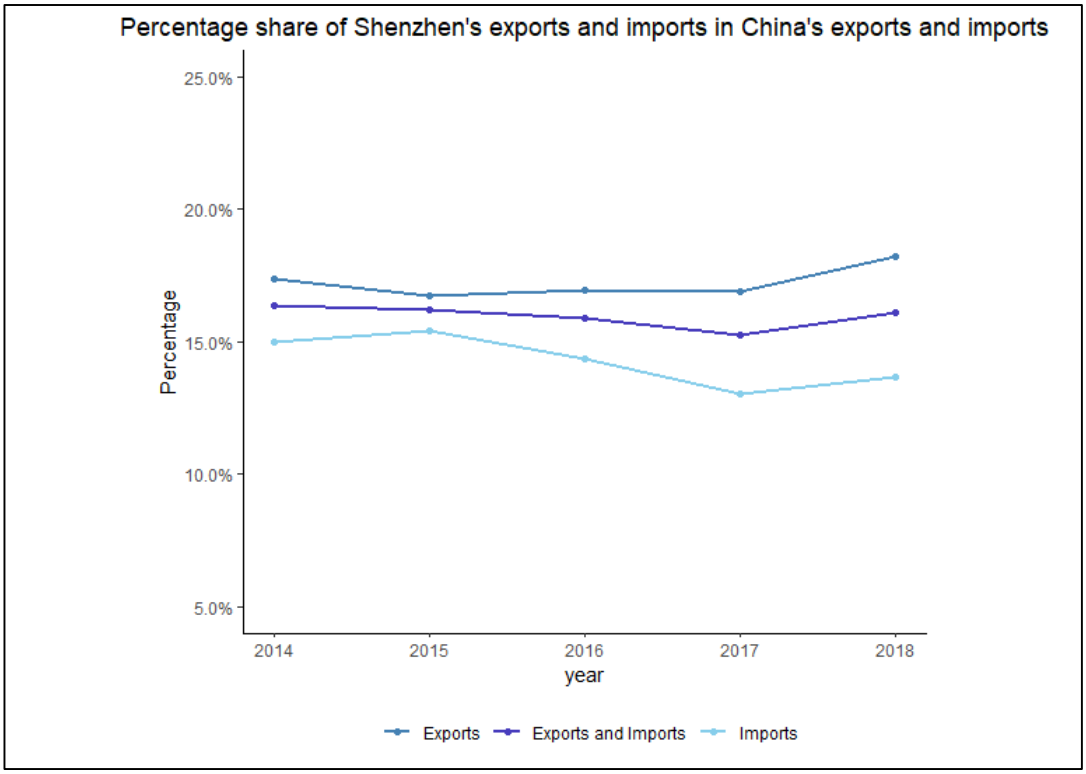


Figure 8.25 Ratio of total import and export of Shenzhen Customs to the total import and export volume of China in 2014-2018

Source: China Customs Administration

8.5.4 Among Enterprises Using Hong Kong as Trade Intermediary, state-owned enterprises have declined and private enterprises have risen

On the one hand, private enterprises established by private capital are continuously evolving. On the other hand, state-owned enterprises have also undergone privatisation. Thus, private enterprises gradually gain better statistics in re-export trade via Hong Kong.

Figure 8.26 shows the dramatic change in trading shares among various ownership types of enterprises in Hong Kong for re-exports. State-owned enterprises show a continuous decline in their export shares from 2000 to 2008. Correspondingly, despite still lagging behind state-owned enterprises in trade share, private enterprises have shown a development momentum that cannot be ignored. From 2000 to 2011, the export trade value of private enterprises increased from nearly zero to approximately 15% of total export trade, surpassing state-owned enterprises in 2012 and ranking second among trading companies in Hong Kong's trade type of business. Meanwhile, Wholly Foreign-Owned enterprises play a very important role in the re-export trade in Hong Kong, with shares at 30% or more of Hong Kong's total exports and even reached 60% in 2008. Therefore, Hong Kong's trade depends to a large extent on the development of foreign-invested enterprises.

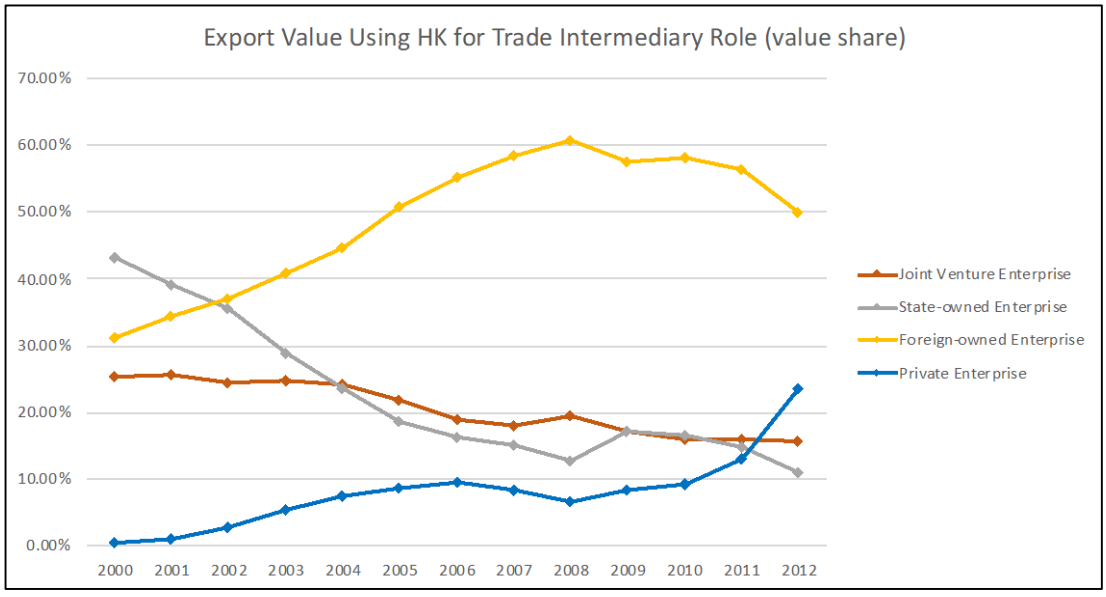


Figure 8.26 Export value shares from Chinese enterprise utilising Hong Kong as a re-export centre

Source: China Customs Database

Figure 8.27 further shows the percentage change of export value for different ownership types of enterprise. The growth rate of private enterprises was higher than other ownership types from 2001 to 2005. Although the financial crisis caused a growth rate of negative 30% in 2008, a positive rapid growth is observed after 2009.

The reason is primarily China's entry into the WTO in December 2001. Various existing trade barriers are dismantled, symbolising the removal of long-term monopolistic protective shield from state-owned enterprises on exports. A more open, free and competitive market thus formed and private enterprises can then directly participate in foreign trade. The release of corporate trade vitality has led to an essential surge in the export shares of private enterprises and maintained a growth rate of more than 20% between 2001 and 2005. One of the essences of Hong Kong producer service development is to pay close attention to the development orientation and to long-term needs of private enterprises in the Mainland, thereby fully utilising their capital strength and establishing stable trade links with them.

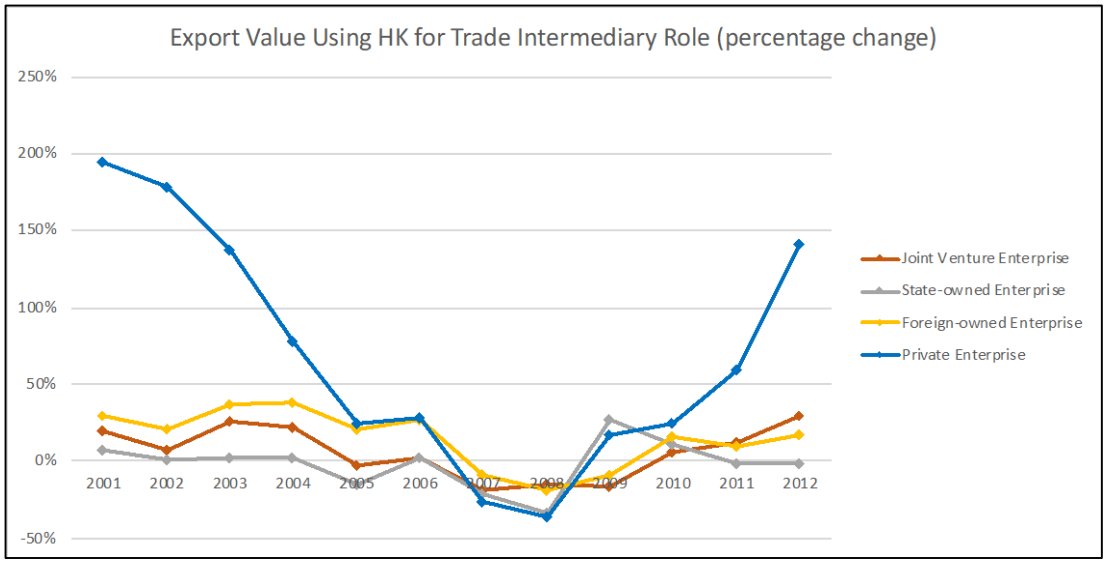


Figure 8.27 Percentage change of export value shares from Chinese enterprises utilising Hong Kong as a re-export centre

Source: China Customs Database

Therefore, Hong Kong needs to continue the following measures: consolidating the established financial service system, legal framework, and human resources; taking advantage of attracting foreign capitals to invest; paying close attention to the import and export demand of private enterprises in Mainland China; and embracing private enterprises and enhancing their role as a 'front shop'.

8.5.5 Value of Mainland exports through Hong Kong has been increasing, and export activities have expanded towards more provinces at inner areas.

An analysis of China Customs Database (2000-2012) shows that the southeast coast had the highest export value in 2000, owing to its intrinsic geographical advantage, namely, adjacency to Hong Kong. This advantage leads to more exports sent via Hong Kong. Nevertheless, after 2000, with the economic prosperity of the Mainland, an increasing number of provinces exported over a billion through Hong Kong, expanding from the southeast coast to the inner areas. Furthermore, an analysis of China Industrial Enterprise Database (1998-2013) shows that over the years, Hong Kong-funded enterprises have moved away from Hong Kong, but the benefits of exploring new markets outweigh the disadvantages of being far away from Hong Kong. Our studies show that more and more provinces in the inner area of the Mainland export through Hong Kong. The number of provinces that have over one billion RMB export value through Hong Kong have also increased.

8.5.6 Hong Kong-funded enterprises gradually moved towards inner areas, together with export growth relocation

The above analysis of China Industrial Enterprise Database also shows that in 2007, only coastal provinces, such as Jiangsu, Fujian, Guangdong and Shandong have a high export value from Hong Kong-funded enterprises. Over the following years, the export of Hong Kong-funded enterprises in the inner provinces had continuously expanded. In 2013, the export of inner provinces, such as Hubei, Hunan, Sichuan and others, reached a level above RMB 200 million and exceeded exports of various coastal provinces. This trend entails that Hong Kong-funded enterprises have gradually moved from the coastal areas to distant inner provinces, comprising a compelling reason calling changes for the undermined ‘Front Shop, Back Factory’ model. The ‘back factory’ has relocated, and thus the ‘front shop’ must follow accordingly to better serve Hong Kong-funded enterprises in the Mainland.

8.5.7 Hong Kong has a simplistic export structure and concentrates in four major industries. Mainland exports focus on manufacturing products.

Mainland exports via Hong Kong are mainly electrical machinery while imports

are mainly jewellery and precious metal. For commodity categories after the third place, compared with imports, Hong Kong exports have a higher chance of becoming primary products. Of the total exports, textile materials and products account for 4.16%. Manufacturing products dominate Hong Kong import goods. These figures reveal a mismatch between the import demand from the Mainland and the re-export structure based on Hong Kong.

Table 8.2 shows that, among the Hong Kong export commodities, electrical machinery has been on an absolute advantageous role with their shares. The second is jewellery and precious metals industry, although its exports were only 6.99% in 2015 and declined further since then to 4.64% in 2018. Optical, photographic, medical or surgical instruments and apparatus rank third while textiles products rank fourth, both at approximately 4% in 2018. Thus, the product types of China exporting through Hong Kong are limited. Electrical machinery accounts for more than two-thirds, while products other than the top four all have shares less than 2%. These figures also show that Hong Kong export structure concentrates on electrical machinery.

The same issue of structural simplicity arises in Hong Kong imports, aligning with the result in Chapter 2.3. Hong Kong imports 71 jewellery and precious metals far above products in other industries, accounting for over 40% in shares. The other top industries are 85 electrical machinery, 74 metal and miscellaneous products, and 39 plastic articles. The proportion of other industries are all less than 2%. The limited imports indicate that the trading system has a weaker anti-shock ability and drives less growth. Hong Kong needs to diversify its import and export products, and improve its limited product component to increase risk resistance.

Table 8.2 China's top four commodity categories through Hong Kong import and export

Exports and import from/to Hong Kong	2014	2015	2016	2017	2018
Exports					
85 - Electrical Machinery and Equipment and Parts Thereof; Sound Recorders and Reproducers, Television Image and Sound Recorders and Reproducers, and Parts and Accessories of Such Articles	43.48%	52.08%	51.55%	50.56%	51.41%
84 - Nuclear Reactors, Boilers, Machinery and Mechanical Appliances; Parts Thereof	14.61%	14.21%	14.84%	15.74%	16.53%
71 - Natural or Cultured Pearls, Precious or Semi-Precious Stones. Precious Metals, Metals Clad with Precious Metal and Articles Thereof; Imitation Jewellery; Coin	14.81%	6.78%	5.21%	4.15%	4.64%
90 - Optical, Photographic, Cinematographic, Measuring, Checking, Precision, Medical or Surgical Instruments and Apparatus	5.45%	5.72%	5.19%	5.12%	4.44%
Imports					
71 - Natural or Cultured Pearls, Precious or Semi-Precious Stones. Precious Metals, Metals Clad with Precious Metal and Articles Thereof; Imitation Jewellery; Coin	9.32%	45.90%	62.94%	12.34%	31.50%
85 - Electrical Machinery and Equipment and Parts Thereof; Sound Recorders and Reproducers, Television Image and Sound Recorders and Reproducers, And Parts and Accessories of Such Articles	11.94%	12.45%	4.60%	12.49%	8.77%
74 - Copper and Articles Thereof	11.57%	8.40%	6.30%	20.12%	15.86%
39 - Plastics and Articles Thereof	11.76%	9.89%	7.31%	12.36%	7.63%

Source: General Administration of Customs of PRC: Monthly Statistical Report Annual Collection (2014-2018) - Total Import and Export Commodity Chapters of Several Countries (Regions)

Figure 8.28 shows the exports shares of major industries[22] in Mainland China. In 2012, the top four industries were as follows: 85 electrical machinery with over 50%; 84 mechanical appliances at approximately 20%; 90 optical, photographic, medical or surgical instruments and apparatus; and 95 toys, games and sports requisites and parts. The latter two account for less than 5% of the total. These figures indicate that

manufacturing products dominate China exports via Hong Kong. From this time trend, the export share of the 85 electrical machinery industry has maintained absolute dominance since 2012 and has continued to gradually increase, reaching 60% in 2018. In second place, 84 mechanical appliance industry has maintained a steady level at approximately 20% of the total exports. The possible reason may be due to a more inelastic demand of mechanical appliances compared with electrical machinery, and lack policies that are favourable to the electronic appliance industry in the past 10 years, and thus the quantity increment is not as rapid as electrical machinery, revealing a steady or a slightly falling export share. The remaining industries have export shares of less than 5%, reflecting the dominance of the export structure of the Mainland by manufacturing products in a relatively simplistic export structure, and also matches Hong Kong export structure. The expansion of electrical machinery and mechanical appliances exports may be due to the rise of the Mainland domestic manufacturing industry, lifting up the export volume. At the same time, the growth of electrical machinery made essential contributions to the economy. Two reasons are considered: firstly, the number of enterprises under the Mainland electrical machinery industry has grown rapidly and contribute a large number of exports; secondly, foreign demand on electrical machinery products such as tape recorders is growing and boosts its shares.

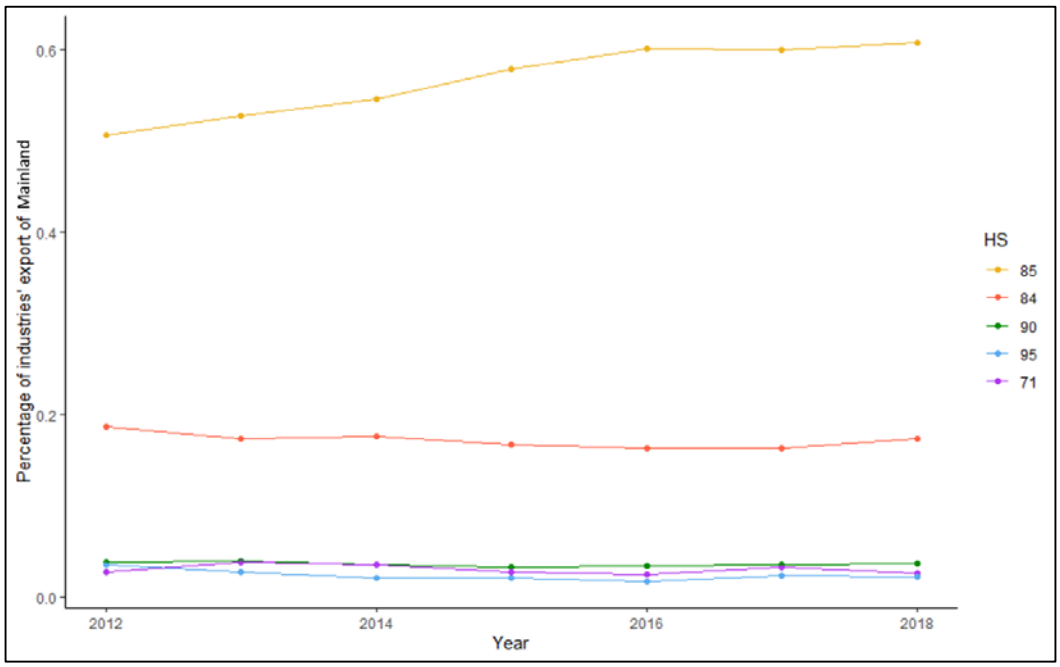


Figure 8.28 Shares of export of major industries in Mainland China in 2012-2018

Source: Hong Kong Census and Statistics Department

End Notes

[1] Single insurance refers to the insurance that the applicant pays the premiums at once, an up-front lump-sum payment. Investment-linked insurance (including both single and annual payment) refers to that the value of a policy at any time when providing that life insurance is based on the investment performance of the corresponding investment fund.

[2] Non-investment-linked insurance (including both single and annual payment) include permanent, deposit, term and other types of insurance.

[3] Annualised insurance, namely, non-single-payment insurance is which the policyholder makes annual payments during the payment period. It is a type of term insurance.

[4] HKD value is converted based on the real-time exchange rate in September 2019. The cumulative new policy premiums for life insurance reached SGD 706 million in 2000 and SGD 3.534 billion in 2018.

[5] The feature distinguishing onshore and offshore insurance is that the essential entity participating in transactions is a non-local resident. For further HK onshore and offshore insurance details, see Appendix E. Mainland investment accounts for over 70% of HK's offshore insurance.

[6] According to the Standard & Poor's rating, AA and BBB ratings refer to having 'very strong' and 'adequate' capacity to meet financial commitments, respectively.

[7] Non-durable consumption goods include food, beverages, perfumery and cosmetics, tobacco, pharmaceuticals, lamps and batteries.

[8] Semi-durable consumption goods include leather and travel goods, footwear, clothing (including knitted and non-knitted) and plastics.

[9] 22 beverages, spirits and vinegar include wine, whisky and other alcoholic drinks, and thus fall into the luxury category.

[10] Three-plus-one enterprises generically receive equipment or invest in building factories, raw materials, and samples from foreign investors, and exporting all products. These three-plus-one enterprises are locally responsible for providing land, factories and labour.

[11] Data source: Hong Kong Monthly Digest of Statistics, December 2019 - Four Key Industries and Other Selected Industries.

[12] *The Outline Development Plan for the Guangdong-Hong Kong-Macao Greater Bay Area* proposes that the GBA should build an international financial hub. HK should play a leading role in the financial field, consolidate and enhance its status as an international financial centre and further advance markets under the Connectivity Mechanism.

[13] The so-called 'Front Shop, Back Factory' model refers to the use of HK and Macao regions to take advantage of their uniqueness in acquiring overseas orders, carry out marketing and foreign trade, and play as a 'shop'; while the PRD region utilises its particular advantages, such as low-cost labour, in production and act as 'factories'.

[14] A reverse takeover refers to a non-listed company that acquires a listed shell company and merges with it to become a subsidiary of a listed company. The original shareholders of the non-listed company obtains 50%–90% equity of the listed company.

[15] Variable interest entities (VIE) refer to the separation of overseas registered listed enterprises from domestic business operating entities. The former commonly control the latter through agreements. Business entities are VIEs of listed entities.

[16] HKEX adopts the Hang Seng industry classification system

[17] Offshore bonds are issued by non-financial companies using financial subsidiaries (excluding banks and brokers). Other offshore centres cover US dollar bonds in the US, UK, Singapore, Ireland, Malaysia, Taiwan and other unspecified offshore markets.

[18] The Outline states that 'Vigorously develop special financial industries: support HK to build a green financial centre in the Greater Bay Area, build an internationally recognised green bond certification institution; support Guangzhou to build a green finance reform and innovation pilot zone, and study and establish an innovative futures exchange with carbon emissions as its first product; support the development of Macao's special financial services such as leasing, explore misplaced development with neighbouring areas, and study the establishment of a RMB denominated securities market, a green financial platform, and a Sino-Portuguese financial service platform in Macao.'

[19] Hong Kong Stock Exchange Report: 'Development Trends of Green Bonds: Global, Mainland China and Hong Kong'

[20] Offshore trade covers 'Merchanting' and 'Merchandising for Offshore Transactions' by enterprises and organisations in HK (excluding external associated enterprises). The goods involved in offshore trading were shipped from external sellers \ directly to external buyers, and these goods do not enter or leave the HK border.

[21] The income from offshore trading refers to the gross profit earned from 'Merchanting' and commissions from 'Merchandising for Offshore Transactions', and not the actual value of the goods involved.

[22] The percentage is calculated from dividing HK's export value by the total export value of all industries in that year.

Chapter 9
New Economic Cooperation Models between Hong Kong and the Greater Bay Area and Policy Recommendations

9.1 Introduction

Hong Kong and the Pearl River Delta (PRD) have collaborated intensively and effectively since China's economic reform and opening up in the late 1970s. However, the once-successful market-driven economic cooperation model of 'Front Shop, Back Factory' encountered severe challenges in the face of the PRD's rapidly rising economic power and Hong Kong's declining role as its bridge to the world in the last two decades. In addition, the global political and economic landscapes present more recent turbulence, such as from the China-US trade war and the COVID-19 pandemic. This study aims to search for new economic cooperation models between Hong Kong and the Greater Bay Area (GBA) and present policy recommendations based on the findings of comparison with other Bay Areas and data analysis and field work in the PRD.

The GBA presents both an opportunity and a challenge for Hong Kong. The GBA, especially its export-oriented industrial structure, is difficult to sustain in the long term in the face of challenges of higher internal operating costs, weak customer demand, difficulty in entering the Mainland market and the uncertainties caused by the trade war between China and the US. For Hong Kong to better integrate with other GBA cities, this study puts forward four economic development models: Hong Kong as science and technology centre in the GBA; Hong Kong Riding on Increasing Affluence of Mainland Consumers and Investors; Leading the GBA to go global and to drive the economic growth from enterprise internationalisation; and Continuously utilising Hong Kong to provide producer services for Mainland enterprises, setting up 'front shops' in the GBA, and improving access to customer groups. Subsequently, relevant policy recommendations of each model for promoting Hong Kong's economic growth and integration with the GBA are presented.

9.2 Comparison with Other GBAs

Our historical and structural analysis examine the growth trajectories of established bay areas in the US and Japan. The strengths, weaknesses, opportunities, and threats

for Hong Kong to reposition itself in the Guangdong-Hong Kong-Macao Greater Bay Area are likewise explored.

9.2.1 Findings

Our comparative analysis clearly identifies the GBA to hold the largest population and land mass. In particular, the GBA has an economy dominated by manufacturing with advanced services lagging behind, shortage of global command and control functions, a high-tech industry dominated by telecom manufacturing, and growing quantity with poor quality in terms of invention patents.

The analysis also identifies an interesting common pattern of economic, geographical and political transformation. The American and Japanese bay areas started with the growth of port functions and related manufacturing activities, followed by the dominance of advanced producer services and eventually transformed themselves into a global and national hub of high-tech industries.

Hong Kong has always functioned as an important gateway connecting the Chinese economy and the rest of the world. Continuing socioeconomic changes at international, national and regional levels indicates that Hong Kong has to reposition itself to fulfil its mandate as a special administrative region under the ‘One Country, Two Systems’.

The greatest strength of Hong Kong lies in software (i.e. world-leading universities, excellent education system, global talents, intellectual property rights protection, abundant venture capital, transparent regulatory system) rather than hardware, which is complementary to Hong Kong’s partners across the border. While Hong Kong can take pride in its efficiency and marketisation, a commercial culture over-emphasising instant returns is detrimental to curiosity-driven and time-consuming scientific R&D.

9.2.2 Policy Recommendations

This common transition is made possible by a heavy investment in basic research, commitment of the federal/national government to cutting edge technology in the defence and military sector, provision of venture capital, attraction of global talents and establishment of a regional institutional framework to foster cross-jurisdictional synergy, coordination and management.

Cultivation of a culture conducive to basic scientific research cannot count on the private sector. Only the government can afford to invest in blue-sky R&D. The future of Hong Kong as a regional hub of R&D and technological innovation is widely recognised and envisioned. Whether or not this future can eventually materialise depends to a great extent upon the determination and long-term rather than short-term investments of the HKSAR government.

Drawing insights from international cases and conducting in-depth analyses of the GBA, our contextual and situational inventory clearly identifies considerable potentials of Hong Kong as a (new) regional hub of technological innovation and services. In light of the abovementioned findings, we therefore propose four potential models of economic collaboration between Hong Kong and the PRD.

9.3 Model 1: Hong Kong Becoming a Science and Technology Centre in the GBA

The cost-driven economic development era of the PRD, which has lasted for several decades since the late 1970s, is undoubtedly approaching the end. Complementarities in comparative advantages of Hong Kong and the PRD are diminishing, urging its relationships to transform from a regional production complex towards a regional innovation system, though with cross-boundary characteristics. One of the key objectives for the Guangdong-Hong Kong-Macao Greater Bay Area development is to become an internationally competitive science and technology centre. Therefore, regardless of the aspect of economic development or governance focus, the GBA is running on the avenue towards innovation-oriented development. Hong Kong bears responsibility for developing itself as a science and technology centre and leading the new waves of industrial upgrading and technological innovation in the GBA, due to, if not more, the presence of world-leading universities, globally competitive talents and internationally recognised producer services. The following are findings and policy recommendations on how to facilitate Hong Kong as the science and technology centre in the GBA.

9.3.1 Findings

Based on the data analysis on the Annual Survey of Industrial Firm, the findings are as follows:

- The high-tech development among the Guangdong cities shows a tremendous variance. Cities in the Eastern PRD have a higher percentage of high-tech industrial firms than those in Western PRD.

- High-tech percentages increase as the distance to ports and Hong Kong decreases. The reason that Eastern PRD has larger high-tech firms may be caused by its proximity to Hong Kong, and thus has better access to and can import high-tech material from the world markets.

Based on the Chinese Patent Data, the findings are as follows:

- Firstly, PRD lags behind YRD in both newly granted and total patents.

- Secondly, limited patents are held by universities and jointly held by universities and enterprises in PRD.

- Thirdly, patents in PRD are concentrated within and across industries.

Based on our above analysis, Hong Kong has numerous advantages to offer for being a science and technology centre in the GBA. However, how are these advantages of Hong Kong perceived and used by the high-tech industries in the GBA? The findings of our firm-based survey and interviews of high-tech firms show that:

- The PRD has departed remarkably from the export-oriented labour-intensive development model during the early reform period towards an emerging high-tech industrial development model equipped with a regionally and almost-completed supply system and mixed target market.

- To a greater degree, Hong Kong- and other overseas-funded high-tech firms use Hong Kong services and technological resources, which domestic-funded firms rarely use.

- Intra-firm organisational linkages between parent firms in Hong Kong and branch offices in the PRD form critical channels for labour mobility, technology transfer and producer service provision.

- Hong Kong's key roles in facilitating high-tech industrial development in the

PRD mainly dwell in market expansion, international operation, corporate financing and producer service provision, while the positive effects of investment and entrepreneurship, technological support and labour supply are yet subordinate.

- The above findings further underpin the extreme importance of Hong Kong's special status as an international window for the PRD and Mainland China and its intra-firm cross-border organisational linkages with the PRD for participation in high-tech industrial development.

9.3.2 Policy Recommendations

9.3.2.1 Strengthening Hong Kong R&D Capability

Based on the findings of data analysis on the Annual Survey of Industrial Firms and Chinese Patent Data Analysis, this study presents the following policy recommendations:

9.3.2.1.a Fortify and enhance higher education and innovation capability

Hong Kong's comparative advantage in innovation lies in its research capability, of which its universities can pioneer. However, insufficient communication between researchers and the industry and lack of entrepreneurial experience and business vision among researchers result in low applicability and transferability of findings. Across the border, cities in Guangdong, such as Shenzhen, lack leading research universities but are close to the market and has an excellent ecosystem for entrepreneurs and start-ups. Enhancing the communication and cooperation among the cities in GBA can complement each other's deficiencies and create innovation synergy.

The government needs to unleash the research potentials from Hong Kong universities and encourage the conversion of R&D into production. The universities can further strengthen the contribution of academic research on innovation through better profit sharing and promoting arrangements with its researchers.

9.3.2.1.b Emulate the US Bayh-Dole Act

The government and universities must grant most of the patent profits to inventors

and stimulate additional motivation for R&D commercialisation in universities. Meanwhile, universities and professors can also reasonably appropriate gains from such strategy, increasing the motivation of patents commercialisation and application by patent holders.

A case in point is the Bayh-Dole Act[1], which pushed scientific and technological transformation in the US. Prior to its establishment, the title (ownership) to any invention created using federal funding was owned by the US Government, and no uniform policy was established among federal agencies for the transfer of the invention into the private sector commercialisation, resulting in a lack of incentive to develop any technology and the decline in the number of inventions despite a boom in funding. The act, by allowing the developer to retain rights to inventions, caused a significant growth in the number of inventions while inviting the involvement of universities in managing the patenting of licensing.

9.3.2.1.c Adjust the University Grants Committee's assessment criteria on funding universities to prompt more social and value-added research projects

The University Grants Committee can adjust the assessment criteria on universities' funds allocation by increasing the weight of research projects with potential positive social effects and contribution beyond the ordinary teaching and scientific research assessment. This increase encourages universities to carry out more socially valuable studies. Furthermore, while balancing the basic research, the committee can also adjust the assessment criteria of university researchers. We suggest a focus on the conversion and application of scientific research progress and encourage professors to participate in other school-enterprise cooperation projects.

9.3.2.1.d Attract, train and cluster world top talents, scholars and entrepreneurs to Hong Kong as an Innovation Centre

Having a free, open and high-quality environment, with universities enjoying the advantage of strong scientific research can attract more talents from all over the world to carry out scientific, innovative and entrepreneurial research and form a world-class innovation centre in Hong Kong. If an excellent career platform can be provided for high-end talents and effectively solve problems including living costs, education for children, and elderly support, then Hong Kong can attract the best talents across the world. Not only should talents be stationed within the city but also sent to the GBA. A

cooperation platform for university students and enterprises in Guangdong can encourage university students to participate in internships and help Hong Kong youth to contribute to the GBA.

9.3.2.2 Making Hong Kong R&D Usable in the GBA

Our firm-based survey shows considerable obstacles for the GBA in using Hong Kong technology and interaction with R&D. The major recommendations in breaking these barriers and building bridges are as follows:

9.3.2.2.a Need to inform PRD high-tech firms of Hong Kong high-tech innovations

Although PRD high-tech firms generally perceive that Hong Kong high-technology innovations are advanced and enjoy immense room for collaboration with Mainland China, they lack a good understanding of such innovations because they do not have the channels or opportunities for contact. Therefore, Hong Kong needs to find ways to let PRD high-tech firms become more familiar with Hong Kong high-technology innovations.

Policies recommended are : 1) encourage Hong Kong high-technology innovations to engage more with the domestic market by adapting to as well as combining with the business models there; 2) facilitate Hong Kong high-technology firms to set up branch offices in the PRD to participate in various promotion activities (e.g. exhibitions and conferences) and interact with domestic firms in Mainland China; and 3) facilitate PRD high-tech firms to set up offices in Hong Kong to engage them with Hong Kong's high-tech innovation system so that they can promote the use of Hong Kong's high-tech innovations to other firms in the PRD.

9.3.2.2.b Further enhancement of Hong Kong R&D environment

Hong Kong's immature innovation system such as a low R&D intensity and public expenditure and an underdeveloped venture capital system for high-tech start-ups has become a key constraint for commercialising such technologies. Given that the PRD has an increasingly huge market and a complete industrial classification, Hong Kong must further enhance its R&D environment to at least complete upstream innovation

activities to better align with the middle and downstream complementarities in the PRD.

Policies recommended are : 1) the Hong Kong SAR government should increase the level of public expenditure on R&D to encourage high-tech entrepreneurship, and existing companies should increase their investment in upstream technology research and development; 2) in addition to applied research, more resources should be given to the Hong Kong Research Grants Council for strengthening Hong Kong's basic research in order to develop applied research; 3) improve the mechanism of universities in Hong Kong in encouraging scholars to transfer their research results to industries, so as to facilitate the commercialization of Hong Kong's high-tech innovation.

9.3.2.2.c Development of 0-1 and 1-100 start-up industries in Hong Kong and High-Tech Firms in the PRD

The results of the study show that the internal organizational connections of the enterprise itself are extremely important for the interactive development of the high-tech industry between Hong Kong and the Pearl River Delta. Therefore, Hong Kong must strive to develop its innovation and local high-tech industries to prevent the hollowing out of the local industries. Hong Kong needs to make good use of its innovative high-tech research to develop its local start-up industries to avoid relying solely on the Pearl River Delta industries.

Policies recommended are : 1) strengthen support for local start-ups and high-tech industries through different policy tools (such as providing more industrial land and financial support for strategic industries); 2) encourage companies to develop and commercialize Hong Kong's hi-tech innovations and improve the local innovation chain; 3) improve the high-tech infrastructure, labour force, and market sales channels for promoting Hong Kong's hi-tech industry; 4) encourage Hong Kong companies to set up factories or branches in the Pearl River Delta Plants to strengthen the ties with Hong Kong's high-tech R&D and producer service industries. .

9.3.2.2.d 'Front Desks, Back Offices' to obtain business in the PRD

The emergence and increasing decoupling effects between Hong Kong's producer services and the PRD's high-technology industries are alarming, against the trends of the former's rapidly rising producer services and the latter's declining role as a unique window to the world. Unlike manufacturing activities that relocated to the PRD since the early 1980s for a lower production cost and vertical disintegration, Hong Kong's

producer services have not formed robust cross-border linkages to cater to both overseas and domestic markets.

Policies recommended are : 1) encourage Hong Kong producer service companies to establish branch offices in the Pearl River Delta to seize the growing domestic market demand in the region and the mainland, thereby forming a "Front Desks, Back Offices" cross-border regional cooperation model in the producer service industry to replace the past economic cooperation model of "Front Shops, Back Offices" economic cooperation model; 2) improve cross-border cooperation to facilitate the flow of data and information; 3) improve cross-border linkages of the producer service industry, for example, mutual recognition of professional qualifications and unified service standards and rules for all cities in the Greater Bay Area.

9.3.2.2.e Smart GBA research and business visa and counters

Improving the cross-border flow of various production resources (e.g. goods, capital, labour force, data, information) can better facilitate Hong Kong to participate in high-tech industrial development in the PRD. The above research findings show that information asymmetry and one-way free flow of labour force (i.e. from Hong Kong to Mainland China) prohibits Hong Kong from sufficiently engaging in the GBA development. Smart information technologies and potential new regulations on cross-border visa arrangement may help to fill in this gap.

Policies recommended are : 1) use the new generation of information and communication (ICT) technology to promote the development of Smart Guangdong-Hong Kong-Macao Greater Bay Area to facilitate the flow of people, goods, capital and information; 2) innovate cross-border Visa application arrangement by using smart technology to provide Mainland’s and overseas businessmen and researchers in the Greater Bay Area with just-in-time online business Visas, and set up the Greater Bay Area Business and Research cross border counter, like the existing APEC (Asia-Pacific Economic Cooperation) counter to facilitate them to come to Hong Kong to do business and research.

9.4 Model 2: Hong Kong Riding on Increasing Affluence of Mainland Consumers and Investors

China has shown increasing importance as a market and a source of capital to Hong

Kong. Hong Kong companies used to consider the Mainland as a production place with low cost of land and labour. As the income and wealth of Mainland consumers and investors grow, Hong Kong companies need to alter their view and consider the Mainland as an important market. Chinese residents in first-and-second-tier cities have increasing need for diversified investment and high-end consumption. Hong Kong can offer a one-stop service centre of wealth management for Mainland investors. Moreover, Hong Kong companies need to diversify the sourcing countries for several important imports of consumption goods to meet the growing demand of Chinese consumers. In this case, Hong Kong can tap into its linkages with former commonwealth countries while Macao has linkages with Portuguese-speaking regions in the world. The followings are findings and policy suggestions on how to meet the demand of Chinese investors and consumers.

9.4.1 Stock Investment

The Hong Kong stock market and the Mainland A-shares market are distinct and have complementary advantages. The attraction for investors also differs, which mutually brings incremental funds. Nonetheless, the Hong Kong stock market is prominent but not sufficiently diversified, while A-shares are diverse but have lacklustre advantages.

9.4.1.1 Findings

Mainland residents participating in overseas investments has limitations on channels and product types, for which the main channels are Shenzhen-Hong Kong Stock Connect and Shanghai-Hong Kong Stock Connect under the Connectivity Mechanism.

The two stock markets have distinct characteristics. Hong Kong H-shares concentrate on the financial and information technology industries while Mainland A-shares cover a broader range and are more dispersed.

Mainland investors mainly invest in private enterprises in the Hong Kong stock market. These investments concentrate on biotechnology and information technology industries. However, Hong Kong and overseas investors primarily focus on the manufacturing industries in A-shares. The two stock markets have complementary

advantages and can bring incremental funds to each other.

9.4.1.2 Policy Suggestions

9.4.1.2.a Further relax product restrictions and investment quotas in the security market

Further relaxation of product restrictions and investment quotas in the security market under the Connectivity Mechanism and expansion of existing trading channels are suggested to optimise asset allocation of investors from both Hong Kong and the Mainland.

9.4.1.2.b Attract more high-quality enterprises from the Mainland to list on the Hong Kong Stock Exchange

Attract more high-quality enterprises from the Mainland to list on the Hong Kong Stock Exchange and more talents that are familiar with the Mainland industries. Thus, industry components can be diversified and the return on investment of Hong Kong stocks can increase, attracting more capital to invest in Hong Kong.

9.4.1.2.c Introduce more talents who are familiar with Mainland enterprises

Introduce more talents who are familiar with Mainland enterprises and securities markets to increase the investment return on Hong Kong stocks, thereby attracting more Mainland investors to invest in Hong Kong.

9.4.2 Insurance Financing

Attracting Mainland capital to invest in the Hong Kong insurance market can optimise the asset allocation of Mainland customers and promote the development of the insurance industry in Hong Kong. The findings and suggestions are as follows:

9.4.2.1 Findings

- The Hong Kong insurance industry has proliferated in recent years, focusing on individual life business, centred on annualised non-linked insurance. The

market is occupied by three giant companies. Products are mostly in US$ and HK$. For offshore insurance, the majority of investments comes from the Mainland.

- Mainland businesses are the core driving force of the development of the Hong Kong insurance industry. Before 2016, the premiums from the Mainland first rose significantly, then later sharply declined due to foreign exchange control. Thus, the business composition subsequently changed. In 2018, the proportion of annualised insurance from the Mainland increased while that of the single premium insurance decreased.

- Hong Kong insurers have advantages and opportunities such as globalised service, high investment returns, high employee and customer quality, low insurance premiums, high sum assured, wide coverage and tax and debt avoidance. Nevertheless, they also have disadvantages and risks on fluctuating returns, high exchange rate risks, troublesome claim-making and increasingly intensive competition from Mainland insurers.

9.4.2.2 Policy Recommendations

9.4.2.2.a Vigorously develop RMB insurance products and improve the investment capacity of RMB assets

RMB insurance products need vigorous development and improvement of the investment capacity of RMB assets. As an essential centre for offshore RMB transactions, Hong Kong developing RMB insurance business can avoid the risks brought by changes in foreign exchange policies in the Mainland and help to grasp such considerable market. After the strengthening of foreign exchange regulations in 2016, Mainland customers increasingly demand for RMB insurance products from Hong Kong. However, leaders of the Hong Kong insurance industry are all Anglo-American insurers and are not experts in investing RMB assets, resulting in low returns. Therefore, we recommend that Hong Kong insurers hire more specialists in RMB asset investment and thus improve their related investment capabilities and competitiveness.

9.4.2.2.b Adjust product types to develop annualised non-linked products (such as permanent life insurance).

Product types must be appropriately adjusted to develop annualised non-linked

products (such as permanent life insurance) to meet the increasing demand for insurance investment from Mainland customers and avoid the instability caused by changes in Mainland policies (such as reduce or terminate the universal life policies) to maintain business growth.

9.4.2.2.c Actively participate in the development of the GBA and deepen the insurance cooperation between Guangdong, Hong Kong and Macao.

Finally, actively participate in the development of the GBA and deepen the insurance cooperation between Guangdong, Hong Kong and Macao. According to a survey conducted by the Hong Kong Trade Development Council in 2018, the Mainland middle-classes intend to invest 27% of their household income in insurance and financial investment in 2017, an increase of approximately 10% over 2013. With the rapid economic development of the GBA, the middle class has increased year by year, and so does their demand for wealth management and health care. The nine Mainland cities in the GBA only have insurance density by US$1,000 and insurance penetration rate by 6%, and therefore the potential for development is enormous. The Outline development plan for the Guangdong-Hong Kong-Macao Greater Bay Area and the Hong Kong Insurance Authority together propose to establish the GBA Insurance Service Centre and to set up Insurance Connect as a long-term goal. These plans can promote the development of the Hong Kong insurance industry in the GBA and attract more Mainland customers.

9.4.3 Import

As one of the world's top five international free ports, Hong Kong has advantages for its re-exports by having low trade barriers and low import and export taxes. Except for a few products, all other imported goods are not taxed and many categories do not require customs clearance. Therefore, as a re-export centre supporting the Mainland, Hong Kong can help Mainland enterprises to import at a lower cost. Furthermore, according to The Mainland and Hong Kong Closer Economic Partnership Arrangements (CEPA), most Hong Kong products exported to the Mainland also enjoy the benefit of zero tariffs.

However, the dependence of the Mainland on Hong Kong re-exports is decreasing

year by year. As the Mainland deepens the reform and opening up, many of its ports continuously improve. Imports in the PRD, YRD, and the Beijing-Tianjin-Hebei Urban Agglomerations together with other regions have expanded substantially whereas those passing through Hong Kong have declined. The efficiency of customs duties in the Mainland has also been enhanced, and the customs clearance process has been simplified. In 2018, the customs clearance time was reduced to 65 hours. In addition, the transformation and upgrade of industries in the PRD and the departure of manufacturing industries away from Hong Kong are the two main reasons responsible for Hong Kong slowing down in import on manufacturing products for the Mainland.

9.4.3.1 Findings

- The growth rate of Hong Kong conducting imports for the Mainland has slowed down in recent years. These imports are mostly manufactured re-export products (electrical machinery, telecommunication and sound recording equipment) and usually transported by land to the PRD.

- The Mainland imports limited consumption goods from Hong Kong, which mainly include food, luxury goods and clothing, trading with partners such as the US, France and Chile. Compared with the Mainland, Hong Kong product sources and types are better diversified, providing its trading advantage on intermediate goods.

- Hong Kong has a distinct advantage in merchandising luxury goods. In particular, the local jewellery retail industry accounts for approximately half of the sales of major Hong Kong-listed jewellery companies.

9.4.3.2 Policy Recommendations

9.4.3.2.a Establish trade with second- and third-tier cities to bring new growing areas to the trade and logistics industry

Hong Kong should not merely focus upon serving consumers in the GBA, but also consider the broad market of the entire Mainland, especially the second- and third-tier cities and several less developed regions. Establishing trade with these regions can bring new growing areas to the trade and logistics industry of Hong Kong.

9.4.3.2.b Adopt a more diversified mode of transportation for trade to further develop in markets outside of the GBA

A more diversified mode of transportation can be adopted to further develop in markets outside of the GBA, thereby expanding the existence of Hong Kong import assistance.

9.4.3.2.c Vigorously explore new product types and trading partners.

Thirdly, vigorously explore new product types and trading partners, especially expanding on products with high demand in the Mainland. Appropriately expand the re-exports in products such as automobiles, pharmaceuticals, medical devices and high-tech products, while maintaining advantageous products such as foods and luxury goods. This strategy can provide Mainland consumers with differentiated and better product options.

9.4.3.2.d Encourage local talents in Hong Kong to work and train in the GBA

Introduce talents who are familiar with the Mainland consumption market and encourage local talents in Hong Kong to work and train i***n*** the GBA. After learning and accumulating experience in the Mainland, recruit them back to Hong Kong to achieve a virtuous cycle with a dynamic exchange.

9.5 Model 3: Hong Kong Leading the GBA to Go Global and Drive New Economic Growth from Enterprise Internationalisation

In the long term, the low-cost strategy from China and the PRD is not sustainable. Enterprises need to actively carry out technological innovation and transfer low value-added production to low-cost countries and regions through the Belt and Road Initiative.

Hong Kong has the advantages of a world-leading business environment, mature modern economy, well-developed financial services industry and a highly internationalised free trade system. According to the '*Outline development plan for the Guangdong-Hang Kong-Macao Greater Bay Area*', the region is to be developed into an essential support zone, investment and financing base and overseas service hub of the Belt and Road Initiative. As a leader of the GBA, Hong Kong can seize the

opportunity to make full use of its advantages, promote the internationalisation of Hong Kong and Mainland enterprises and help more companies to better integrate into the global market system, thereby further enhancing global competitiveness. This strategy can drive the GBA to achieve industrial transformation and upgrading, and form a new impetus for economic growth.

9.5.1.1 Findings

Data are derived from the Securities Data Corporation and the Orbis Global Enterprise Database. Study and analysis of the acquisition model and post-acquisition performance of bidders and Hong Kong bidders in Mainland China reveals the following:

- From the industrial distribution of acquiring company side, Hong Kong financial enterprises account for over half of the total, and the Mainland enterprises are diverse in industry distribution. However, in terms of acquired companies, the situation is the opposite. Hong Kong enterprises acquire a wider and more balanced industry whereas the Mainland values and focuses on manufacturing enterprises.

- The proportion of the high-tech industry is low for Hong Kong acquirers at less than 10% in recent years. By contrast, that of the Mainland acquirers is high. In terms of the degree of technology of the acquired companies, the recent proportions of high-tech companies acquired by Hong Kong companies are low at less than 20% while those of Mainland companies are high.

- From the geographical location of the acquired company, Hong Kong enterprises have a higher proportion of overseas companies and more international advantages than the Mainland, which mainly acquire companies in China. However, judging from the total number of companies located in overseas countries, Hong Kong has gradually weakened its advantages in recent years and may soon be outbalanced by the Mainland.

- Hong Kong has a high proportion of targets in the countries and regions along the Belt and Road. By contrast, most of the acquisition targets of the Mainland enterprises are within China. In addition, the target overseas companies acquired by Hong Kong are mostly Southeast Asian countries while those of the

Mainland are mainly European companies.

- From the perspective of the efficiency on acquisitions, Hong Kong enterprises have a high degree of completion, which is more professional and mature in merging business. By comparison, Mainland enterprises have lower completion of their M&A activities.

Study of the subsidiaries of the Mainland and Hong Kong worldwide reveals the following:

- The proportion of overseas subsidiaries of Hong Kong enterprises is much higher than that of Mainland enterprises, and the degree of internationalisation is high.
- The overseas subsidiaries of Hong Kong enterprises are highly concentrated in the Commonwealth countries and regions, with clear location preferences. Mainland enterprises have no apparent regional preferences.
- Hong Kong subsidiaries in the countries along the Belt and Road account for a higher proportion of the global total than the Mainland. However, due to the large number of overseas Hong Kong subsidiaries, those located in the countries of the Belt and Road account for fewer proportions than that of the Mainland.

9.5.2 Policy Recommendations

9.5.2.1 Encourage a more diversified industrial development pattern

Hong Kong should adjust its industrial policies to encourage a more diversified industrial development pattern. More professionals from different industries can be employed to cultivate the suitable ones under Hong Kong characteristics, to enable the economy to avoid risks arising from uneven industrial development.

9.5.2.2 Commit to nurturing high-tech industries and enhancing the competitiveness of science and technology

Hong Kong should be committed to nurturing high-tech industries and enhancing

the competitiveness of science and technology. Policies can be formulated to encourage Hong Kong enterprises to invest in high-tech industries, and to help Mainland high-tech enterprises to enter the international market. More high-tech industry and innovative talents can be encouraged to work in Hong Kong.

9.5.2.3 Commit to financial technology (fintech) to ensure the status and advantages of international financial centres.

Hong Kong must commit to fintech to ensure the status and advantages of international financial centres. The government can adopt policies to reform funding and financing, reduce the approval process and supervision methods and support local financial enterprises in technological innovation, especially through the acquisition of overseas or Mainland financial technology enterprises. These strategies can help quickly make up for shortcomings and strengthen financial enterprises.

9.5.2.4 Participate in international economic cooperation and competition in the construction of the Belt and Road

Hong Kong must enhance its ability to participate in international economic cooperation and competition in the construction of the Belt and Road. The high degree of specialisation and internationalisation of its enterprises can be utilised in the global M&A activities to help Hong Kong and Mainland enterprises promote the effective connection of international and domestic markets, promote the development of high-end and high-value-added enterprises, and participate in higher-level international economic cooperation and competition.

9.5.2.5 Encourage companies to expand the scope of overseas investment and strengthen their international advantages.

The Hong Kong government needs to encourage companies to expand into a broader overseas market and strengthen their international advantages. By expanding the scope of overseas investment, Hong Kong enterprises can enhance their overseas holding capacity, enjoy more dividends brought about by internationalisation and evade the regional risks of investing overseas. At the same time, Hong Kong can also help Mainland enterprises to enter overseas markets and enhance their international competitiveness with their advantages.

9.6 Continuously Utilise Hong Kong to Provide Producer Services for Mainland Enterprises by Setting up a 'Front Store' in the GBA and Improving Access to Customer Groups

In the last model, we suggest that Hong Kong continue to reinforce its role as a financial centre and provide producer services to the Mainland enterprises. Hong Kong can attract high-quality companies to list in Hong Kong Exchange and talents who have expertise in the Mainland industries. Hong Kong must also strive to offer offshore bond financing for the Mainland enterprises and establish itself as a green financial centre.

While moving up (technologically) and moving out (to Belt and Road Initiative) are key to sustainable growth in China, the Mainland has to rely on low-cost manufacturing for a significant period of time to absorb its surplus labour. Hong Kong can continue to utilise its advantage in producer services and further provide value-added services to both the overseas buyers and the manufacturers in the Mainland. The following are several findings and policy suggestions.

9.6.1 Findings

Below are the development trends and policy recommendations for Hong Kong services to Mainland enterprises in financial and trade services through Hong Kong assistance to Mainland corporate finance and their trade data.

Hong Kong can help Mainland enterprises overcome financial difficulties.

Given the restrictions of China's listing system, Mainland enterprises, especially a large number of private ones, face difficulties in listing and financing:

- The number of Mainland companies listed in Hong Kong has considerably increased. Among the number of Mainland companies listed in Hong Kong, private enterprises account for more than half and have become the main force for Hong Kong listings.

- The number of private enterprises issued by the Mainland in the Hong Kong stock market, market value and the transaction amount are increasing.

- The new listing system has attracted several of the 'new economy companies' in the Mainland to seek financing in Hong Kong. However, over the years, Hong Kong's capital market listed companies have concentrated on financial services

and real estate, while biomedical and technology companies are rare. Their share is far behind those of the New York Stock Exchange and Nasdaq. The Hong Kong stock market is listed for more companies in different industries. Financing has much room for improvement.

- The Hong Kong stock market is unevenly distributed, and dominated by the financial industry. For a long time, the finance, real estate and consumer goods services industries has been well-developed and employ numerous talents with rich market experience. Nonetheless, Hong Kong lacks understanding of new technology companies and the information technology industry. In terms of valuation, the Hong Kong market has a higher level of understanding of the Mainland finance, new energy and consumer goods industries, but not of other fields.

- In terms of debt financing, the demand for Mainland enterprises is enormous, especially on infrastructure construction projects in the GBA.

- Green bond financing for green projects was in high demand during the industrial upgrading of the Guangdong-Hong Kong-Macao Greater Bay Area. However, these places have no uniform green definition. In addition, green bonds can increase the cost burden of enterprises.

In terms of providing trade logistics services for Mainland enterprises, analysis of data from the Hong Kong Bureau of Statistics and China Customs data reveal that:

- Trade is Hong Kong's most important pillar industry, accounting for more than 20% of national GDP. Mainland China is the most important import and export destination and the second-largest destination for offshore trade.

- In recent years, Hong Kong status as an entrepot for the Mainland has gradually declined, and its role of 'front store' has weakened.

- The import and export shares of the YRD is on the rise, while those of the PRD is on the decline. Overall, the geographic location of companies exporting through Hong Kong shows a trend of expansion from the southeast coast to the inland areas, and the proportion of state-owned enterprises has declined in recent years. In addition, Hong Kong-funded enterprises gradually moved inland, where the growth of Hong Kong-invested enterprises' exports increased.

- Hong Kong mainly exports a single product category, mainly concentrated on the manufacturing industries such as electromechanical, audio-visual equipment and their parts and accessories. At the same time, the export structure of the Mainland is also dominated by the export of manufacturing products.

9.6.2 Policy Recommendations

On perspectives of Hong Kong's provision of financial services to Mainland enterprises:

9.6.2.1 Understand private companies in the GBA to meet their needs

Hong Kong Financial Services Industry can strengthen the understanding of the GBA companies, especially private companies, to meet their development status and financing needs.

9.6.2.2 Improve the current listing system

The current listing system needs improvement and high-quality companies from diverse industries must be encouraged to list in Hong Kong.

9.6.2.3 Attract talents in all fields

Hong Kong capital markets are in demand of talents who understand the Mainland industries, especially those in high-tech and clean energy. Hong Kong can attract talents in all fields by offering fast-tract talent immigration channels. Hong Kong can also attract the top talents through its high-quality education and medical system and by providing favourable housing assistance.

9.6.2.4 Support the GBA companies in offshore debt financing

Hong Kong can support the GBA companies in offshore debt financing. In the construction of the Guangdong-Hong Kong-Macao Greater Bay Area, infrastructure such as transportation, communications, electrical and pipelines are important parts of its development. The huge scale of infrastructure construction has brought enormous demand for infrastructure financing. The funds of the Hong Kong offshore market can

be a useful supplement to the debt financing if the domestic financing cost is high.

9.6.2.5 Clearly define green standards and provide green bond issuers subsidies

Hong Kong needs to communicate and cooperate with the Mainland to clearly define green standards. Subsequently, the Hong Kong government can provide green bond issuers subsidies to reduce overall financing costs.

The following are suggestions for Hong Kong's provision of trade services to Mainland enterprises:

9.6.2.6 Strengthen the core competitiveness of 'front store' and further provide value-added services

Hong Kong needs to understand the needs of Mainland enterprises and further provide value-added services. In certain cases, the value-added services[2] provided by trading companies for their traditional businesses have exceeded the original scope of exporters. For example, several companies design and produce parts for suppliers' factories for the production of manufactured goods. Most of the added value of these companies comes from their design team, and their competitive advantage lies in their ability to design products that prosper in the target market.

9.6.2.7 Continue to assist overseas buyers in coordinating production activities in the Mainland

Hong Kong companies assist overseas customers to inspect the products of manufacturers, ensure that the products meet the importing standards and supervise the production schedule to ensure on-time delivery. Hong Kong companies can also assist buyers in coordinating production activities when orders from overseas buyers surge or require prompt delivery.

9.6.2.8 Know more about Mainland products and overseas demand

The export commodities structure is too simplistic. Hong Kong needs to know

more about Mainland products and overseas demand, and actively seek more competitive sources of supply and diversified sales[3].

9.6.2.9 Establish more connection poles in the PRD

Follow the Hong Kong enterprises that previously moved to the Mainland. Subsequently, establish more connection poles in the PRD region and embrace the demand of their private enterprises.

9.7 Conclusion

Hong Kong and the Pearl River Delta (PRD) have collaborated intensively and effectively since China's economic reform and opening up in the late 1970s. However, the once-successful market-driven economic cooperation model of 'Front Shop, Back Factory' encountered severe challenges in the face of the rapidly rising economic power of PRD and the declining role of Hong Kong as its bridge to the world in the last two decades, and the more recent turbulence in the global political and economic landscapes, such as the China-United States (US) trade war and the Coronavirus 2019 (COVID-19) pandemic. Of the traditional Four Key Industries, namely financial services, tourism, trading and logistics, and professional and producer services, only financial services and professional services are the industries that Hong Kong still has an edge, in addition to new high-tech industry that is rapidly developing in the PRD.

This Strategic Public Policy Research Study aims to search for new economic cooperation models between Hong Kong and the Greater Bay Area (GBA) and presents policy recommendations based on the research findings. The five key research objectives are as follows: 1) to comprehensively review recent development trends of the PRD and its economic cooperation with Hong Kong; 2) to examine the competitiveness of the PRD with Yangtze River Delta (YRD) in the new economic environment of China and the world; 3) to explore what and where industries need to be promoted in Hong Kong and the PRD to enhance its competitiveness; 4) to examine new economic cooperation models between Hong Kong and the PRD in the new local and global economic environment; and 5) to identify opportunities and barriers of these new economic cooperation models. To achieve these objectives, we first review socioeconomic development, high-tech innovation, and cross-jurisdictional synergy of world-leading bay areas, and thus draw lessons for the development of the Guangdong-

Hong Kong -Macao Greater Bay Area. A SWOT analysis of Hong Kong in fostering new economic development of the GBA under the "One Country, Two Systems" framework is then carried out. By identifying the strengths, weakness, opportunities and threats of Hong Kong, we examine four potential economic cooperation models: (1) Model 1 - Hong Kong becoming a science and technology centre in the GBA; (2) Model 2 - Hong Kong riding on increasing affluence of Mainland consumers and investors; (3) Model 3 - Hong Kong leading the GBA to go global and driving new economic growth from enterprise internationalisation; and (4) Model 4 - Continuously utilising Hong Kong to provide producer services for Mainland enterprises, setting up 'front desk' in the GBA and improving access to customer groups.

Model 1 is analysed by examining the high-tech development in Guangdong and the innovation performance and participation of Hong Kong in the high-tech industrial development in Dongguan and Shenzhen. The Annual Survey of Industrial Firms, patent analysis, firm-level questionnaire survey and in-depth interviews in the PRD are used as bases. For Model 2, Hong Kong's producer service performance on stock investment, insurance financing and importation is analysed. Hong Kong's overseas investment and international holding capacity and its contribution to the construction of the Belt and Road are examined for Model 3. Hong Kong's equity and bond financing and trade and logistics performance are evaluated for Model 4.

This study began in 15 April 2018, and since then numerous events have affected Hong Kong and the GBA. Thus, this study has been heavily affected by the social unrest in 2019 and then by COVID-19 in 2020 to the present. We have exerted our best efforts to complete this research project despite these severe obstacles, especially in carrying out fieldworks and questionnaire surveys in the Pearl River Delta in late 2020 and early 2021 as affected by COVID-19 travel restrictions. The summary of the major findings of the study are as follows:

9.7.1 Comparative Study of Bay Areas

Our comparative analysis with other major Bay Areas in the world -- New York, San Francisco and Tokyo -- suggests that the sheer size of the GBA stands out, although its gross domestic product (GDP) per capita still lags behind. Economic, geographical and political transformation in the American and Japanese bay areas all started with the growth of port functions and related manufacturing activities, followed by the

dominance of advanced producer services. Eventually, the bay areas transformed into global and national hubs of high-tech industries. This development path appears to be the direction that GBA is aiming for.

Hong Kong has always functioned as an important gateway connecting the Chinese economy and the rest of the world under the "One Country, Two Systems". The greatest strength of Hong Kong lies in software (i.e. from its world-leading universities, excellent education system, global talents, intellectual property rights protection, abundant venture capital, transparent regulatory system, professional conduct rather than hardware, which is complementary to Hong Kong's partners across the border.

9.7.2 Model 1 - Hong Kong becoming a science and technology centre in the GBA

The degree of high-tech development considerably varies among the Guangdong cities. Those in the Eastern PRD have a higher percentage of high-tech industrial firms. Such percentage increases as the distance to ports and Hong Kong decreases. The PRD lags behind the YRD in both newly granted and total patents. Moreover, PRD has much fewer patents held by universities as well as patents jointly held by universities and enterprises. Rather, patents in the PRD are more concentrated within and across industries.

Recently, the PRD has departed remarkably from the export-oriented and labour-intensive development model during the early reform period towards an emerging high-tech industrial development model equipped with a regional and almost-complete supply system and mixed target market. Intra-firm organisational linkages between parent firms in Hong Kong and branch offices in the PRD form critical channels for labour mobility, technology transfer and producer service provision. Hong Kong's key roles in facilitating high-tech industrial development in the PRD now mainly relate to market expansion, international operation, corporate financing and producer service provision, whereas its previous advantages in investment and entrepreneurship, technological support and labour supply are declining.

Despite its R&D advantages due to having the largest number of internationally high-ranking universities in the GBA, the use of Hong Kong's R&D by high-tech firms in the PRD is limited. In particular, Hong Kong -funded and other overseas-funded

high-tech firms still partially use Hong Kong's knowledge and technological resources, but domestic-funded firms are using much less of such support. The main reasons are its higher cost in comparison to that in Mainland China, insufficient alignment with Mainland China's market demands, limited penetration into the awareness of business communities in the PRD, unawareness of Hong Kong's innovation and technology and the rapid development of the PRD in competition with Hong Kong. The decoupling of Hong Kong technologies with domestic-funded and other overseas-funded firms in the PRD are apparent and alarming, and thus merit considerable attention.

For Model 1, we propose the following policy recommendations : 1) fortify and enhance higher education and innovation capacity; 2) emulate the US Bayh-Dole Act to provide more incentives for university researchers to engage in technology transfer; 3) adjust the assessment criteria of the University Grants Committee for funding universities to prompt more socially value-added research projects; 4) attract, train and cluster world top talents, scholars and entrepreneurs for Hong Kong as an Innovation Centre; 5) promote Hong Kong's high-technology innovations to PRD high-tech firms; 6) further enhance Hong Kong's R&D environment by increasing public expenditure on R&D, giving more resource to the Research Grants Council for strengthening Hong Kong's basic research in order to develop applied research, and improving the mechanism of universities to encourage scholars to commercialize their research results; 7) develop 0-1 and 1-100 start-up industries in Hong Kong and high-tech firms in the PRD; 8) facilitate 'Front Desks, Back Offices' to obtain producer service businesses in the PRD; and 9) implement smart "just-in-time" GBA business and high-tech visa and cross-border counters to facilitate Mainland businessmen to come to Hong Kong to do business and understand our high-tech innovation and Mainland scholars and researchers to carry out collaborative research in developing Hong Kong as a GBA international innovation hub.

9.7.3 Model 2 - Hong Kong riding on increasing affluence of Mainland consumers and investors

Mainland residents who wish to participate in overseas investments have limited channels. At present, the Shenzhen-Hong Kong Stock Connect and Shanghai-Hong Kong Stock Connect under the Connectivity Mechanism are the main channels. Hong Kong H-shares concentrate on the financial and information technology industries.

Mainland A-shares are more dispersed and cover a broader range of industries. Mainland investors concentrate on private enterprises in the Hong Kong stock market, specifically in biotechnology and information technology industries. In turn, Hong Kong and overseas investors primarily invest in the manufacturing industries in A-shares. Thus, the two stock markets have complementary advantages and can bring incremental funds to each other.

Majority of the source of income of the Hong Kong insurance industry comes from Mainland residents. The best sellers are individual life businesses, centred and annualised non-linked insurance. Before 2016, the premium income from the Mainland significantly increased, then later declined sharply due to foreign exchange control. Thus, the business composition changed subsequently. In 2018, the proportion of annualised insurance from the Mainland proliferated, while that of single premium insurance declined. Hong Kong insurers have advantages of high investment returns, high employee and customer quality, low insurance premiums, wide coverage and tax and debt avoidance. Nevertheless, the disadvantages and risks are fluctuating returns, high exchange rate risks, troublesome claim-making and increasingly intensive competition from Mainland insurers.

The growth rate of Hong Kong imports from overseas to the Mainland has slowed down in recent years. These imports are mostly manufactured intermediate products (e.g. electrical machinery, telecommunication and sound recording equipment), that are commonly transported by land to the PRD. Consumption goods imported from Hong Kong to the Mainland hold little shares. These goods include food, luxury products and clothing, from trading partners such as the United States (US), France and Chile. Compared with other ports in the Mainland, Hong Kong's imported product sources and types are more diversified. In addition, Hong Kong has a distinct advantage in merchandising luxury goods.

For Model 2, we propose the following policy recommendations : 1) further relax product restrictions and investment quotas in the security market under the Connectivity Mechanism, and broaden existing trading channels that aim to optimise asset allocation of investors from both Hong Kong and the Mainland; 2) attract more high-quality enterprises from the Mainland to list on the Hong Kong Stock Exchange and more talents that are familiar with the Mainland industries; 3) introduce more talents who are familiar with Mainland enterprises and securities markets to increase

the investment return on Hong Kong stocks; 4) vigorously develop RMB insurance products and improve the investment capacity of assets; 5) appropriately adjust product types to develop annualised non-linked products (such as permanent life insurance); 6) actively participate in the development of the GBA and deepen the insurance cooperation between Guangdong, Hong Kong and Macao; 7) go beyond serving consumers in the GBA, but also consider the broad market of the entire Mainland, especially the second- and third-tier cities and less developed regions; 8) adopt a more diversified mode of transportation and thereby further develop in markets outside of the GBA; 9) vigorously explore new product types and trading partners, especially expanding on products with high demand in the Mainland; and 10) introduce talents who are familiar with the Mainland consumption market and encourage local talents in Hong Kong to work and be trained in the GBA.

9.7.4 Model 3 - Hong Kong leading the GBA to go global and driving new economic growth from enterprise internationalisation

For this model, we compare the M&A events in Hong Kong and the Mainland. In terms of industry distribution, Hong Kong's acquirers are mostly from financial industries (accounting for over 50%) and have low percentages of high-tech firms. By comparison, the Mainland's acquirers are more diverse in industry and have high percentage of high-tech firms. Hong Kong's acquirees have a wider range of industries and less concentrated than the Mainland's acquirees. In terms of geographical location, Hong Kong's acquirees are more internationalised and have a high proportion of targets in the Belt and Road countries and regions. By contrast, most of the Mainland's acquirees are local firms. In addition, the overseas acquirees of Hong Kong are mostly from Southeast Asian countries, while that of the Mainland are mainly from European companies. In terms of efficiency on acquisitions, Hong Kong enterprises have a high degree of completion, showing a higher professionalism and maturity in M&A. By studying subsidiaries of Mainland and Hong Kong enterprises, we find that the latter is much higher than that of the former, showing a high degree of internationalisation. Overseas subsidiaries of Hong Kong enterprises are highly concentrated in the Commonwealth countries and regions, and their location preferences are clear. Mainland enterprises have no apparent regional preferences. Among the acquirers targeting the companies in the Belt and Road, Hong Kong has a larger share than the

Mainland.

For Model 3, we propose the following policy recommendations : 1) adjust industrial policies to encourage a more diversified industrial development pattern; 2) Hong Kong to commit to nurture high-tech industries and enhance the competitiveness of science and technology; 3) Hong Kong to commit to financial technology innovation to ensure the status and advantages of international financial centres; 4) broaden Hong Kong participation in international economic cooperation and competition in the construction of the 'Belt and Road'; and 5) the Hong Kong government needs to encourage companies to expand into a broader overseas market and strengthen their international advantages.

9.7.5 Model 4 - Continuously utilising Hong Kong to provide producer services for Mainland enterprises

We analyse the listed companies in Hong Kong and find that more than half are private enterprises in the Mainland. Over the years, listings in the Hong Kong exchange have been dominated by financial and real estate companies while some high-tech companies (e.g. biomedical) are rare. Their share is far behind the New York Stock Exchange and Nasdaq. In terms of valuation, the Hong Kong market has a high level of understanding of the Mainland finance, new energy and consumer goods industries, but not of other industries. In terms of debt financing, the demand for Mainland enterprises is enormous, especially on infrastructure construction projects in the GBA. Bond financing for green projects had a huge demand during the industrial upgrading of the Guangdong-Hong Kong-Macao Greater Bay Area. However, these areas have no uniform green definition.

The import and export shares of the YRD in China's total is on the rise, while those of the PRD is on the decline. Overall, the geographic location of companies exporting through Hong Kong shows an expansion trend from the southeast coast to the inland areas, and the proportion of state-owned enterprises has been declining in recent years. Hong Kong-funded enterprises also gradually moved inland, where the growth of their exports come from.

Based on the current findings on the opportunities and obstacles facing the four new economic cooperation models, the following policy recommendations for building bridges and removing barriers between Hong Kong and the GBA are proposed.

For Model 4, we propose the following policy recommendations : 1) Hong Kong Financial Services Industry must strengthen understanding of the GBA companies, especially private ones, to meet their development status and financing needs: 2) improve the current listing system and encourage high-quality companies from diverse industries to list in Hong Kong; 3) attract talents in all fields; 4) Hong Kong to support the GBA companies in offshore debt financing; 5) Hong Kong needs to communicate and cooperate with the Mainland to clearly define green standards; 6) follow the Hong Kong enterprises to the Mainland and establish more connection poles in the PRD region by using the 'Front Desks, Back Offices' model; and 7) Hong Kong to embrace the demands of Mainland private enterprises.

9.7.6 Concluding Remarks

This SPPR study was awarded on 16 January 2018 and completed on 24 October 2021. Soon after the award of the study, the State Council announced the *Guangdong-Hong Kong-Macao Greater Bay Area (GBA) Outline Development Plan* on 18 February 2018, elevating the coordination of the development between Hong Kong and the Greater Bay Area to the national level. This indicates the need to speed up the development of the GBA and the urgent need to promote new economic cooperation models between Hong Kong and the Greater Bay Area in replace of the old ones which is also the main objective of this Study. Chapter 10 of the *GBA Outline Development Plan* highlights new economic development platforms of Qianhai in Shenzhen, Nansha in Guangzhou, Hengqin in Zhuhai and other areas in the 9 PRD cities in the GBA as cooperation zones to promote such development. All the 9 PRD cities have been very active in getting ready for the new economic cooperation between Hong Kong and the GBA by developing land, infrastructure and innovative policies to promote such cooperation. Unfortunately, Hong Kong was heavily affected by the social unrest in 2019 and then by COVID-19 in 2020 to the present. It was 23 November 2020 that the Guangdong-Hong Kong-Macao Greater Bay Area Development Office was established by the HKSAR government as focal point of liaison between the HKSAR government and the mainland authorities for the development of Hong Kong and the Greater Bay Area. Free flow of people is one of the most important factors in economic development. But, unfortunately, because of COVID-19, free flow of people has been halted by the travel restriction between Hong

Kong and the Mainland. As a result, economic activities and possible economic cooperation have been greatly reduced to a minimum. COVID-19 will eventually be over and travel restriction will be lifted. We hope that the findings on the opportunities and barriers of different economic cooperation models and policy recommendations of this study can help to formulate policies to get ready for better economic cooperation between Hong Kong and the GBA when travel restriction is lifted and economic activities become normal. Opportunities are for those who are prepared and it is better late than never.

End Notes

[1] The Bayh–Dole Act is designed to address intellectual property rights generated by federally funded research in 1980. The key change of Bayh–Dole was that inventors were allowed to fully acquired ownership of the federally funded inventions.

[2] Value-added services include, but are not limited to, HK companies assisting overseas customers in product inspection, which ensures that products meet procurement standards, and monitoring production schedules for on-time delivery. When overseas buyers' orders surge or if they need swift delivery, HK companies can also assist in coordinating production activities.

[3] For example, HK can use its advantages on international trade and finance to first recognise the demand from New Zealand, and then communicate with the Mainland manufacturers to discuss their capability of production to meet such demand. In playing a 'matchmaking' role, HK can access a new market.

Public Dissemination

In regard to public dissemination activities, although under severe travel restrictions as influenced by the social unrest in Hong Kong in 2019 and the following COVID-19 pandemic since January 2020, the research team has tried their best to disseminate their research findings through both face-to-face meetings and internet channels.

1. Guangdong-Hong Kong-Macao Greater Bay Area Information Net

http://www.dupad.hku.hk/hkprd/greaterbay/

The Guangdong-Hong Kong-Macao Greater Bay Area Information Net is part of the Strategic Public Policy Research (SPPR) on Guangdong-Hong Kong-Macao Greater Bay Area of the University of Hong Kong. This Information Net is based on the Hong Kong-Macao-Pearl River Delta Development Information Net jointly developed by the Task Force on Hong Kong-Pearl River Delta Development of the University of Hong Kong and the Centre for Studies of Hong Kong, Macao and Pearl River Delta of Sun Yat-Sen University. It primarily serves for two objectives. First, it aims to provide a one-stop easy-to-use service for business executives, SMEs, government officials, academics, and anyone in Hong Kong, Macao, the PRD and the world who are looking for information and reference materials on the development of the Guangdong-Hong Kong-Macao Greater Bay Area. Second, it strives to become an information hub for Guangdong-Hong Kong-Macao Greater Bay Area related research materials and database developed by public bodies, universities and research organizations throughout the Guangdong-Hong Kong-Macao Greater Bay Area.

The information portal covers all spatial units of the Guangdong-Hong Kong-Macao Greater Bay Area. The content for each spatial unit is divided into six main sections and each of these sections includes statistic table and hyperlinks into the related content providers. The six main sections are: 1) Statistic Data; 2) Tourism, Transport and Maps; 3) Planning, Infrastructure and Construction; 4) Socio-Economic Development; 5) Government; and 6) Research. This portal can help to search for information that are needed for doing business in Hong Kong, Macao, and the PRD - the Guangdong-Hong Kong-Macao Greater Bay Area.

In addition to that, it as well offers information of other Bay Areas around the world, including the Yangtze River Delta, Tokyo Bay Area, San Francisco Bay Area, New York Bay Area, Greater London, and Randstad Region. The Bibliography of Guangdong-Hong Kong-Macao Greater Bay Area compiled by Professor Anthony G.O. Yeh (PI) is also updated and

uploaded to the website for keeping users up to date with the latest research on the development of the Greater Bay Area.

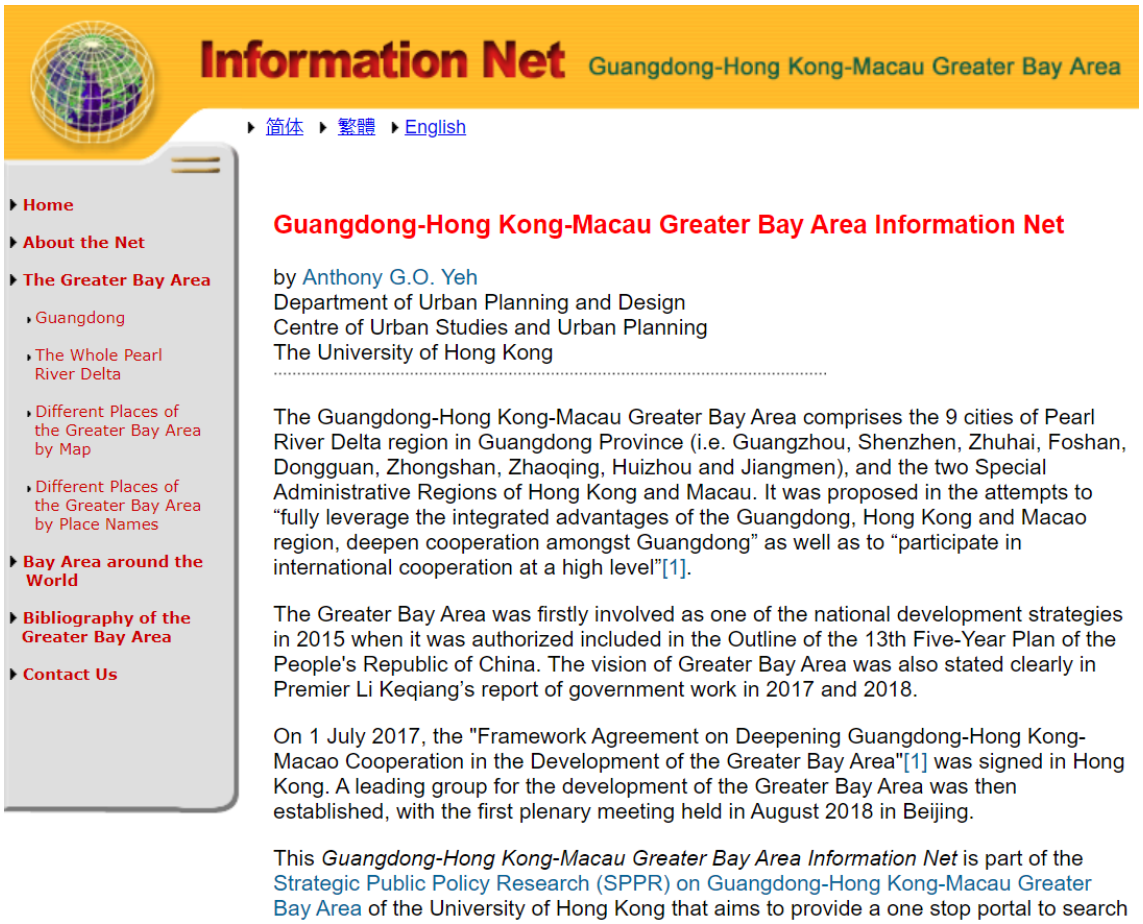


Figure 10.1 Homepage of the Guangdong-Hong Kong-Macao Greater Bay Area Information Net

2. Conference Organized in Guangzhou in December 2019

https://mp.weixin.qq.com/s/mbP9eODCy-zzuqVQ-2QA1w

The *International Summit Forum on Innovation and Regional Development for the Guangdong-Hong Kong-Macao Greater Bay Area* was co-organized by the research team on 13-14 December 2019 under the SPPR funding, partnered with the Guangdong Academy of Sciences and The Institute of Strategic Studies for the Guangdong-Hong Kong-Macao Greater Bay Area. Professor Anthony Yeh (PI) presented a keynote speech titled "Future Development of a Changing Guangdong-Hong Kong-Macao Greater Bay Area", to report and discuss tentative research findings of the Strategic Public Policy Research. Professor Anthony Yeh (PI) delivered a keynote speech at the conference and Professor George Lin (Co-I) hosted the round table discussion on new trends and policy responses on the Guangdong-Hong Kong-Macao Greater Bay Area development. Scholars, urban planners, government officials, and business

leaders from Mainland China, Hong Kong, Macau, USA, Canada, among others, attended the conference. The issues raised and discussed at the conference have been widely published in Chinese news media.



Figure 10.2 Group photo of the keynote speakers of the International Summit Forum on Innovation and Regional Development for the Guangdong-Hong Kong-Macao Greater Bay Area

3. Webinar Organized in Hong Kong in July 2021

http://www.dupad.hku.hk/cusup/gba/webinar/index.html#top

The research group organized a Webinar in Hong Kong under the SPPR funding on 9 July 2021 to present and discuss the major findings of the Strategic Public Policy Research. The Webinar provided a platform for industry and business leaders and professionals to share their insights on new economic cooperation models between Hong Kong and the Greater Bay Area and policies needed for promoting such development. Group members present 4 keynote speeches to disseminate research findings and policy implications. 9 distinguished panelists, from the Law Society of Hong Kong, Li & Fung Development (China) Ltd, British Chamber of Commerce, Hong Kong Trade Development Council, Hong Kong Institute of Planners, Chartered Institute of Logistics, Hong Kong General Chamber of Commerce, Hong Kong Academy of Sciences, and Federation of Hong Kong Industries, attended the panel discussion

and gave constructive insights to our research achievements and policy implications. There were over 270 participants form universities, government departments, businesses, and professional institutes across Hong Kong, Mainland China, Singapore, USA, UK, and Australia joining the Webinar.



Figure 10.3 Panel discussion on research findings and policy recommendations

4. Other Activities

4.1 Conferences Attended

1) HKUST IPP Seminar: Guangdong-Hong Kong-Macao Bay Area Development - A 'One Country, Two Systems' Perspective, organized by Hong Kong University of Science and Technology, Hong Kong, 11 May 2018 **(Y. Xie)**
2) *One Belt One Road and the Greater Bay Area: Economic Development Seminar*, organized by Chinese University of Hong Kong, Hong Kong, 25-27 July 2018 **(J. Li)**
3) Keynote speech on "The Planning and Development of the Guangdong-Hong Kong- Macau Greater Bay Area in Future Social and Economic Environment (未來社會經濟環境下的粵港澳大灣區發展與規劃)", *China Production Park International Summit and Guangdong-Hong Kong-Macau Greater Bay Area Industrial Development Forum*, organized by Shenzhen Nanshan Science/Production Park Development Association and the Shenzhen Government, Nanshan Science Park International Convention Centre Shenzhen, 15 November 2018 (**A.G.O. Yeh**)
4) Panelist speech titled "From a colonial outpost to a Chinese world city: Reluctant transformation of Hong Kong." *2018 Shanghai International Public Relations Summit*, Shanghai, China. 25 November 2018. (**G.C.S. Lin**)
5) Panelist speech titled "Towards a new borderland of innovation? Cross-border technological innovation in the Guangdong-Hong Kong-Macao Greater Bay Area." *Global Mayors' Forum*, Guangzhou Urban Planning Association, Guangzhou. 7 December 2018. (**G.C.S. Lin**)
6) Consultative Meeting of the Qianhai Shenzhen-Hong Kong Modern Service Industry Cooperation Zone (前海蛇口自貿片區暨前海深港合作區諮委會), Shenzhen Kylin Villa, 14 December 2018

(**A.G.O. Yeh**)

7) Plenary speech titled "Seeing and planning the new Chinese city: Competing discourses, hybrid settlement transition, and challenges for urban studies and planning." *2018 International Conference on Future Urban Development*, Peking University Shenzhen Graduate School, Shenzhen, China. 15 December 2018 (**G.C.S. Lin**)

8) Keynote speech on analytical issues in measuring polycentric urban development in China at the workshop on *Conceptualising, Identifying and Analysing Polycentric Urban Regions*, Delft, the Netherlands. 29 January 2019. (**X. Liu**)

9) Public lecture titled "Remaking urban landscape in China's large cities: State-society nexus and the reproduction of space amidst accelerate urbanization". University of California at Berkeley, Center for Chinese Studies. Berkeley, CA, USA. 4 March 2019. (**G.C.S. Lin**)

10) Plenary speech titled "Advancing Urbanism Inside-out: State, Society and Space in the Making of Chinese Urban Landscape." *International Conference on Comparative Urbanism: Global Perspectives*, Urban Studies Institute, Georgia State University, Atlanta, Georgia, USA. 8 March 2019. (**G.C.S. Lin**)

11) Plenary speech titled "China's urban renewals: Experiences and Prospects." *Public Forum on China's Urban Renewals: Development and Transformation*, Peking University-Lincoln Institute of Land Policy Center for Urban Development and Land Policy, Beijing, China, 20 April 2019. (**G.C.S. Lin**)

12) Panelist speech titled "Innovation for the Guangdong-Hong Kong-Macao Greater Bay Area: Challenges and Opportunities", *Inaugural Meeting for the Guangdong-Hong Kong-Macao Greater Bay Area Strategic Research Institute*. Guangzhou,Guangdong Academy of Sciences, China, 18 May 2019. (**G.C.S. Lin**)

13) Panelist speech titled "Comparison of Technological Innovation between China and the United States", Book *Launch for China-US Innovation Competitiveness Evaluation Report*, Institute for Global Innovation & Development, East China Normal University, Shanghai, China, 19 May 2019. (**G.C.S. Lin**)

14) Plenary speech titled "Managing redevelopment of Chinese cities: State-society relations and the reproduction of space amidst accelerate urbanization." *International Conference on Urban and Public Policy*, Zhengzhou, China, 15 June 2019. (**G.C.S. Lin**)

15) Plenary speech titled "Managing Cross-jurisdictional cooperation and regional economic development for the Yangtze River Delta: Challenges and Opportunities", *Summit Forum on Economic Integration of the Yangtze River Delta: Theories and Practices*, Shanghai, China, 18 June 2019. (**G.C.S. Lin**)

16) Keynote Paper, "From Mega-Cities to Mega-City Regions in China in a New Wave of Urbanization and Economic Transition: Issues and Challenges", *2019 International Conference on China Urban Development - Paradigm Shift to Quality Urbanization*, organized by Peking University—Lincoln Institute Center for Urban Development and Land Policy, College of Urban & Environmental Sciences, Peking University, Department of Urban Planning and Design, The University of Hong Kong, The Regional Science Association of China, Yingjie Exchange Center, Peking University, Beijing, China , 28-29 June 2019. (**A.G.O. Yeh**)

17) Public Lecture titled "Advancing Chinese Urbanism inside out: State, Society and the (Re)Making of China's Urban Landscape." *The Confucius Institute*, State University of New York at Albany, Albany, New York, USA, 11 October 2019. (**G.C.S. Lin**)

18) Plenary speech titled "Governing the growth and (re)development of cities in the Greater Bay Area: Mega-events, urban renewals, and a new model of Chinese urbanism", *International Conference on Social Policy and Governance Innovation*, Lingnan University, Hong Kong, 22 October 2019. (**G.C.S. Lin**)

19) Plenary speech titled "Sustaining the growth and (re)development of Chinese cities: State, Society

and Space in China's Great Urban Transformation." *International Conference on Geoinformatics in Sustainable Ecosystem and Society*, Guangdong Academy of Sciences, Guangzhou, China, 22 November 2019. (**G.C.S. Lin**)

20) Plenary speech titled "Future Development of a Changing Guangdong-Hong Kong-Macao Greater Bay Area", *International Summit Forum: Innovation, Creativity, and Regional Development for the Guangdong-Hong Kong-Macao Greater Bay Area*. Guangzhou Institute of Geography, Guangzhou, China,14 December 2019. (**A.G.O. Yeh**)

21) Public Lecture titled "Seeing and Planning the Cities of the Greater Bay Area: Extended Urbanism, Intensive (Re)Development, and Regional Innovation", *School of Graduate Studies, Lingnan University*, Hong Kong, 15 September 2020. (**G.C.S. Lin**)

22) Invited Webinar titled "Muddling on Pathways of Urban Redevelopment: Same Rhyme, Different Tones, and Diverse Trajectories of Chinese Urbanism", *MIT/Harvard/BU Urban China Zoom Series*, MIT Sustainable Urbanization Lab, 1 February 2021. (**G.C.S. Lin**)

23) Plenary speech titled "Regional Synergetic Development of the Guangdong-Hong Kong-Macao Greater Bay Area", *Huangpu River Innovation Forum: Regional (Urban) Forum*. Ministry of Science and Technology, Shanghai Municipal Government, Shanghai, China, 2 June 2021. (**A.G.O. Yeh**)

4.2 Media Coverages

1) HKU Professor Anthony Yeh responds to Outline Development Plan for the Greater Bay Area，The University of Hong Kong, 19 February 2019 **(A.G.O. Yeh)** (https://www.hku.hk/press/news_detail_19119.html)

2) Wenweipo (葉嘉安倡「前枱後室」拓大灣區) **(A.G.O. Yeh)** (http://paper.wenweipo.com/2019/02/20/FS1902200004.htm

3) Yahoo (港大教授葉嘉安倡香港利用資訊通訊科技 加強人流聯繫) **(A.G.O. Yeh)** (https://hk.news.yahoo.com/%E6%B8%AF%E5%A4%A7%E6%95%99%E6%8E%88%E8%91%89%E5%98%89%E5%AE%89%E5%80%A1%E9%A6%99%E6%B8%AF%E5%88%A9%E7%94%A8%E8%B3%87%E8%A8%8A%E9%80%9A%E8%A8%8A%E7%A7%91%E6%8A%80-%E5%8A%A0%E5%BC%B7%E4%BA%BA%E6%B5%81%E8%81%AF%E7%B9%AB-062213243.html)

4) Mingpao (【大灣區】港大學者倡推區內交通卡口岸增設專用「商貿通道」) **(A.G.O. Yeh)** (https://news.mingpao.com/ins/%E6%B8%AF%E8%81%9E/article/20190219/s00001/1550553469828/%E3%80%90%E5%A4%A7%E7%81%A3%E5%8D%80%E3%80%91%E6%B8%AF%E5%A4%A7%E5%AD%B8%E8%80%85%E5%80%A1%E6%8E%A8%E5%8D%80%E5%85%A7%E4%BA%A4%E9%80%9A%E5%8D%A1-%E5%8F%A3%E5%B2%B8%E5%A2%9E%E8%A8%AD%E5%B0%88%E7%94%A8%E3%80%8C%E5%95%86%E8%B2%BF%E9%80%9A%E9%81%93%E3%80%8D)

5) 灼見名家 (葉嘉安：大灣區是香港發展最後機會) **(A.G.O. Yeh)** (https://www.master-insight.com/%E8%91%89%E5%98%89%E5%AE%89%EF%BC%9A%E5%A4%A7%E7%81%A3%E5%8D%80%E6%98%AF%E9%A6%99%E6%B8%AF%E7%99%BC%E5%B1%95%E6%9C%80%E5%BE%8C%E6%A9%9F%E6%9C%83/)

6) Ta Kung Pao (葉嘉安倡灣區電子簽證「通關易」) **(A.G.O. Yeh)** (http://www.takungpao.com.hk/news/232109/2019/0220/249453.html)

7) Ta Kung Pao （葉嘉安：大灣區應轉型知識型經濟）**(A.G.O. Yeh)** (http://www.takungpao.com.hk/hongkong/text/2019/0219/249355.html)

8) Sing Tao (【大灣區規劃】倡推「前枱後室」接單模式 葉嘉安：不能活在過去式) **(A.G.O. Yeh)**

(https://www.singtaousa.com/%E5%8D%B3%E6%99%82/%E5%8D%B3%E6%99%82%E6%B8%AF%E8%81%9E/2188154-%E3%80%90%E5%A4%A7%E7%81%A3%E5%8D%80%E8%A6%8F%E5%8A%83%E3%80%91%E5%80%A1%E6%8E%A8%E3%80%8C%E5%89%8D%E6%9E%B1%E5%BE%8C%E5%AE%A4%E3%80%8D%E6%8E%A5%E5%96%AE%E6%A8%A1%E5%BC%8F-%E8%91%89%E5%98%89%E5%AE%89/)

9) Hong Kong Standard, "Transport ideas pushed amid worries HK will lose shine" **(A.G.O. Yeh)** (http://www.thestandard.com.hk/section-news.php?id=205160)

10) Nanfang Daily (葉嘉安：深港可探索從"前店後廠"轉到"前臺後室"的新經濟合作模式) **(A.G.O. Yeh)** (http://qh.sz.gov.cn/sygnan/qhzx/zthd_1/2020zwh/wygd/content/post_8339294.html)

11) Sing Tao (施政報告丨學者：港珠缺有組織科研聯繫 內資企業難接觸香港成果) **(A.G.O. Yeh)** (https://std.stheadline.com/realtime/article/1756793/%E5%8D%B3%E6%99%82-%E6%B8%AF%E8%81%9E-%E6%96%BD%E6%94%BF%E5%A0%B1%E5%91%8A-%E5%AD%B8%E8%80%85-%E6%B8%AF%E7%8F%A0%E7%BC%BA%E6%9C%89%E7%B5%84%E7%B9%94%E7%A7%91%E7%A0%94%E8%81%AF%E7%B9%AB-%E5%85%A7%E8%B3%87%E4%BC%81%E6%A5%AD%E9%9B%A3%E6%8E%A5%E8%A7%B8%E9%A6%99%E6%B8%AF%E6%88%90%E6%9E%9C)

12) Guangzhou Academy of Social Sciences (葉嘉安：香港在粵港澳大灣區新經濟結構下的挑戰) **(A.G.O. Yeh)** (http://www.gzass.gd.cn/gzsky/hy_20201127/contents/3493/1122.html)

13) Nanfang Daily (為粵港澳大灣區創新與區域發展出謀獻策) **(A.G.O. Yeh and G.C.S. Lin)** (http://epaper.southcn.com/nfdaily/html/2019-12/16/content_7838554.htm)

14) Southcn.com (粵港澳大灣區創新與區域發展國際論壇舉辦) **(A.G.O. Yeh and G.C.S. Lin)** (http://economy.southcn.com/e/2019-12/16/content_189816568.htm)

15) 廣東新聞聯播 (粵港澳大灣區創新與區域發展論壇在穗舉行) **(A.G.O. Yeh and G.C.S. Lin)** (https://m.itouchtv.cn/article/ff9bab1f3b9d94796ab4b5e3f2a830a4?shareId=N2wMDSRz&from=singlemessage)

16) 香港經濟轉型升級迫在眉睫 (An imminent transformation and upgrading of Hong Kong's economy) **(Z. Tao)** (https://www.fbe.hku.hk/in-the-media/an-imminent-transformation-and-upgrading-of-hong-kong-s-economy)

17) 保險業的發展機遇與戰略 (Hong Kong insurance industry: Opportunities and strategies) **(Z. Tao)** (https://www.fbe.hku.hk/in-the-media/hong-kong-insurance-industry-opportunities-and-strategies)

18) 雙循環下香港青年事業發展新機遇 **(Z. Tao)** (https://www1.hkej.com/dailynews/commentary/article/2710596)

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Glossary of Terms

Beijing-Hong Kong-Macao Highway　京港澳公路

Beijing-Tianjin-Hebei Urban Agglomeration　京津冀城市群

Belt and Road Initiative/One Belt One Road “一帶一路” 戰略

China-United States Trade War　中美貿易戰

China Securities Regulatory Commission　中國證監會

Closer Economic Partnership Arrangement (CEPA) 更緊密經貿關係安排

COVID-19 Pandemic　2019 新型冠狀病毒病

Development Plan for City Clusters in the Guangdong-Hong Kong-Macao Bay Area
粵港澳大灣區城市群發展規劃

Dual Circulation 「雙循環」戰略

Dongguan Songshan Lake Science and Technology Industrial Park
東莞松山湖高科技產業園區

Enterprise Internationalization　企業國際化

Framework Agreement on Deepening Guangdong-Hong Kong-Macao Cooperation in the Development of the Greater Bay Area 2017
《深化粵港澳合作　推進大灣區建設框架協議》 2017

Foreign Direct Investment (FDI) 外商直接投資

Four Pillar Industry　四大支柱行業

Four Pillars and Six Industries　四大支柱及六大產業

Front Desks, Back Offices　前台後室

Front Shops, Back Factories　前店後廠

Global Production Networks (GPNs) 全球生產網絡

Green Bond　綠色債券

Green Financial Centre　綠色金融中心

Growth Enterprise Market (GEM) 創業板市場

‘Going Global’「走出去」戰略

Guangdong-Hong Kong-Macao Greater Bay Area (GBA) 粵港澳大灣區

Guangdong-Hong Kong-Macao Greater Bay Area (GBA) Outline Development Plan
粵港澳大灣區發展規劃綱要

Guangdong-Macao-Hengqin Science and Technology Innovation Poles
粵澳橫琴科技創新極

Guangzhou-Shenzhen-Hong Kong Innovation Corridor　廣深港澳科創走廊

Guangzhou-Zhuhai-Macao Science and Technology Innovation Corridors
廣珠澳科技創新走廊

Hang Seng Industry Classification System　恆生行業分類系統

Hong Kong Stock Exchange　香港聯交所

Hong Kong-Zhuhai-Macau Bridge　港珠澳大橋

IPO Funding　上市融資

Key Research Institute　重點研究所

Made in China 2025 中國製造 2025

Mergers and Acquisitions (M&A) 併購

NASDAQ　納斯達克指數

National Plan on New Urbanization (2014-2020)
國家新型城鎮化規劃(2014-2020 年)

New York Bay Area　紐約灣區

New York Stock Exchange　紐約證券交易所

Offshore Bond Financing Centre　離岸債券融資中心

Offshore Debt Financing　離岸債務融資

Pearl River Delta (PRD) 珠江三角洲

Qianhai Shenzhen-Hong Kong Modern Service Industry Cooperation Zone
前海深港現代服務業合作區

San Francisco Bay Area　三藩市灣區

Shanghai-Hong Kong Stock Connect　滬港通

Shenzhen-Hong Kong Innovation and Technology Co-operation Zone
深港科技創新合作區

Shenzhen-Hong Kong Stock Connect　深港通

The National 12th Five-Year Plan　國家《十二五規劃綱要》

The National 13sth Five-Year Plan　國家《十三五規劃綱要》

The National 14th Five-Year Plan　國家《十四五規劃綱要》

The Outline Development Plan for the Guangdong-Hong Kong-Macao Greater Bay Area 2019《粵港澳大灣區發展規劃綱要》2019

The Outline of the Plan for the Reform and Development of the PRD (2008-2020)　珠江三角洲地區改革發展規劃綱要(2008-2020 年)

The Study on the Action Plan for Livable Bay Area of the Pearl River Estuary
環珠江口宜居灣區建設重點行動計劃

Tokyo Bay Area　東京灣區

U.S. Bayh-Dole Act　美國拜杜法案

University Grants Committee　大學教育資助委員會

Venture Capital Investment　風險投資

World Trade Organization (WTO) 世界貿易組織

Yangtze River Delta (YRD) 長江三角洲

Hong Kong-Shenzhen Innovation & Technology Park/Lok Ma Chau Loop
港深創新科技園/落馬洲河套區